

Shatter Clustering: Seed Percentile Sensitivity Analysis

Sweep over $t_{\text{percentile}}$ with Default $\alpha_{\text{MAD}} = 0.5$, $\alpha_{\text{accept}} = 0.5$, $r = \text{adaptive}$

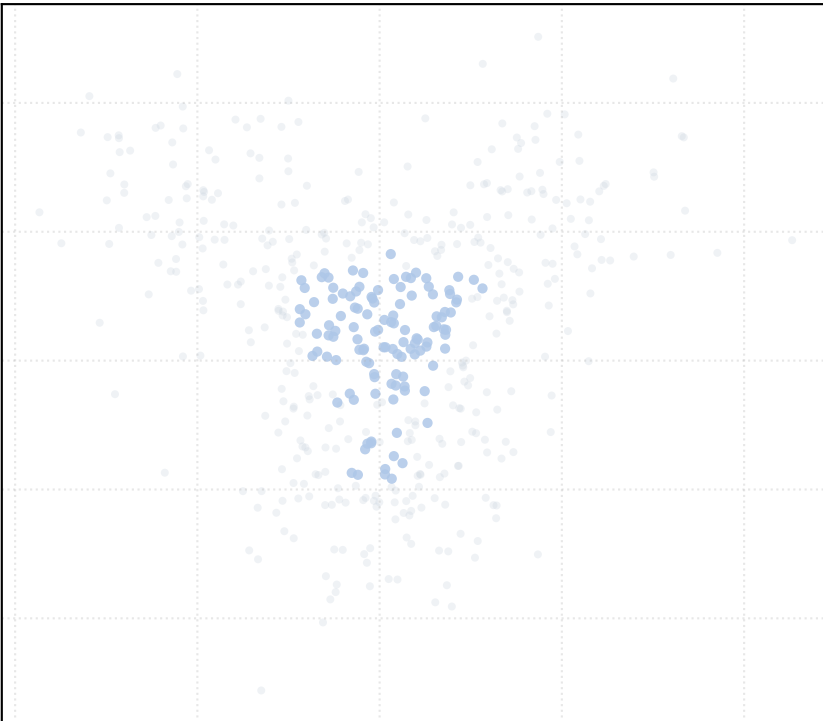
Experimental Configuration

- Total Evaluated Datasets: 30
- Seed Percentiles Tested: [0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5]
- Total Executions: 240 (8 runs per dataset)
- Fixed Default Parameters: $\alpha_{\text{MAD}} = 0.5$, $\alpha_{\text{accept}} = 0.5$, $\text{quant} = 0.15$
- Target Output PDF: `benchmark_report_t_percentile.pdf`

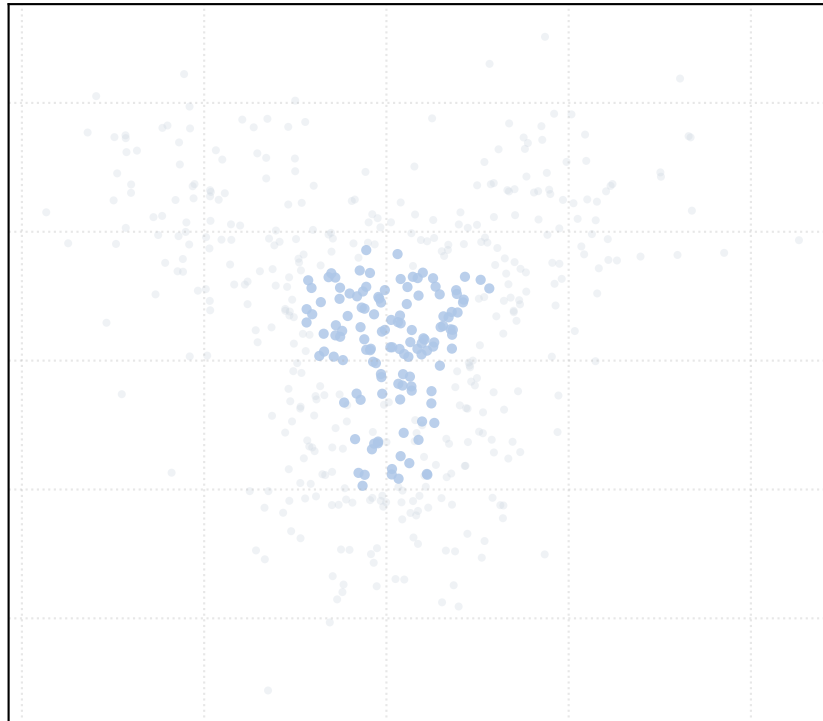
Parameter	Value / Range	Algorithmic Role
t_percentile	{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50}	Top percentile of densest vectors selected as seed centers
alpha_MAD	0.50 (Default)	MAD multiplier for surface compatibility
alpha_accept	0.50 (Default)	Log2 fold-change acceptance cutoff
mean_dist_quant	0.15 (Default)	Quantile for adaptive search radius r
min_intersection	1 (Default)	Overlapping vectors to trigger merge

Dataset: bifurcation_2way_noise_high | Seed Percentile Sweep (t_percentile)

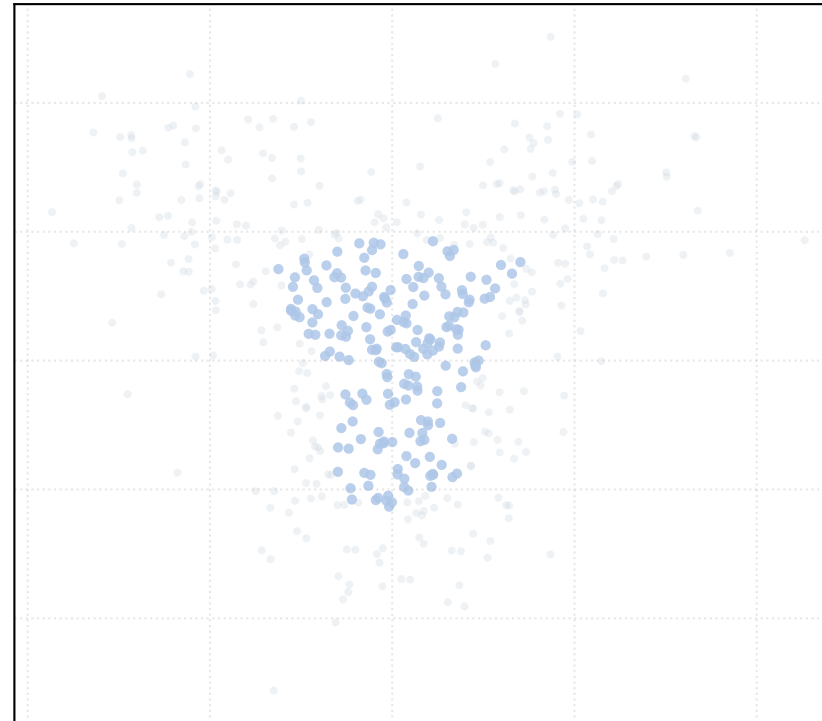
t_percentile = 0.05 [1 Clusters]



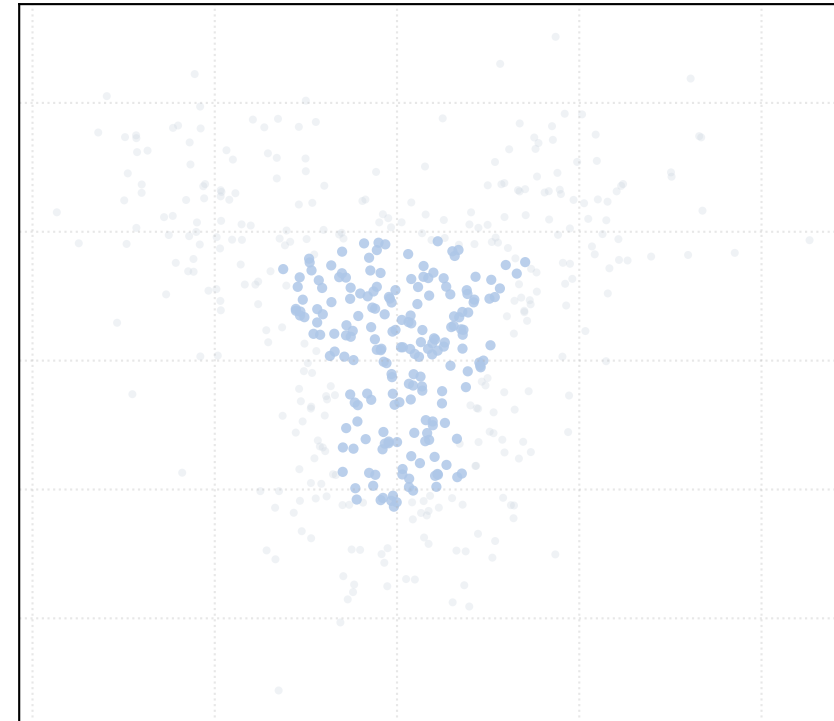
t_percentile = 0.1 [1 Clusters]



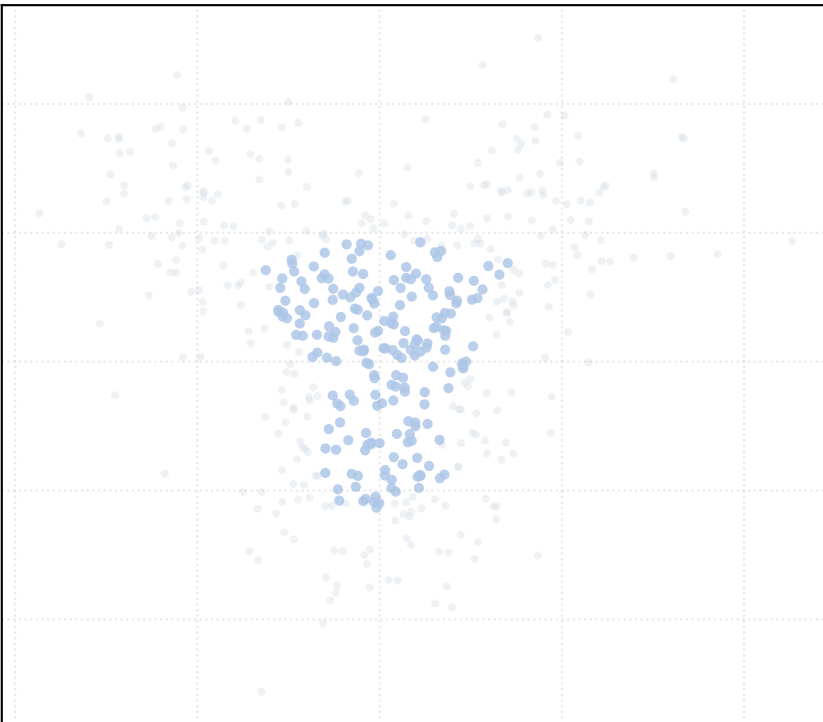
t_percentile = 0.15 (Default) [1 Clusters]



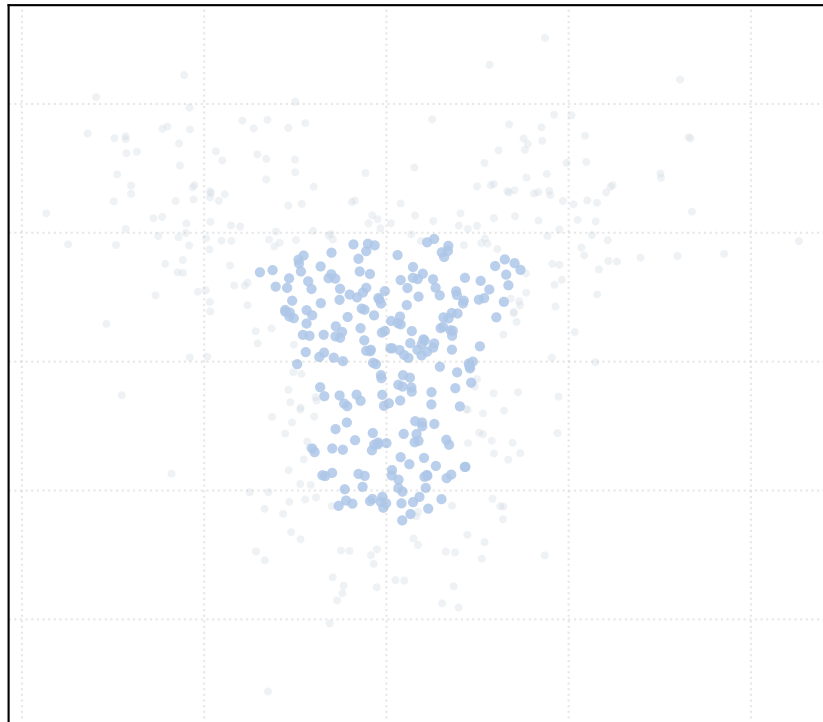
t_percentile = 0.2 [1 Clusters]



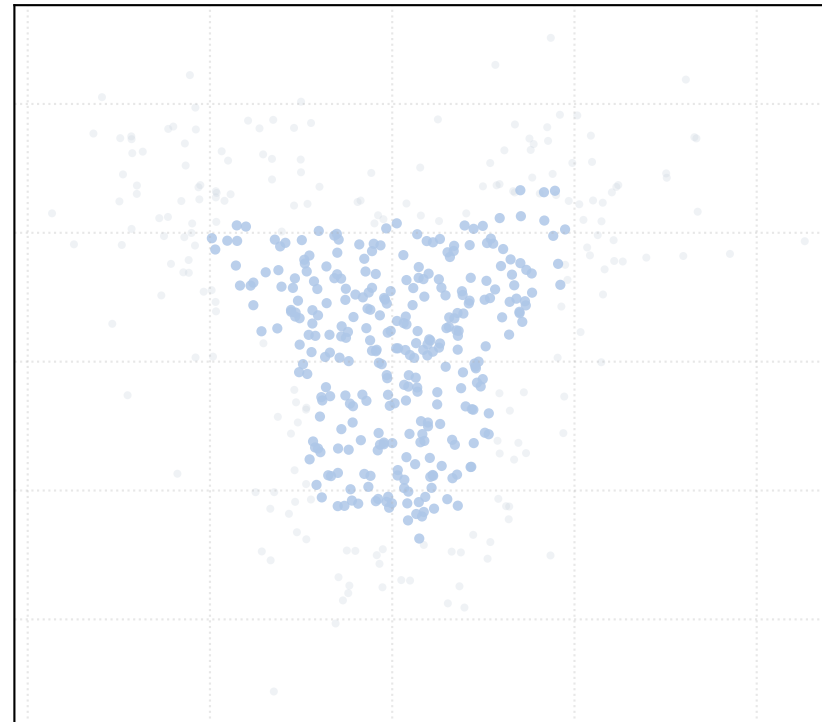
t_percentile = 0.25 [1 Clusters]



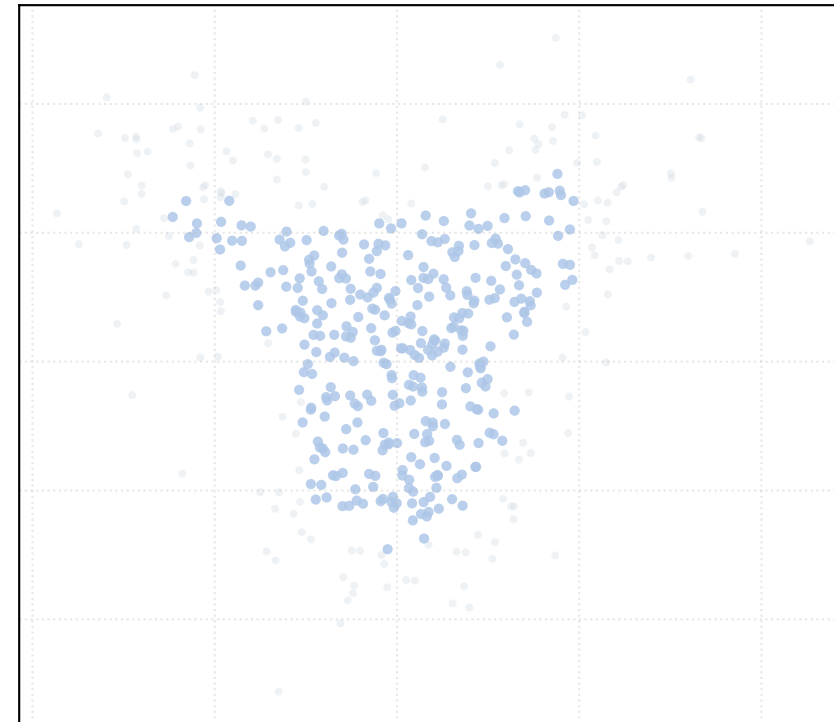
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

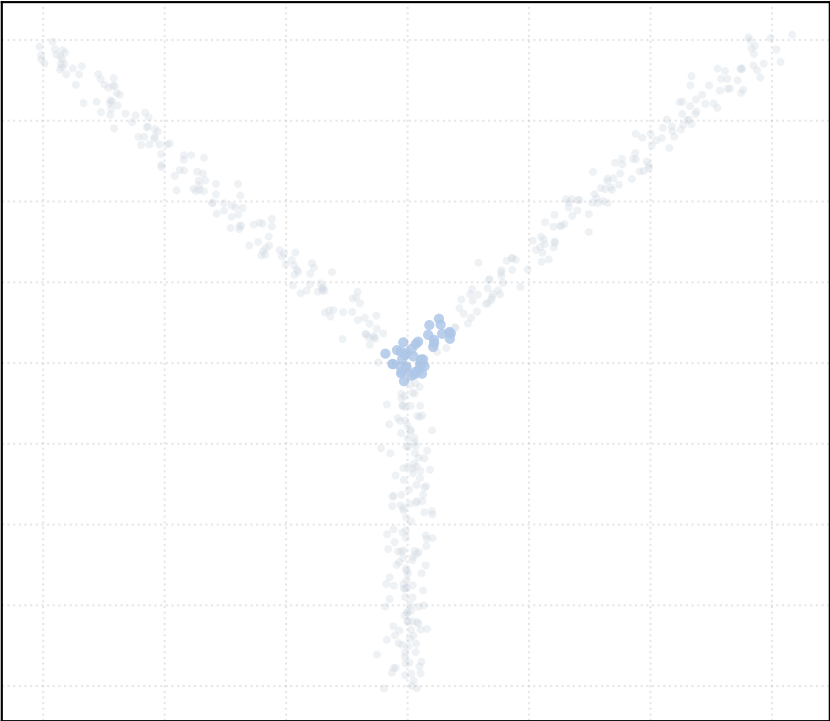


t_percentile = 0.5 [1 Clusters]

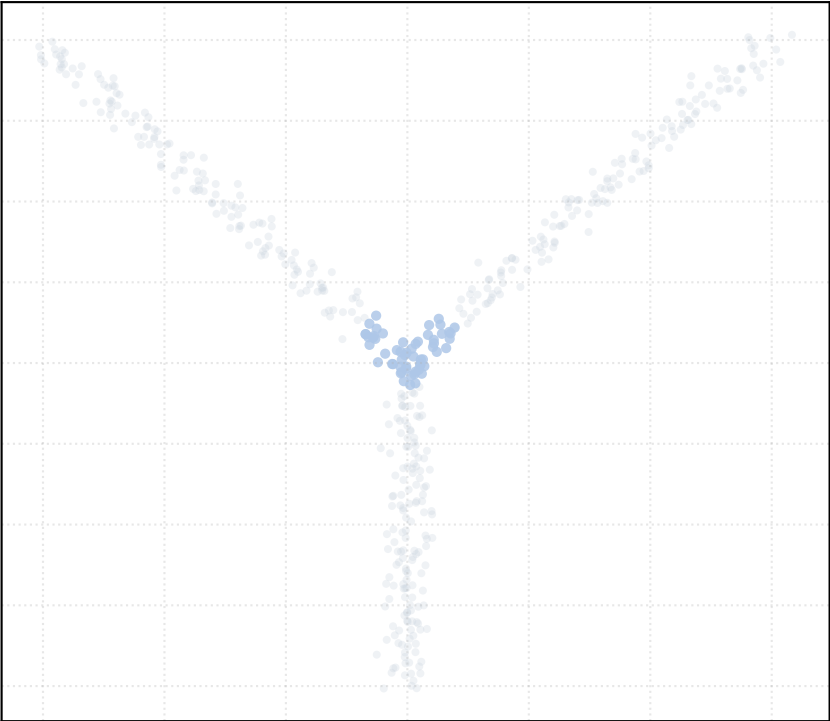


Dataset: bifurcation_2way_noise_low | Seed Percentile Sweep (t_percentile)

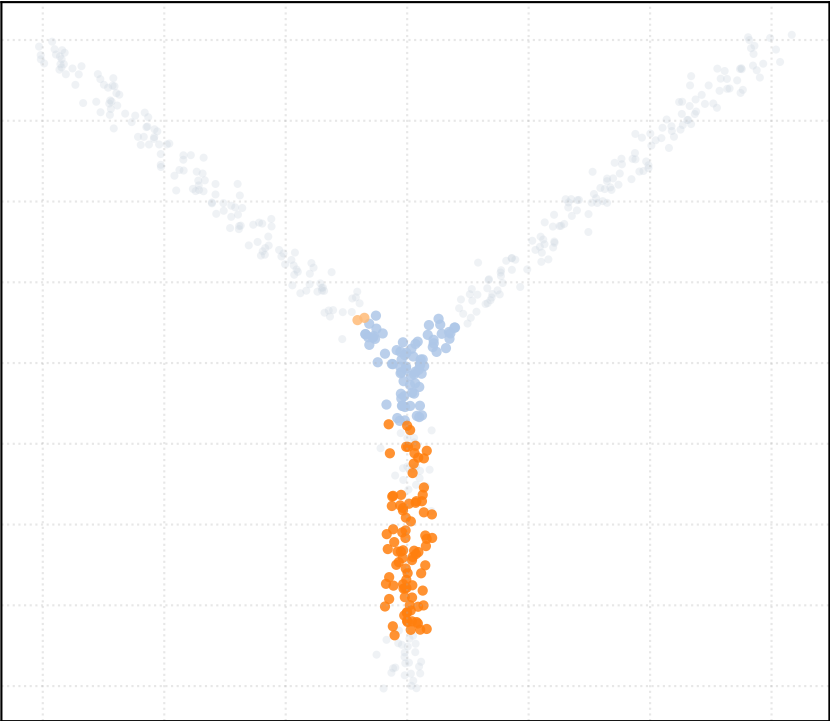
t_percentile = 0.05 [1 Clusters]



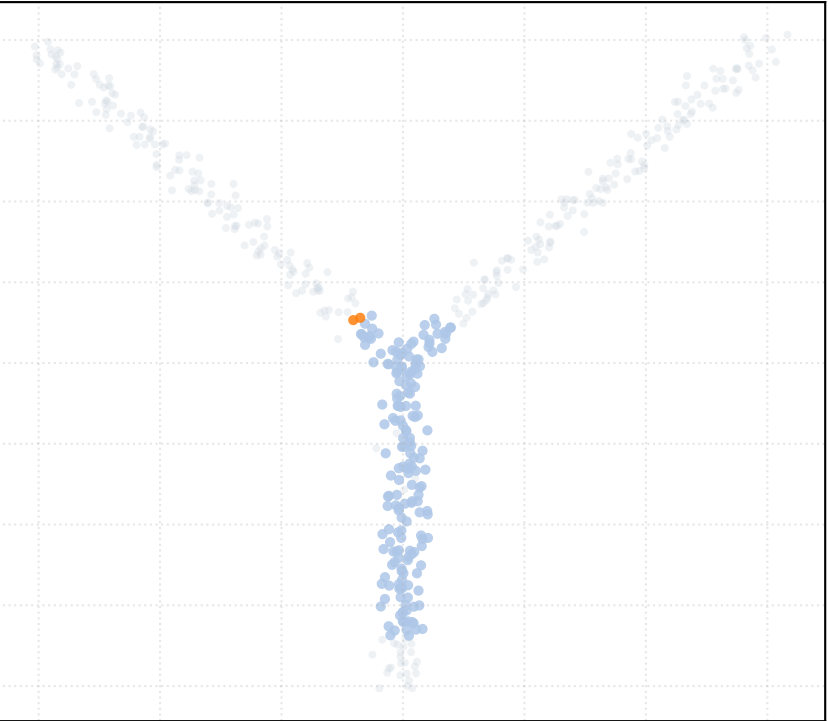
t_percentile = 0.1 [1 Clusters]



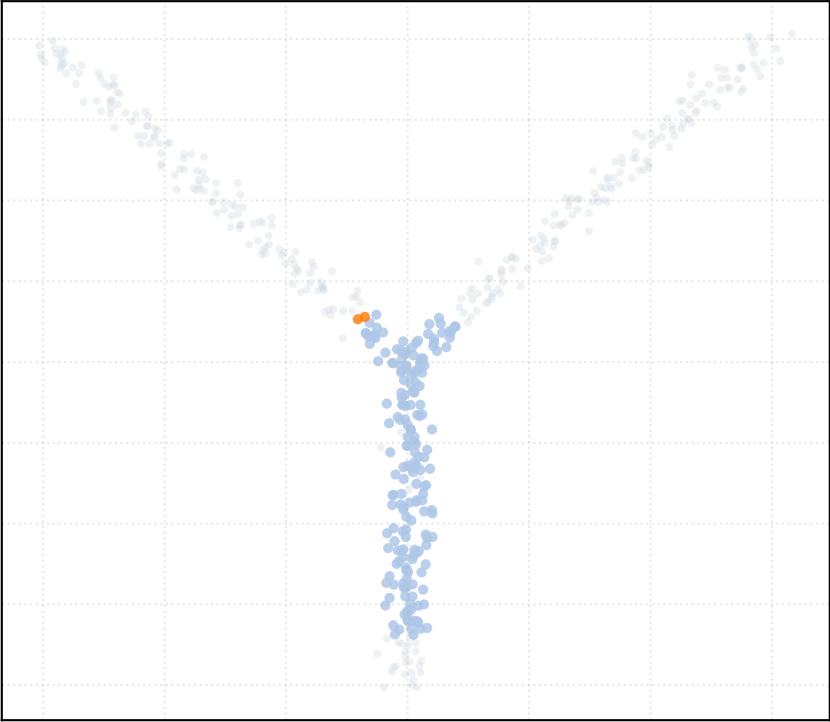
t_percentile = 0.15 (Default) [3 Clusters]



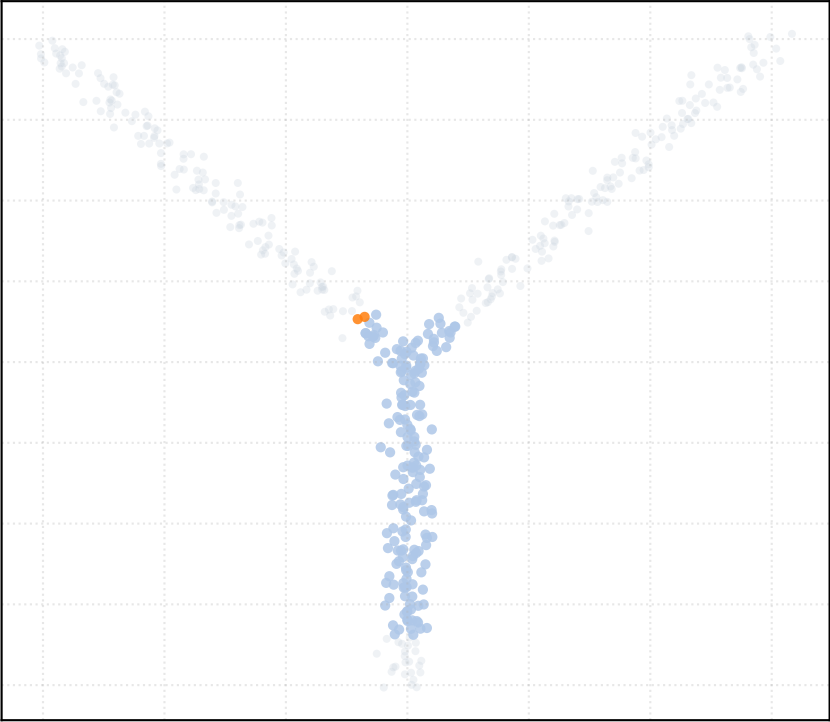
t_percentile = 0.2 [2 Clusters]



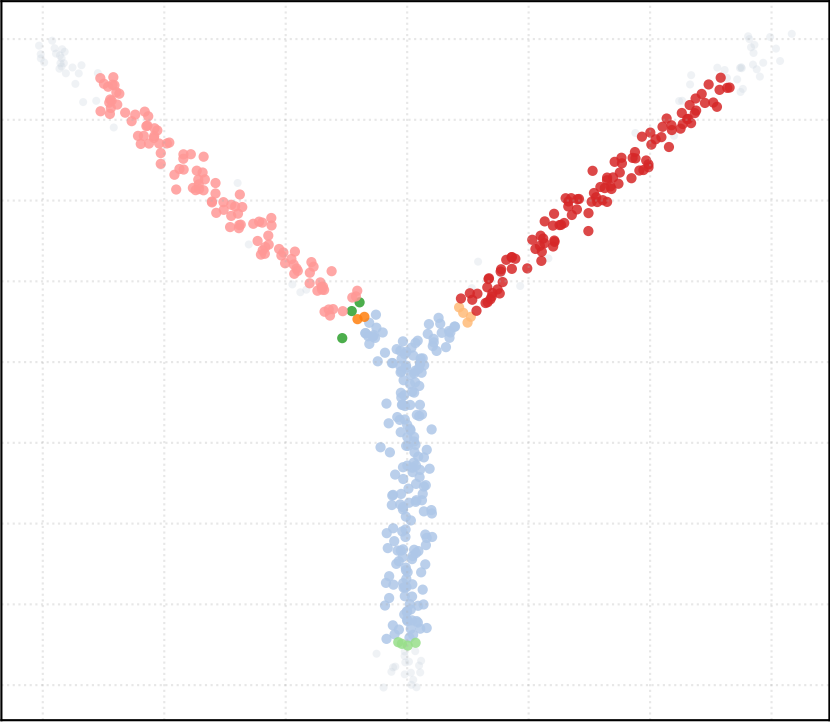
t_percentile = 0.25 [2 Clusters]



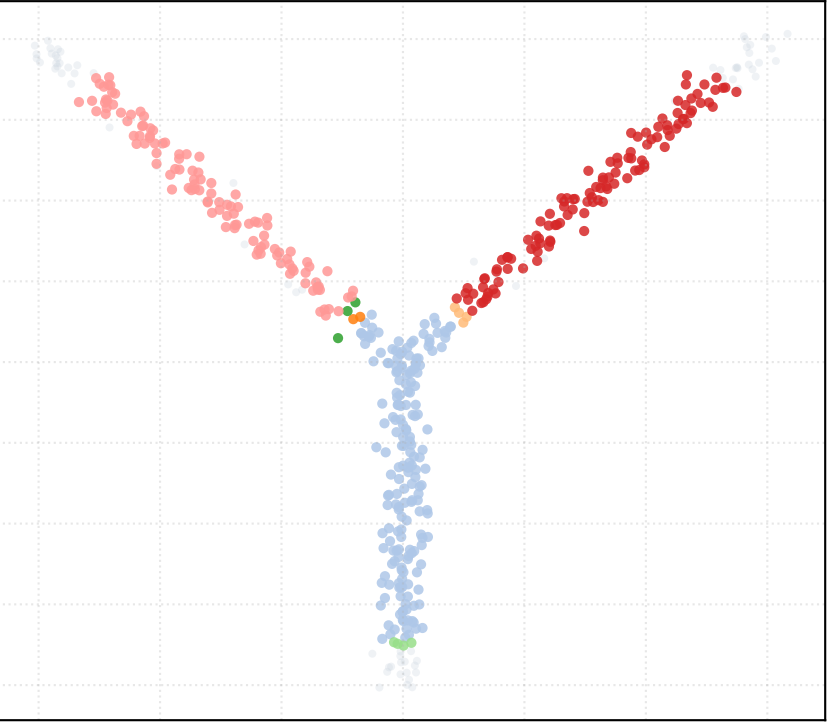
t_percentile = 0.3 [2 Clusters]



t_percentile = 0.4 [7 Clusters]

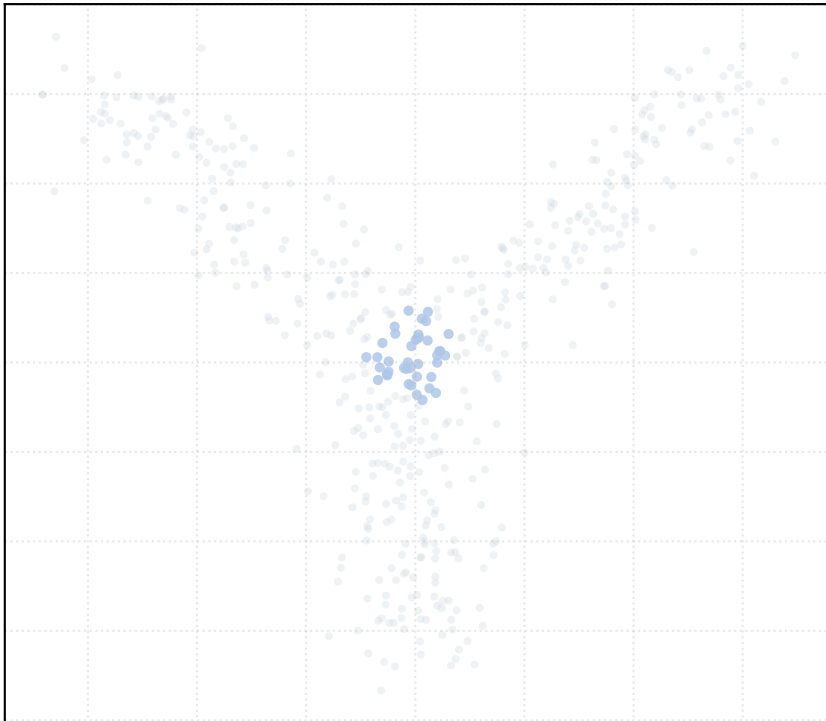


t_percentile = 0.5 [7 Clusters]

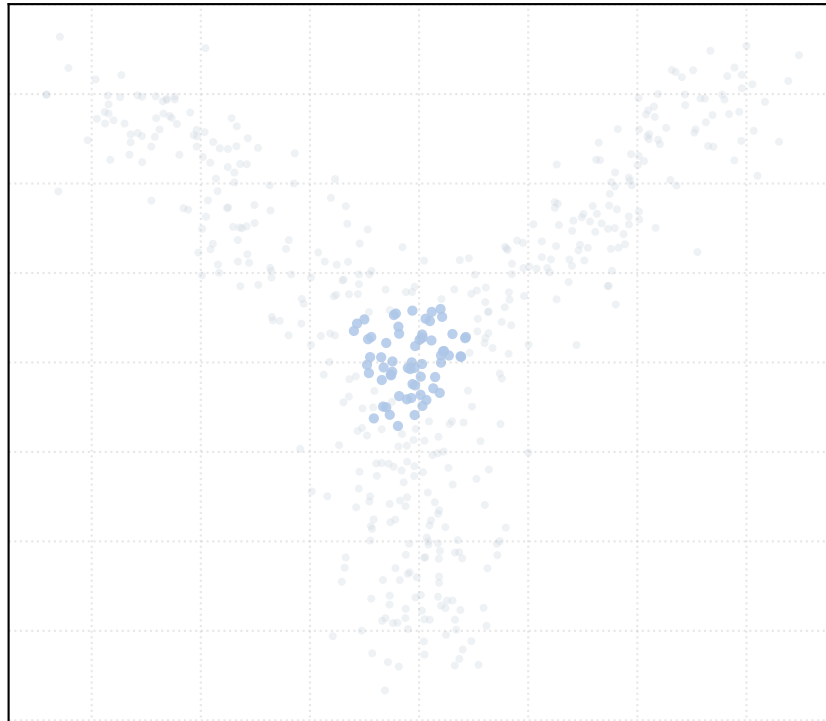


Dataset: bifurcation_2way_noise_medium | Seed Percentile Sweep (t_percentile)

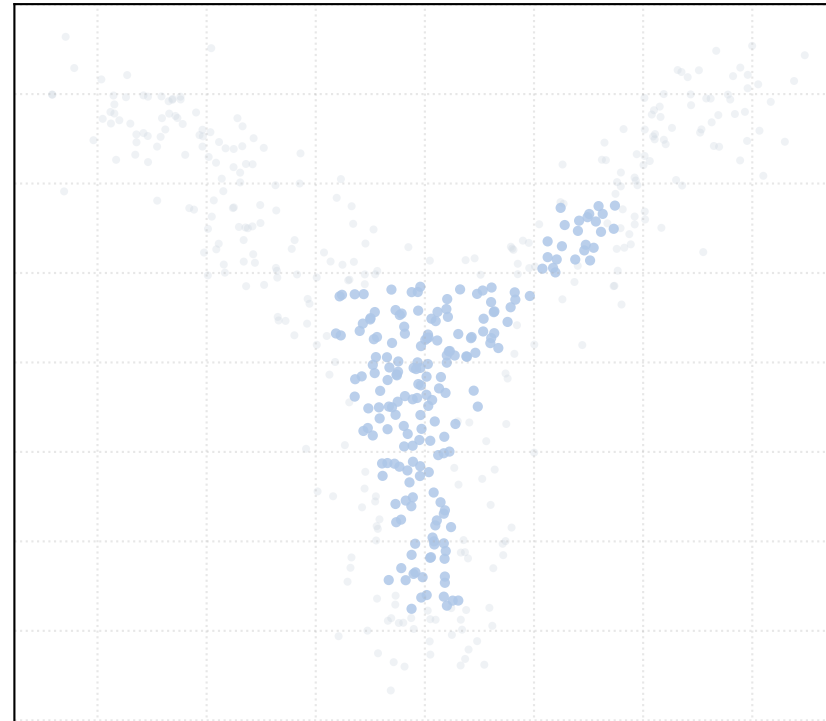
t_percentile = 0.05 [1 Clusters]



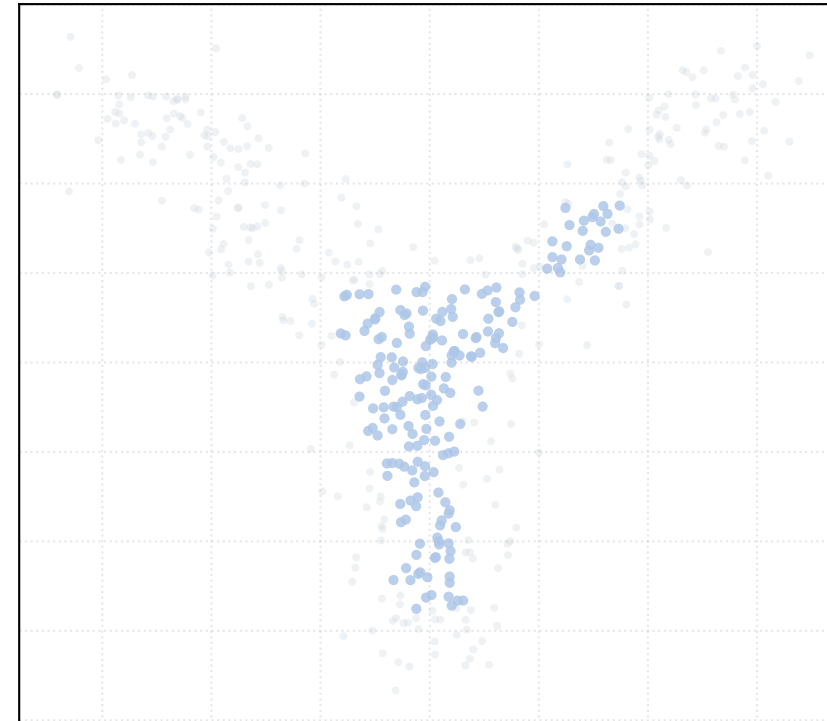
t_percentile = 0.1 [1 Clusters]



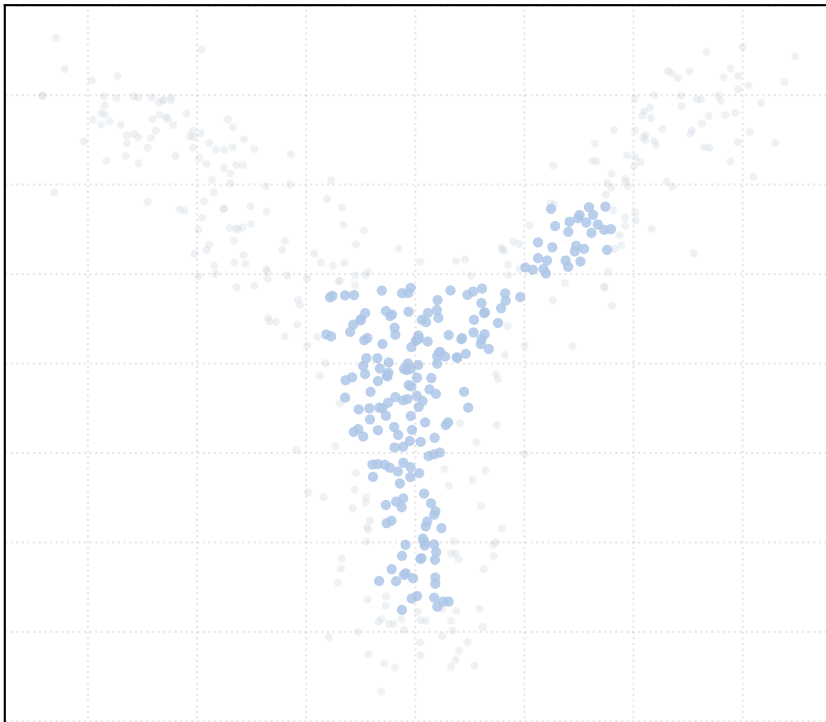
t_percentile = 0.15 (Default) [1 Clusters]



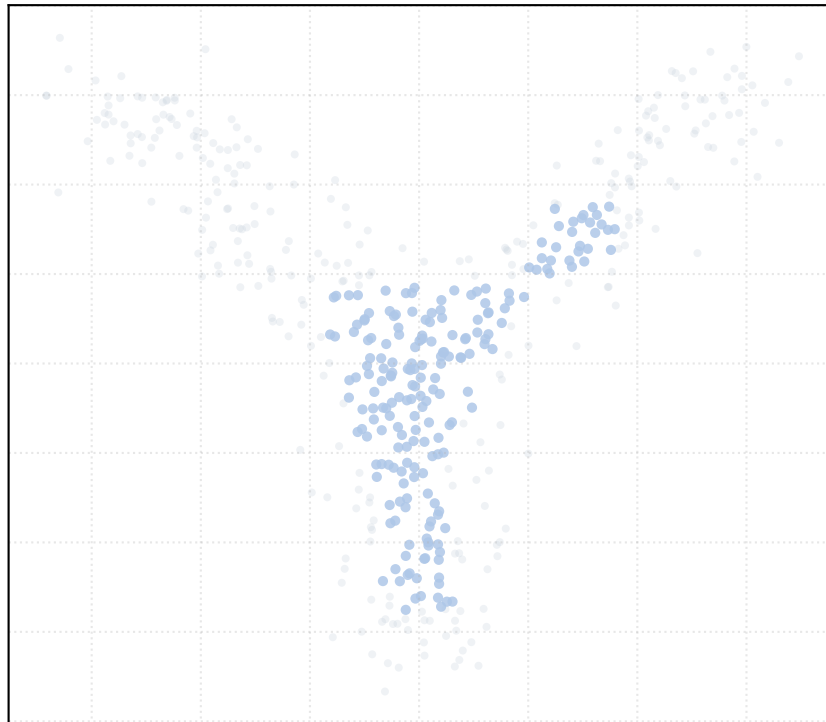
t_percentile = 0.2 [1 Clusters]



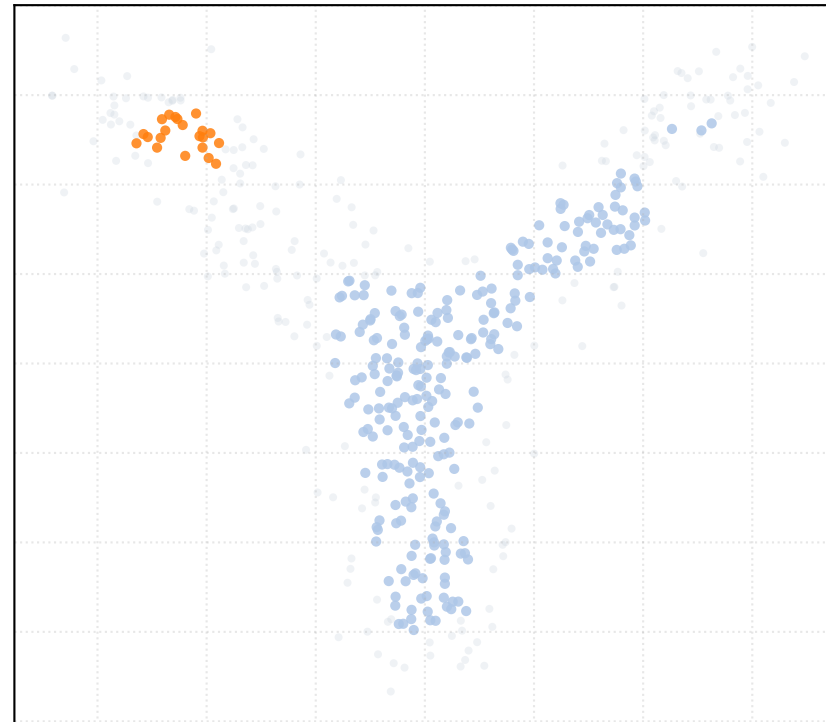
t_percentile = 0.25 [1 Clusters]



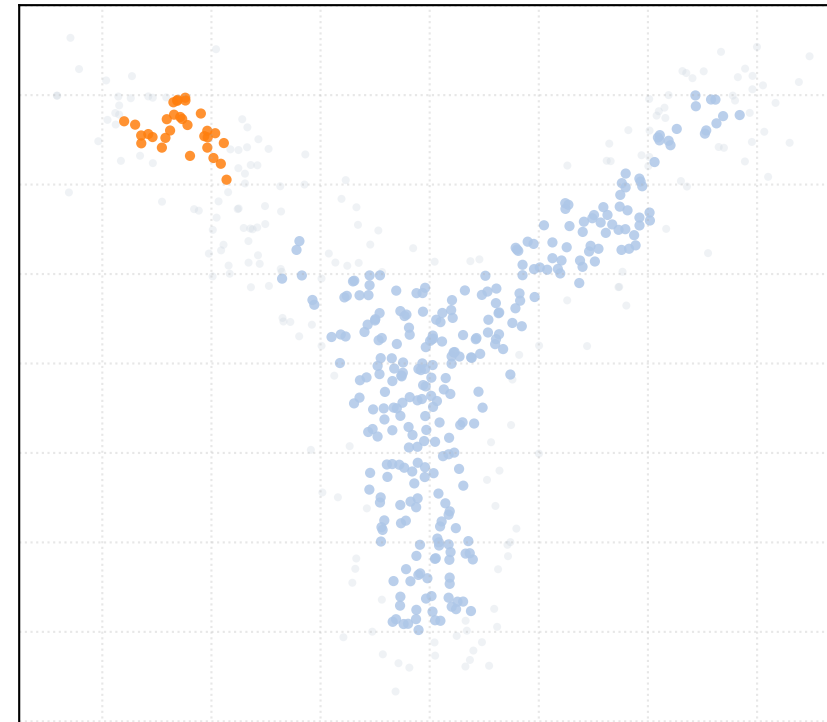
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [2 Clusters]

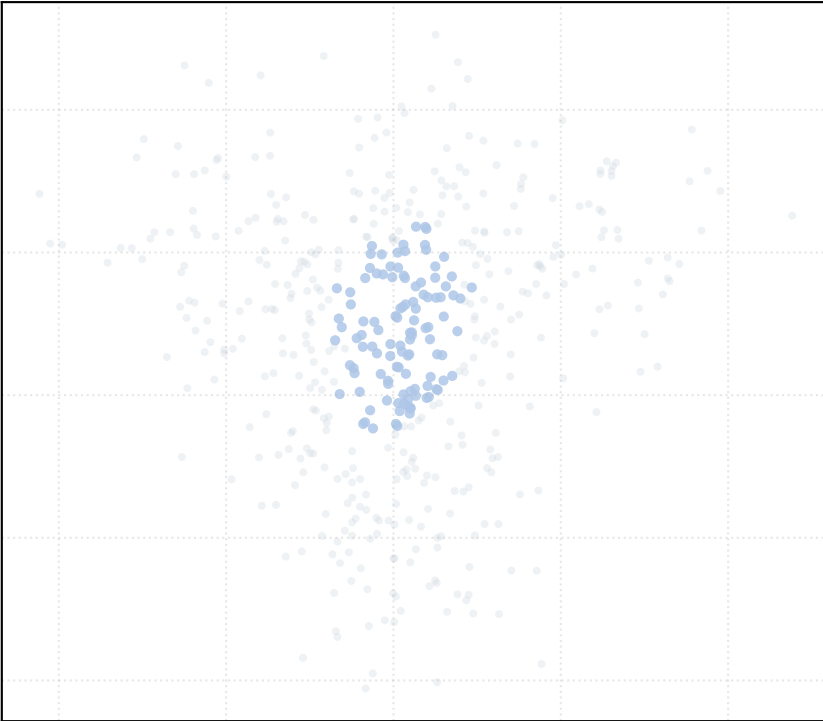


t_percentile = 0.5 [2 Clusters]

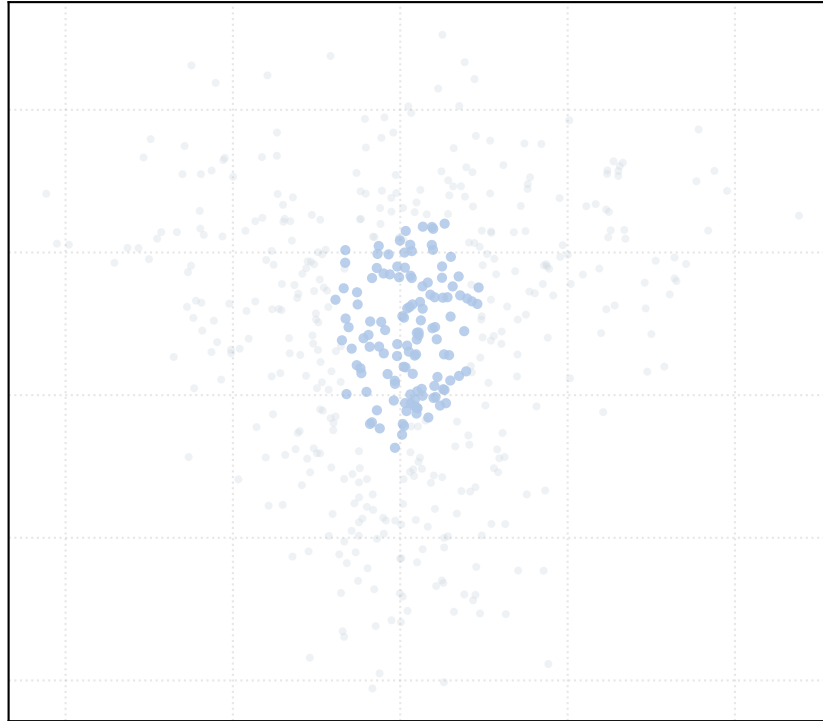


Dataset: bifurcation_3way_noise_high | Seed Percentile Sweep (t_percentile)

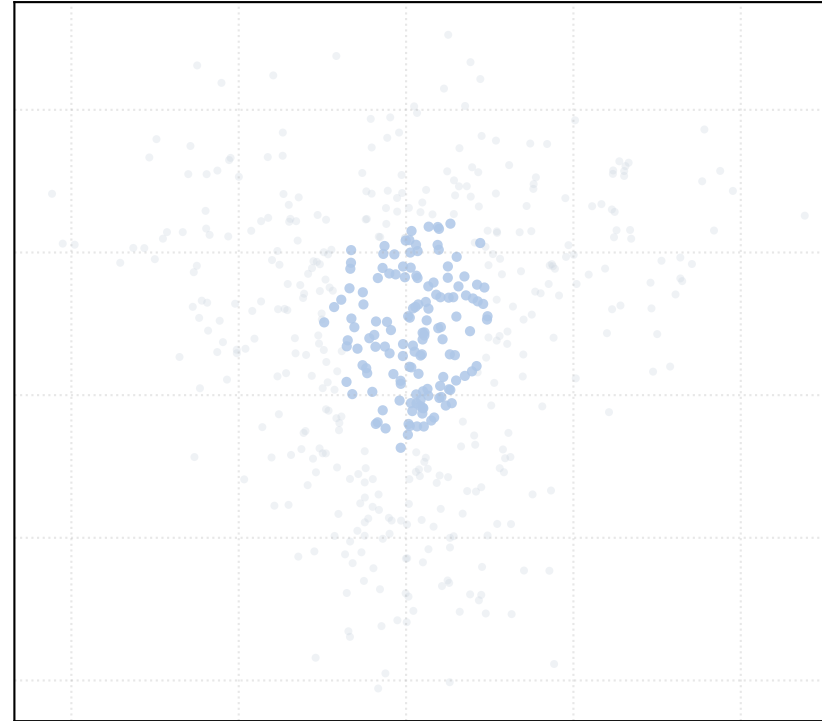
t_percentile = 0.05 [1 Clusters]



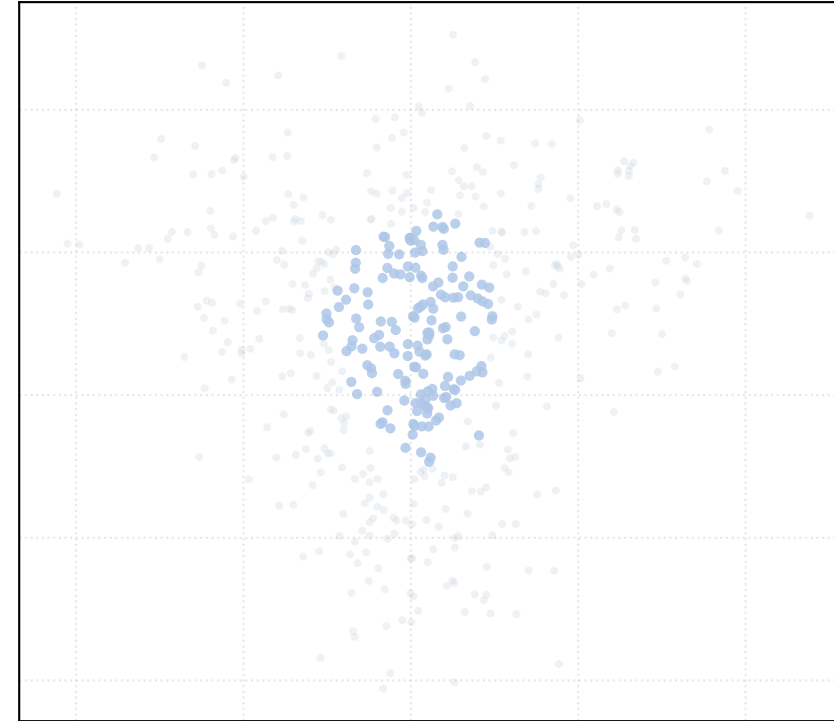
t_percentile = 0.1 [1 Clusters]



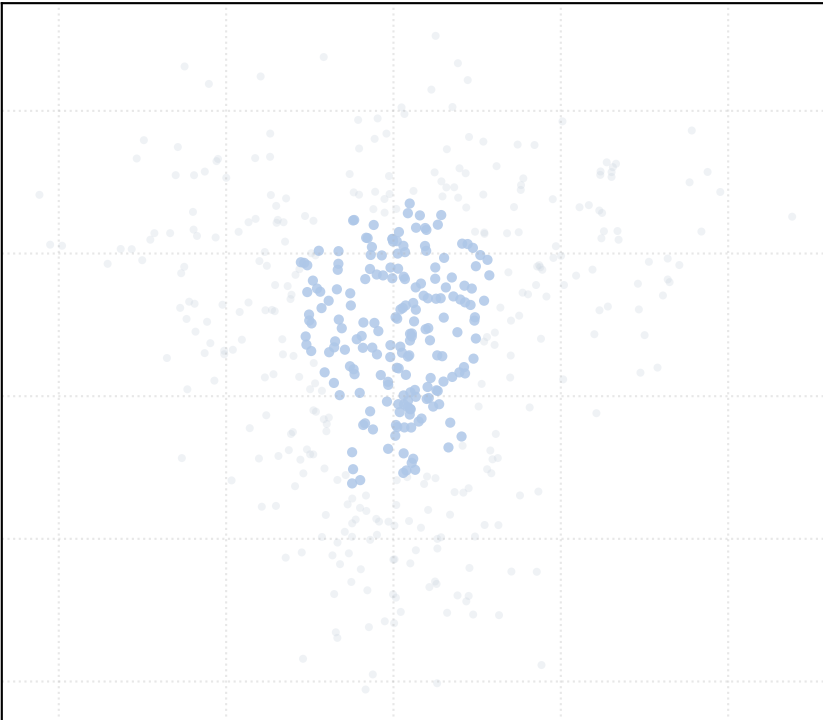
t_percentile = 0.15 (Default) [1 Clusters]



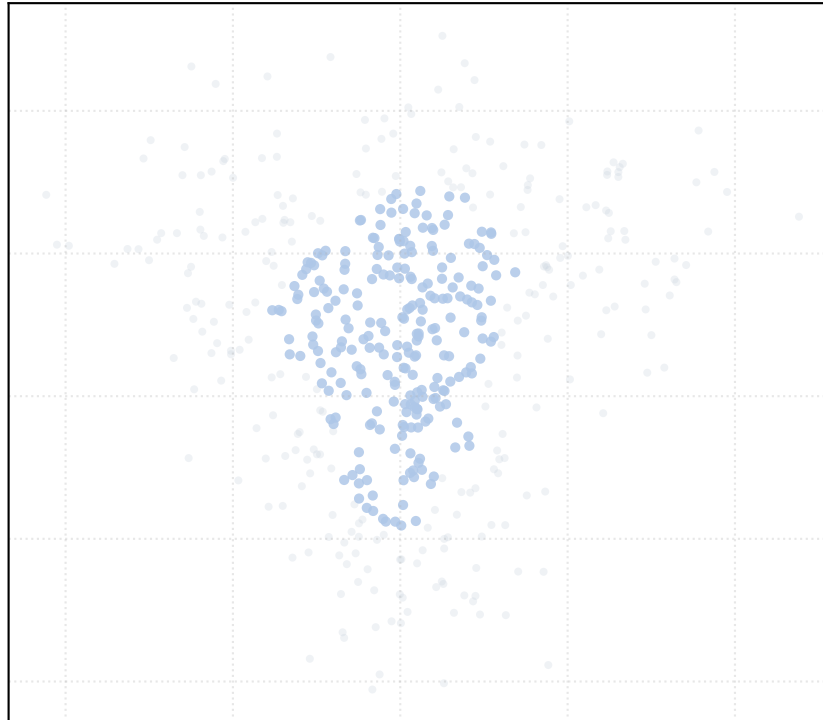
t_percentile = 0.2 [1 Clusters]



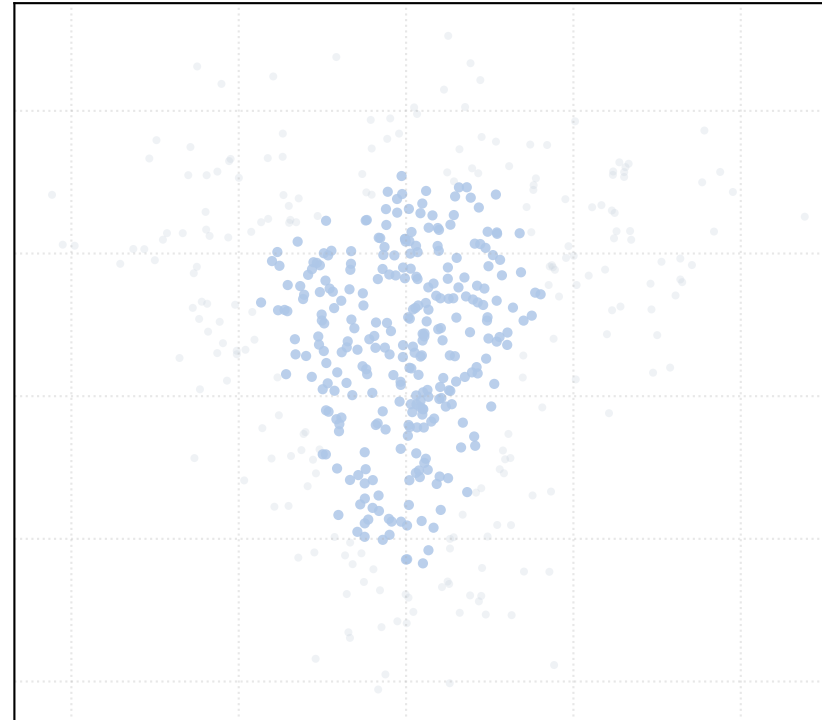
t_percentile = 0.25 [1 Clusters]



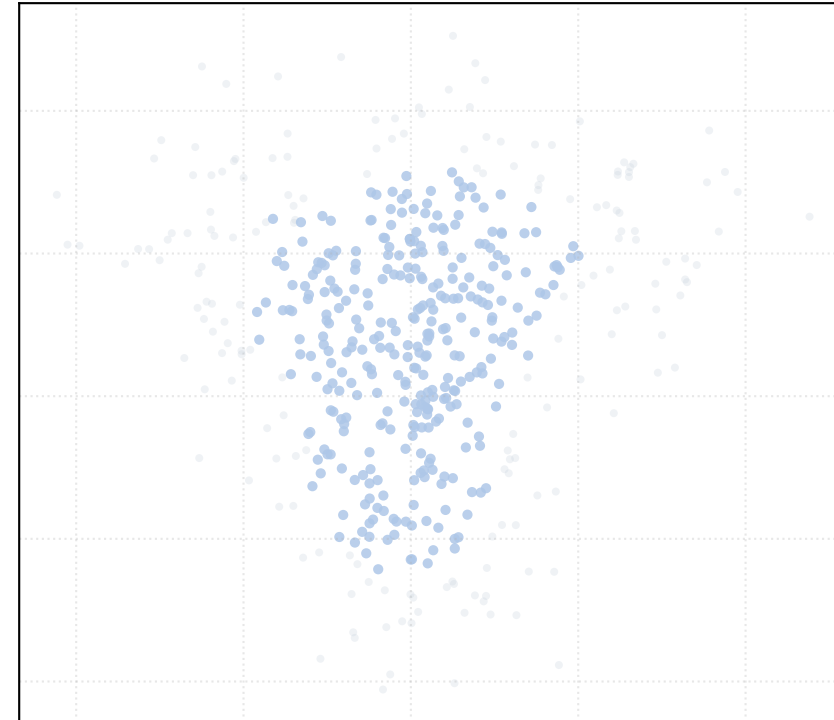
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

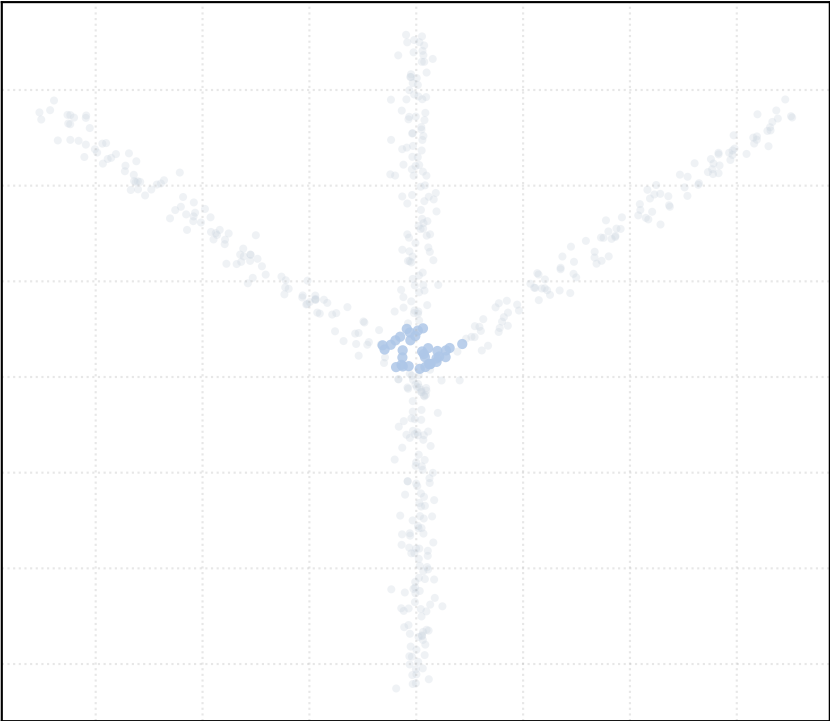


t_percentile = 0.5 [1 Clusters]

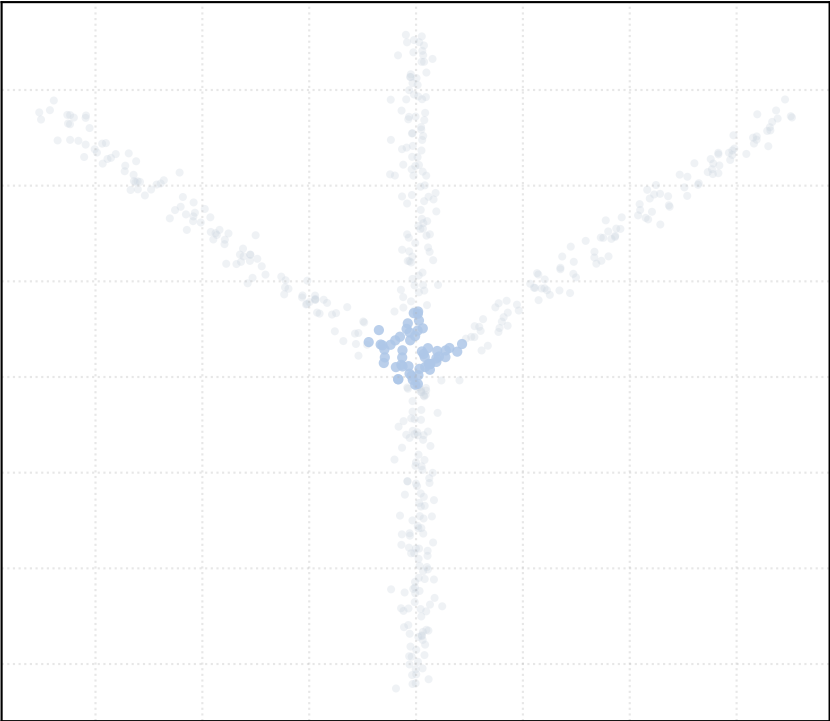


Dataset: bifurcation_3way_noise_low | Seed Percentile Sweep (t_percentile)

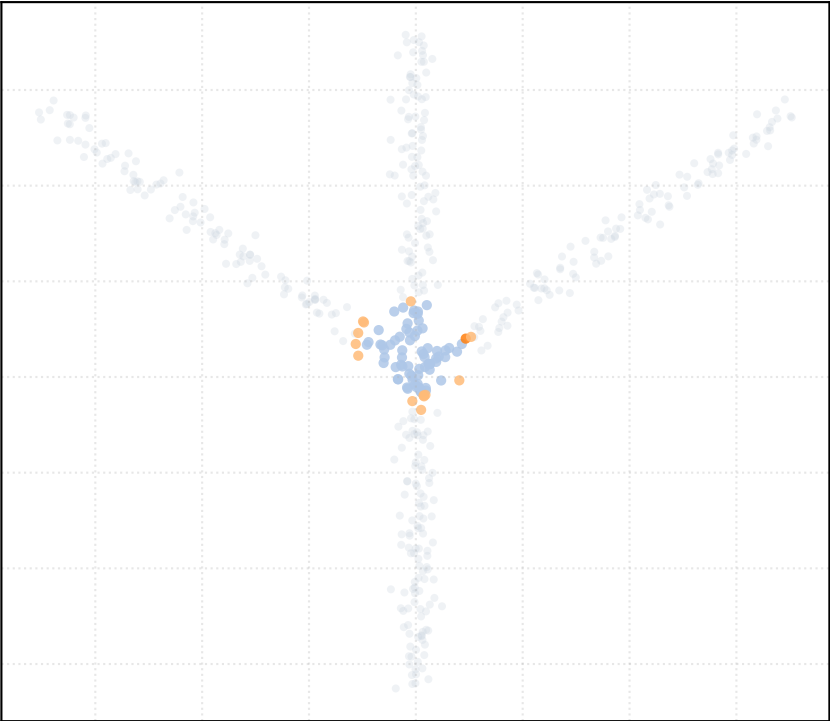
t_percentile = 0.05 [1 Clusters]



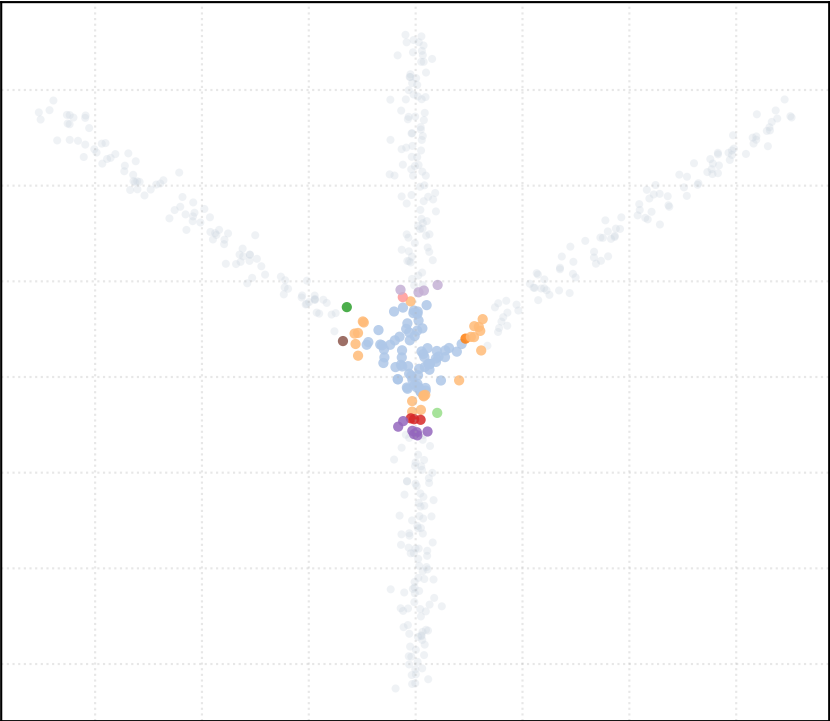
t_percentile = 0.1 [1 Clusters]



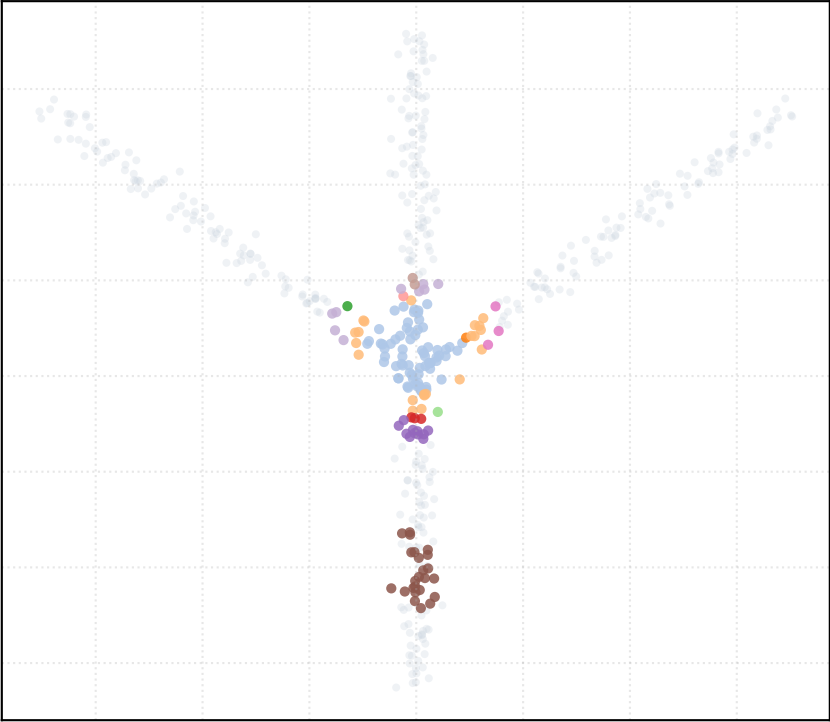
t_percentile = 0.15 (Default) [3 Clusters]



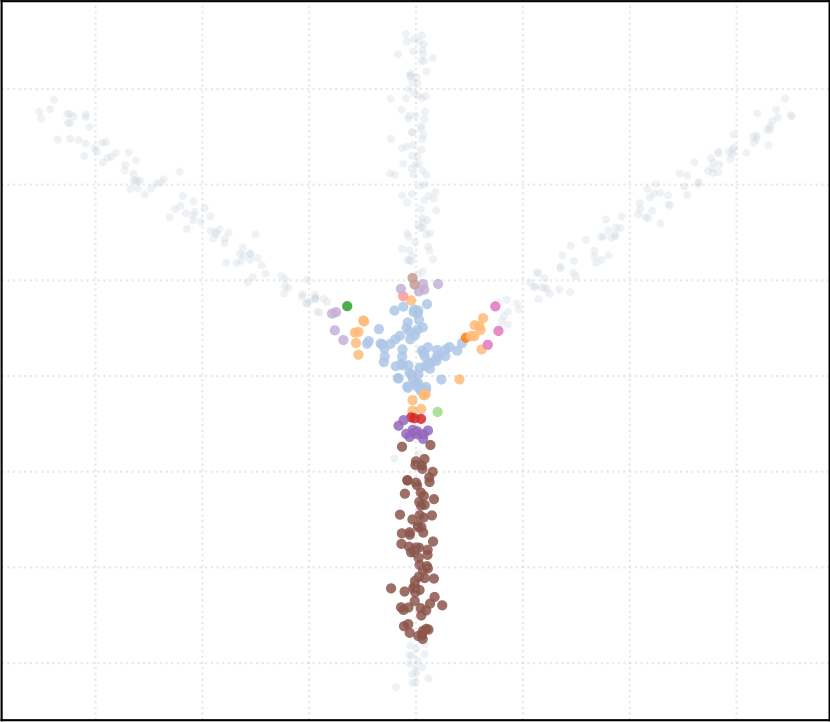
t_percentile = 0.2 [10 Clusters]



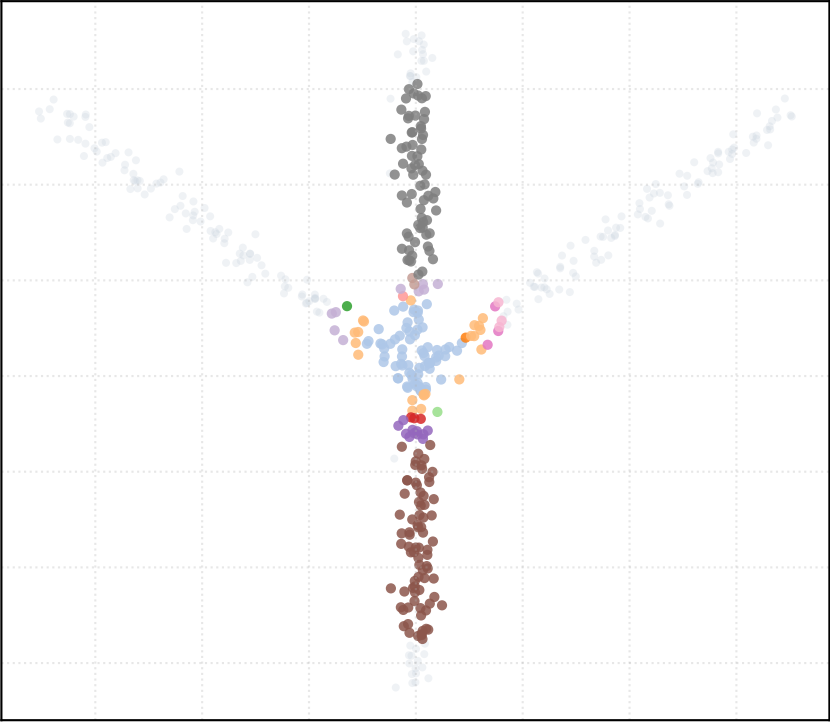
t_percentile = 0.25 [12 Clusters]



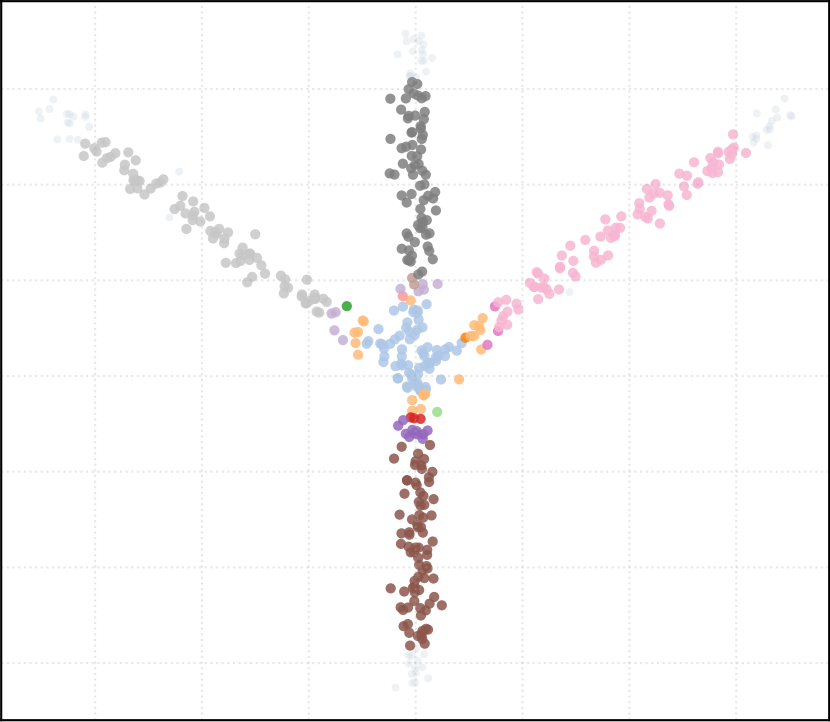
t_percentile = 0.3 [12 Clusters]



t_percentile = 0.4 [14 Clusters]

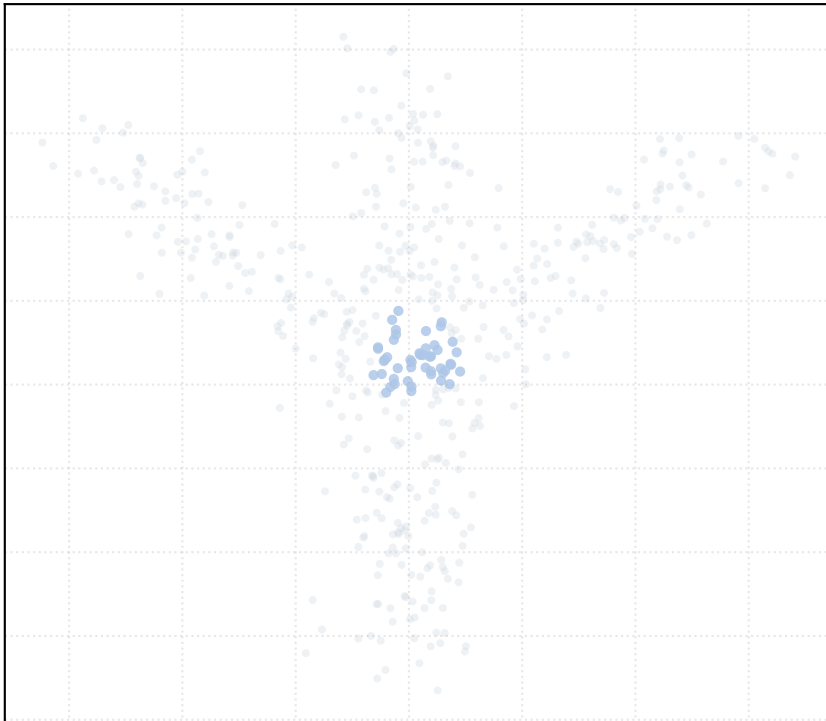


t_percentile = 0.5 [15 Clusters]

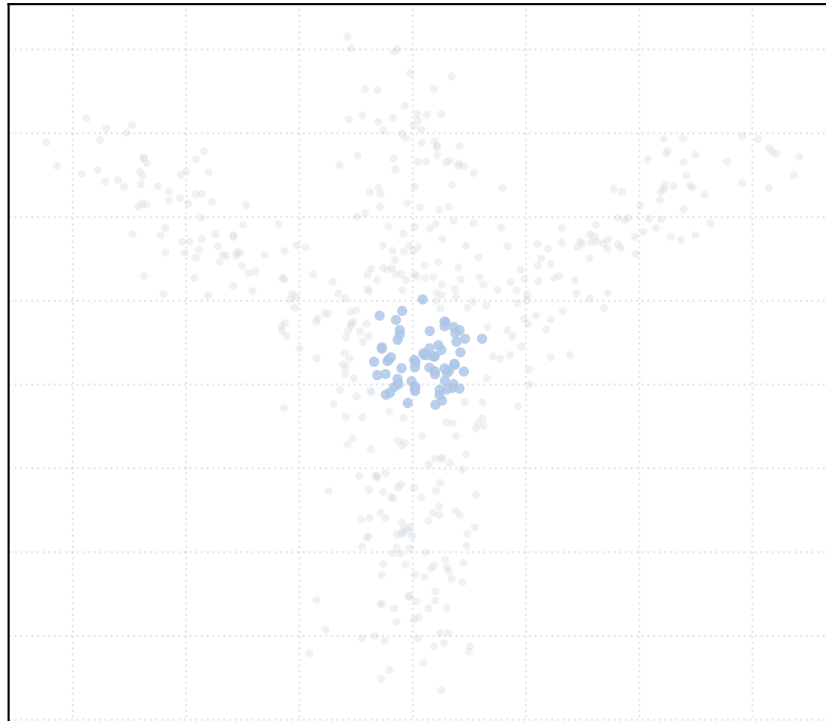


Dataset: bifurcation_3way_noise_medium | Seed Percentile Sweep (t_percentile)

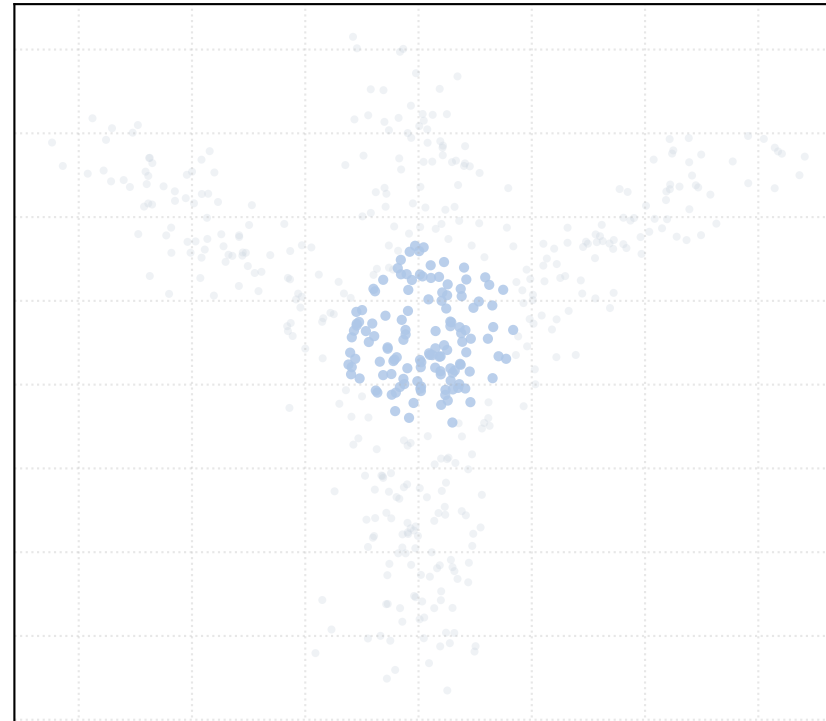
t_percentile = 0.05 [1 Clusters]



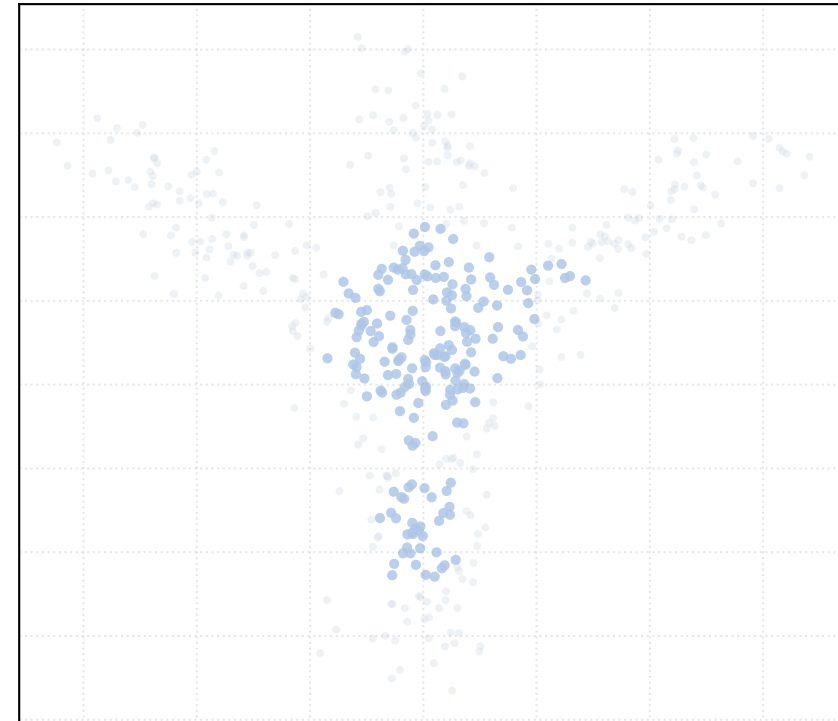
t_percentile = 0.1 [1 Clusters]



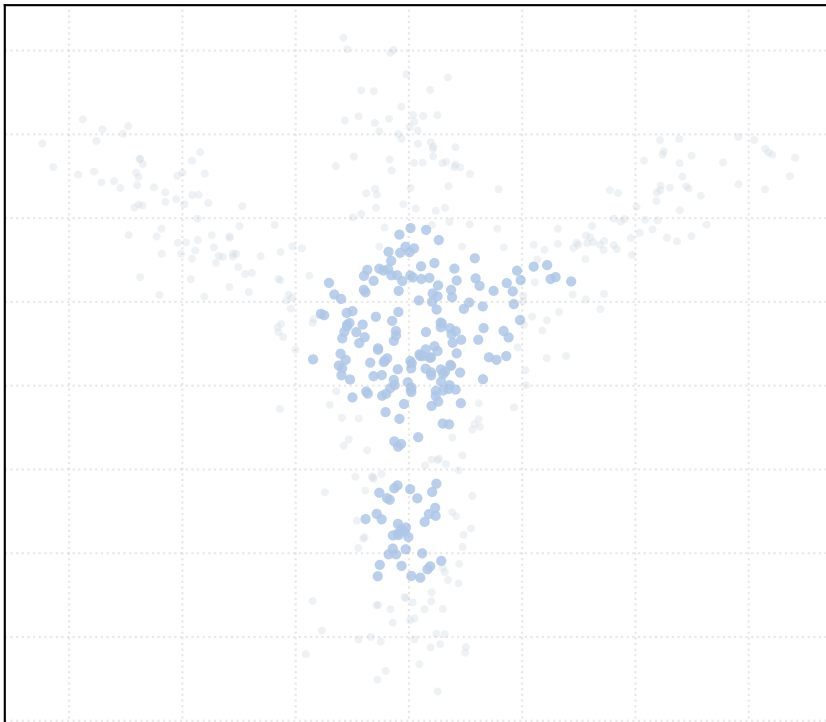
t_percentile = 0.15 (Default) [1 Clusters]



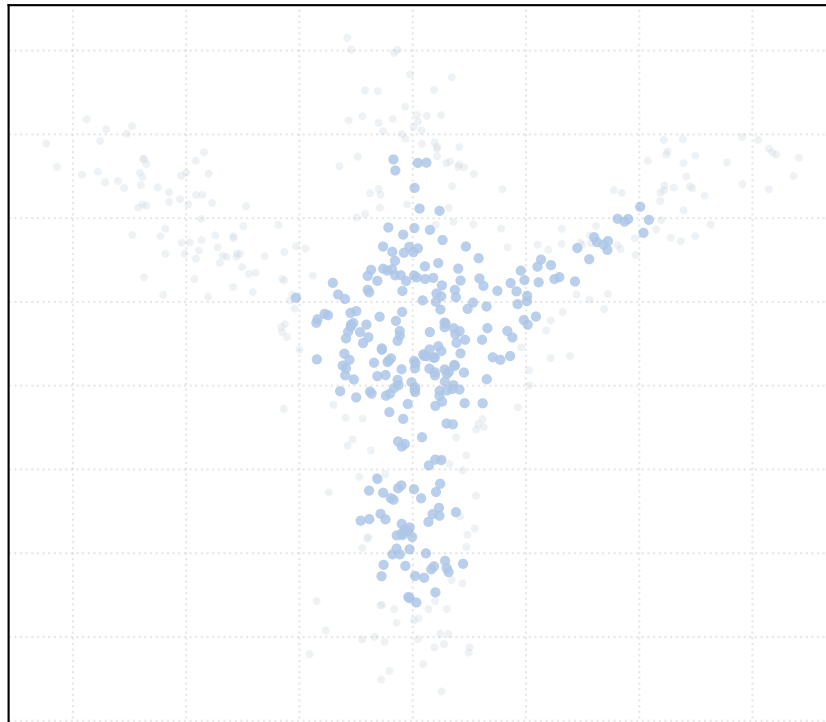
t_percentile = 0.2 [1 Clusters]



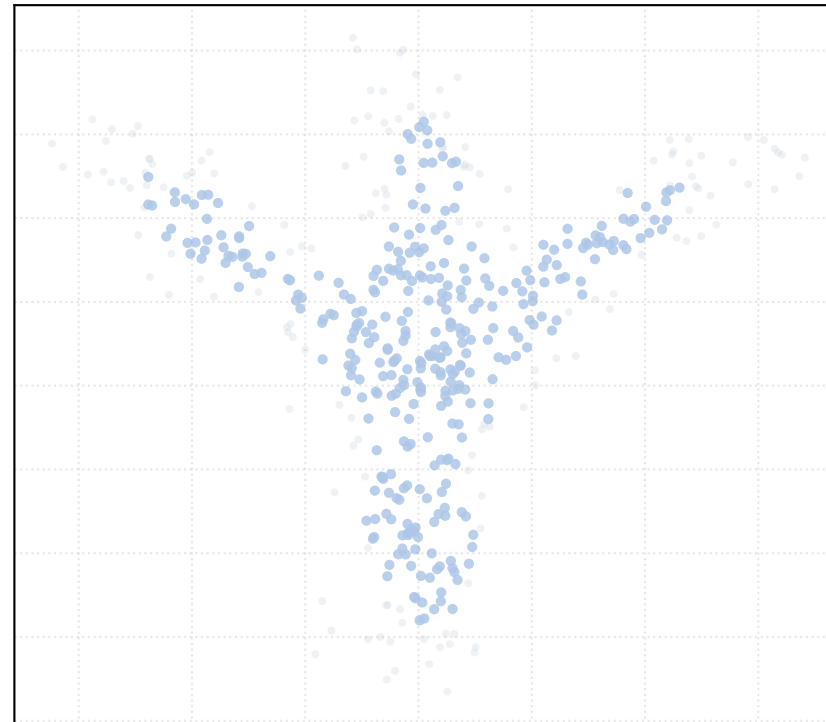
t_percentile = 0.25 [1 Clusters]



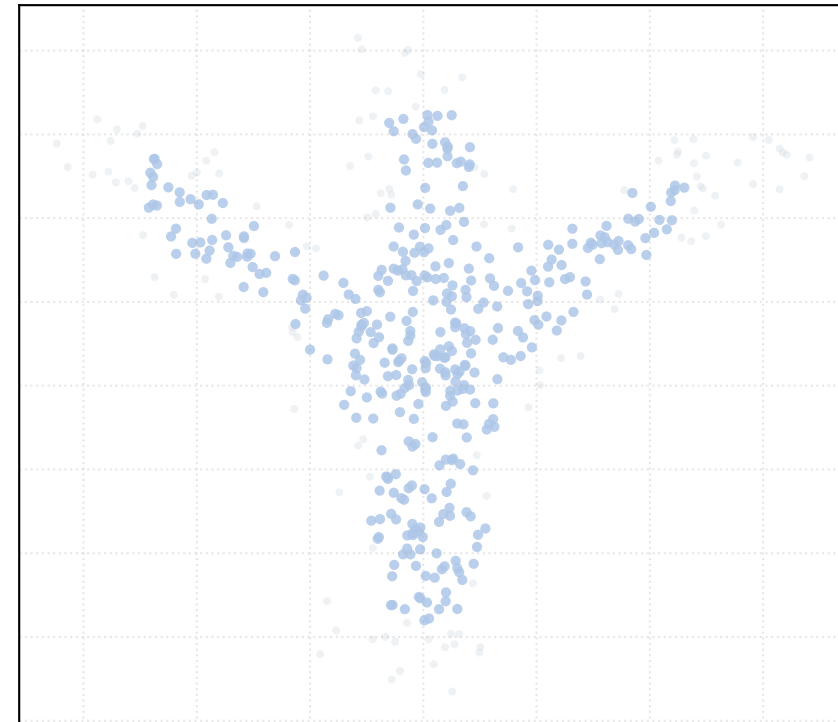
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

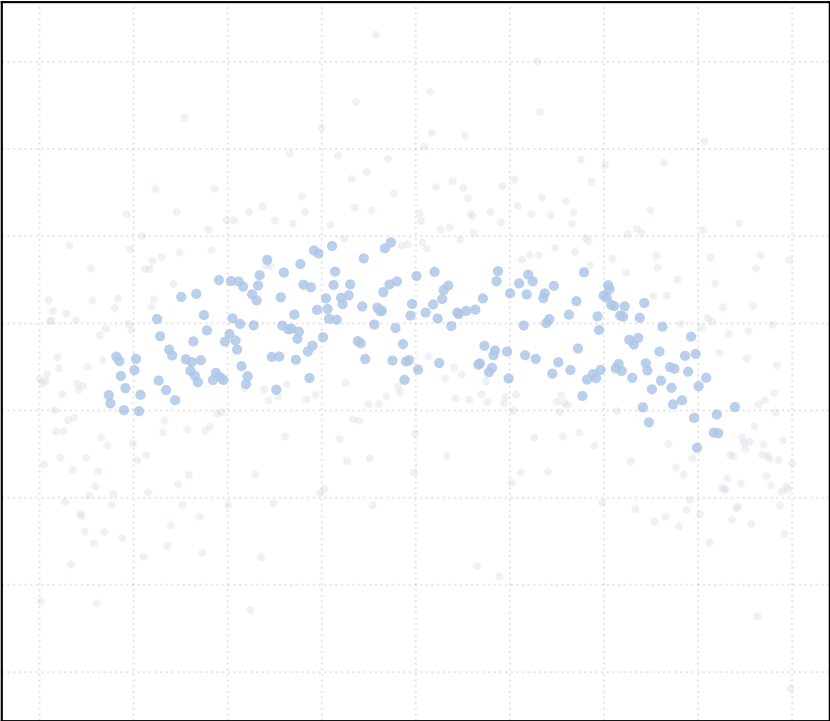


t_percentile = 0.5 [1 Clusters]

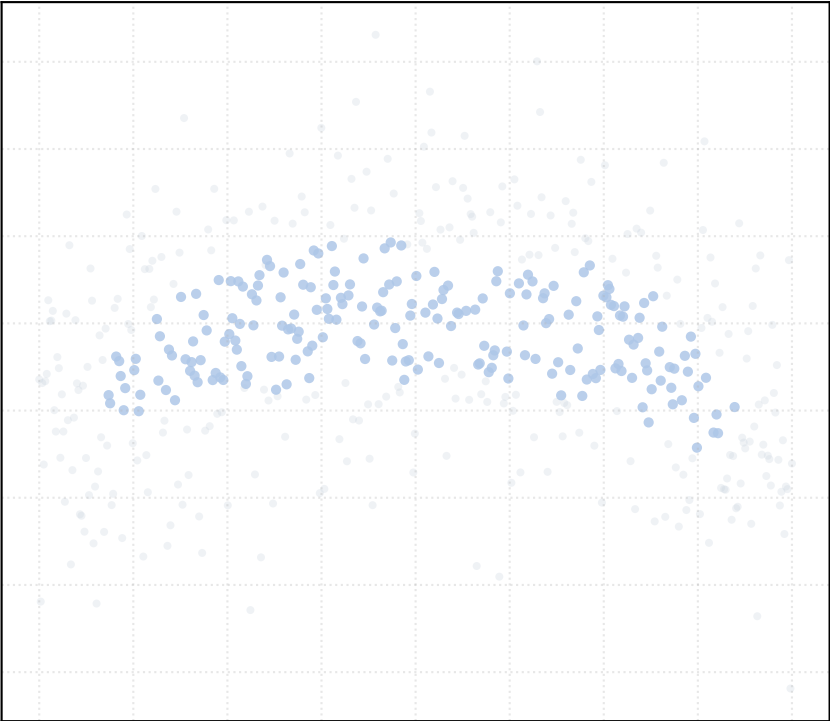


Dataset: circular_arc_noise_high | Seed Percentile Sweep (t_percentile)

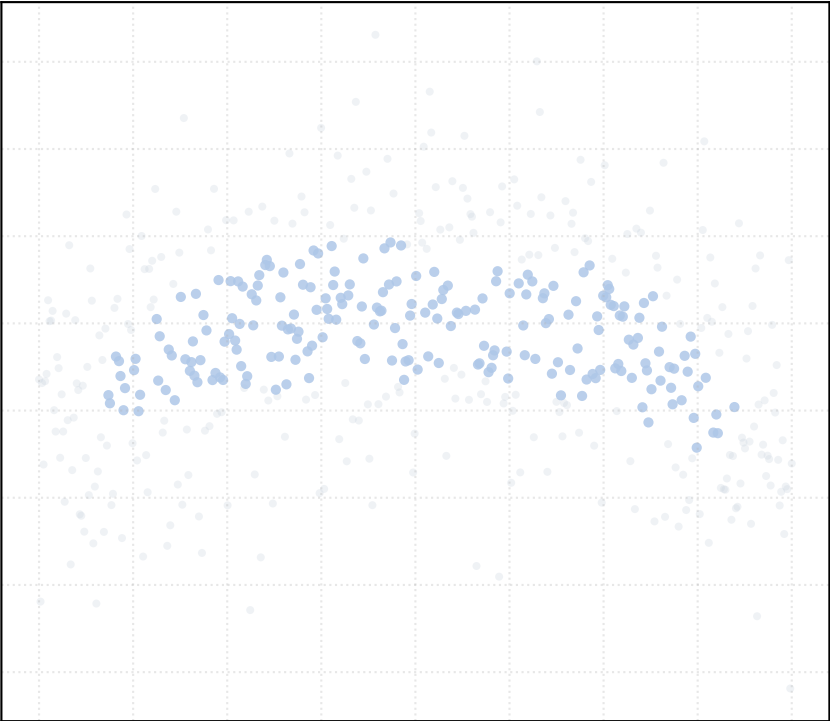
t_percentile = 0.05 [1 Clusters]



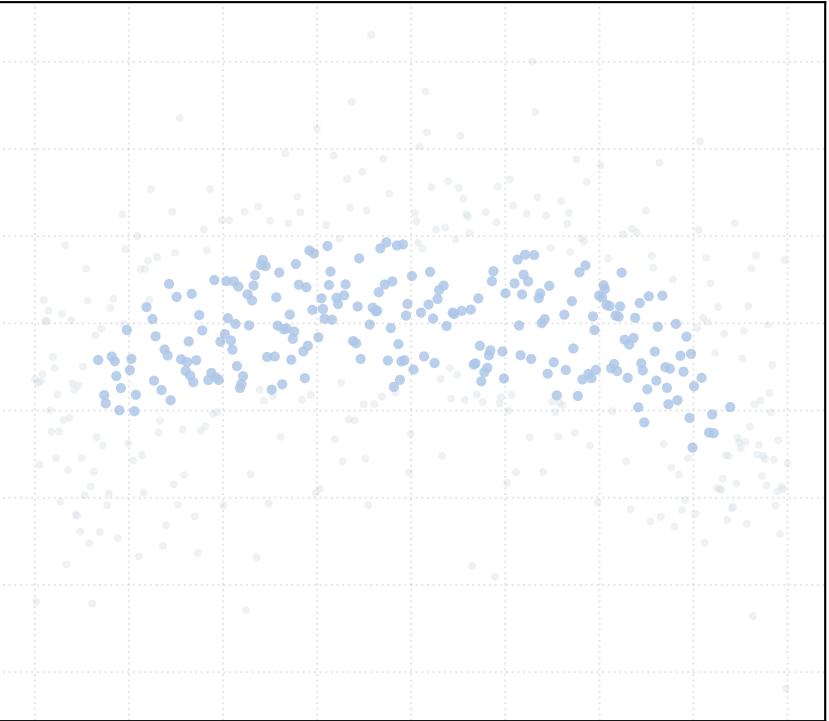
t_percentile = 0.1 [1 Clusters]



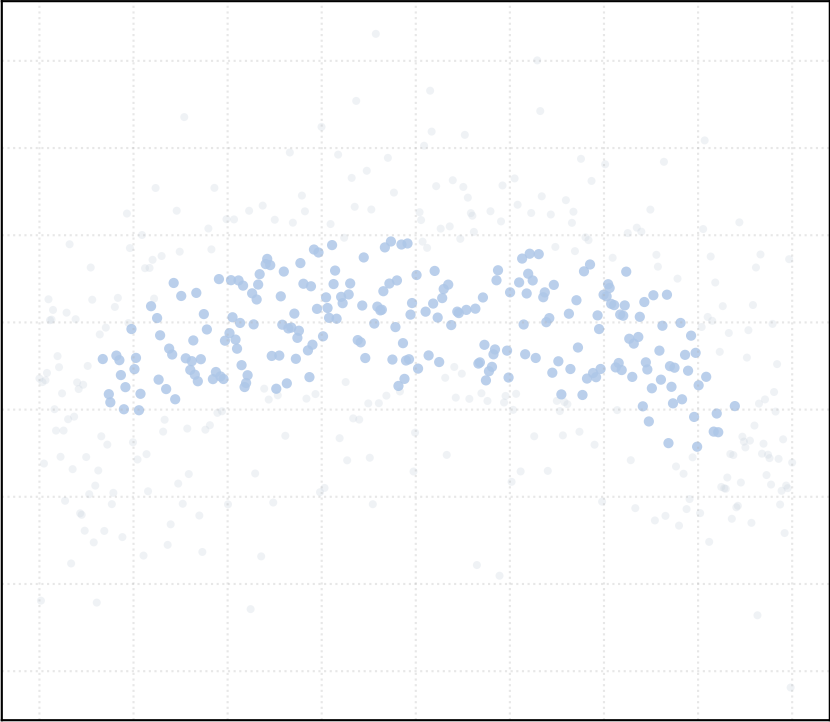
t_percentile = 0.15 (Default) [1 Clusters]



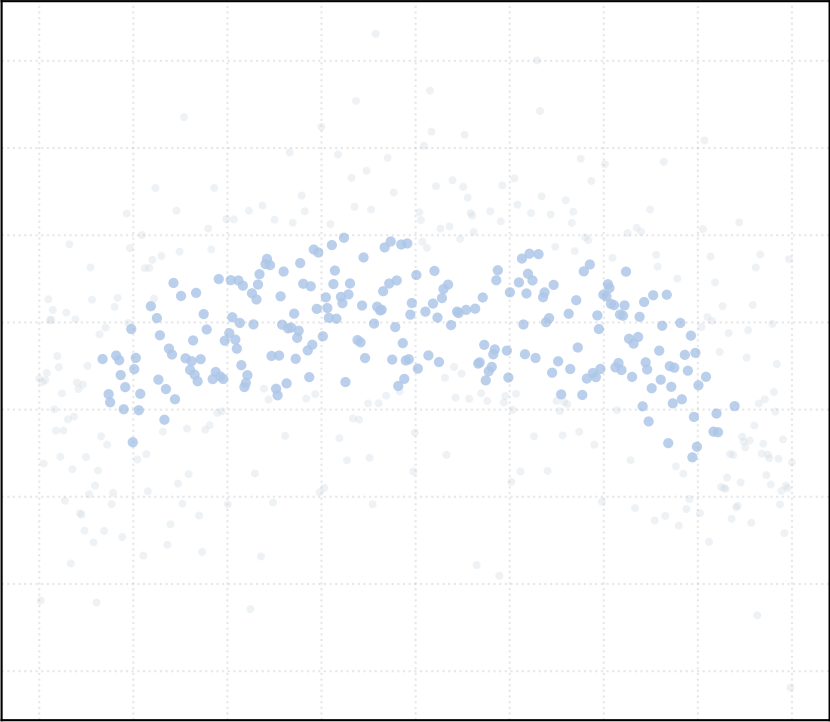
t_percentile = 0.2 [1 Clusters]



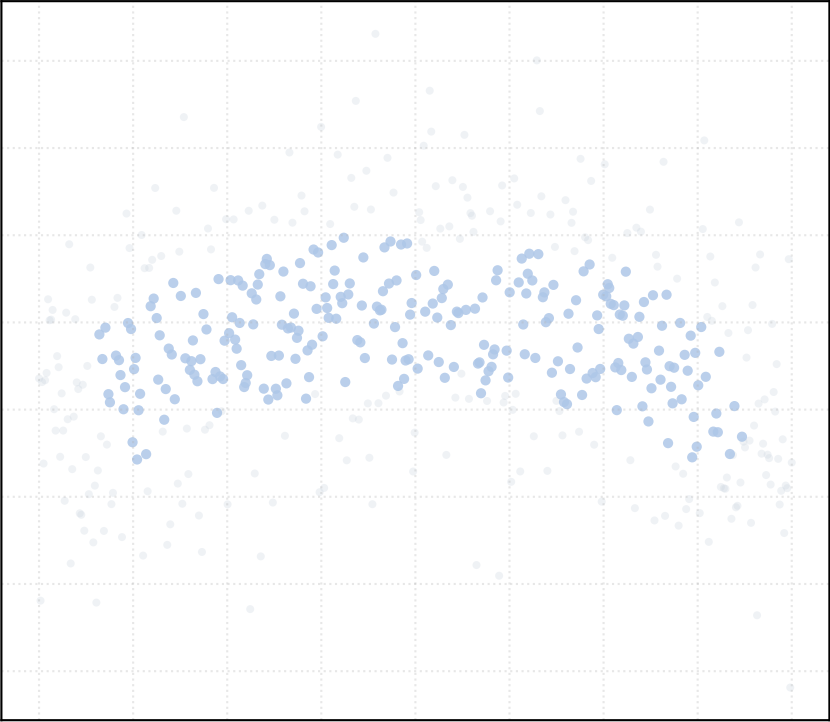
t_percentile = 0.25 [1 Clusters]



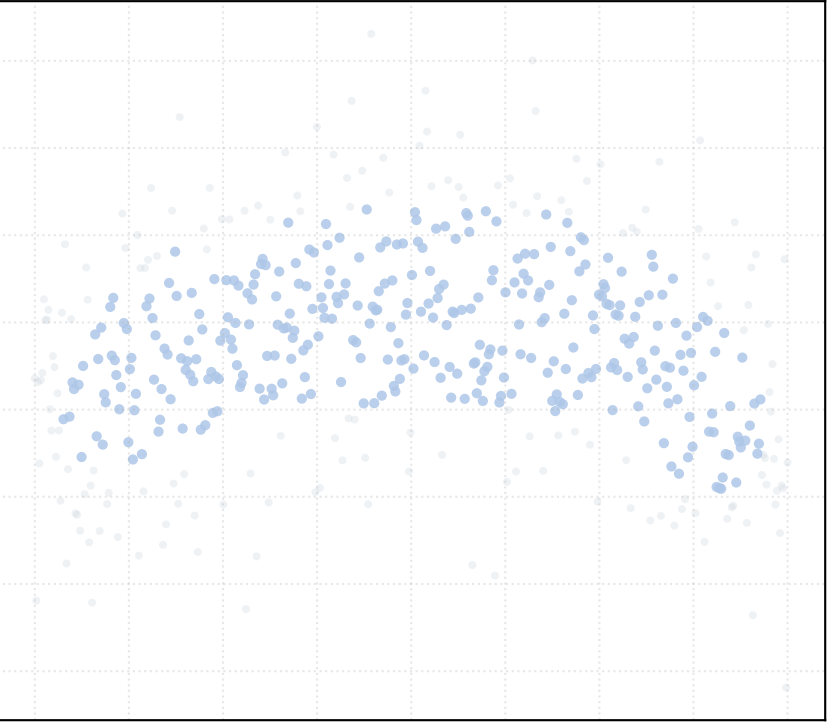
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

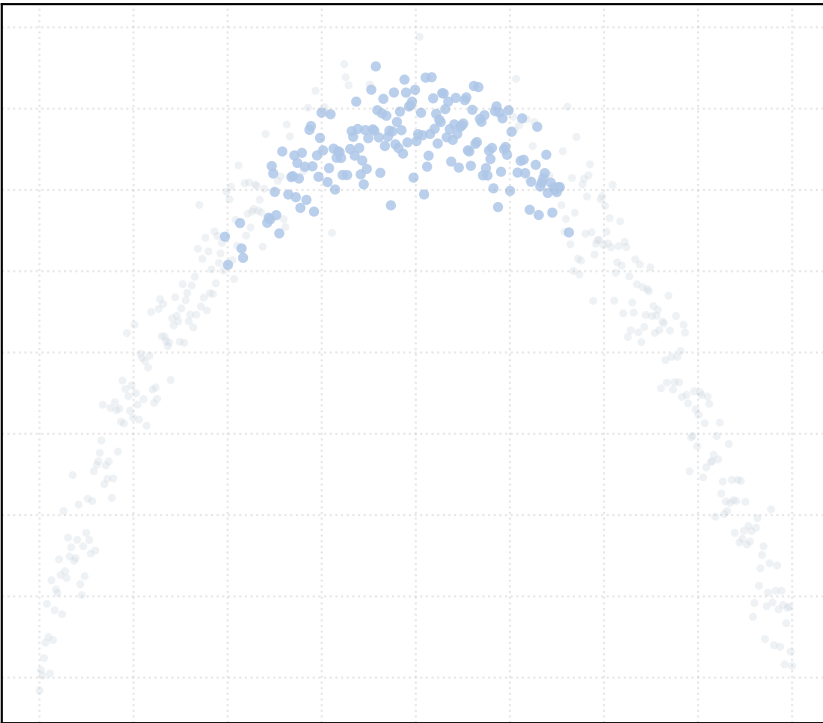


t_percentile = 0.5 [1 Clusters]

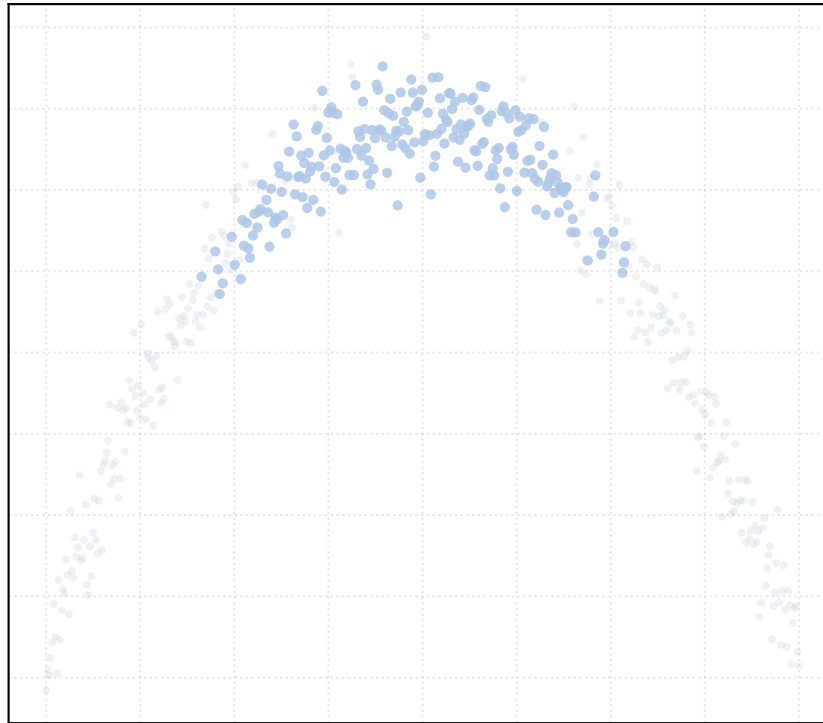


Dataset: circular_arc_noise_low | Seed Percentile Sweep (t_percentile)

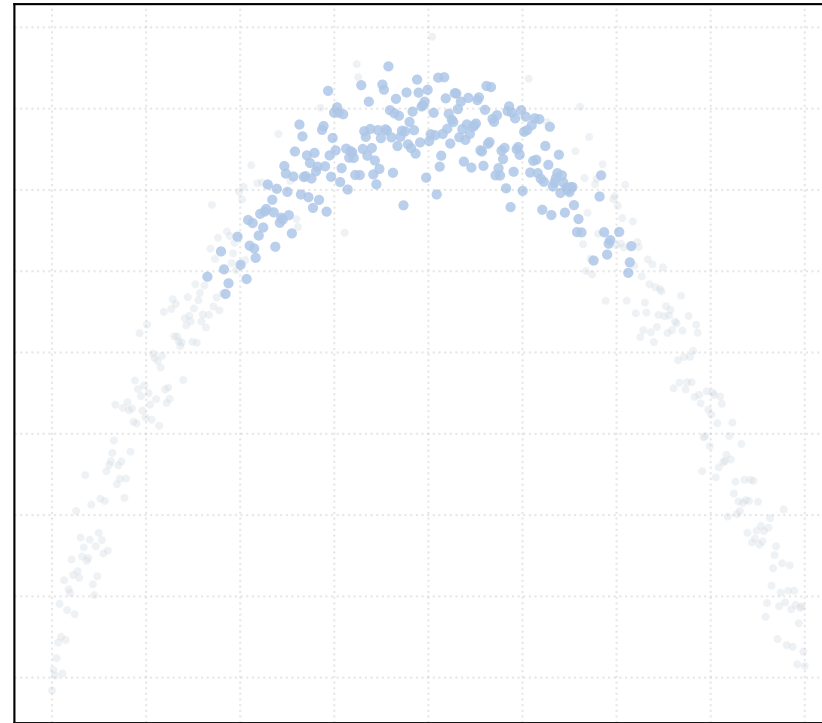
t_percentile = 0.05 [1 Clusters]



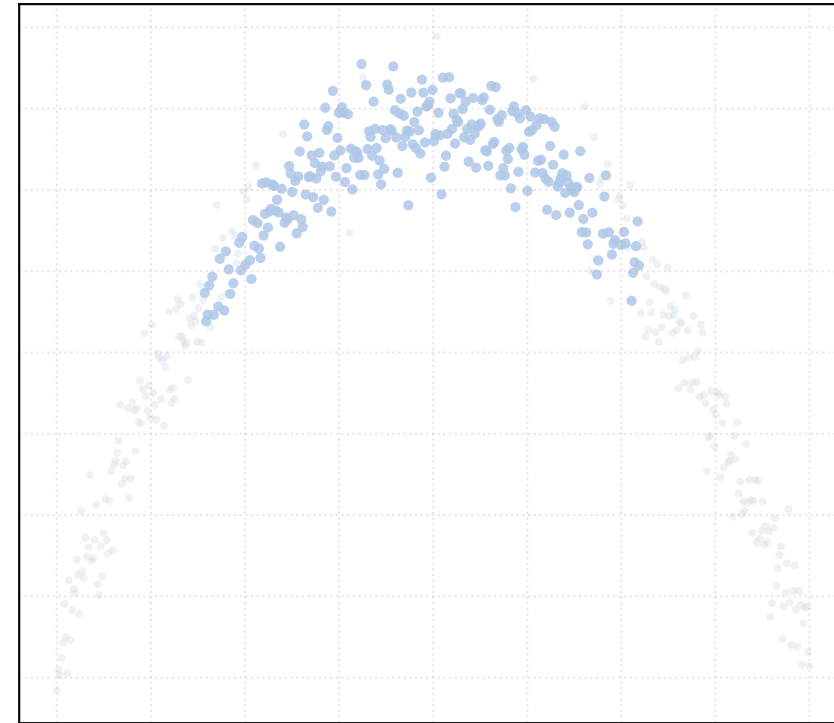
t_percentile = 0.1 [1 Clusters]



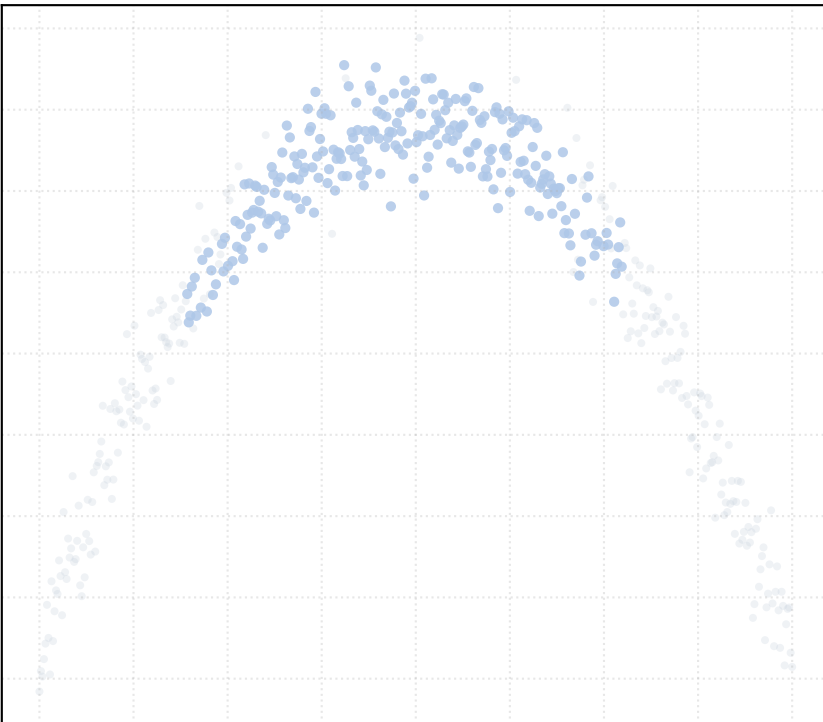
t_percentile = 0.15 (Default) [1 Clusters]



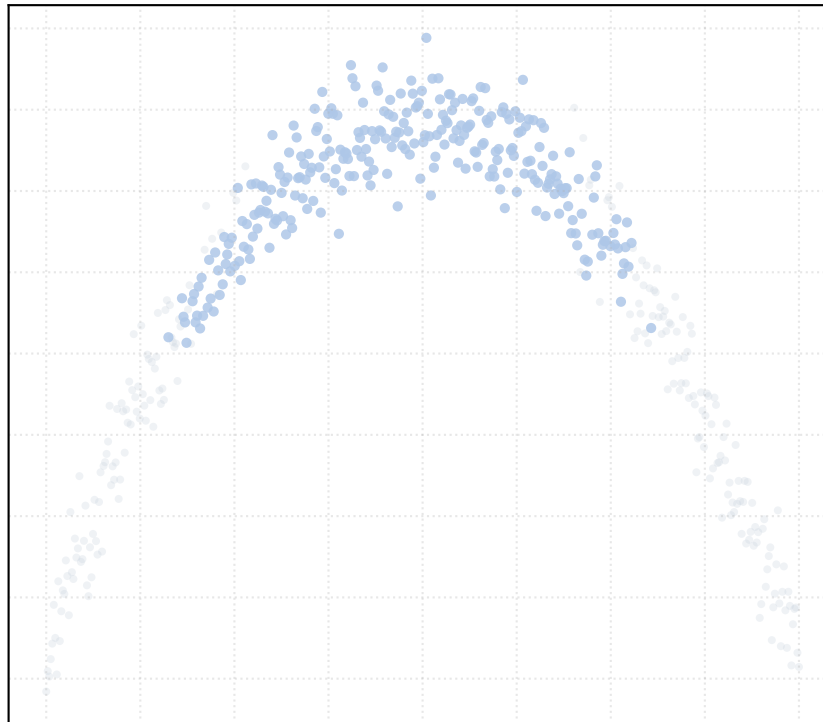
t_percentile = 0.2 [1 Clusters]



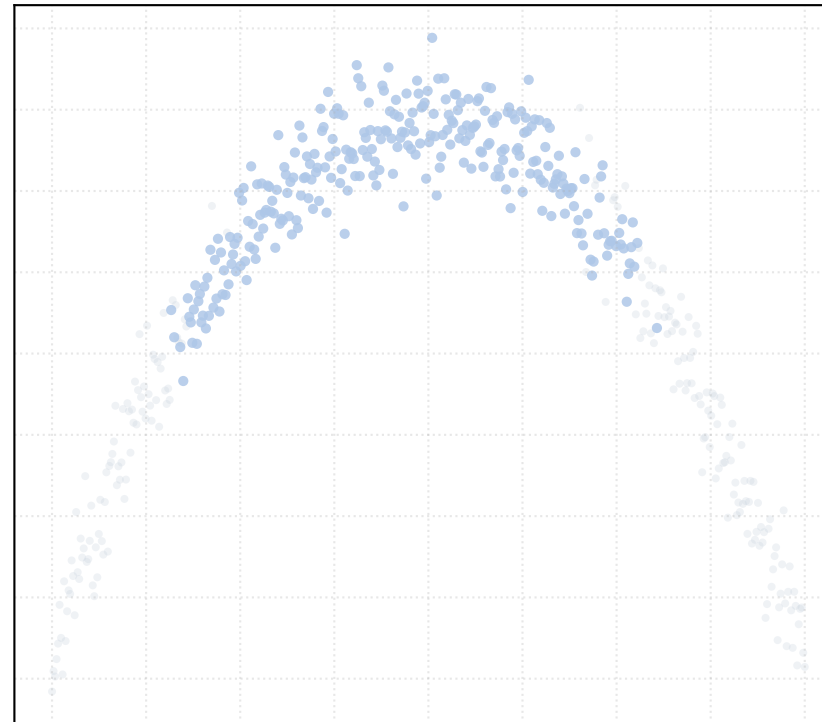
t_percentile = 0.25 [1 Clusters]



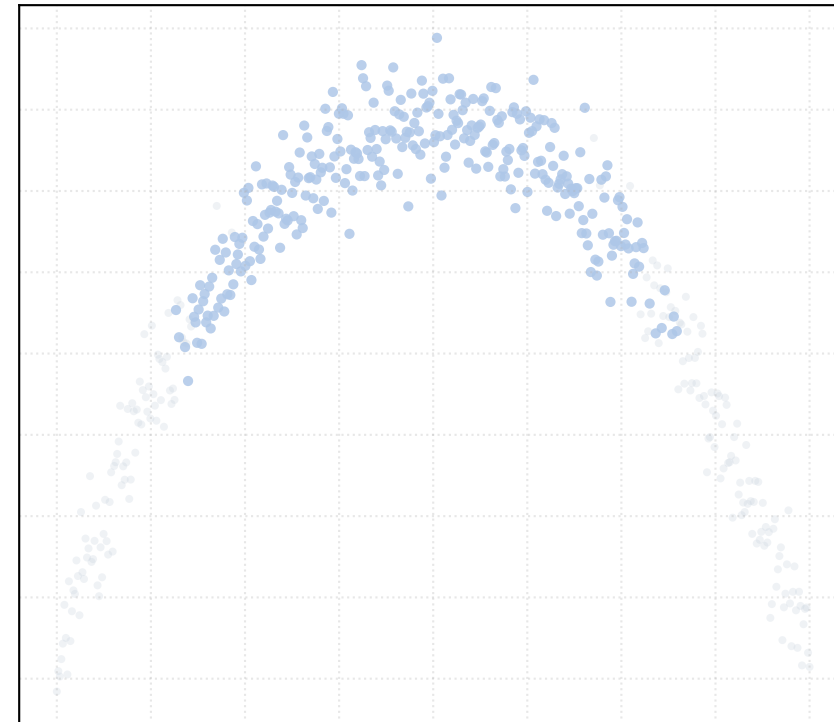
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

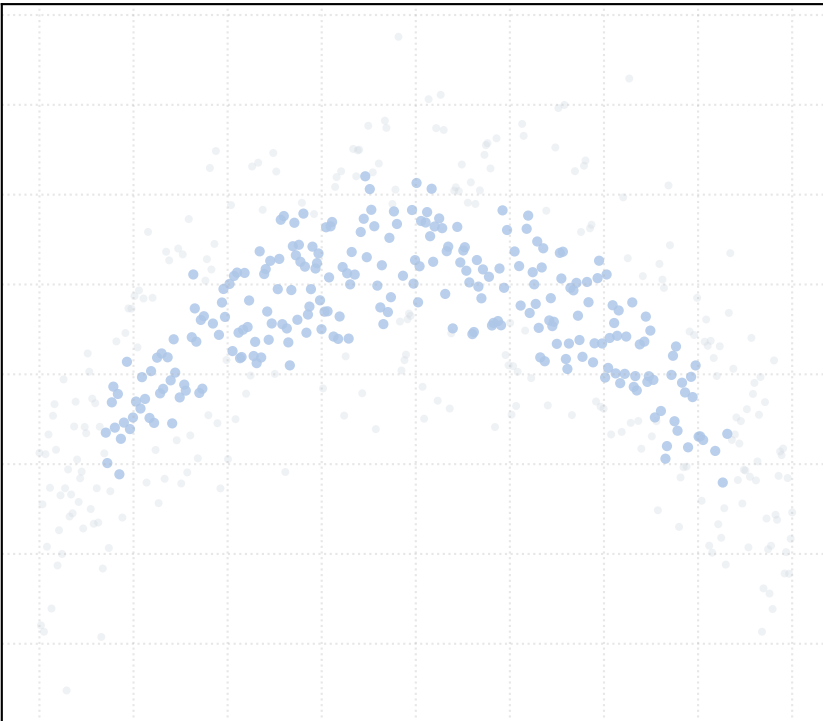


t_percentile = 0.5 [1 Clusters]

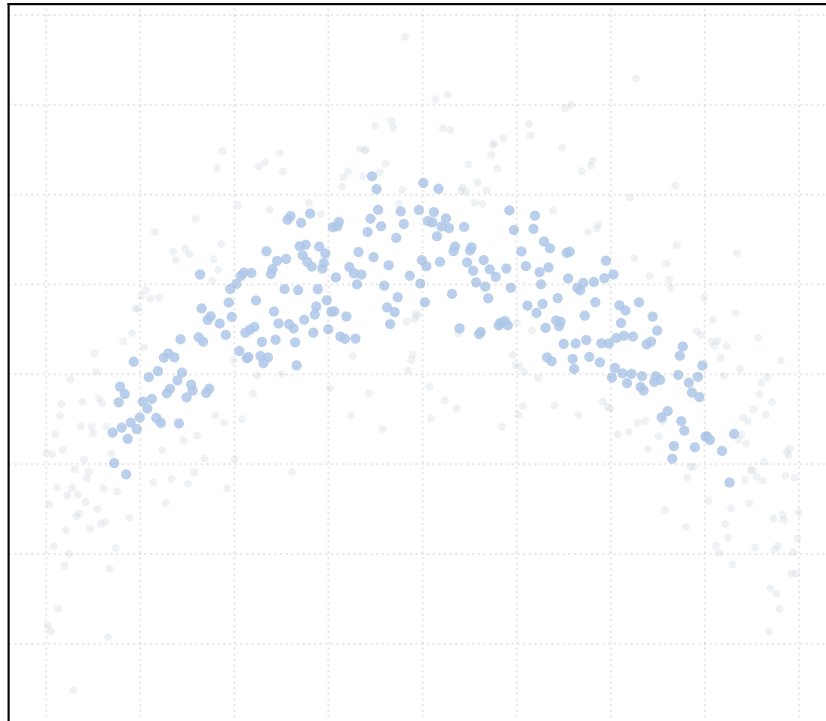


Dataset: circular_arc_noise_medium | Seed Percentile Sweep (t_percentile)

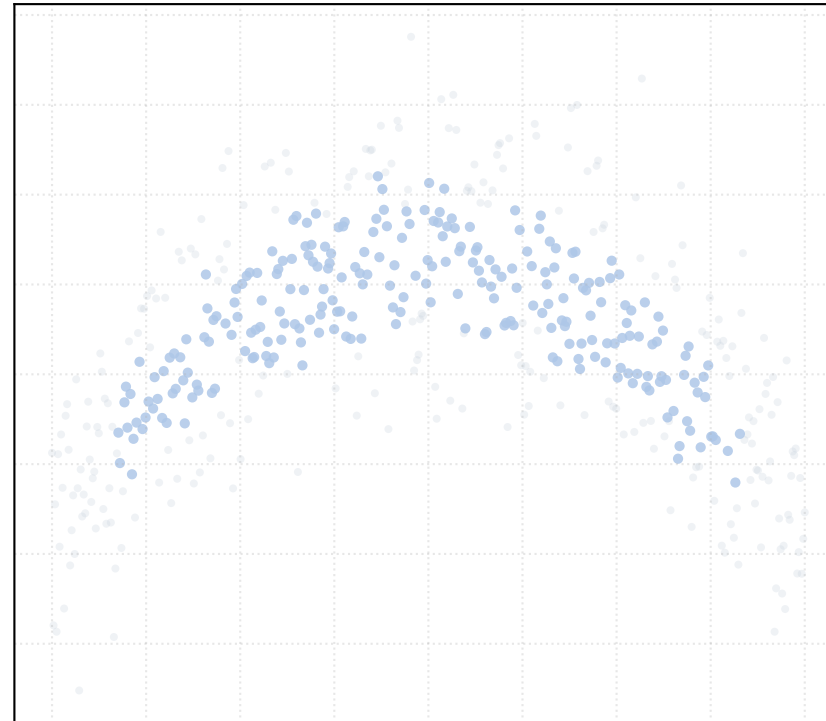
t_percentile = 0.05 [1 Clusters]



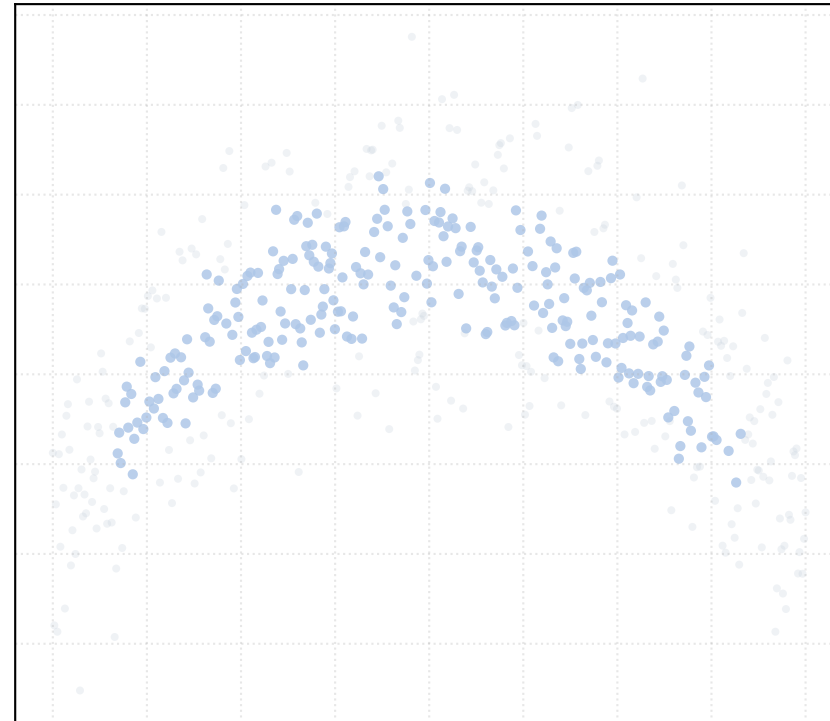
t_percentile = 0.1 [1 Clusters]



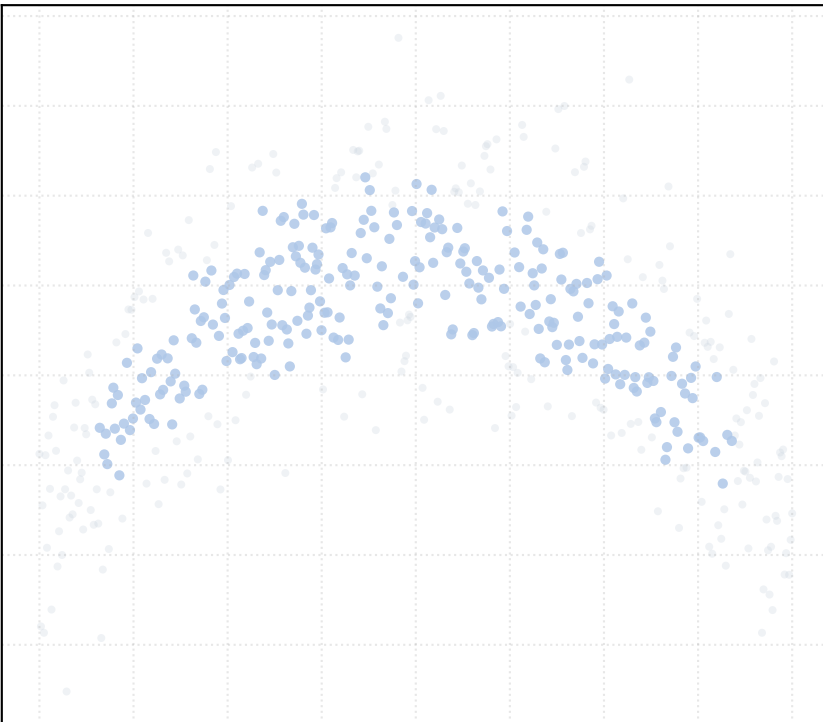
t_percentile = 0.15 (Default) [1 Clusters]



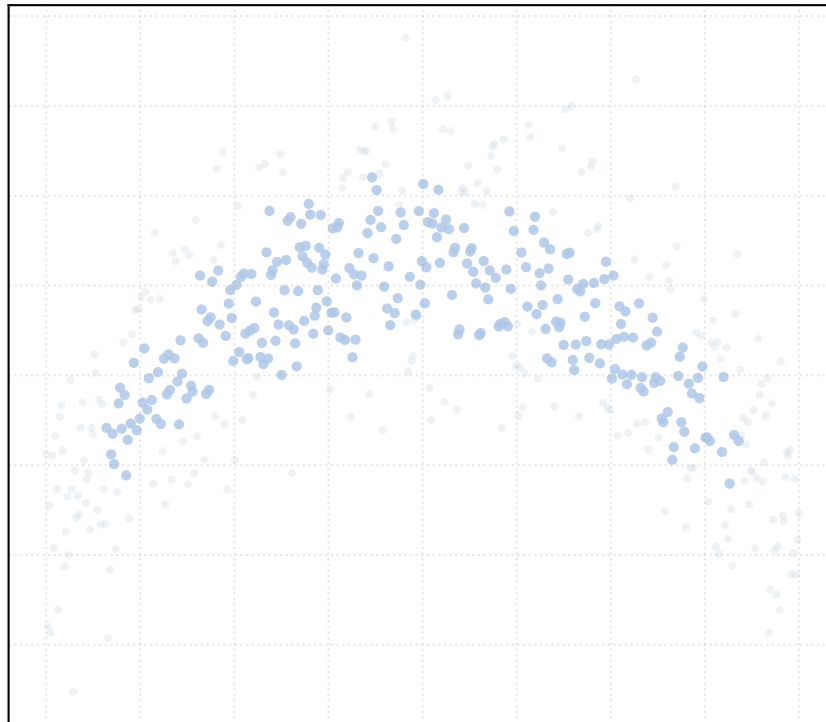
t_percentile = 0.2 [1 Clusters]



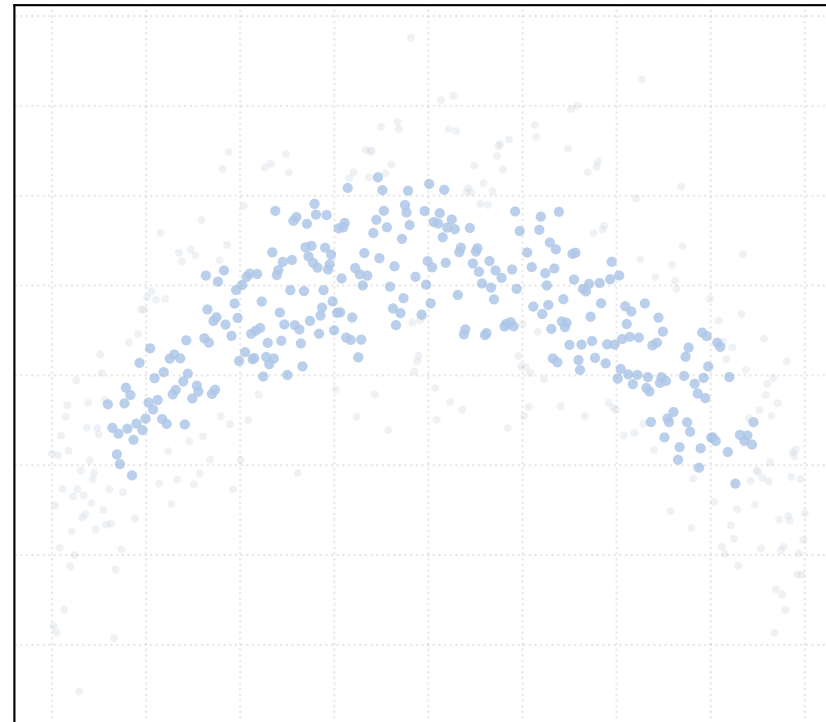
t_percentile = 0.25 [1 Clusters]



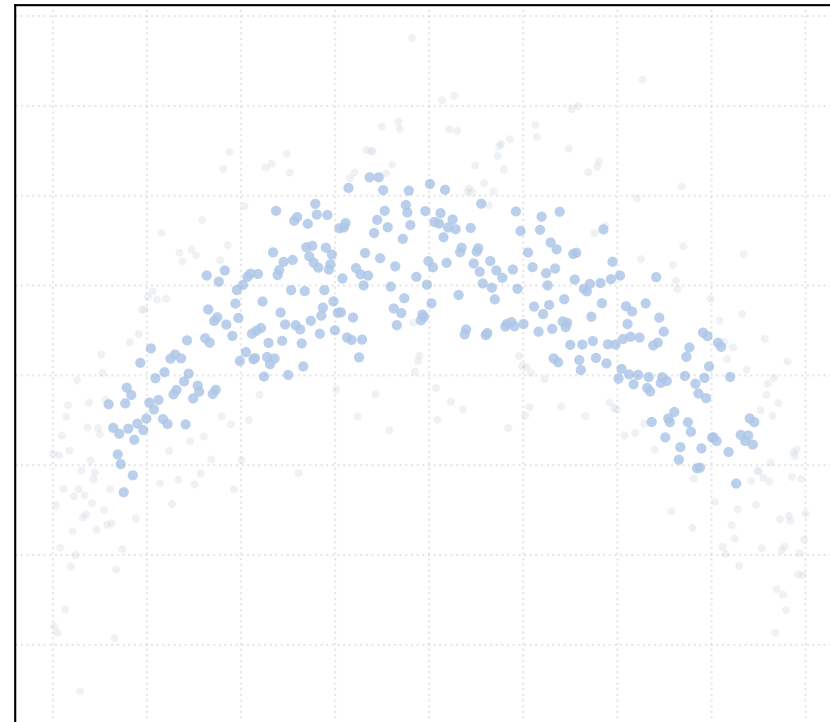
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

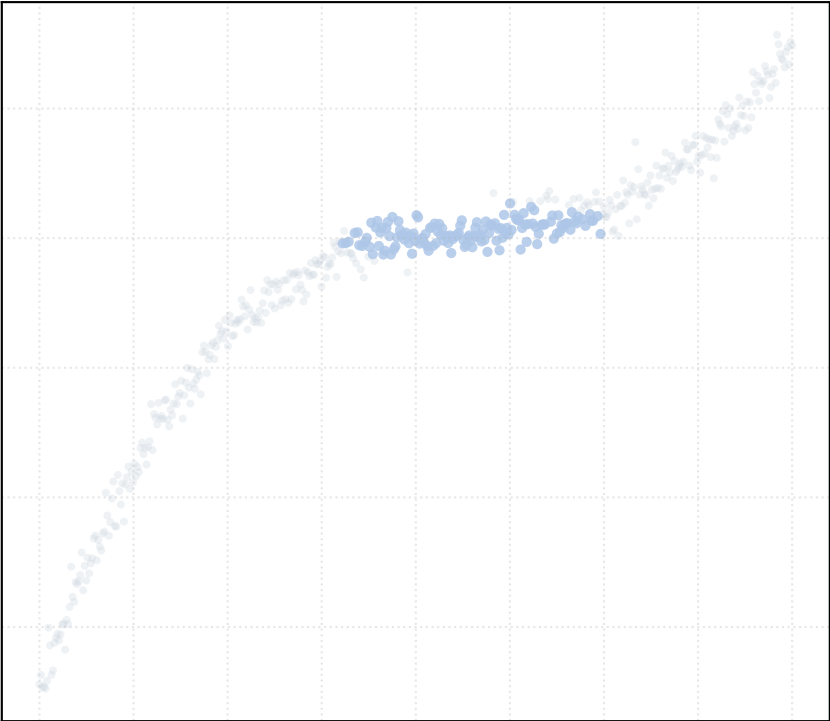


t_percentile = 0.5 [1 Clusters]

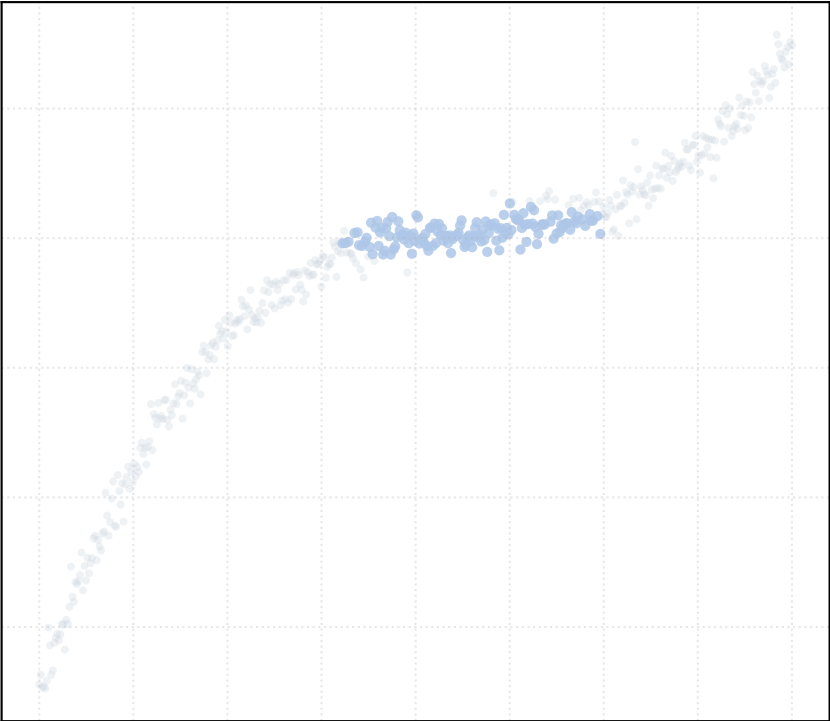


Dataset: cubic_noise_high | Seed Percentile Sweep (t_percentile)

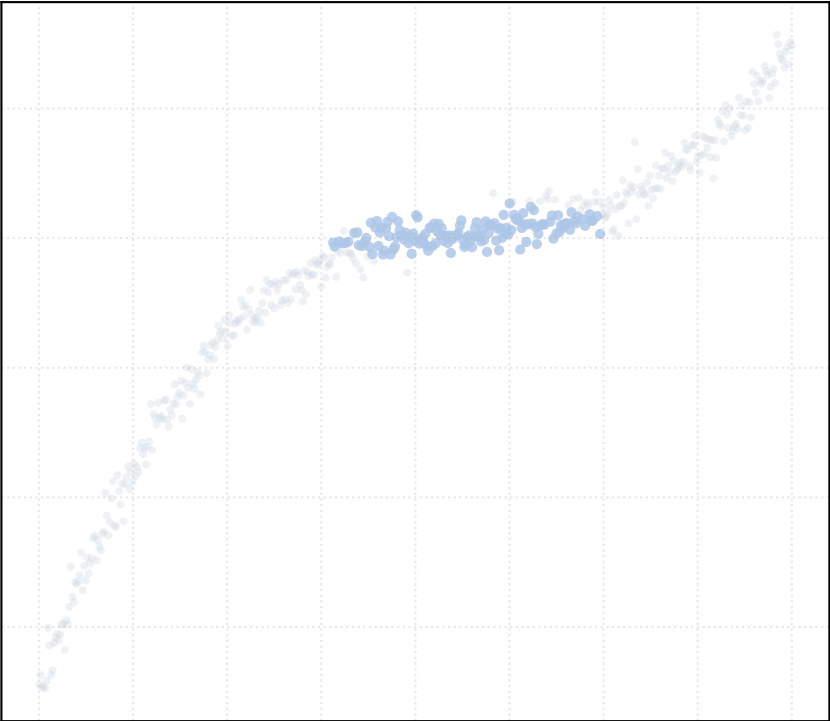
t_percentile = 0.05 [1 Clusters]



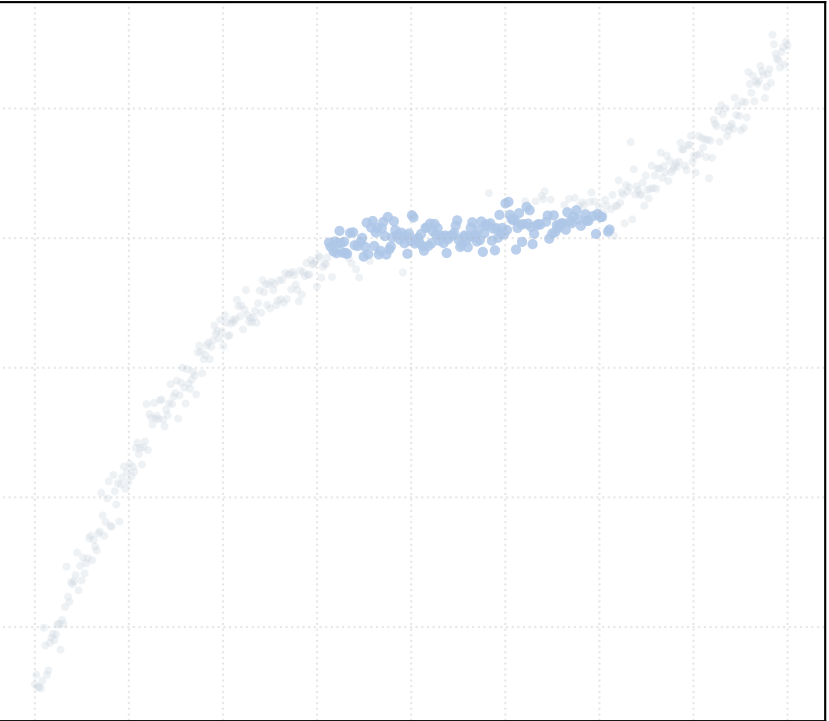
t_percentile = 0.1 [1 Clusters]



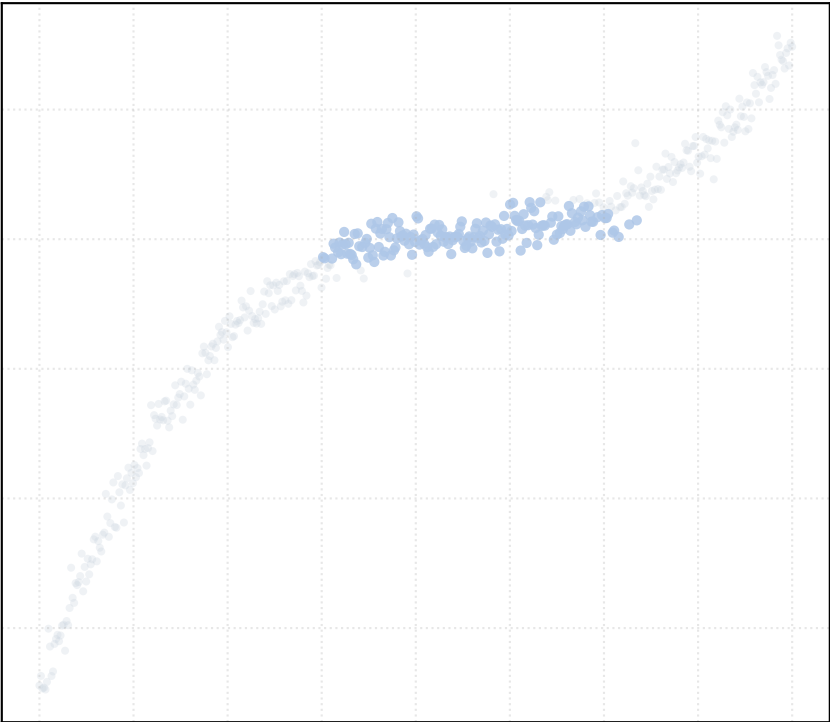
t_percentile = 0.15 (Default) [1 Clusters]



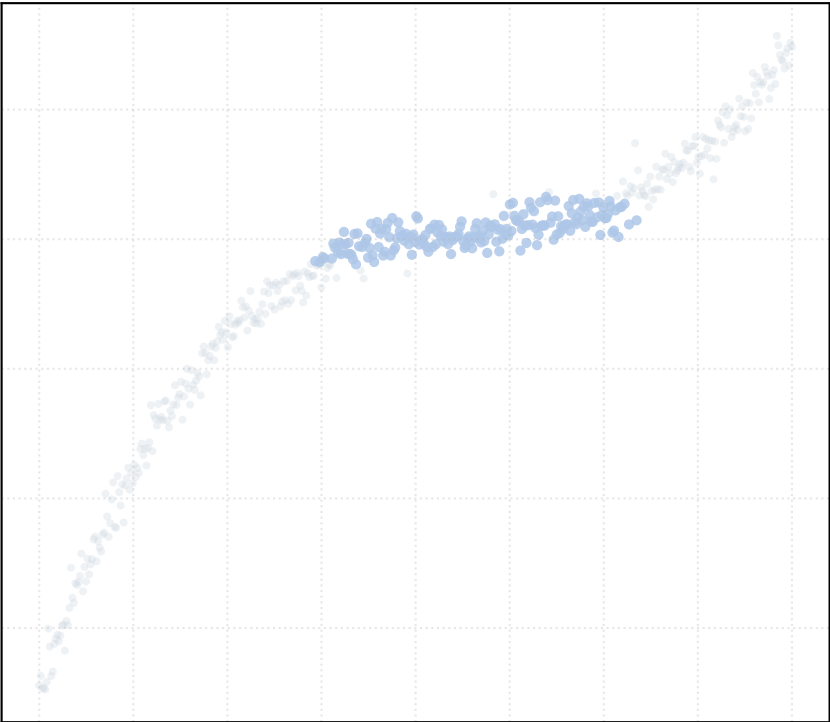
t_percentile = 0.2 [1 Clusters]



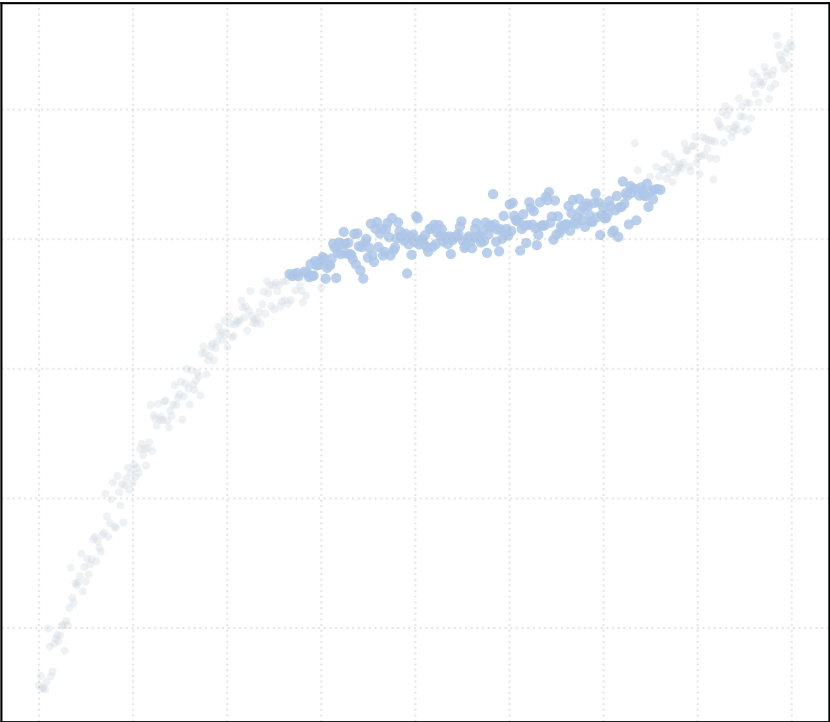
t_percentile = 0.25 [1 Clusters]



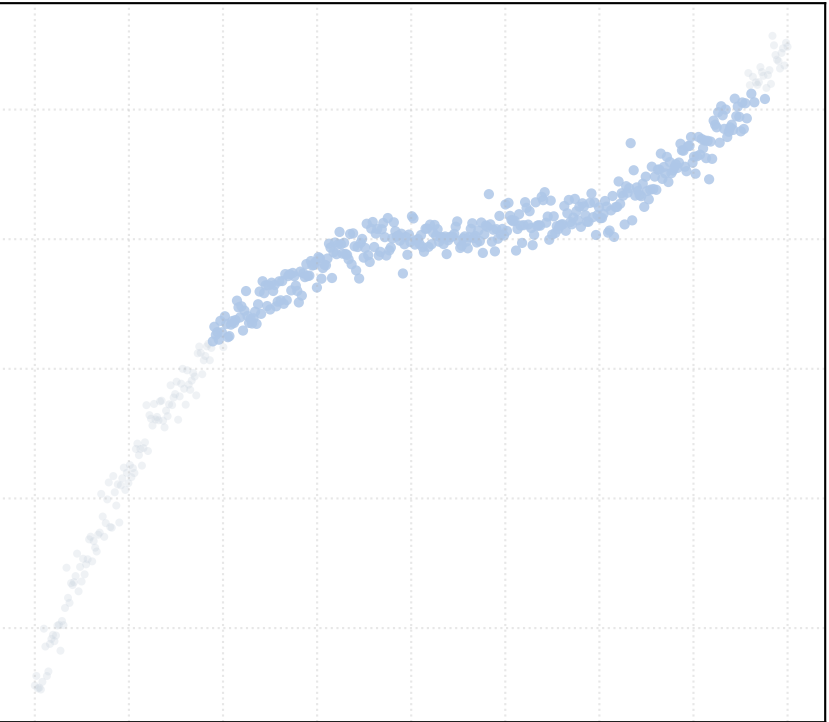
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

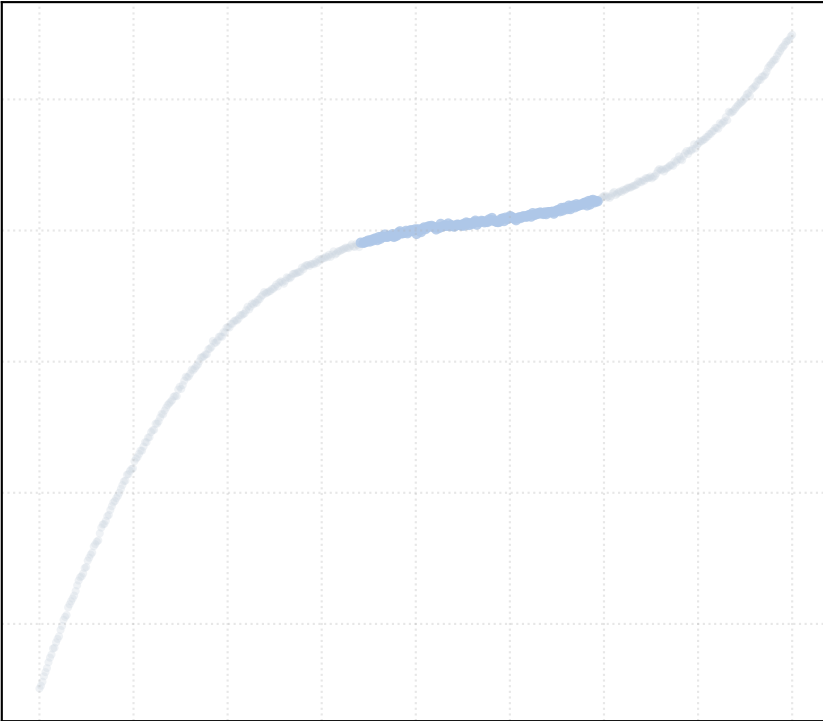


t_percentile = 0.5 [1 Clusters]

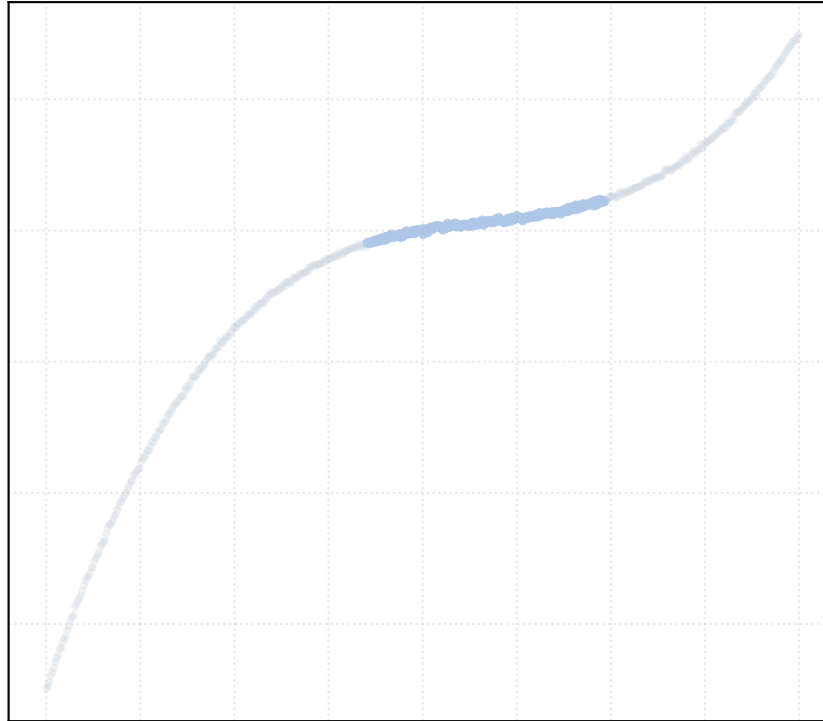


Dataset: cubic_noise_low | Seed Percentile Sweep (t_percentile)

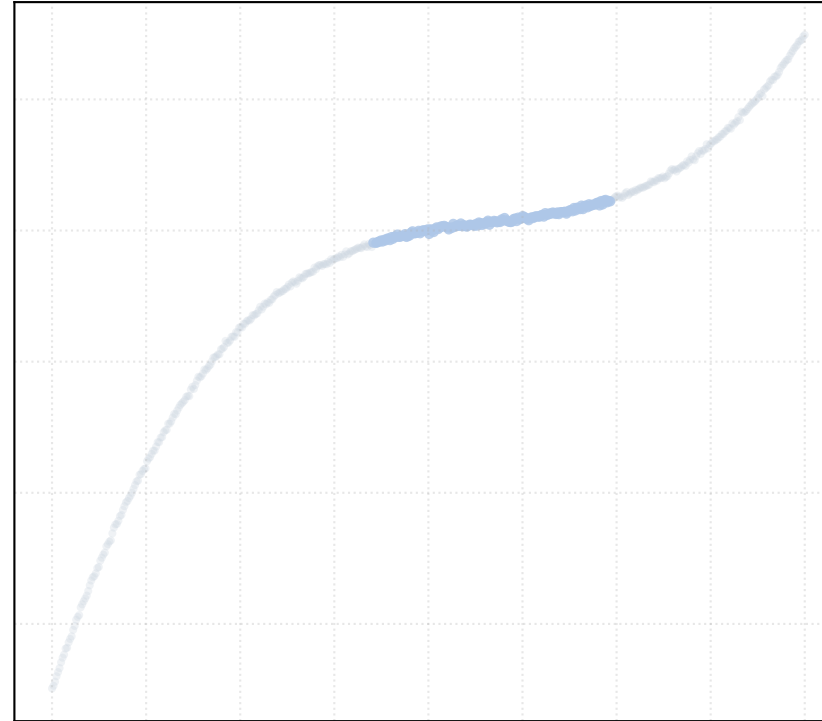
t_percentile = 0.05 [1 Clusters]



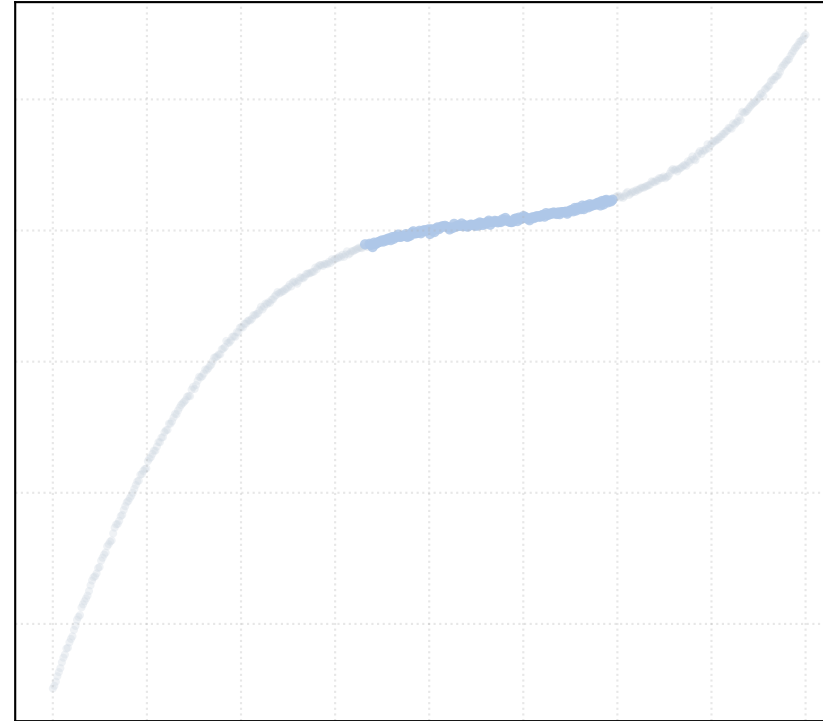
t_percentile = 0.1 [1 Clusters]



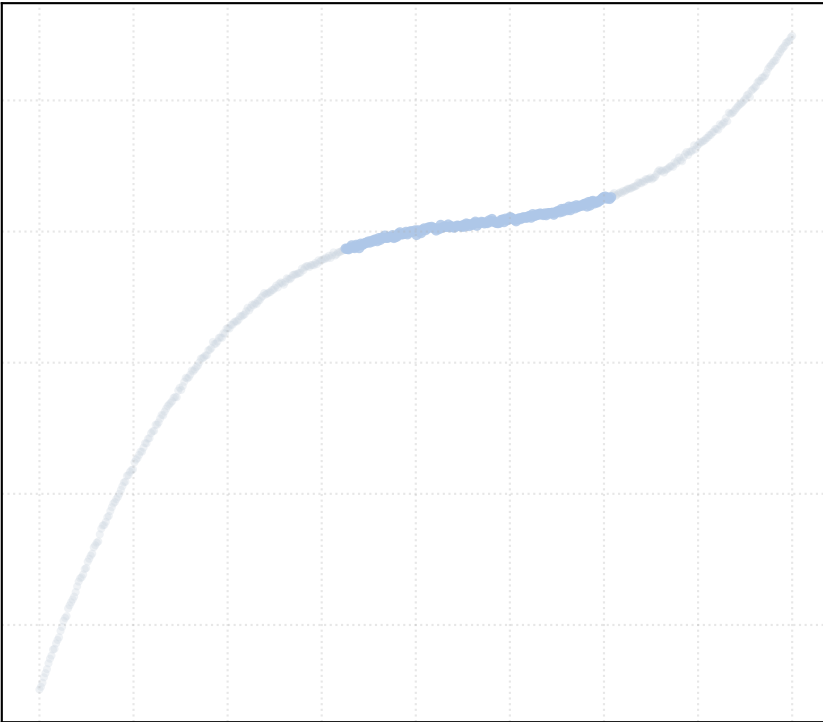
t_percentile = 0.15 (Default) [1 Clusters]



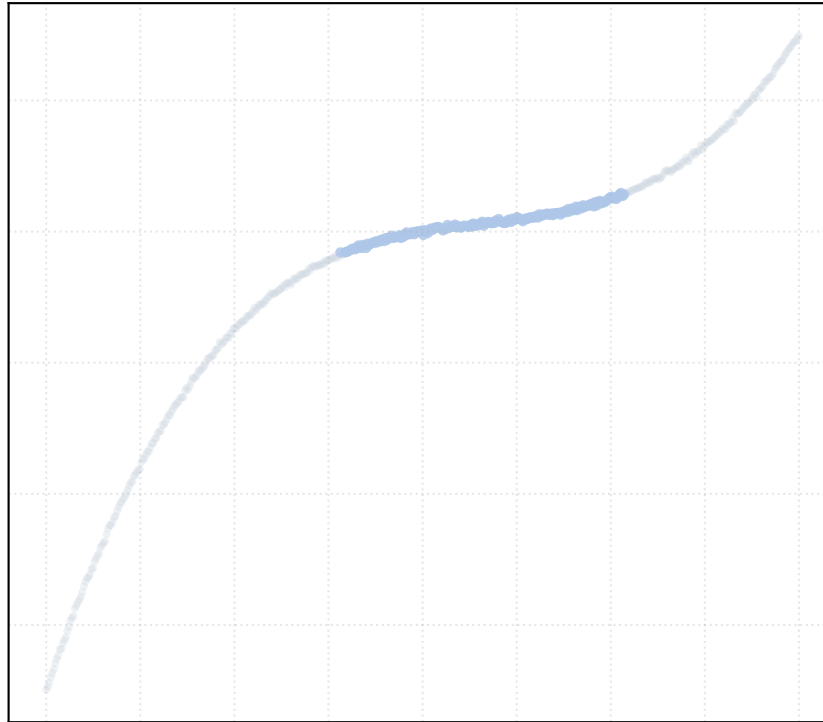
t_percentile = 0.2 [1 Clusters]



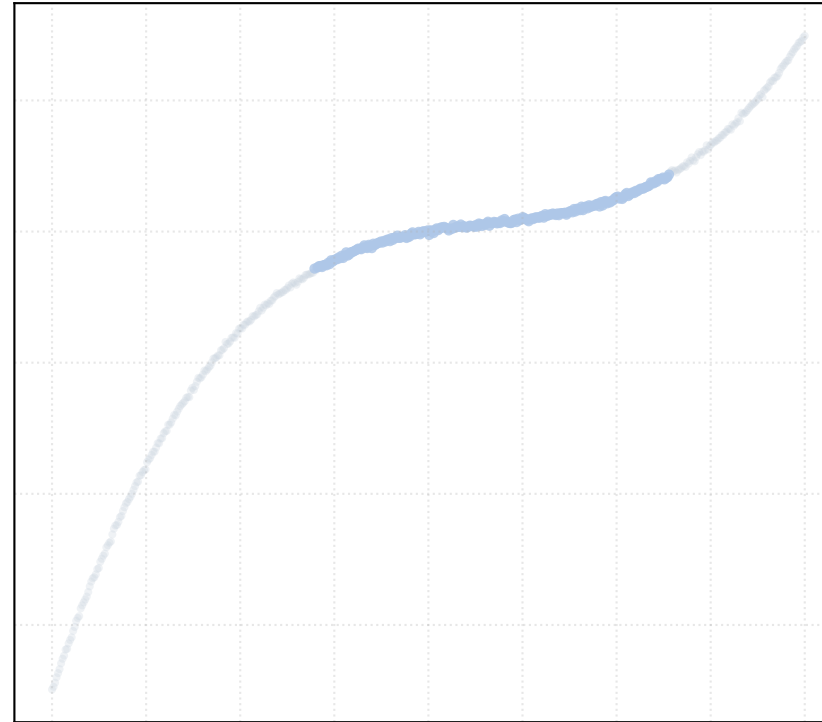
t_percentile = 0.25 [1 Clusters]



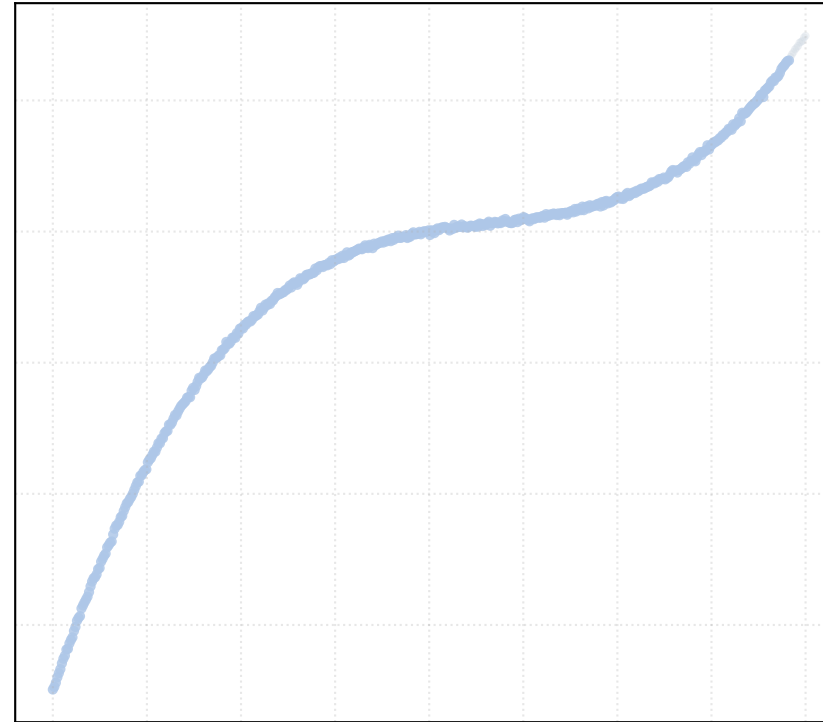
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

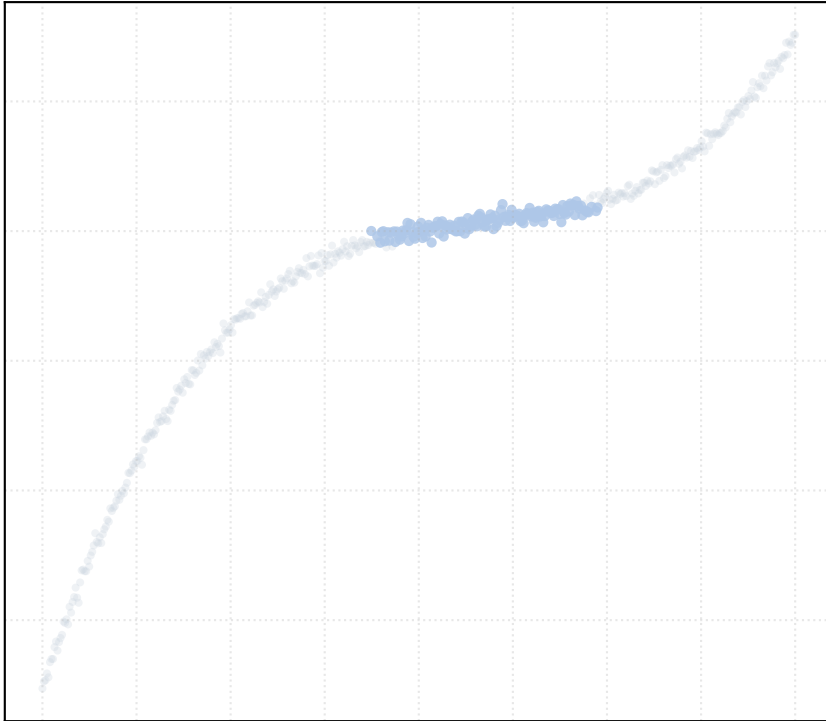


t_percentile = 0.5 [1 Clusters]

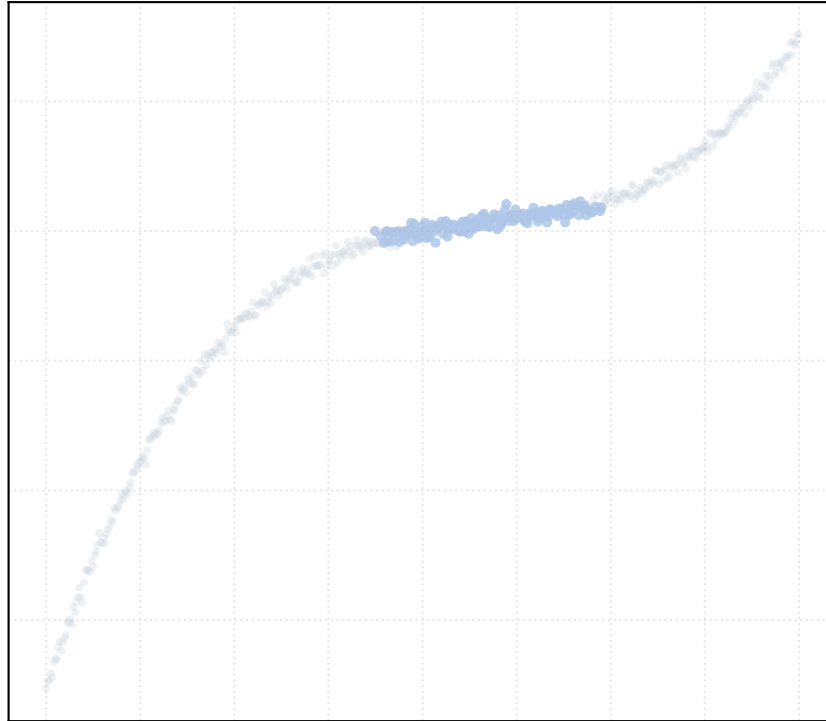


Dataset: cubic_noise_medium | Seed Percentile Sweep (t_percentile)

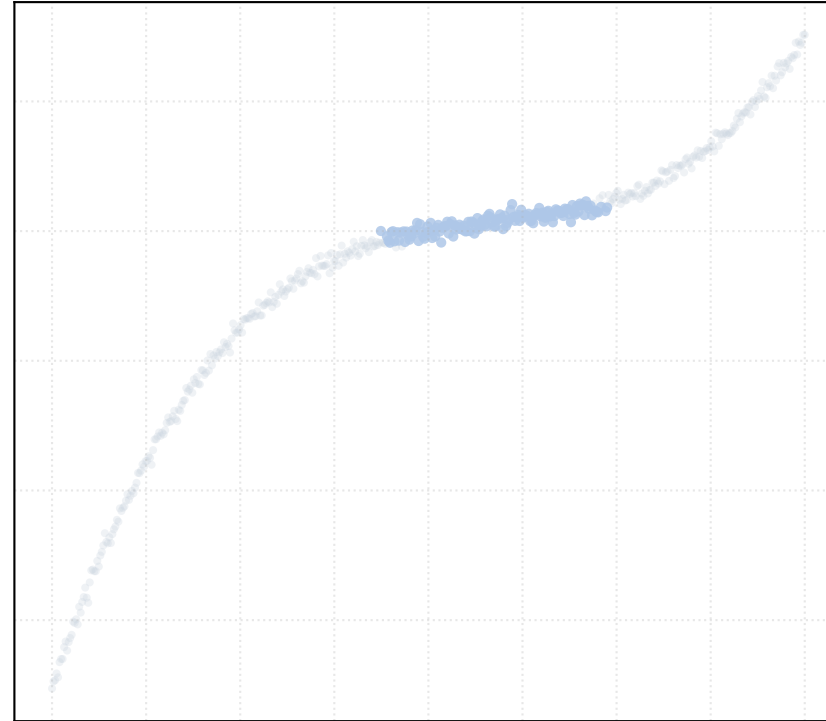
t_percentile = 0.05 [1 Clusters]



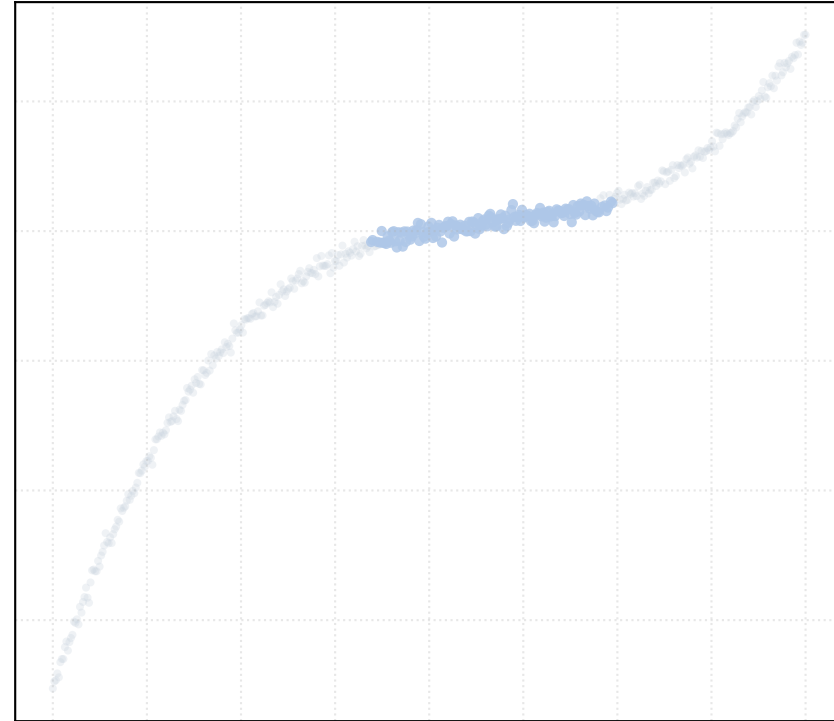
t_percentile = 0.1 [1 Clusters]



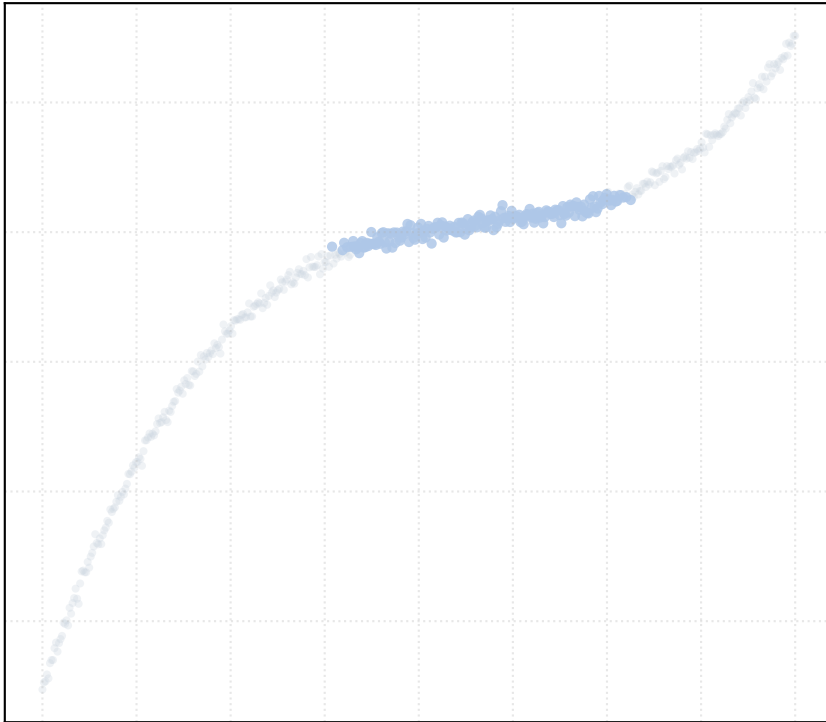
t_percentile = 0.15 (Default) [1 Clusters]



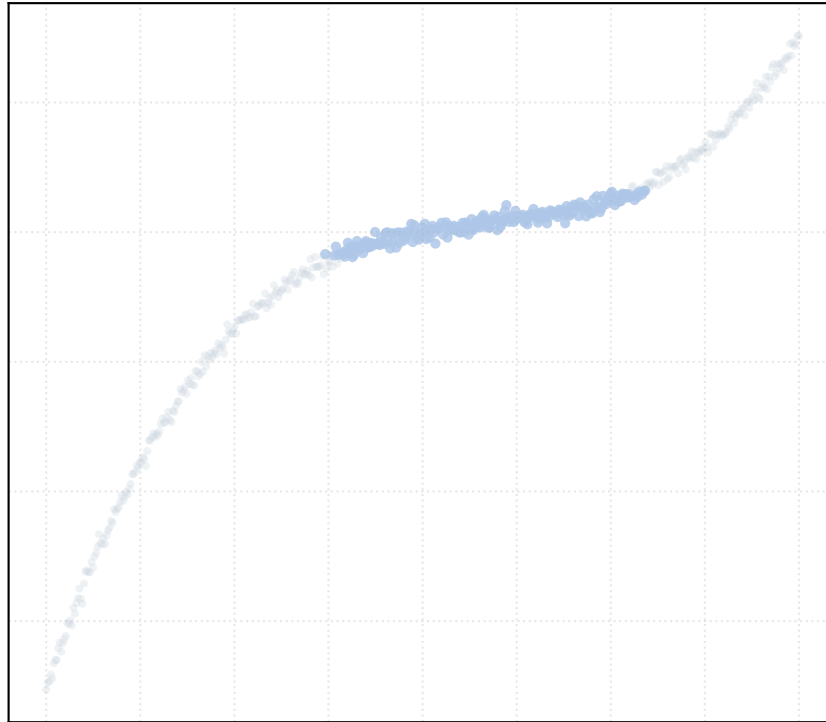
t_percentile = 0.2 [1 Clusters]



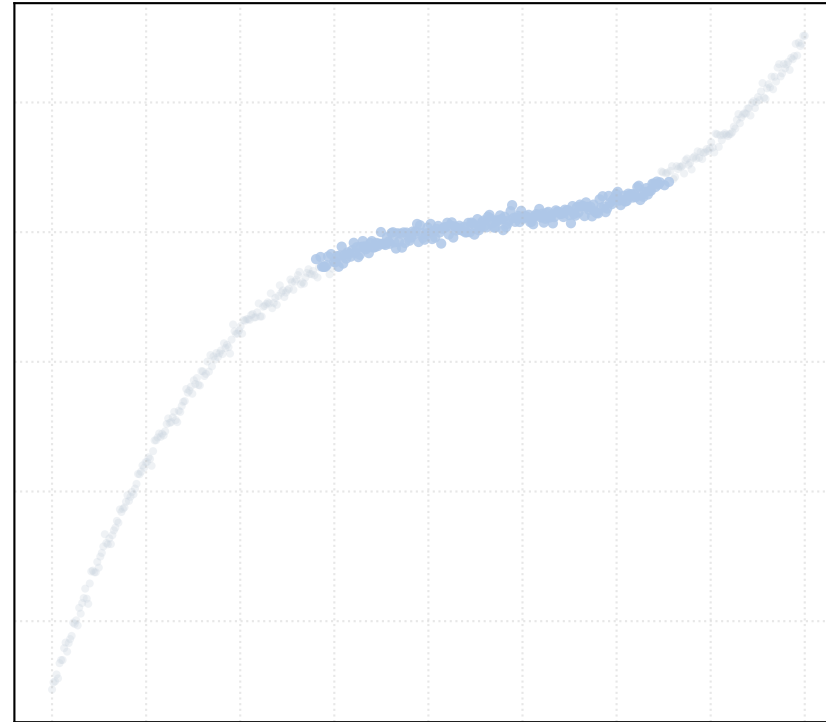
t_percentile = 0.25 [1 Clusters]



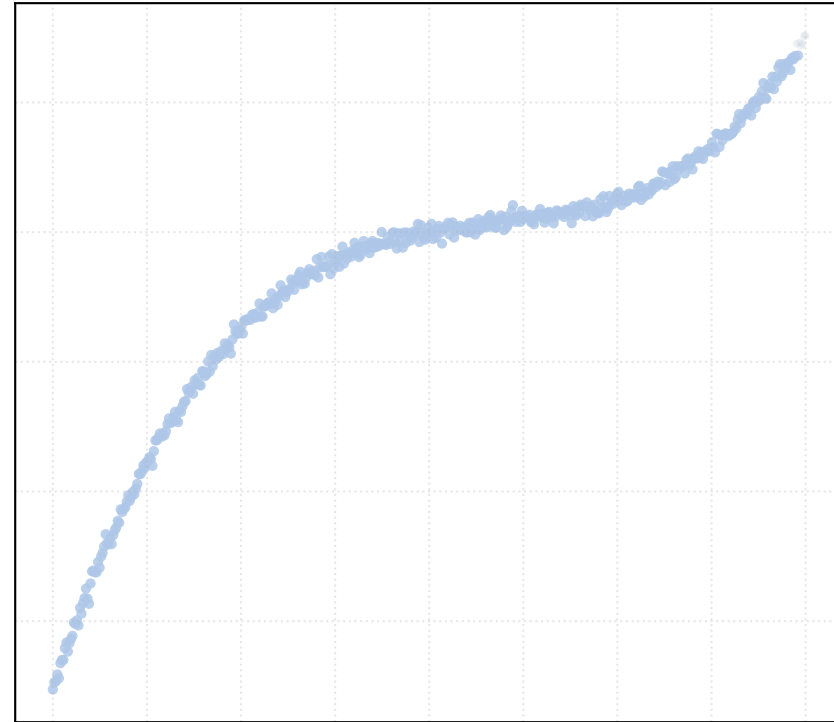
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

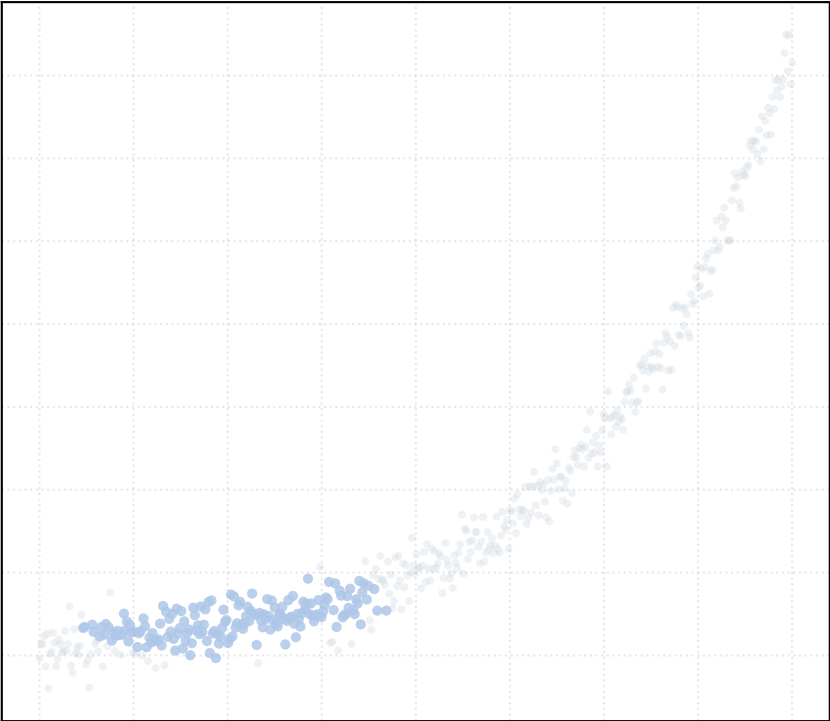


t_percentile = 0.5 [1 Clusters]

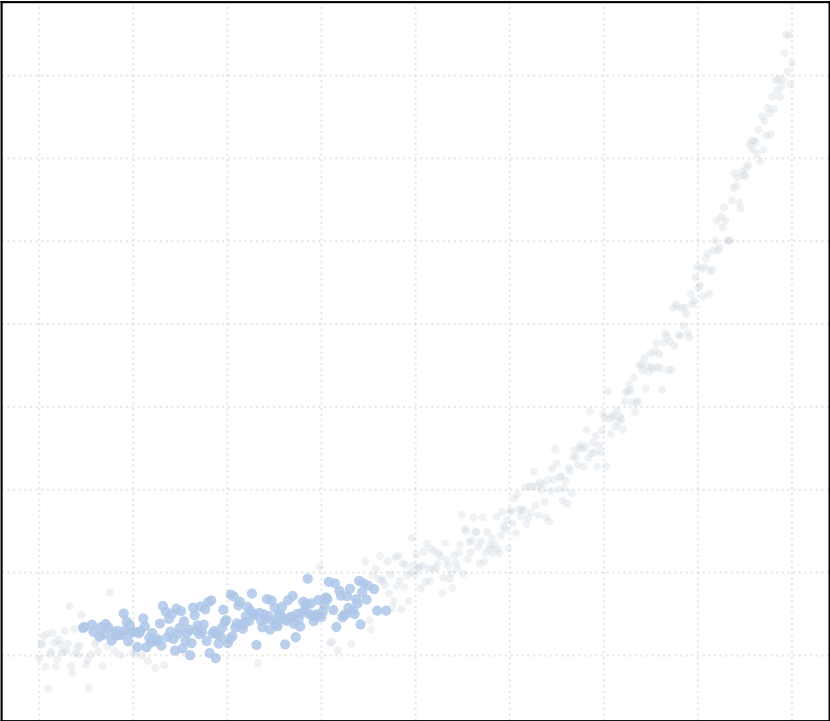


Dataset: exponential_noise_high | Seed Percentile Sweep (t_percentile)

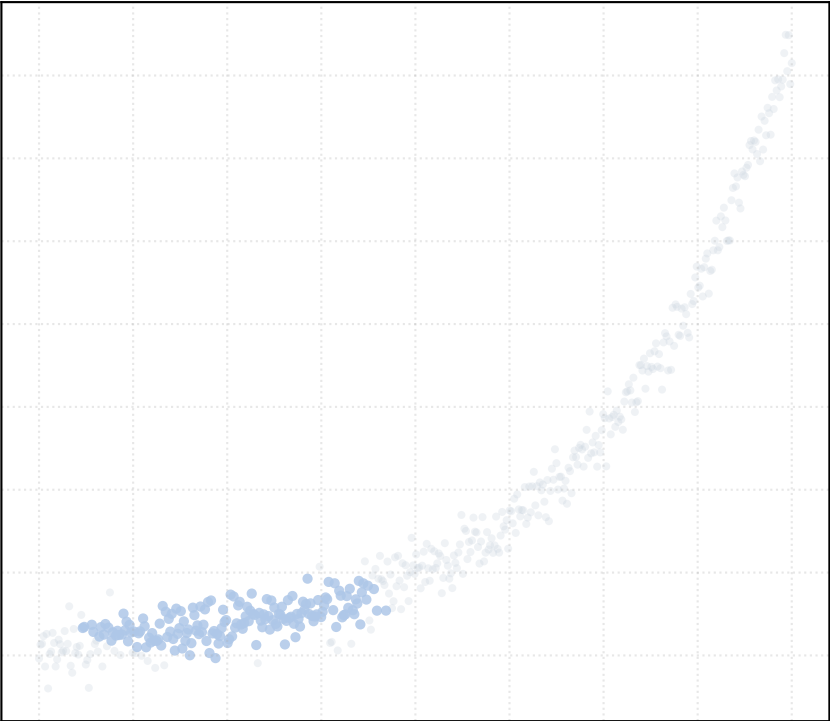
t_percentile = 0.05 [1 Clusters]



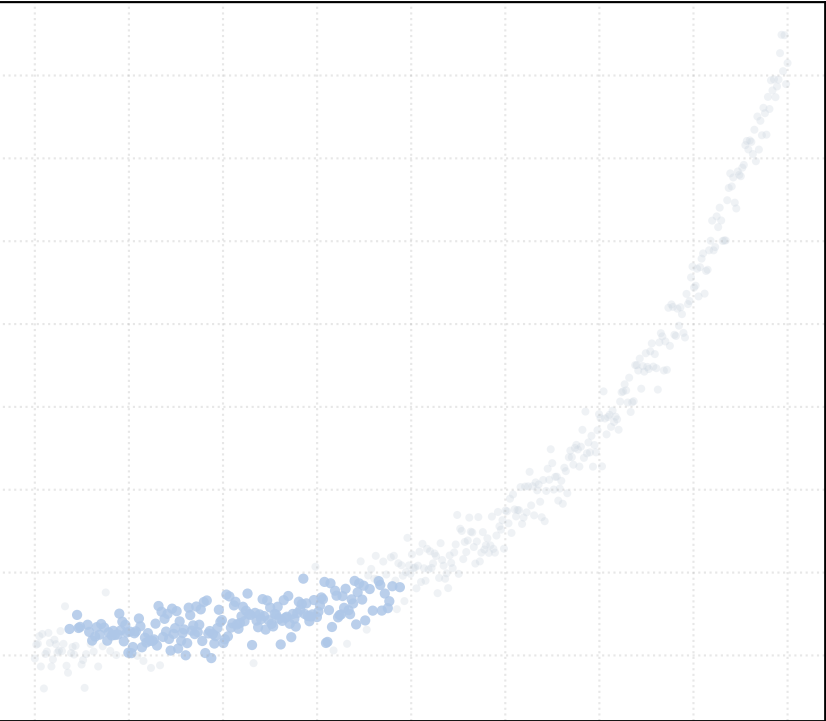
t_percentile = 0.1 [1 Clusters]



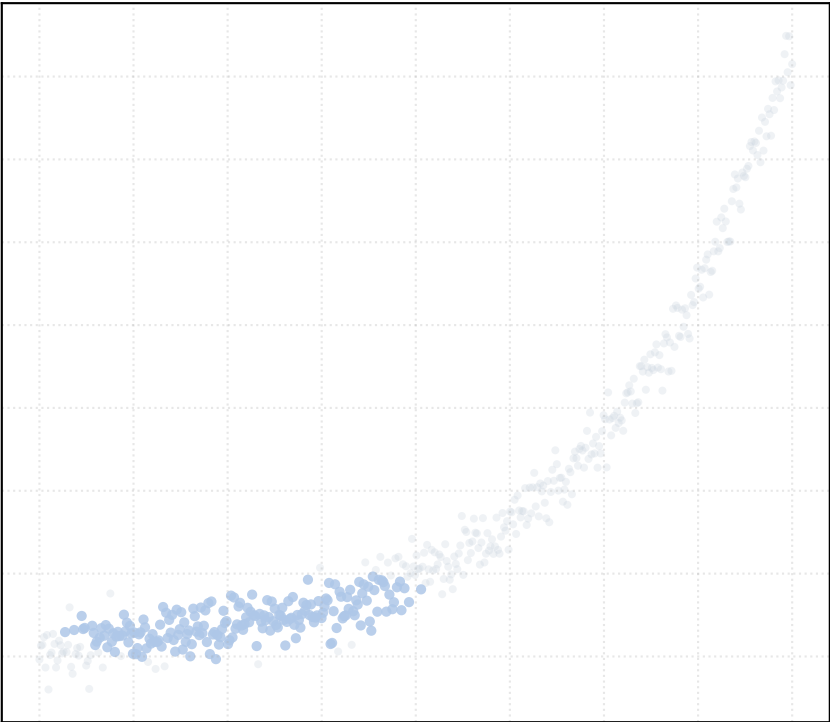
t_percentile = 0.15 (Default) [1 Clusters]



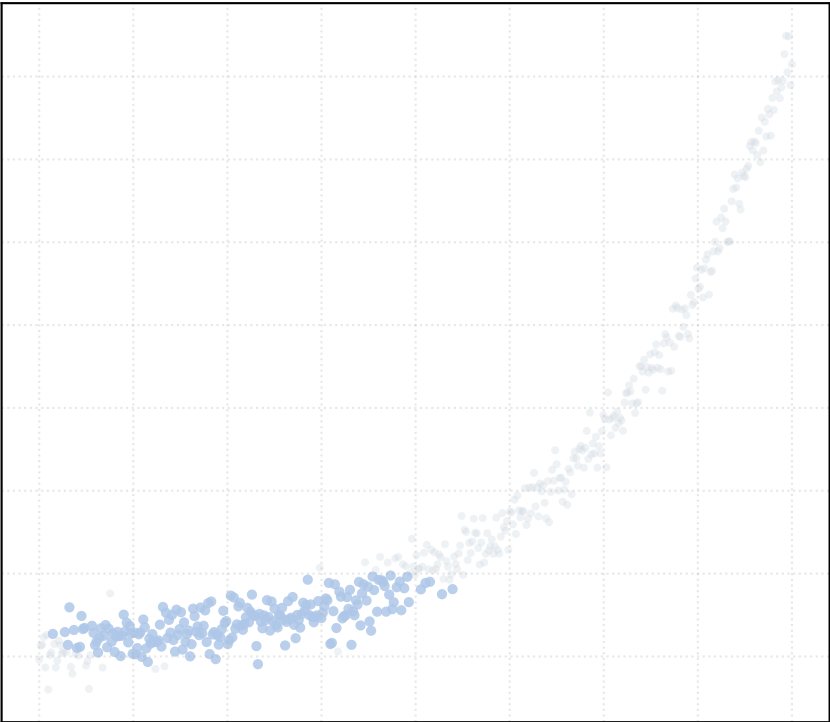
t_percentile = 0.2 [1 Clusters]



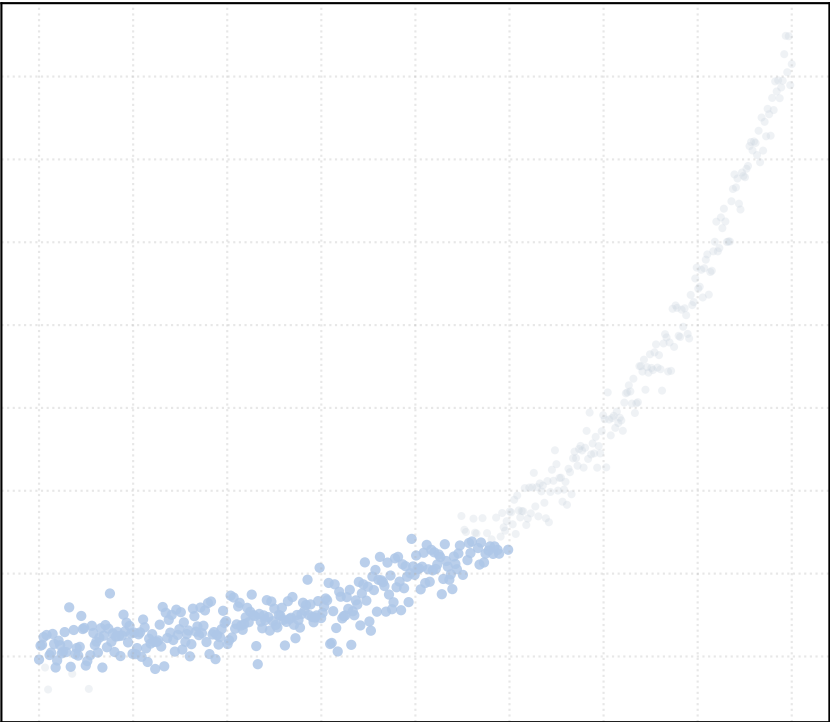
t_percentile = 0.25 [1 Clusters]



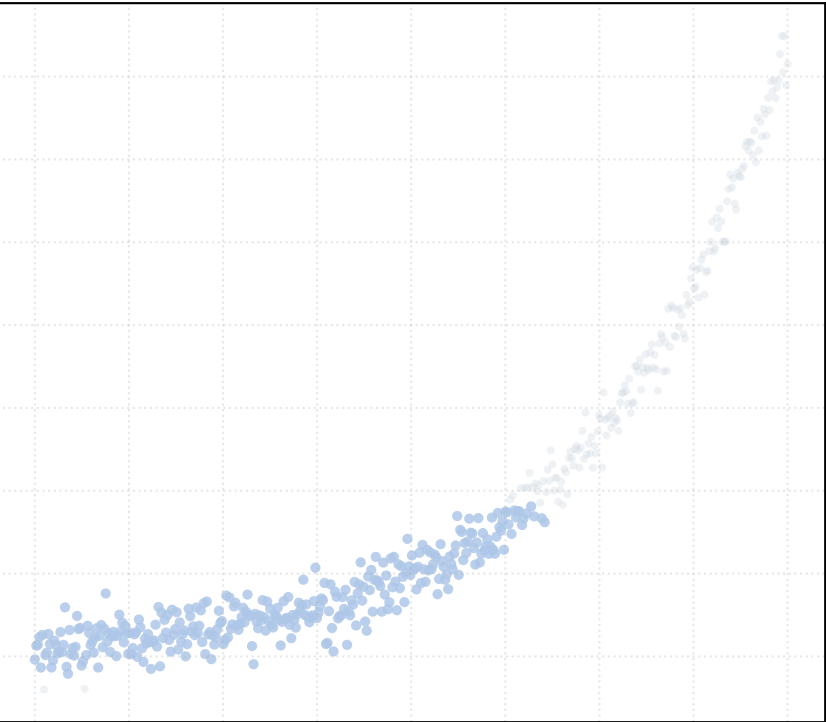
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

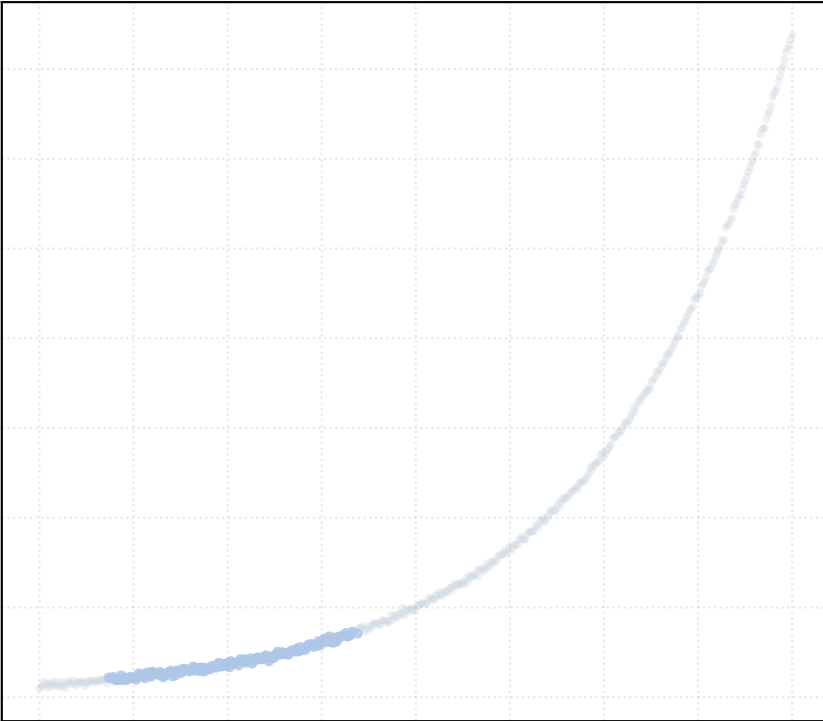


t_percentile = 0.5 [1 Clusters]

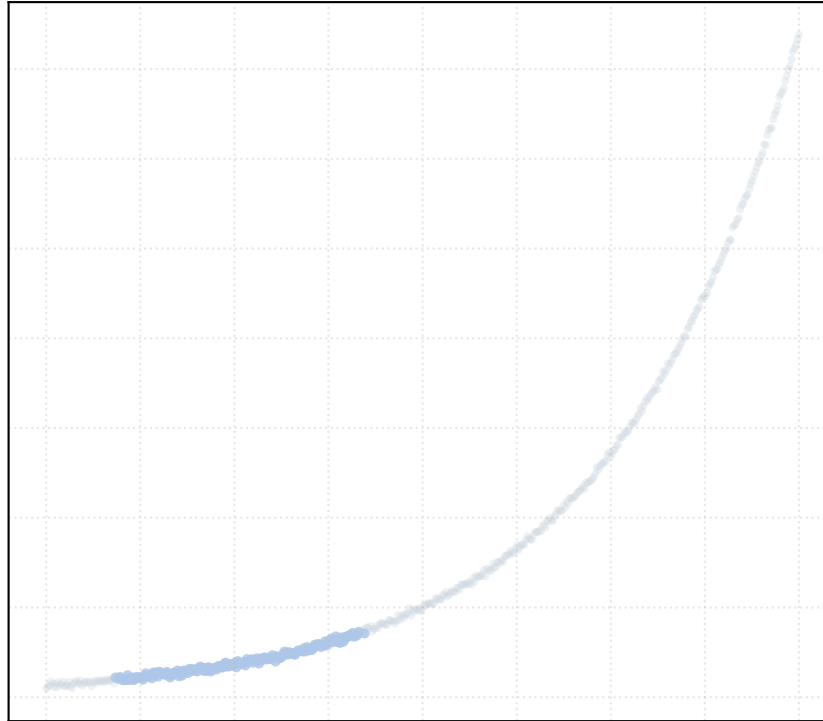


Dataset: exponential_noise_low | Seed Percentile Sweep (t_percentile)

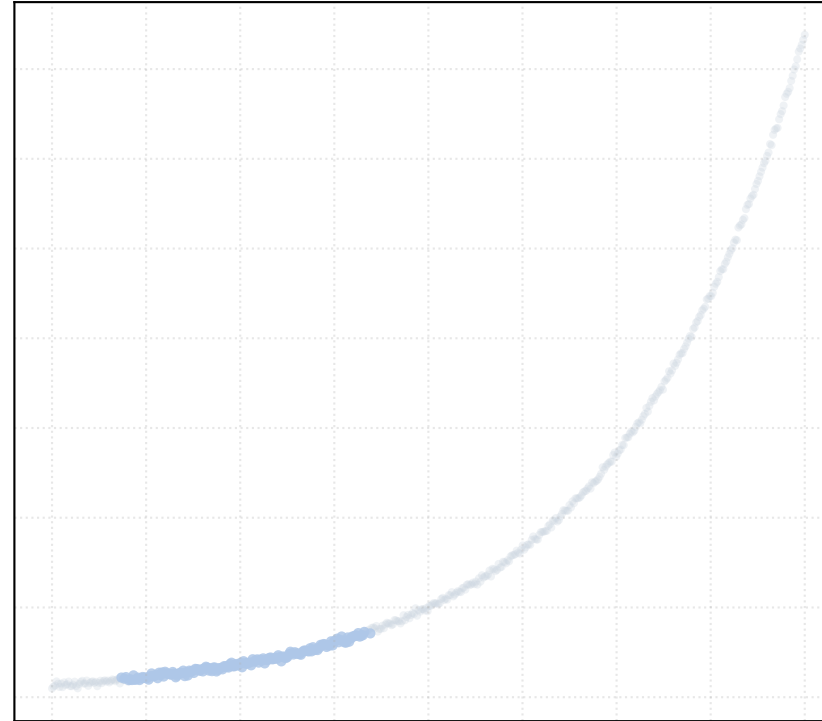
t_percentile = 0.05 [1 Clusters]



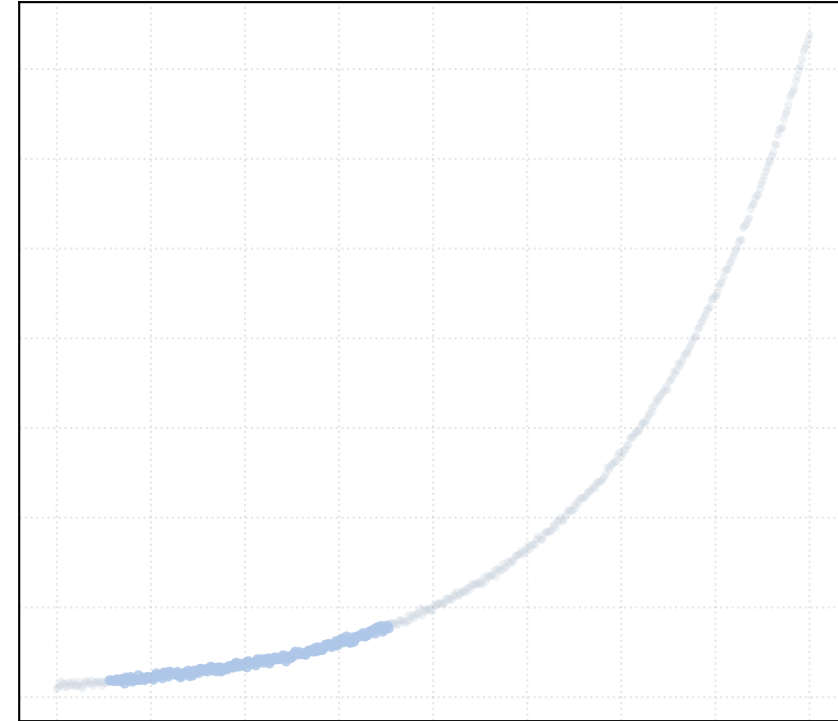
t_percentile = 0.1 [1 Clusters]



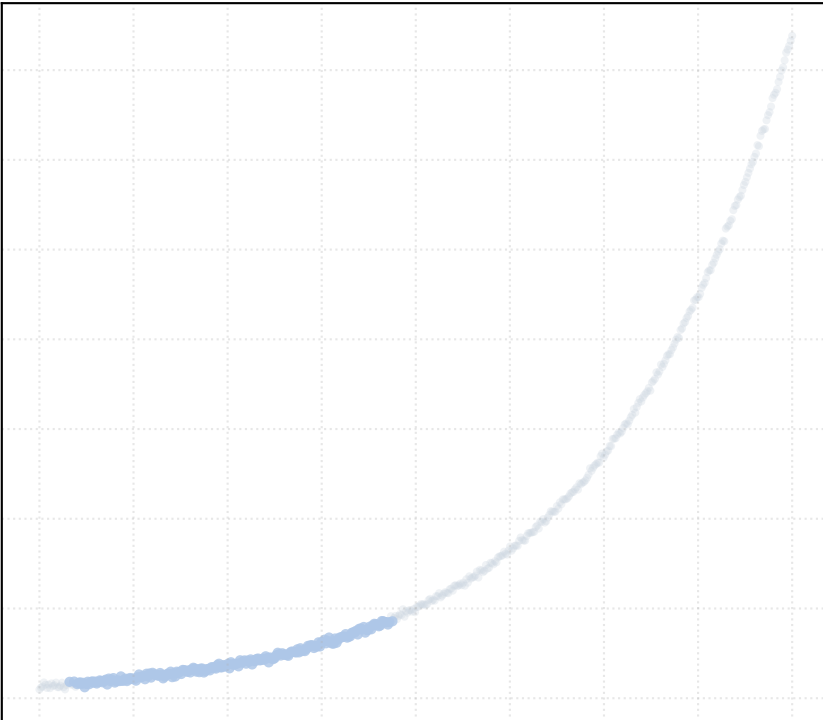
t_percentile = 0.15 (Default) [1 Clusters]



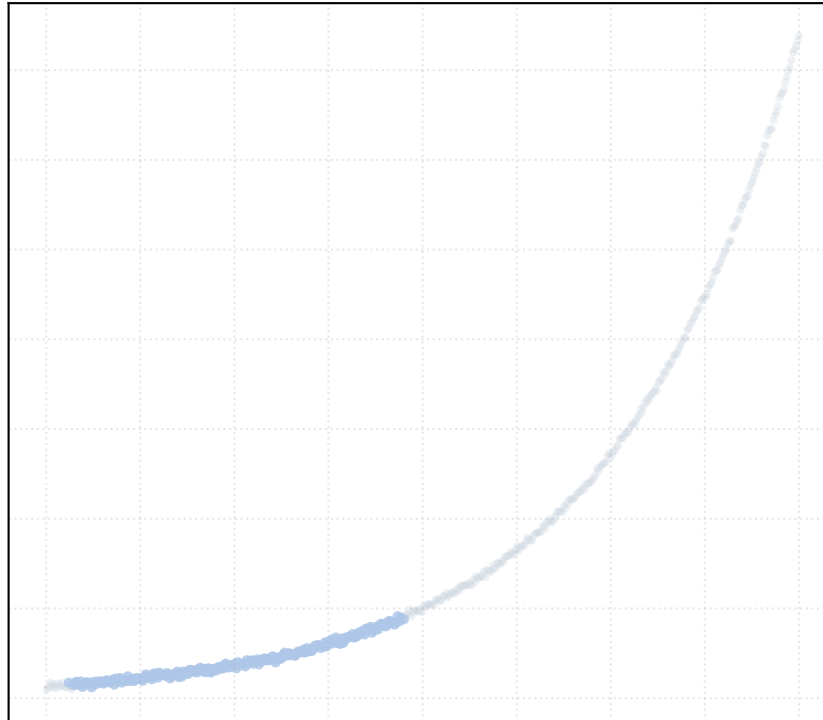
t_percentile = 0.2 [1 Clusters]



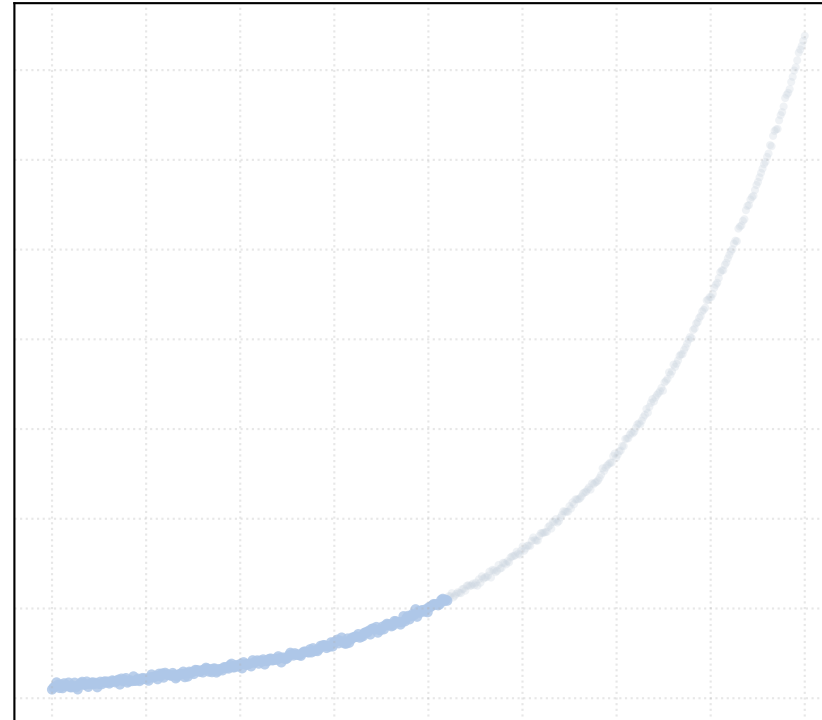
t_percentile = 0.25 [1 Clusters]



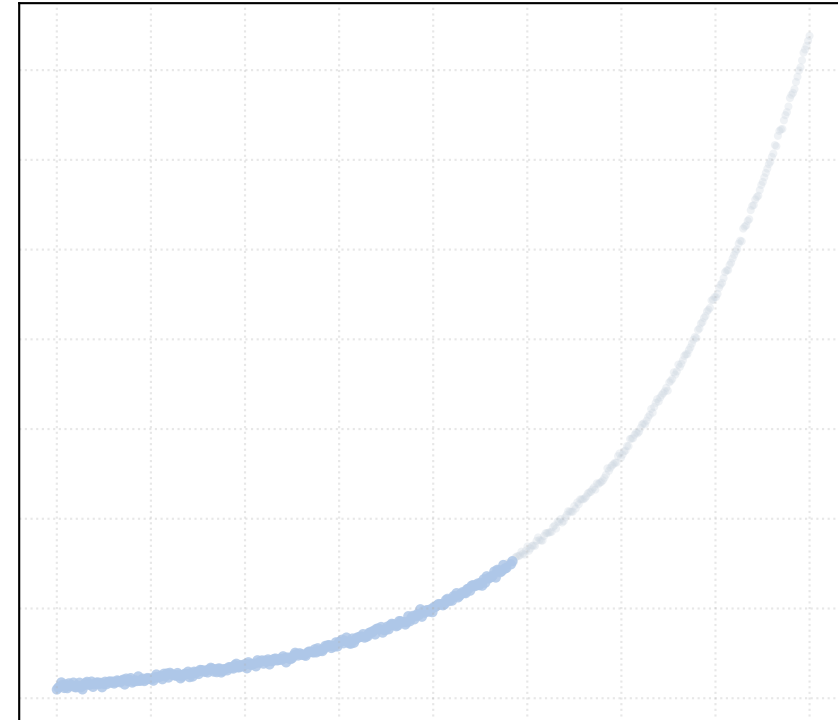
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

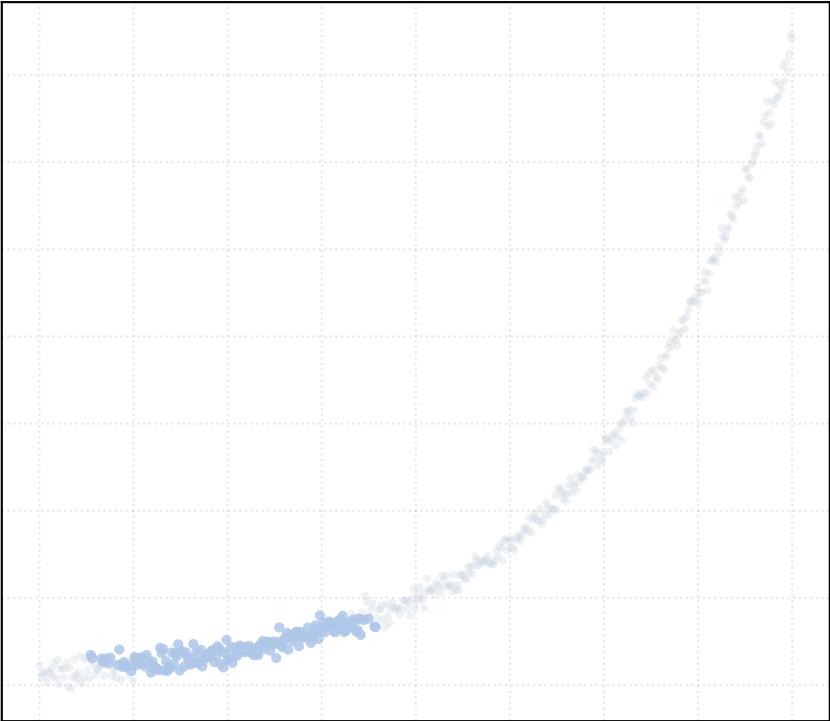


t_percentile = 0.5 [1 Clusters]

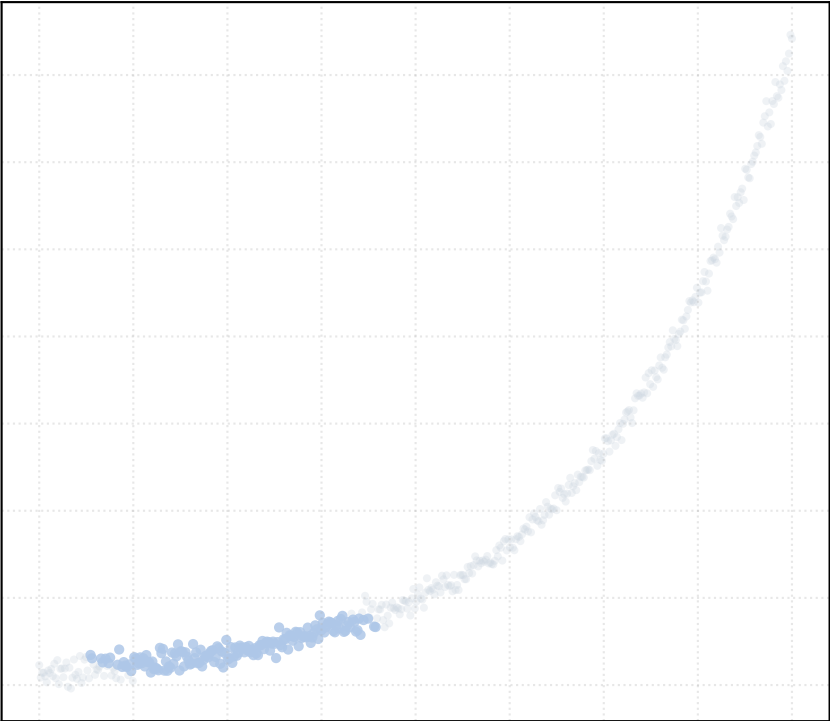


Dataset: exponential_noise_medium | Seed Percentile Sweep (t_percentile)

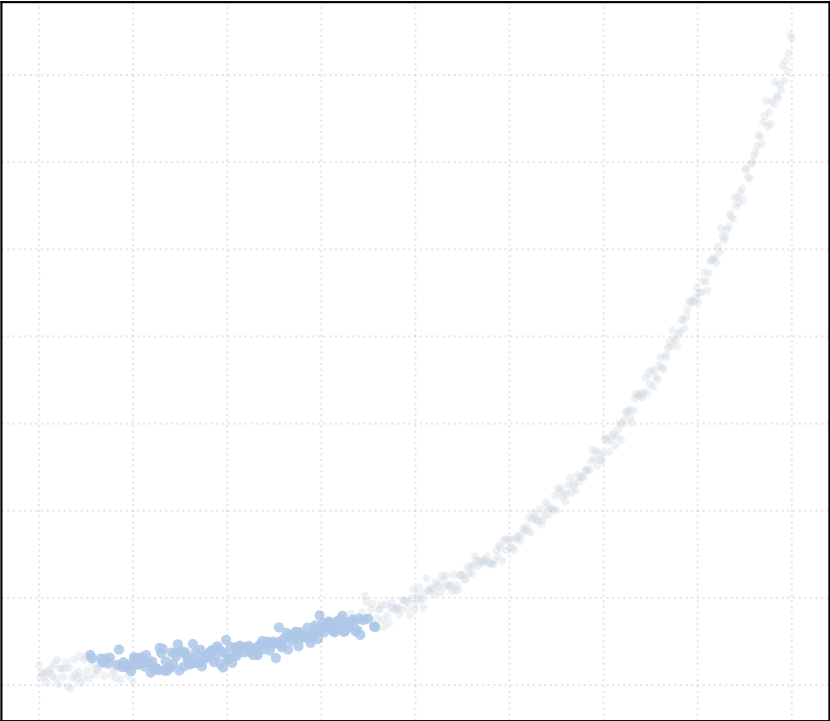
t_percentile = 0.05 [1 Clusters]



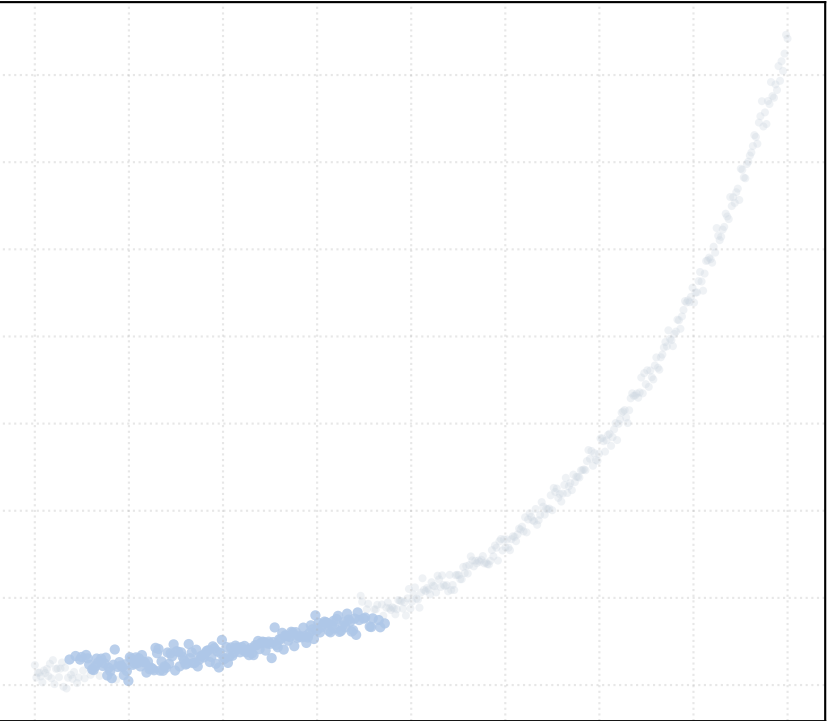
t_percentile = 0.1 [1 Clusters]



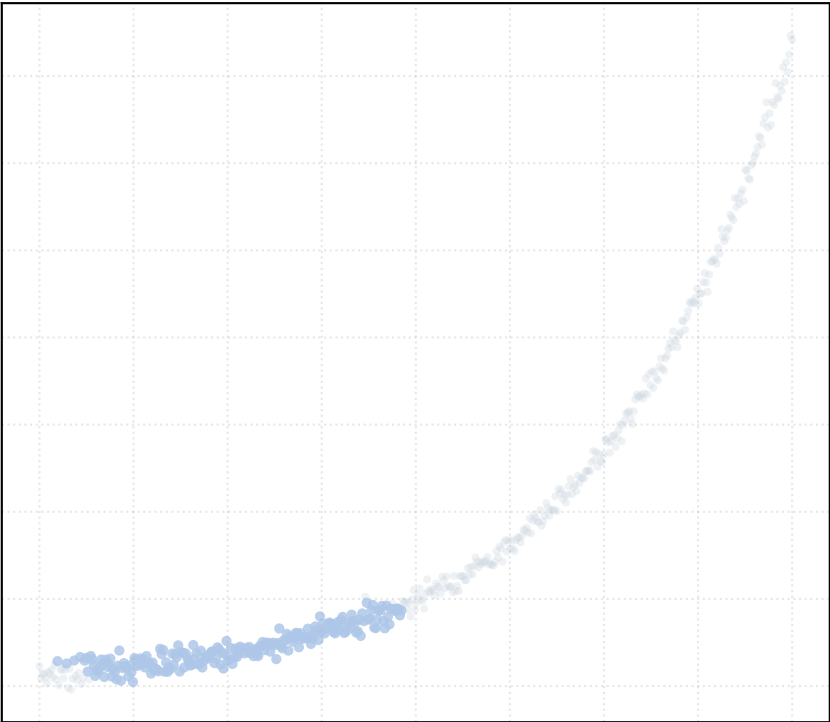
t_percentile = 0.15 (Default) [1 Clusters]



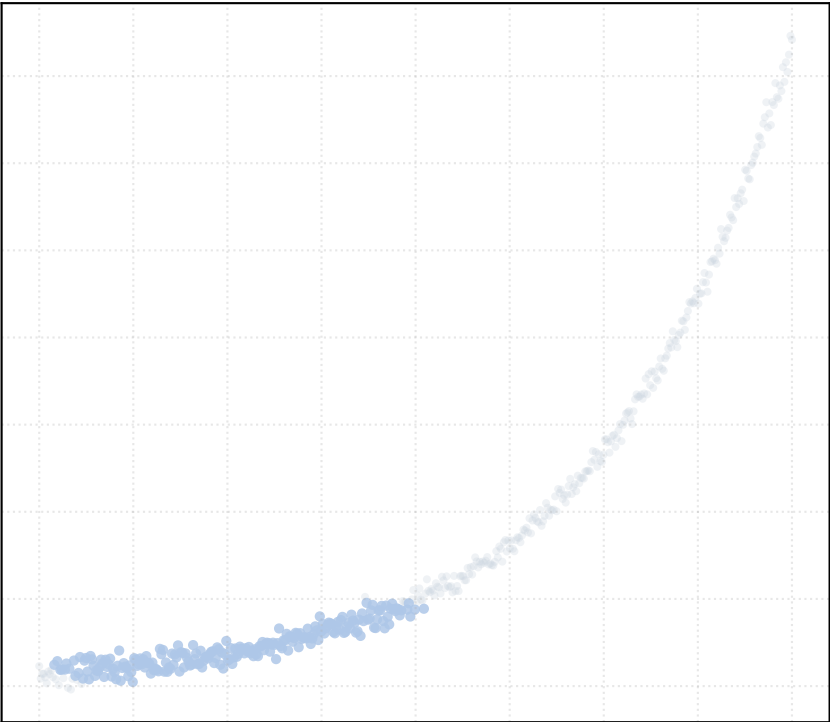
t_percentile = 0.2 [1 Clusters]



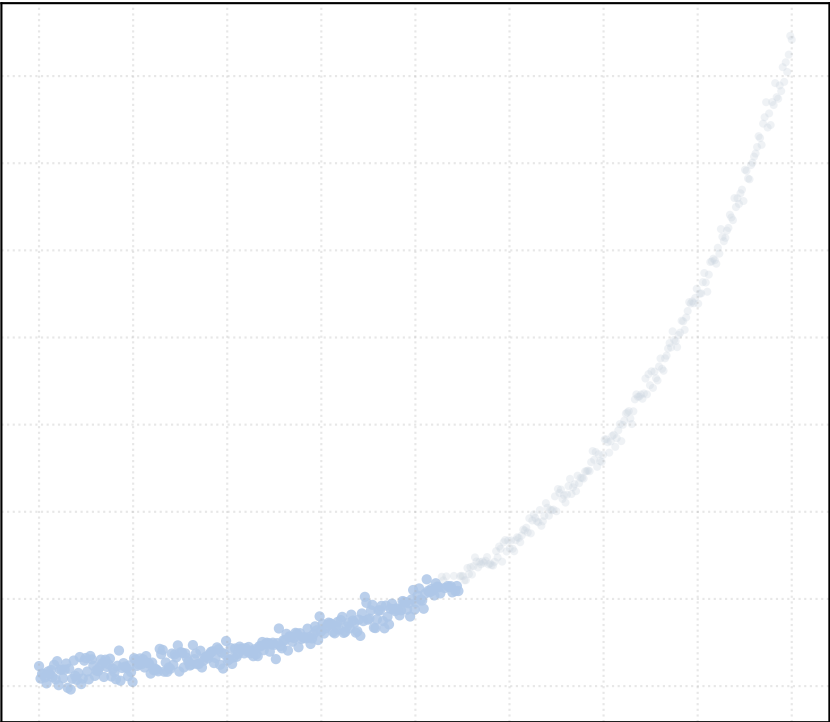
t_percentile = 0.25 [1 Clusters]



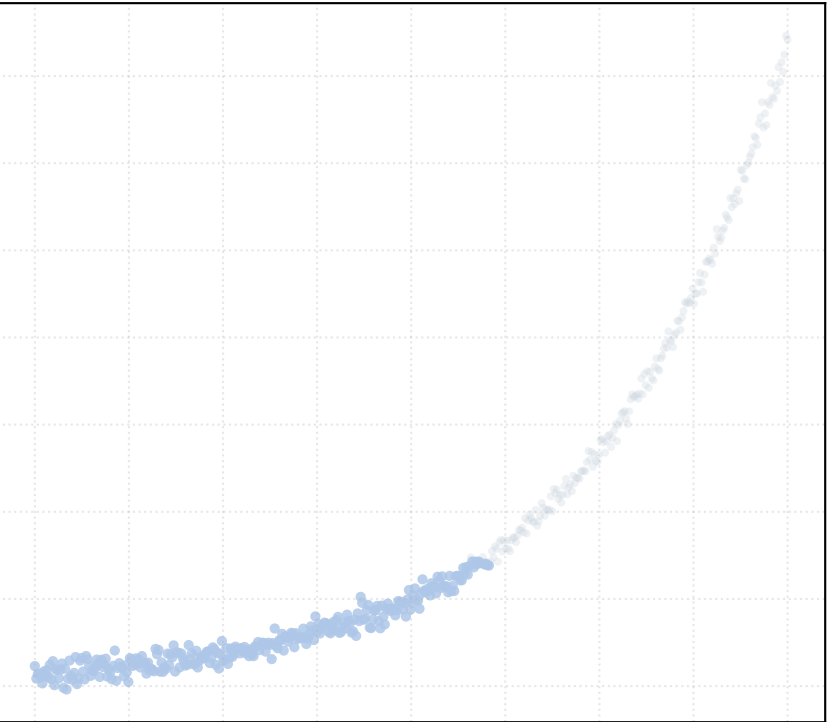
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]



t_percentile = 0.5 [1 Clusters]

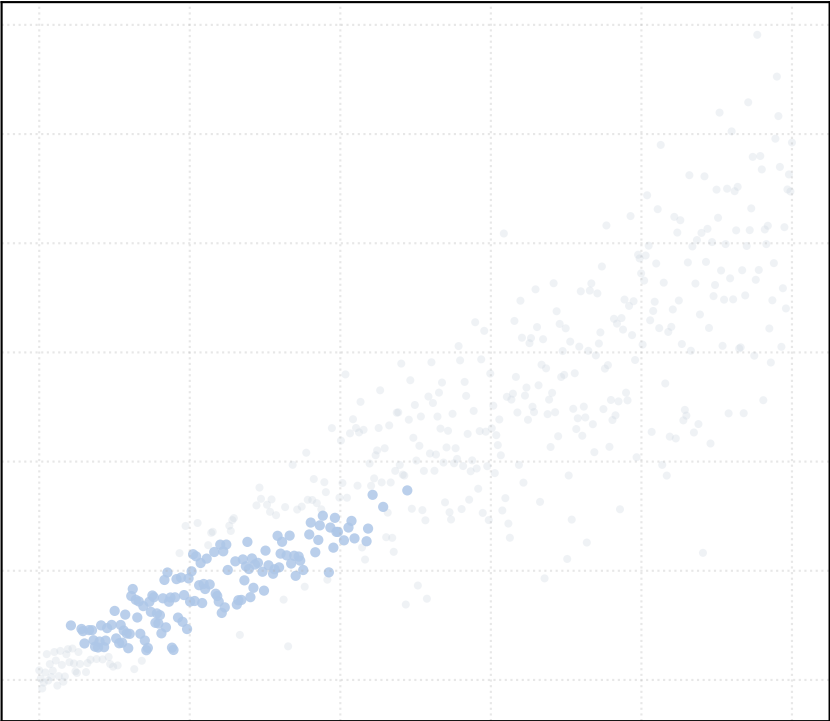


Dataset: heteroscedastic_noise_high | Seed Percentile Sweep (t_percentile)

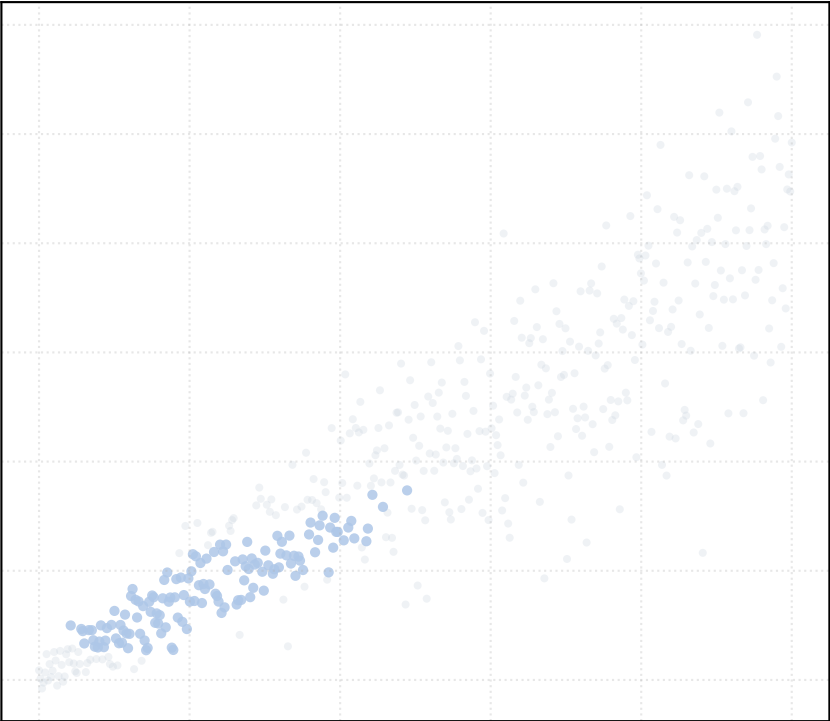
t_percentile = 0.05 [1 Clusters]



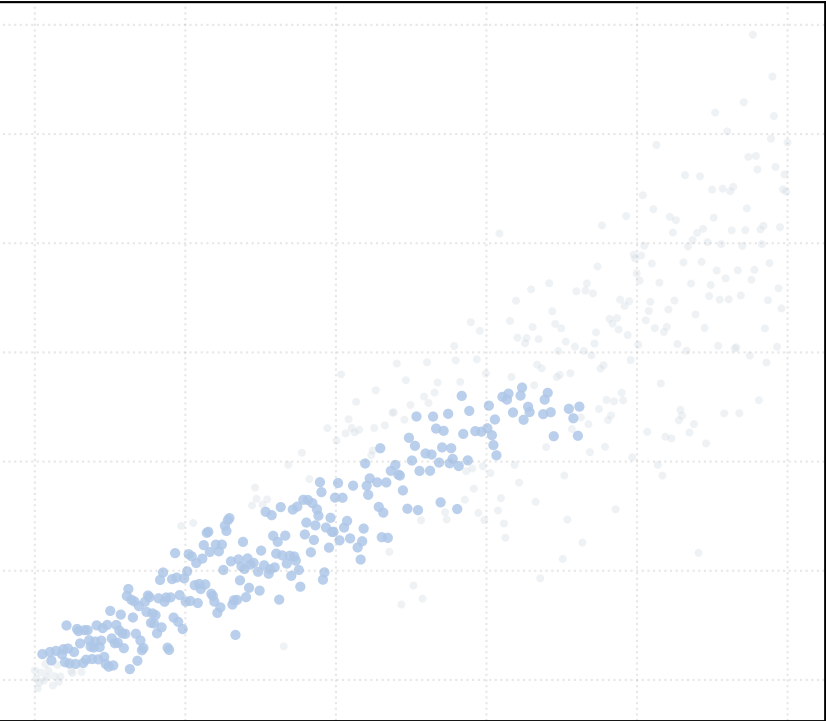
t_percentile = 0.1 [1 Clusters]



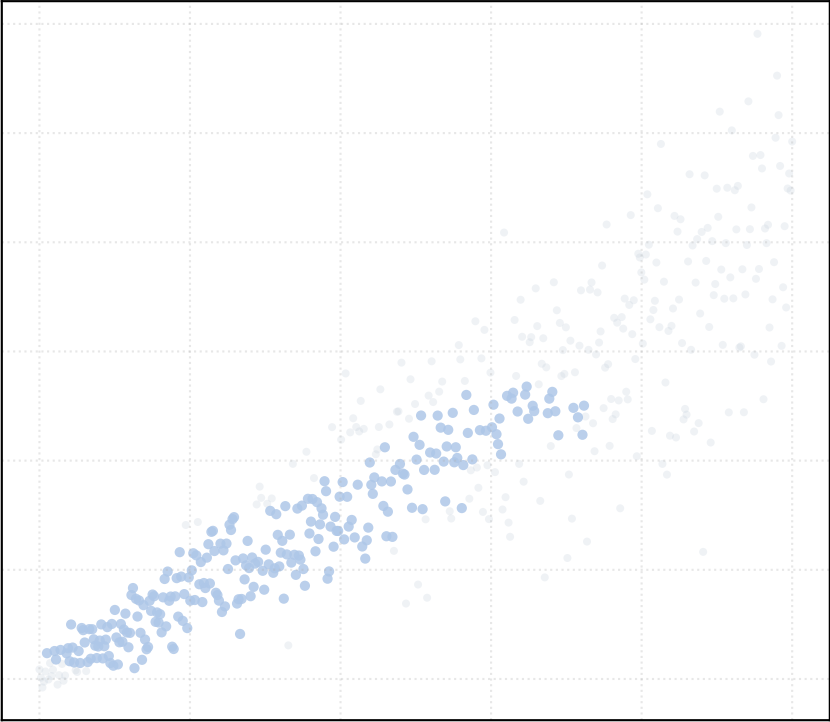
t_percentile = 0.15 (Default) [1 Clusters]



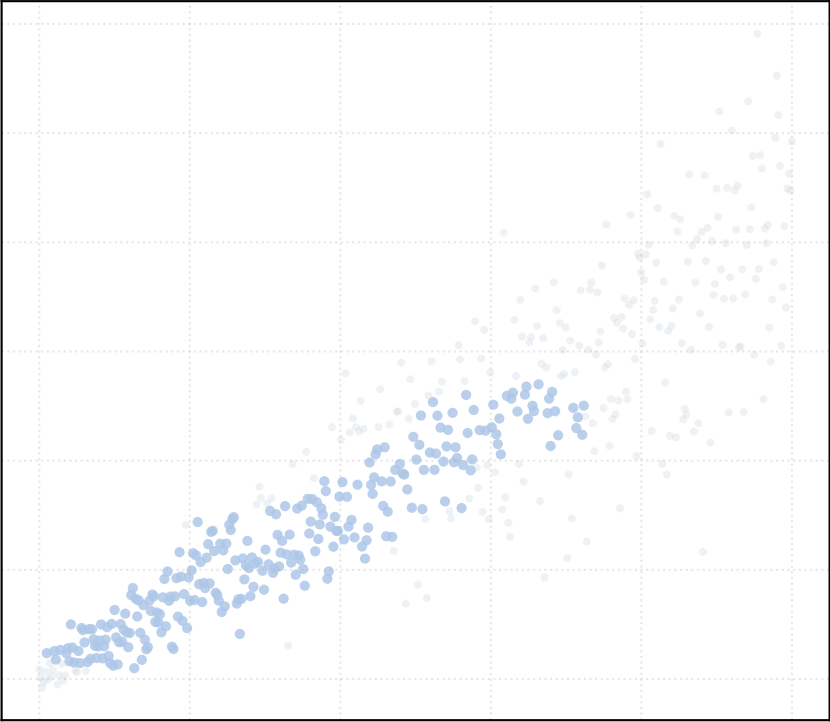
t_percentile = 0.2 [1 Clusters]



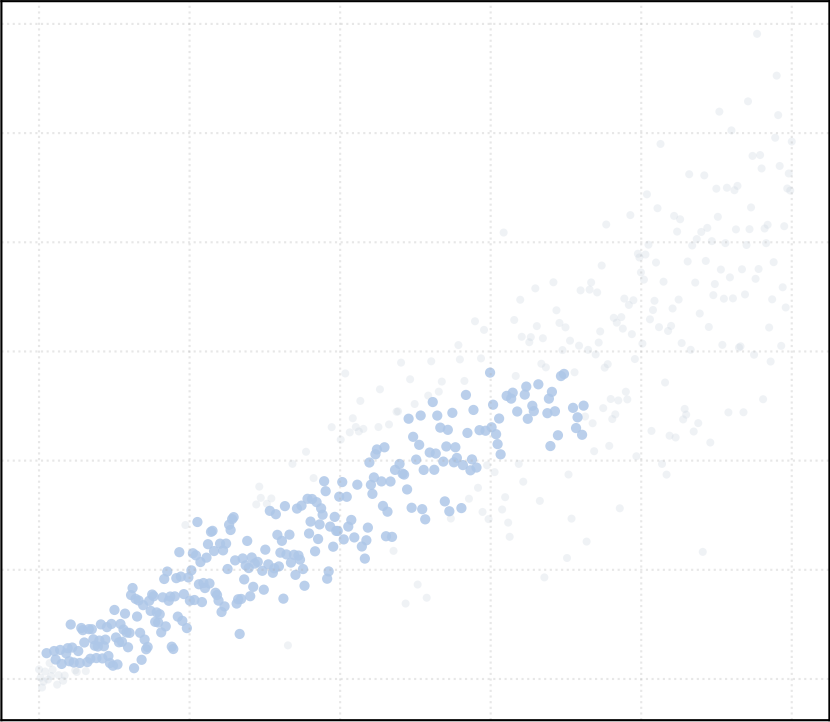
t_percentile = 0.25 [1 Clusters]



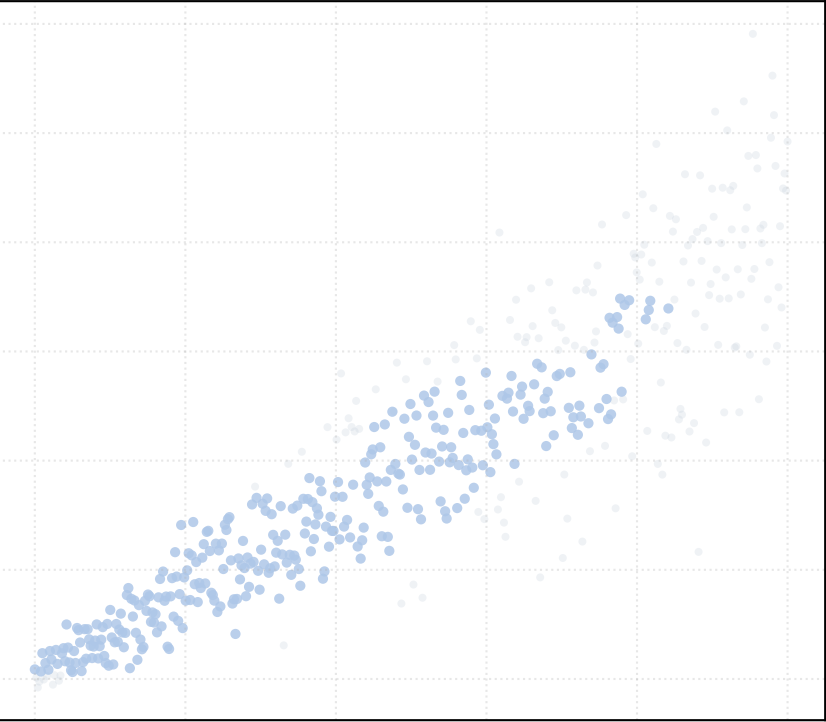
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

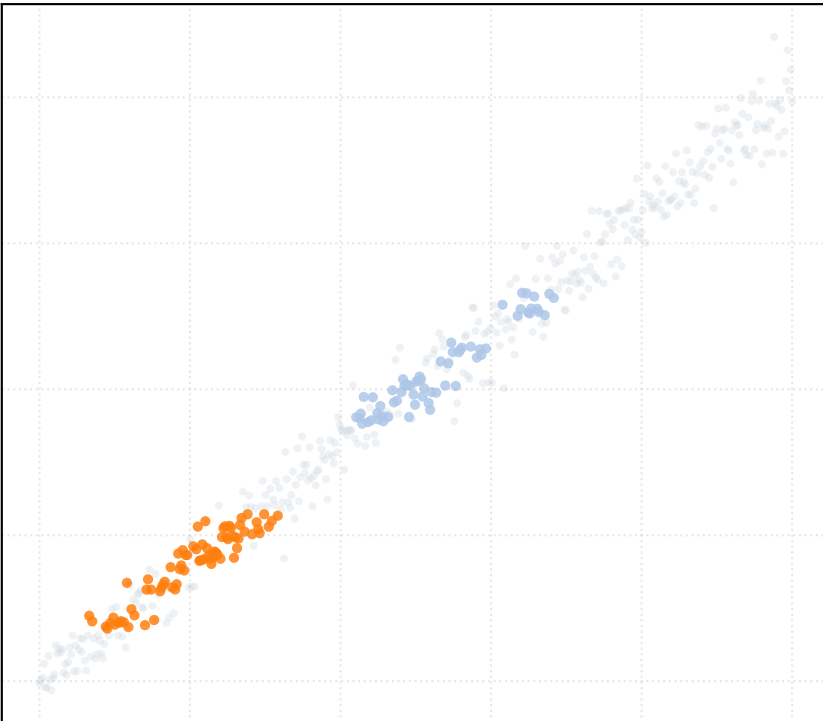


t_percentile = 0.5 [1 Clusters]

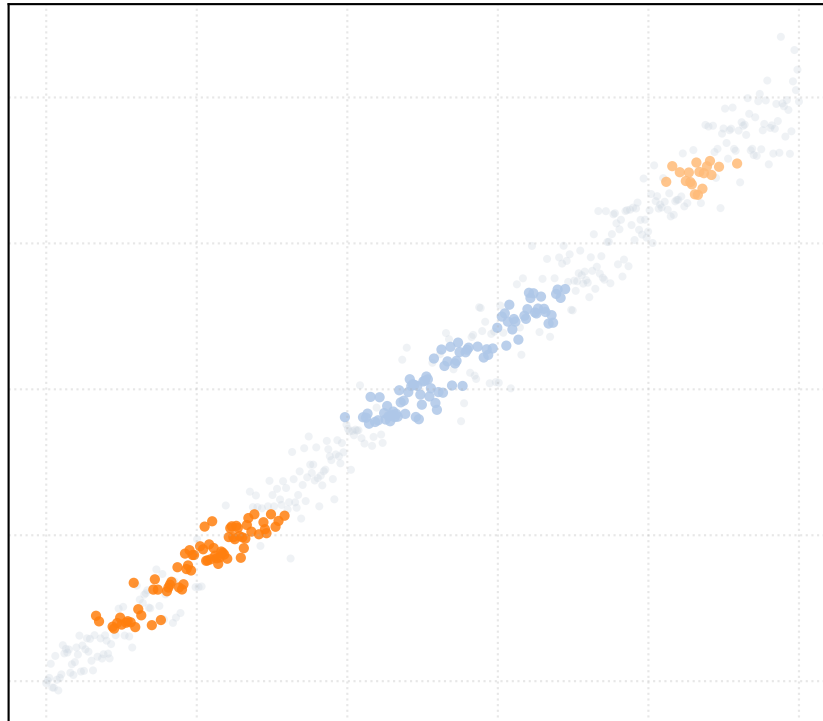


Dataset: heteroscedastic_noise_low | Seed Percentile Sweep (t_percentile)

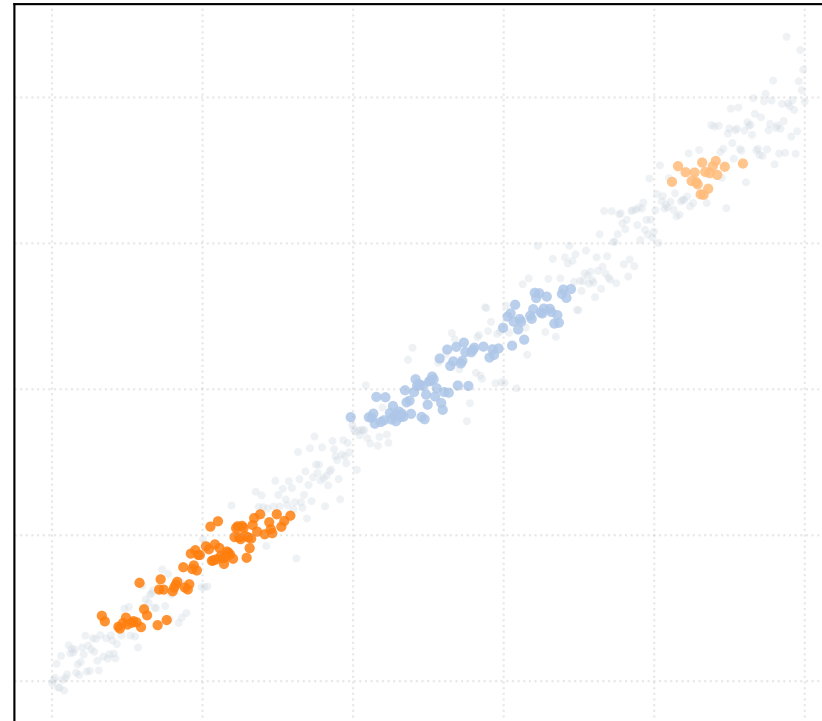
t_percentile = 0.05 [2 Clusters]



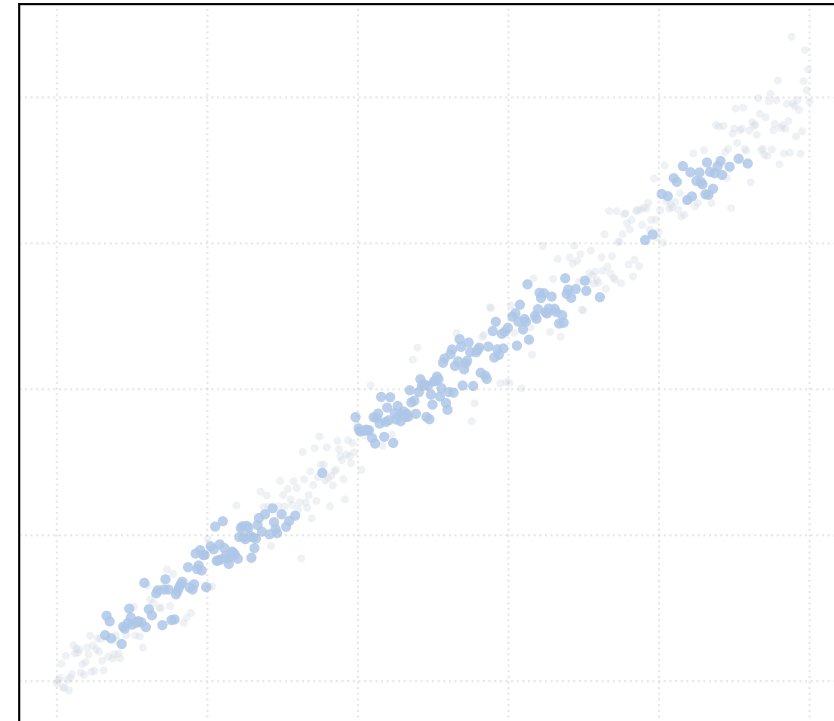
t_percentile = 0.1 [3 Clusters]



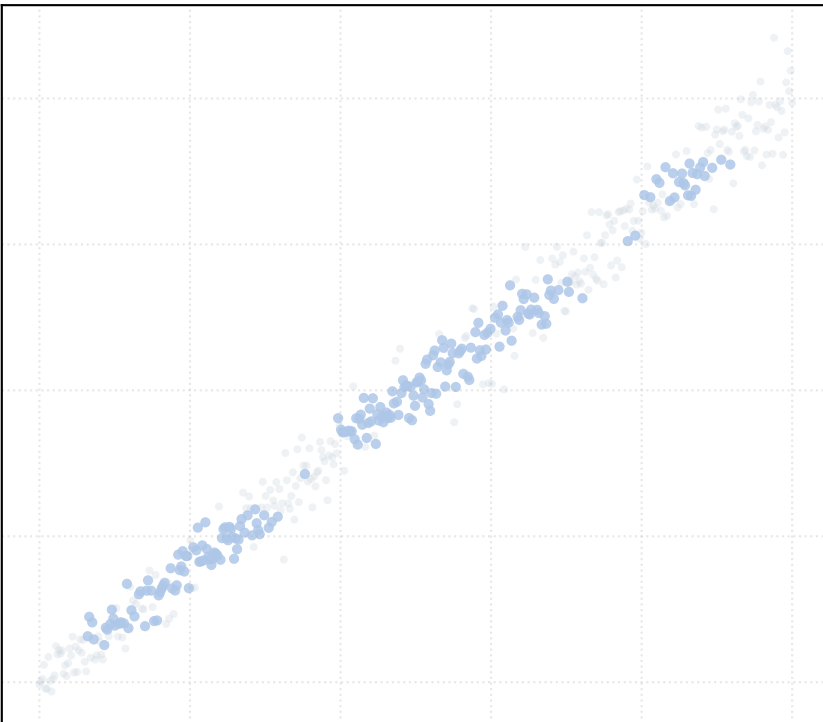
t_percentile = 0.15 (Default) [3 Clusters]



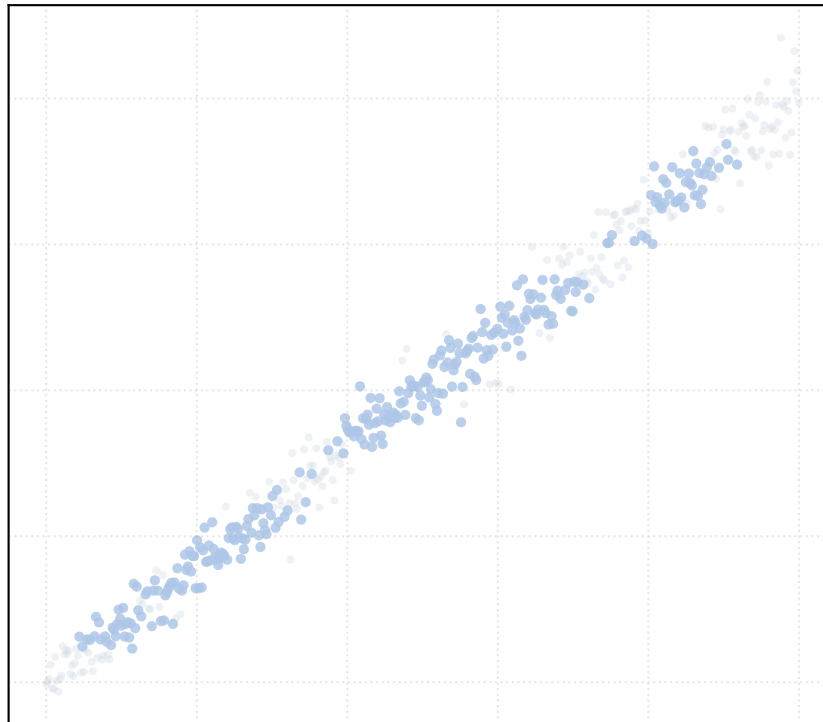
t_percentile = 0.2 [1 Clusters]



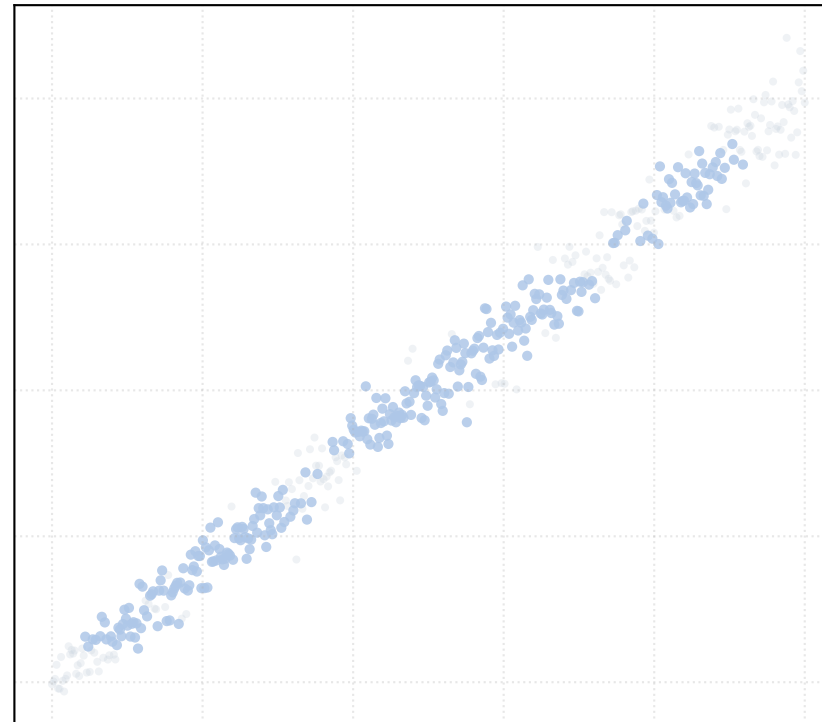
t_percentile = 0.25 [1 Clusters]



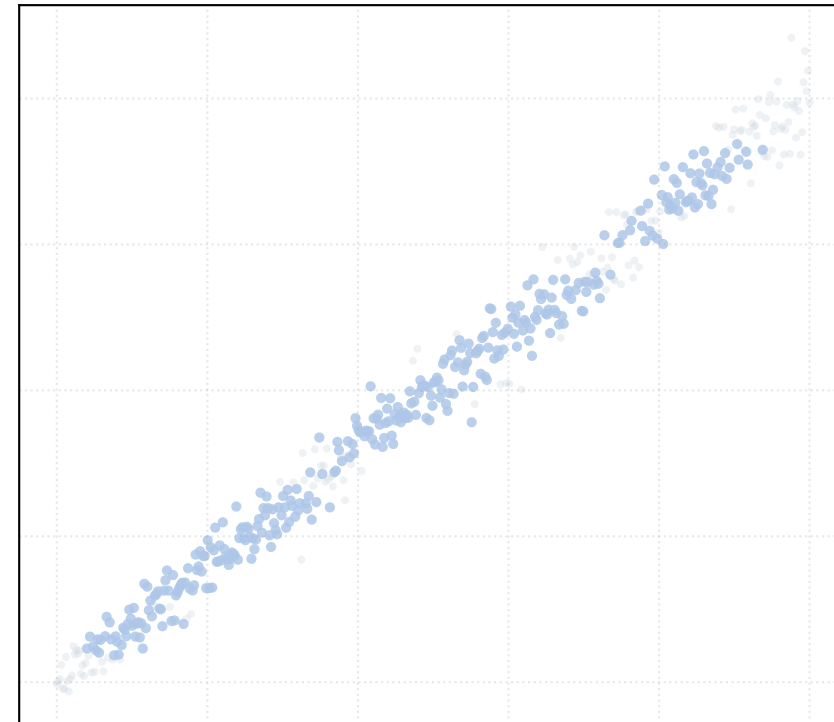
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

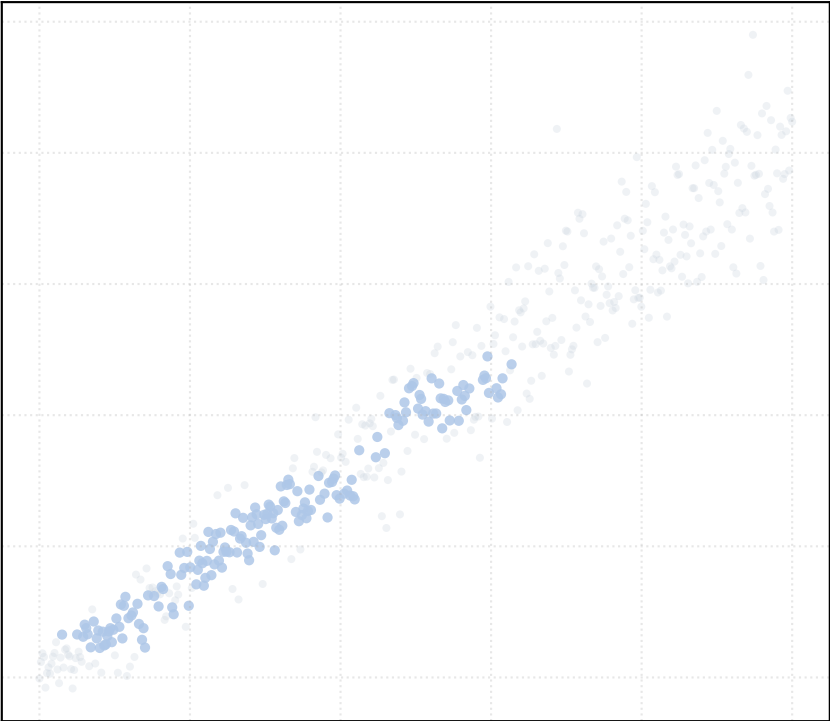


t_percentile = 0.5 [1 Clusters]

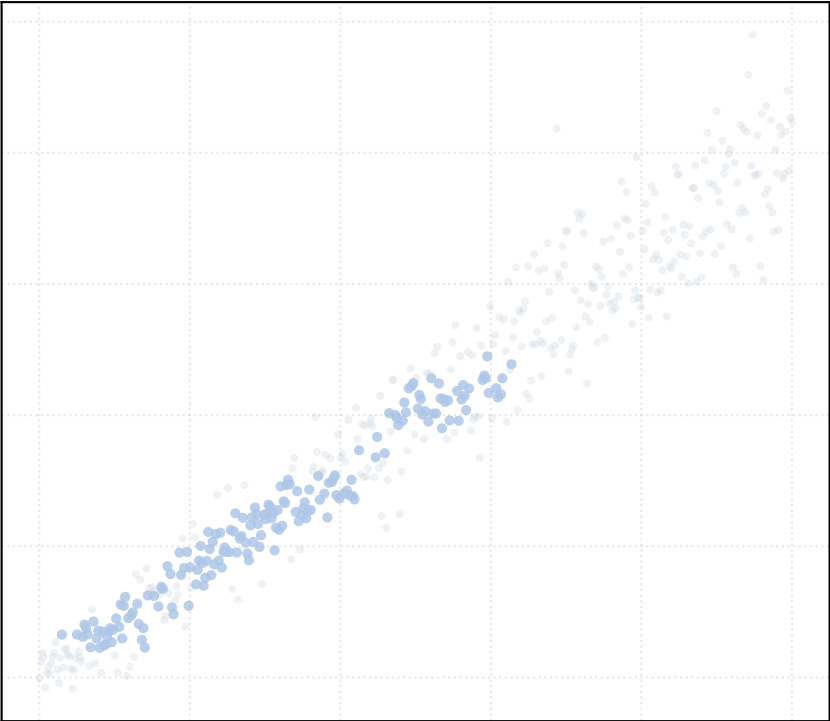


Dataset: heteroscedastic_noise_medium | Seed Percentile Sweep (t_percentile)

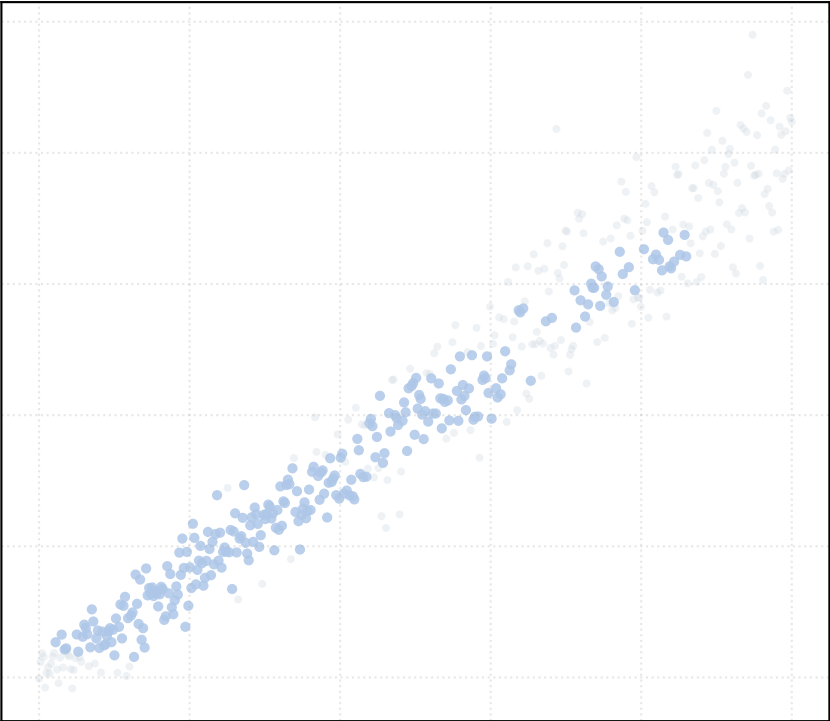
t_percentile = 0.05 [1 Clusters]



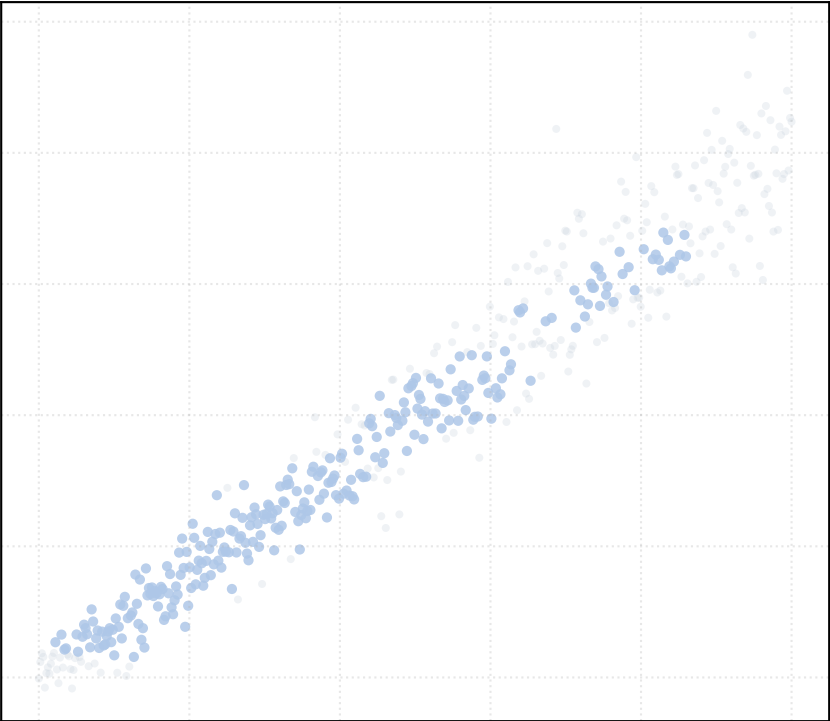
t_percentile = 0.1 [1 Clusters]



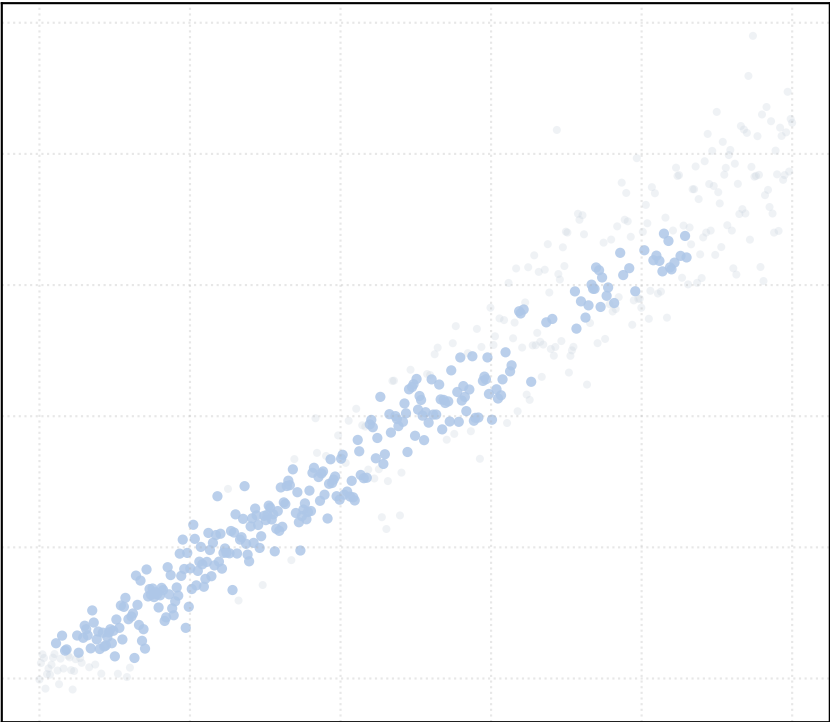
t_percentile = 0.15 (Default) [1 Clusters]



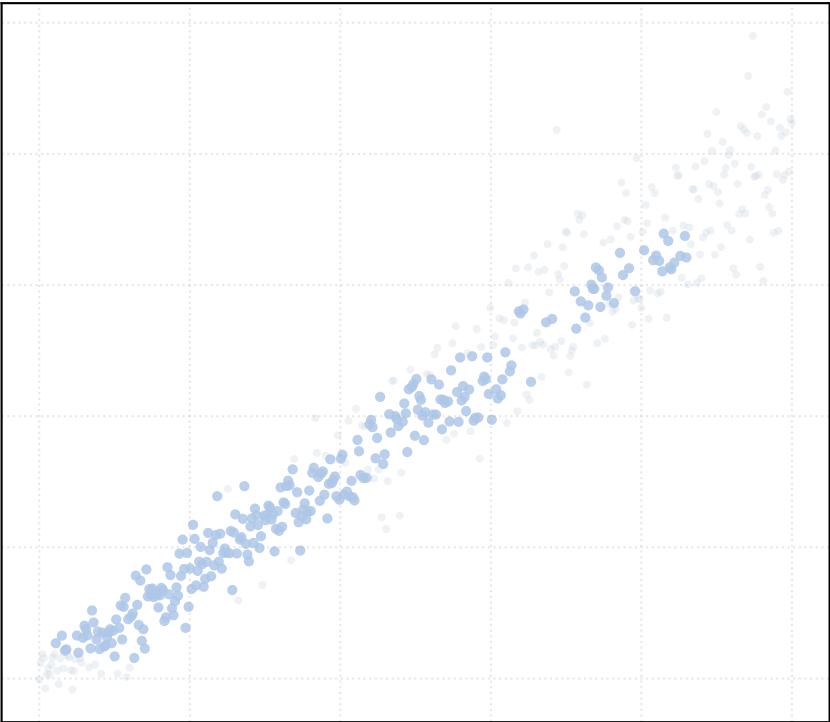
t_percentile = 0.2 [1 Clusters]



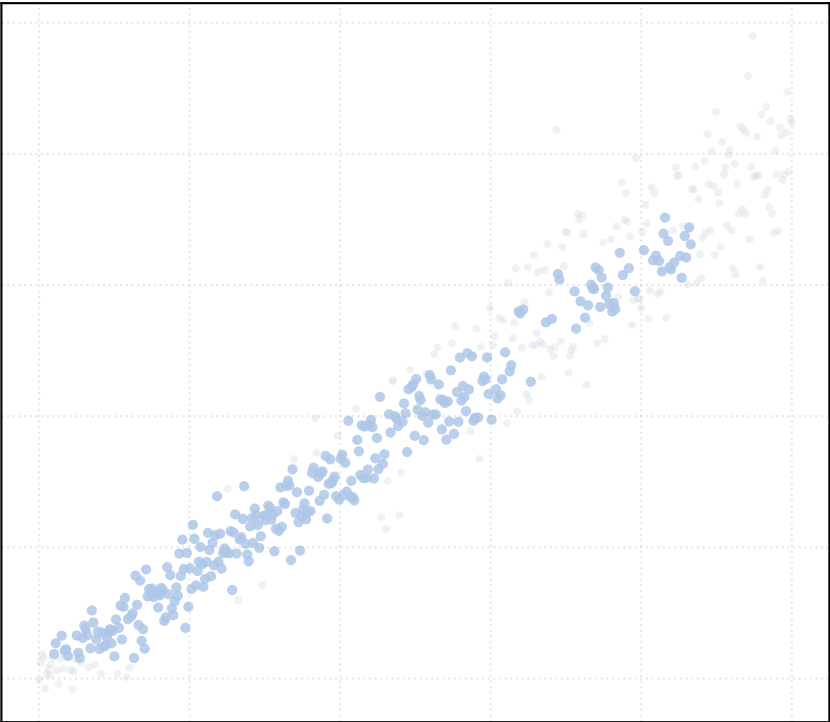
t_percentile = 0.25 [1 Clusters]



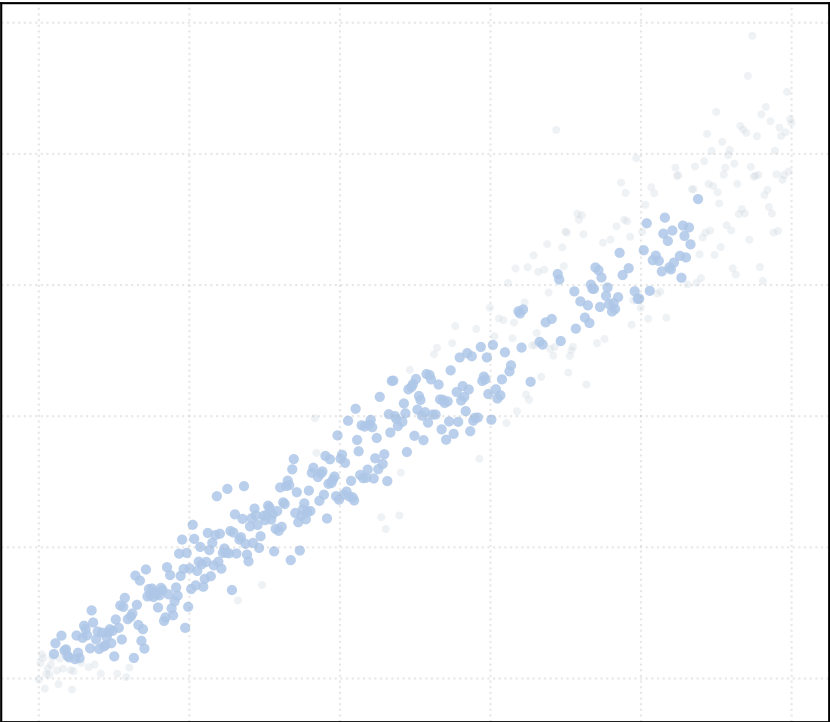
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

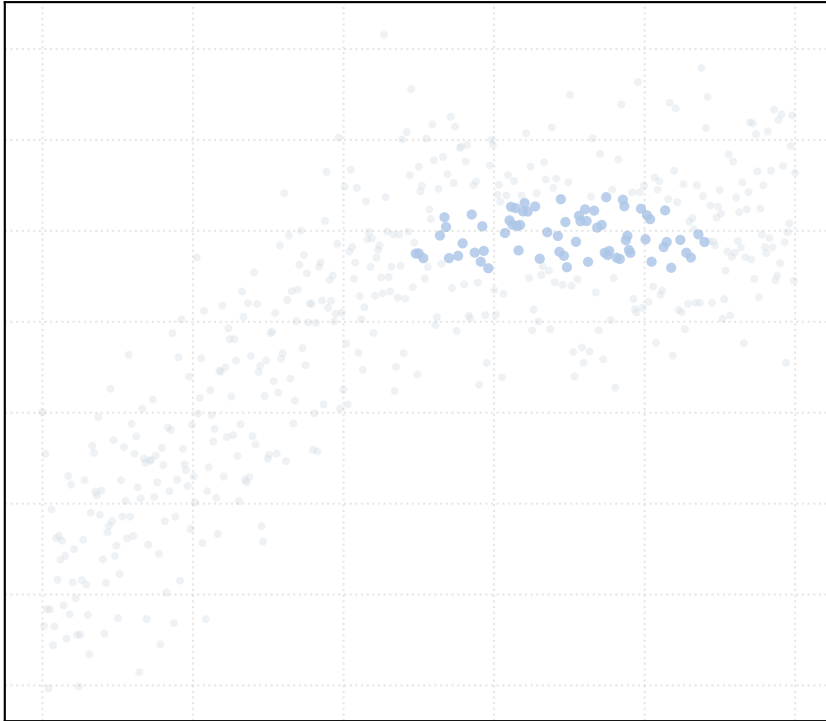


t_percentile = 0.5 [1 Clusters]

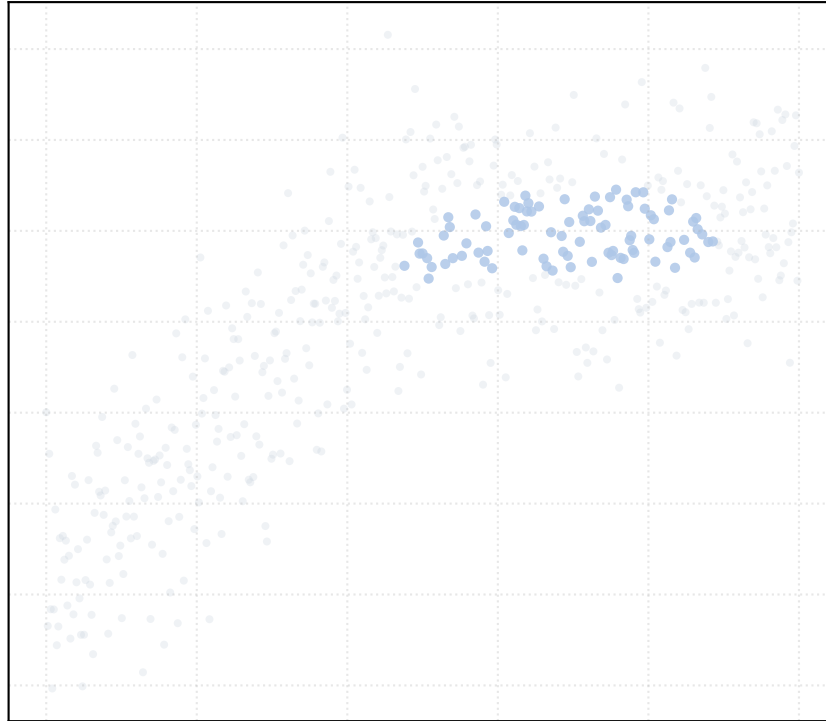


Dataset: kinked_curve_noise_high | Seed Percentile Sweep (t_percentile)

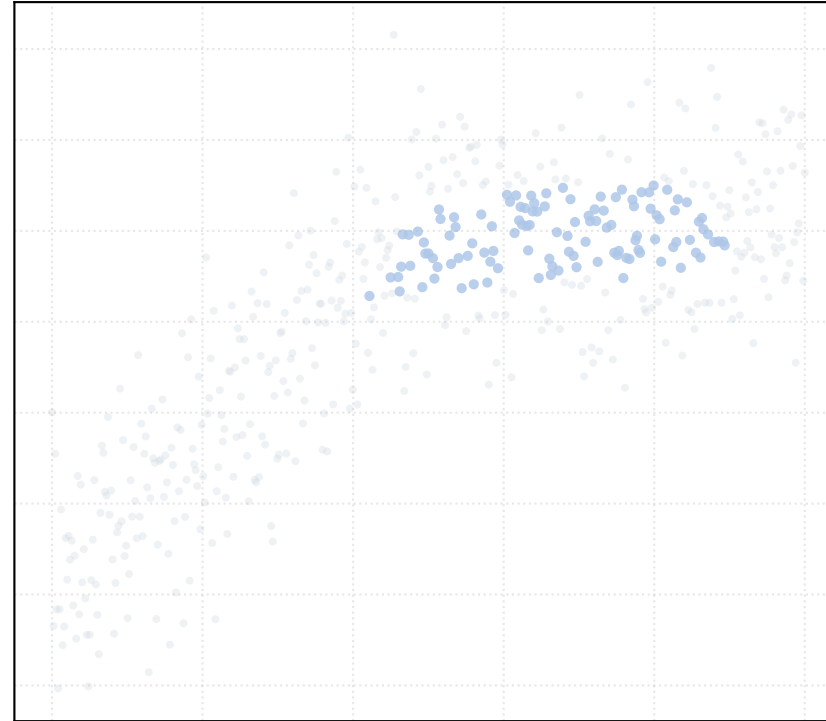
t_percentile = 0.05 [1 Clusters]



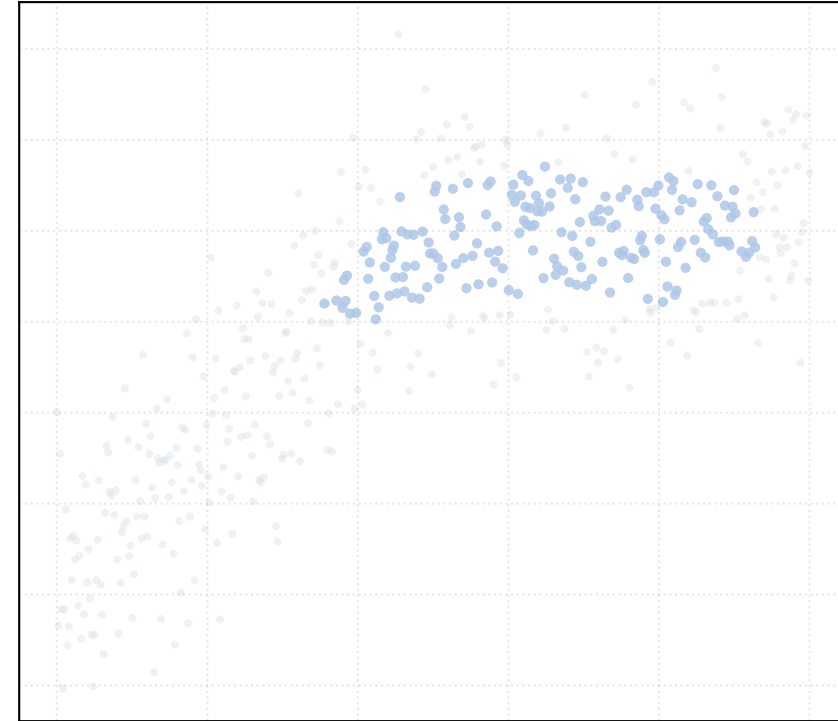
t_percentile = 0.1 [1 Clusters]



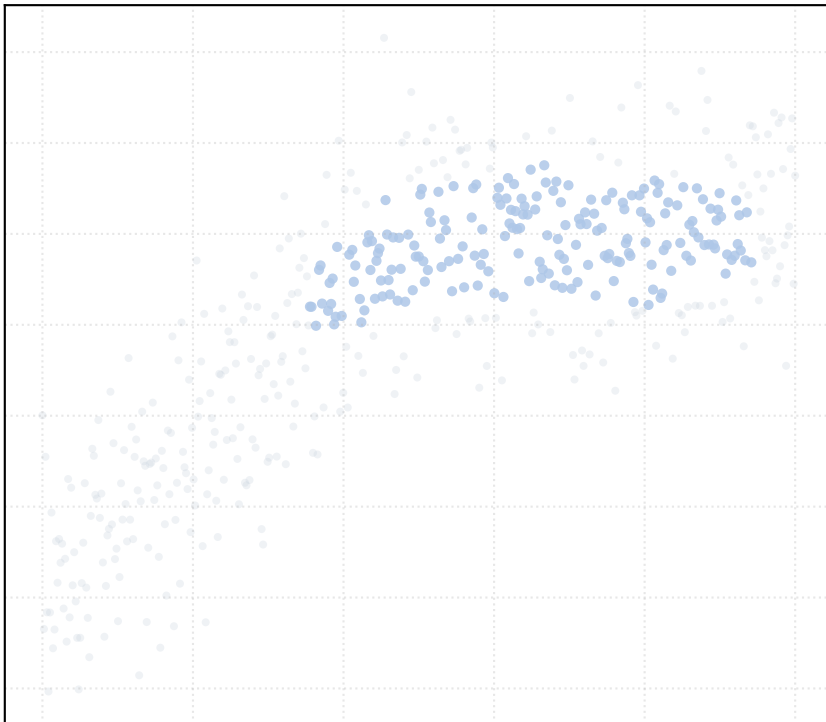
t_percentile = 0.15 (Default) [1 Clusters]



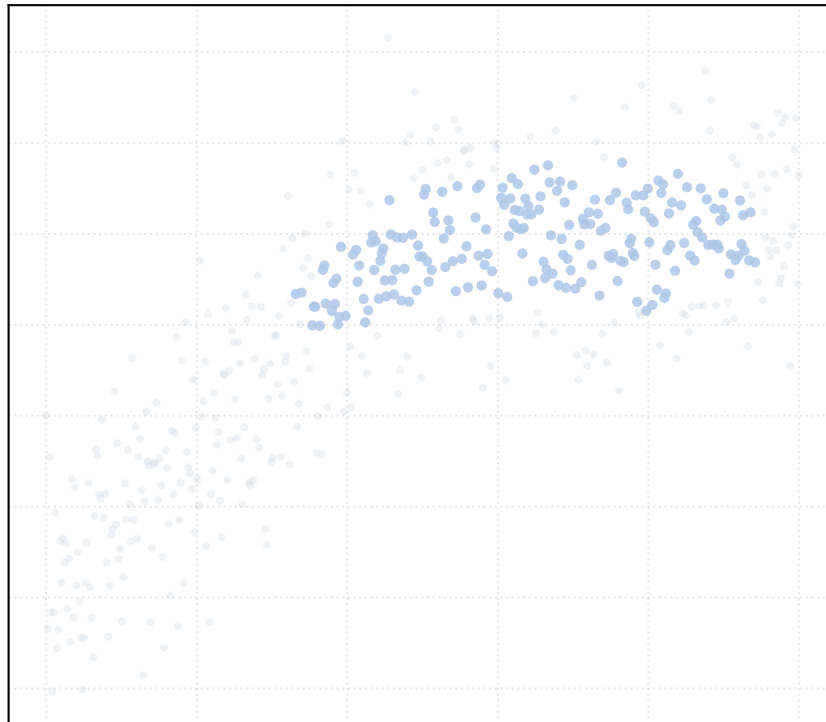
t_percentile = 0.2 [1 Clusters]



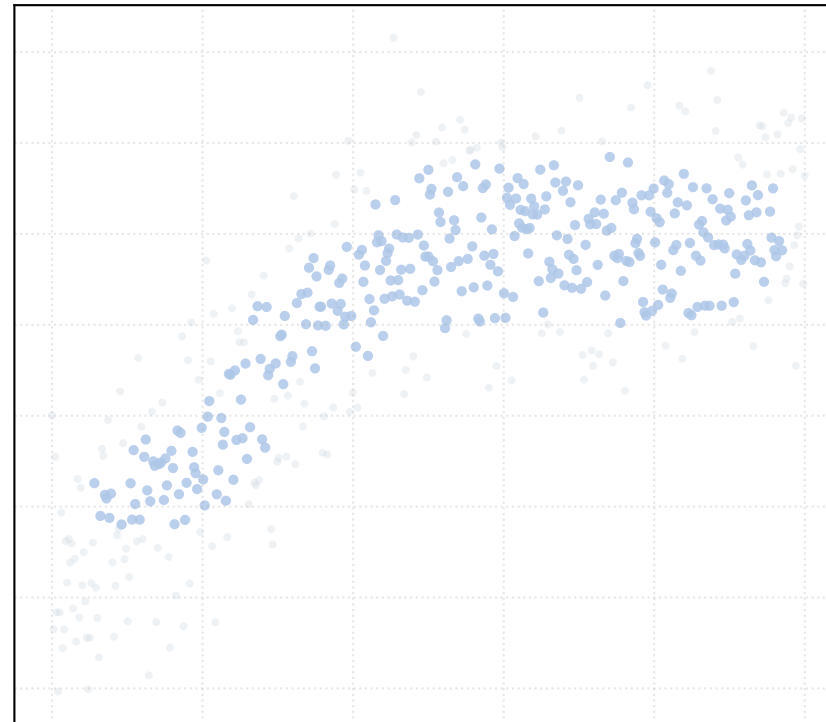
t_percentile = 0.25 [1 Clusters]



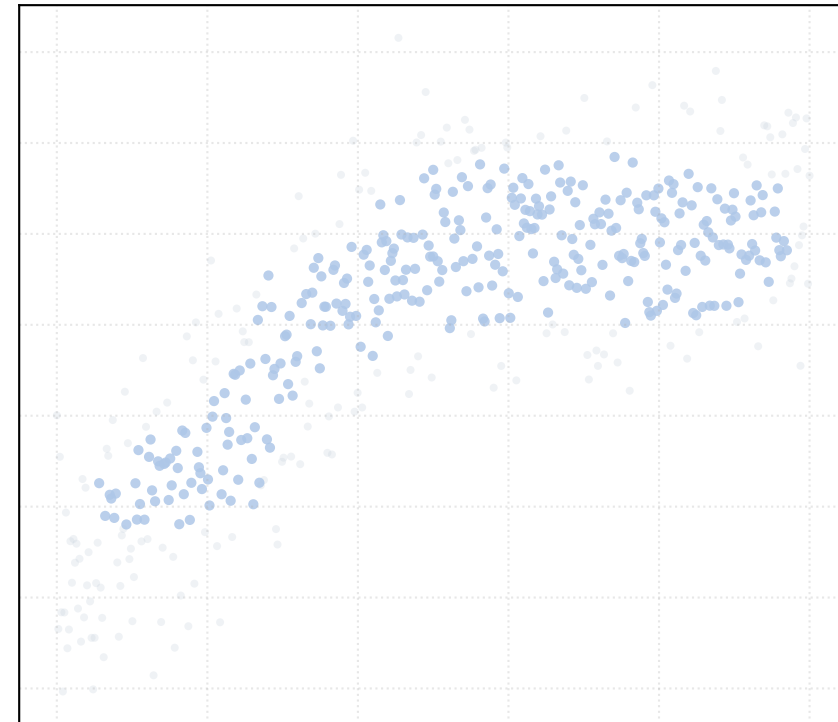
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

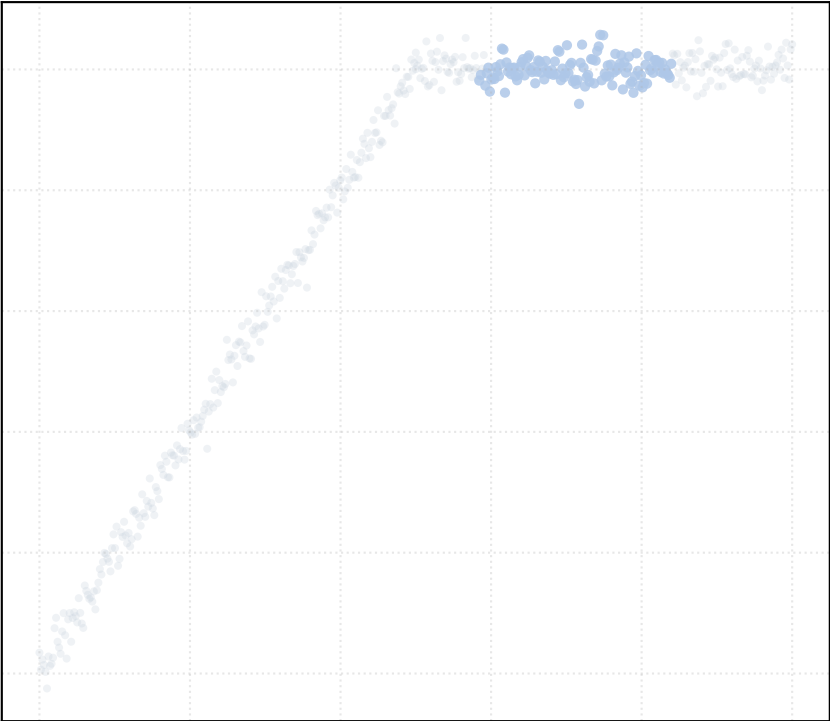


t_percentile = 0.5 [1 Clusters]

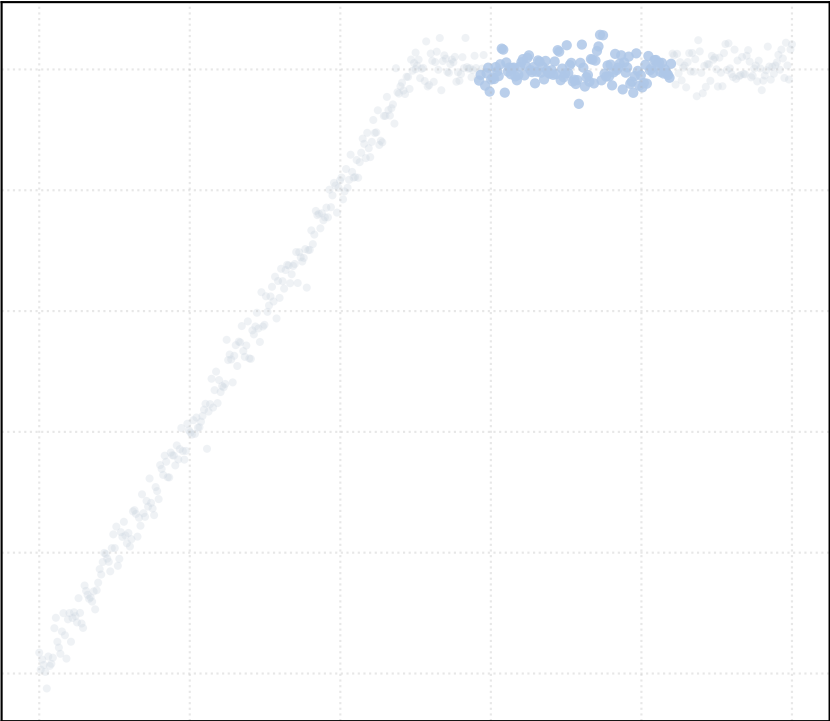


Dataset: kinked_curve_noise_low | Seed Percentile Sweep (t_percentile)

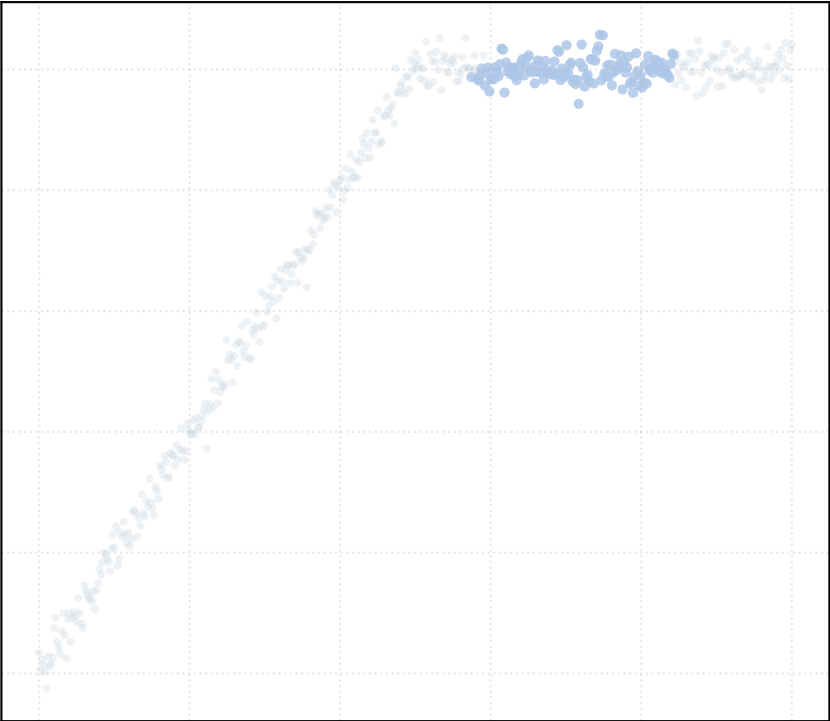
t_percentile = 0.05 [1 Clusters]



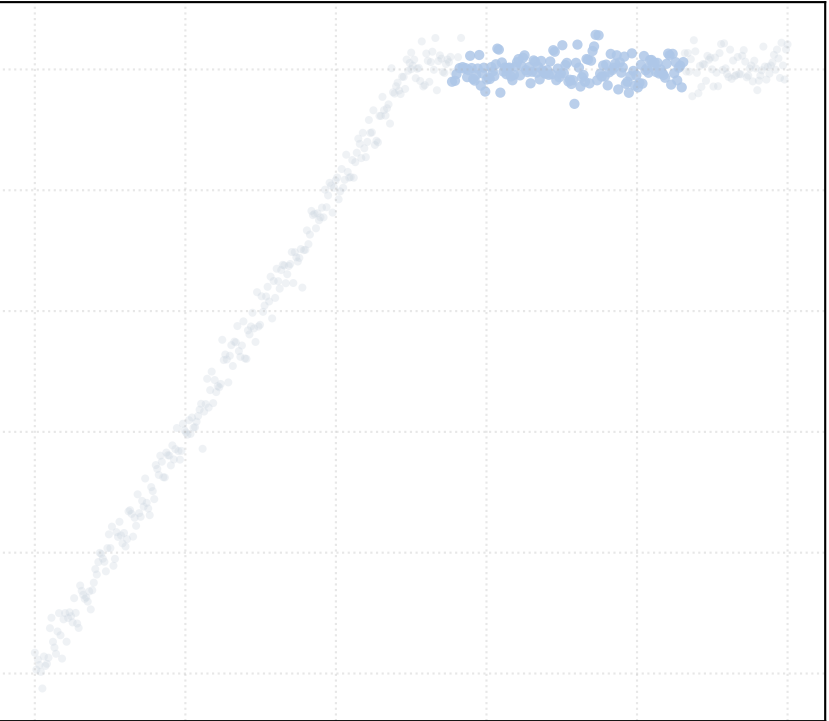
t_percentile = 0.1 [1 Clusters]



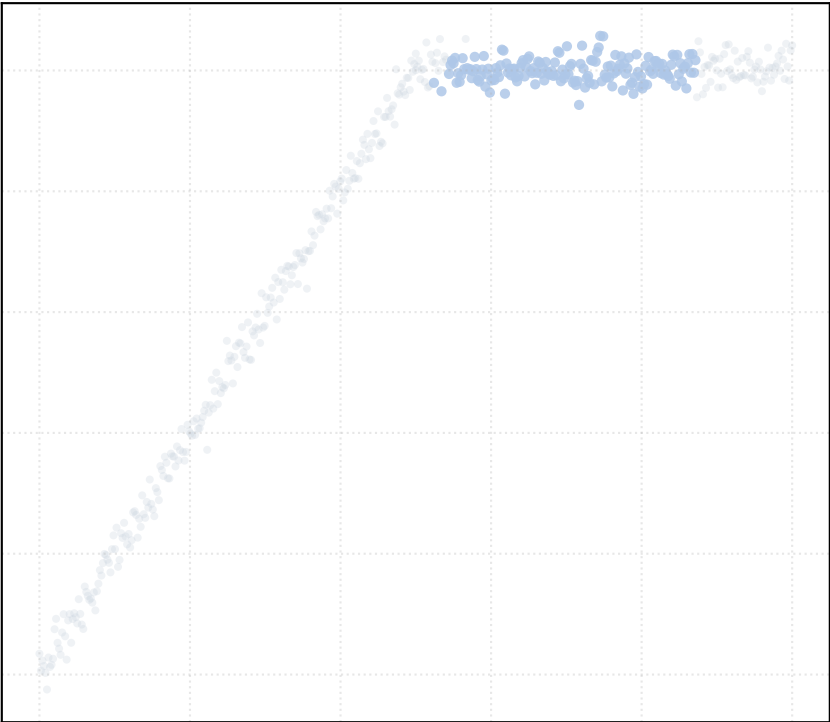
t_percentile = 0.15 (Default) [1 Clusters]



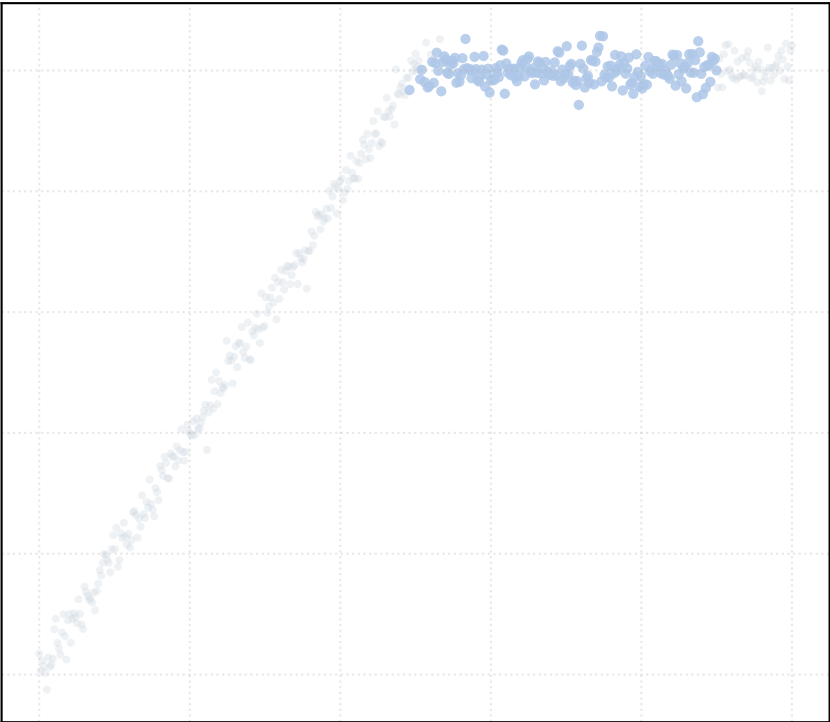
t_percentile = 0.2 [1 Clusters]



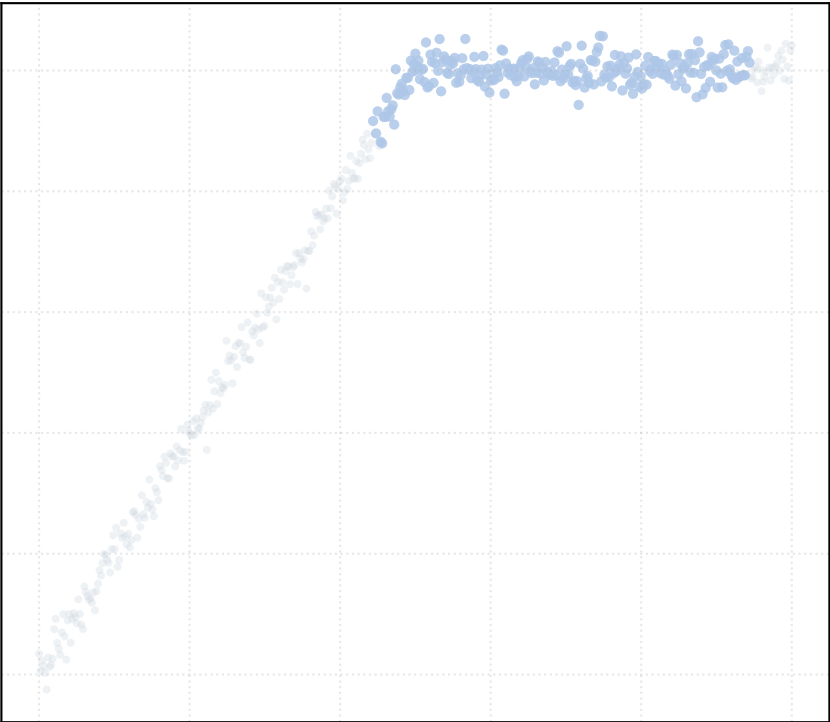
t_percentile = 0.25 [1 Clusters]



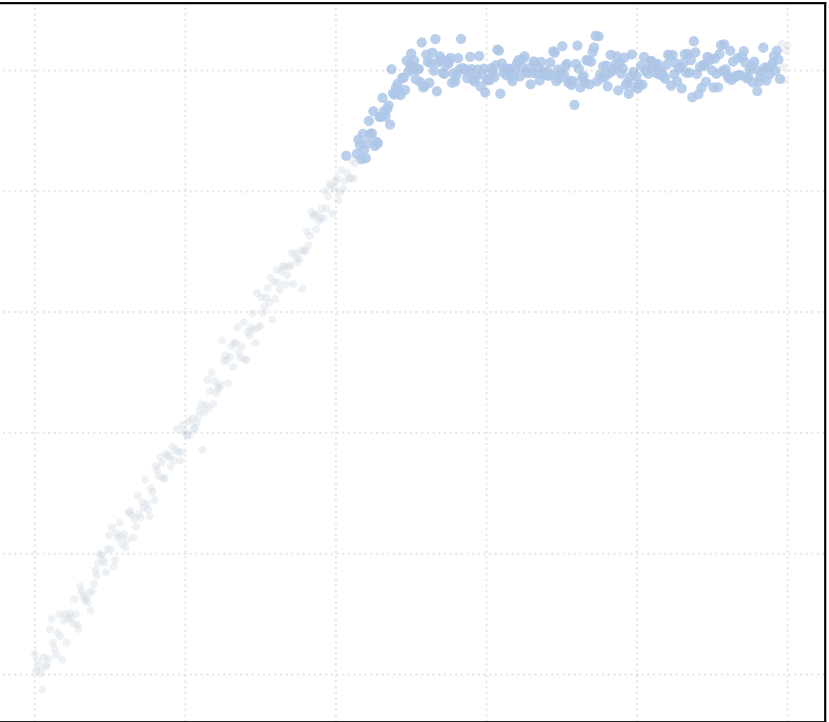
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

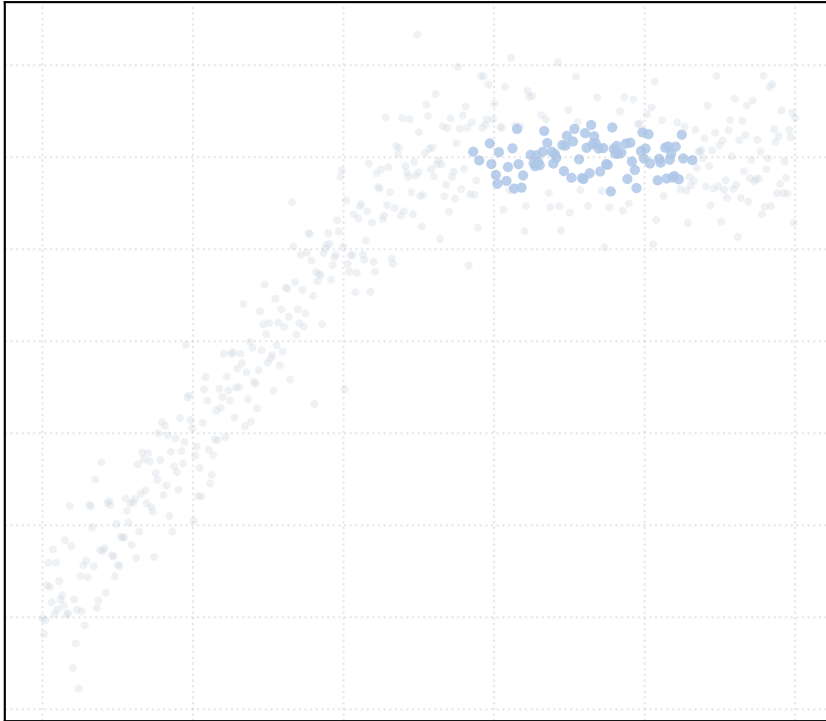


t_percentile = 0.5 [1 Clusters]

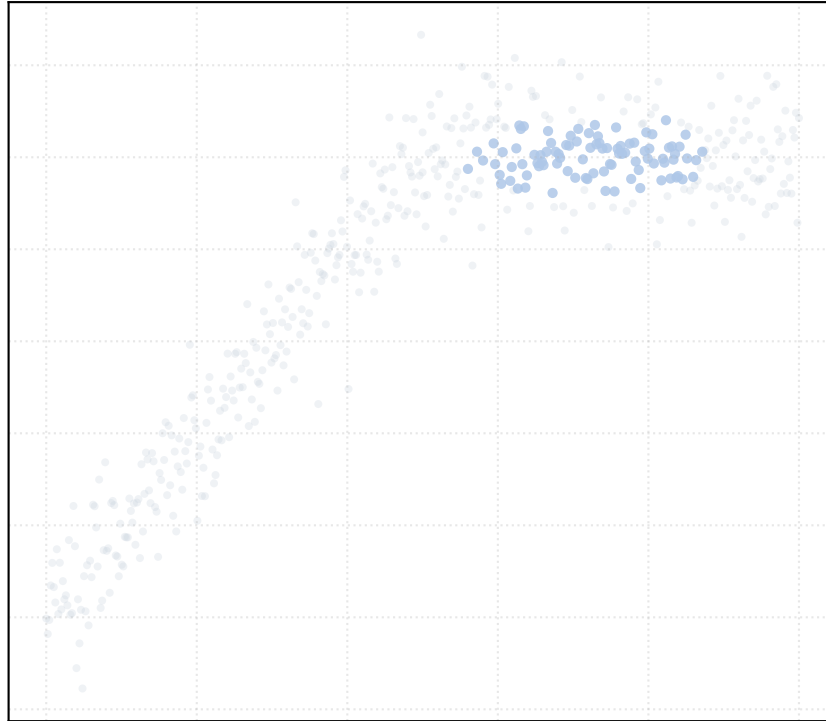


Dataset: kinked_curve_noise_medium | Seed Percentile Sweep (t_percentile)

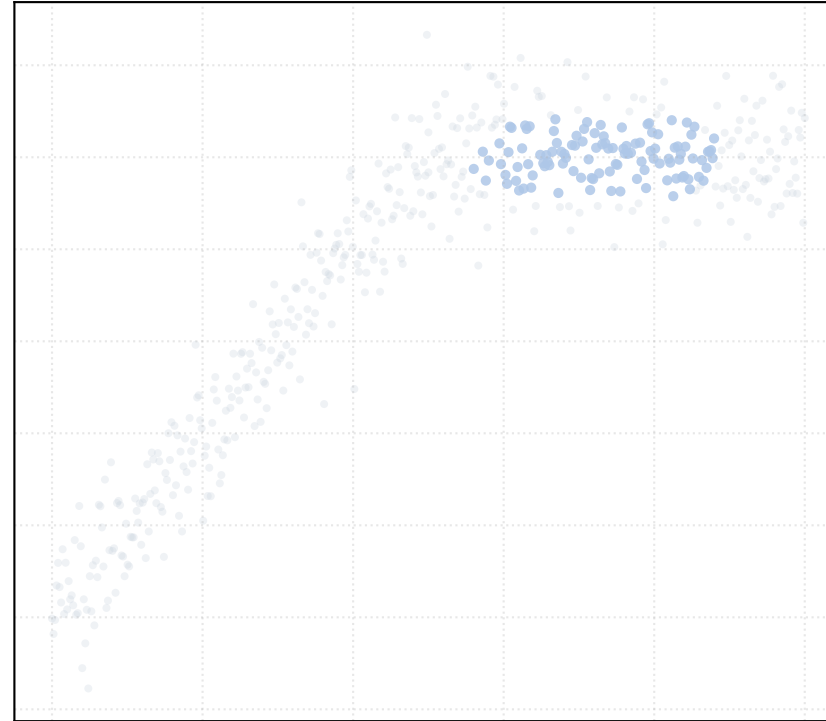
t_percentile = 0.05 [1 Clusters]



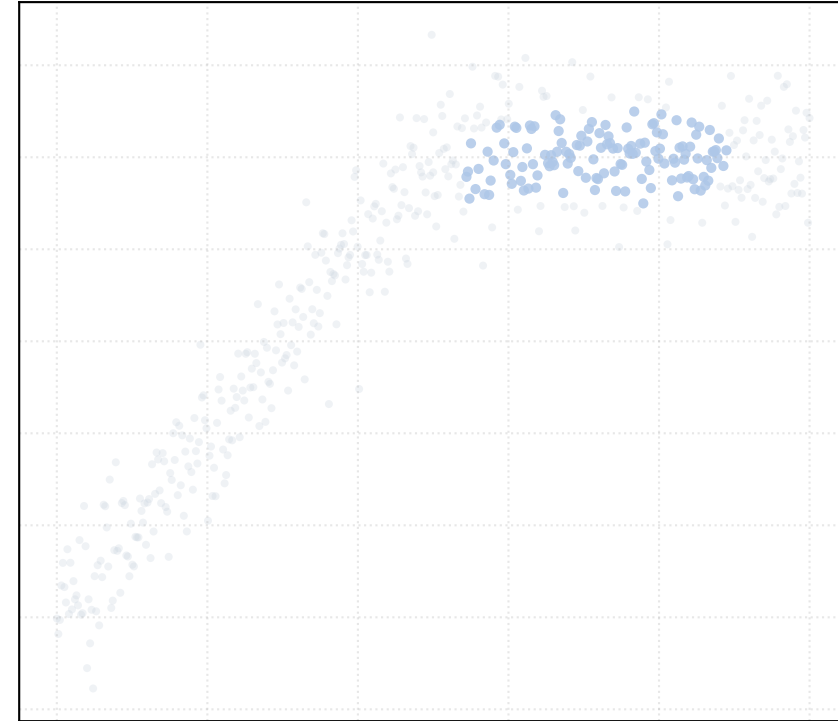
t_percentile = 0.1 [1 Clusters]



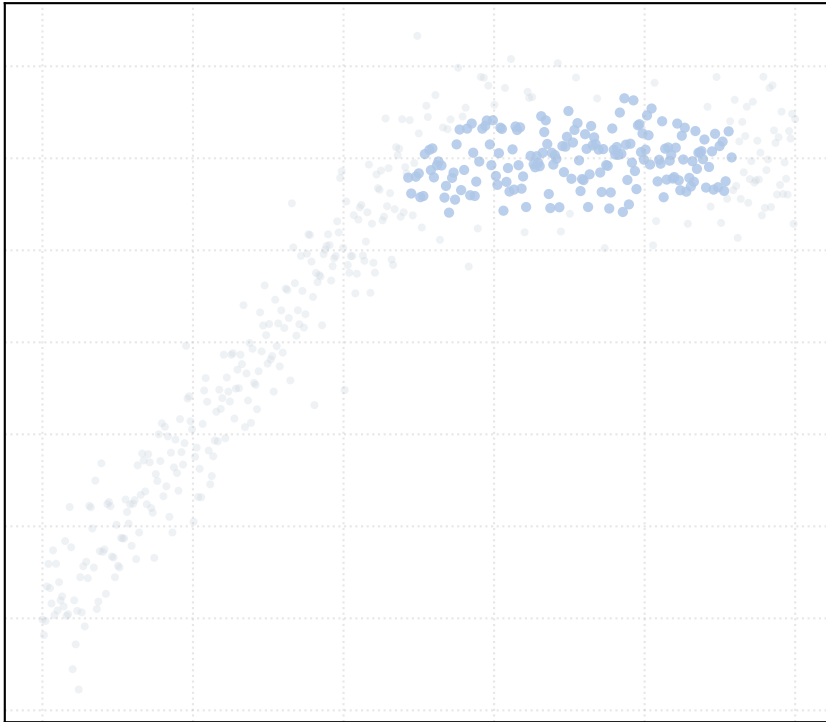
t_percentile = 0.15 (Default) [1 Clusters]



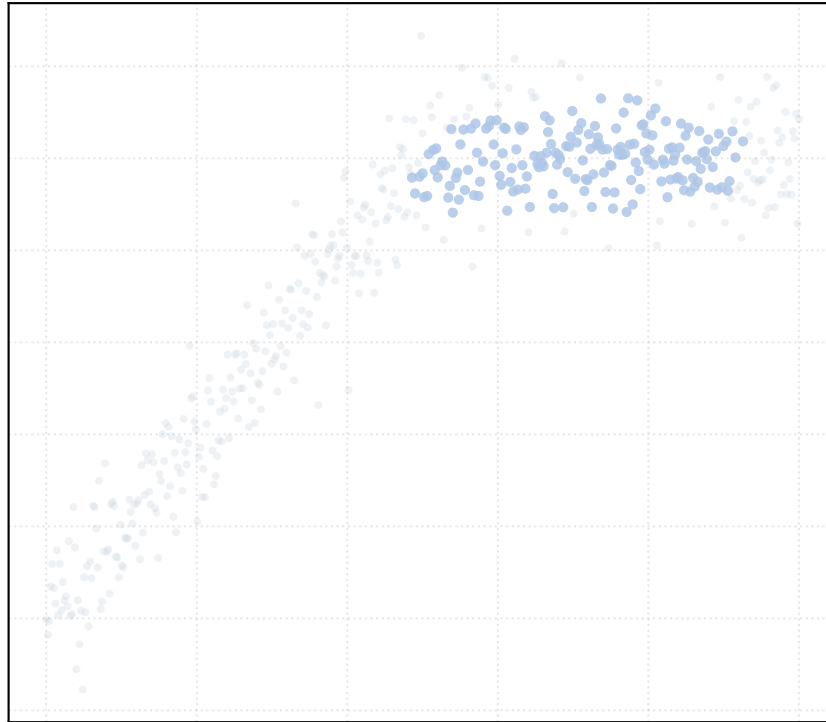
t_percentile = 0.2 [1 Clusters]



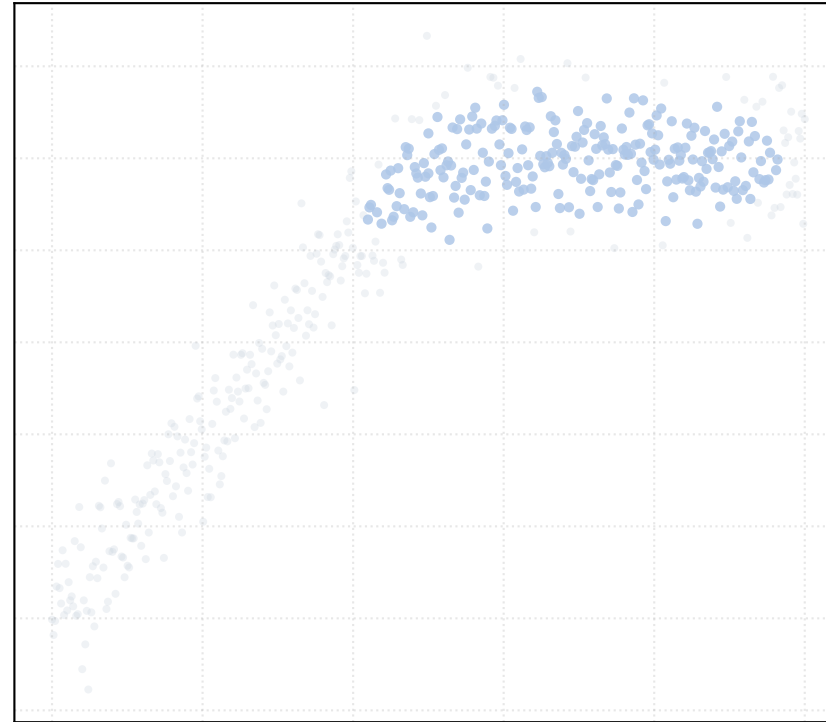
t_percentile = 0.25 [1 Clusters]



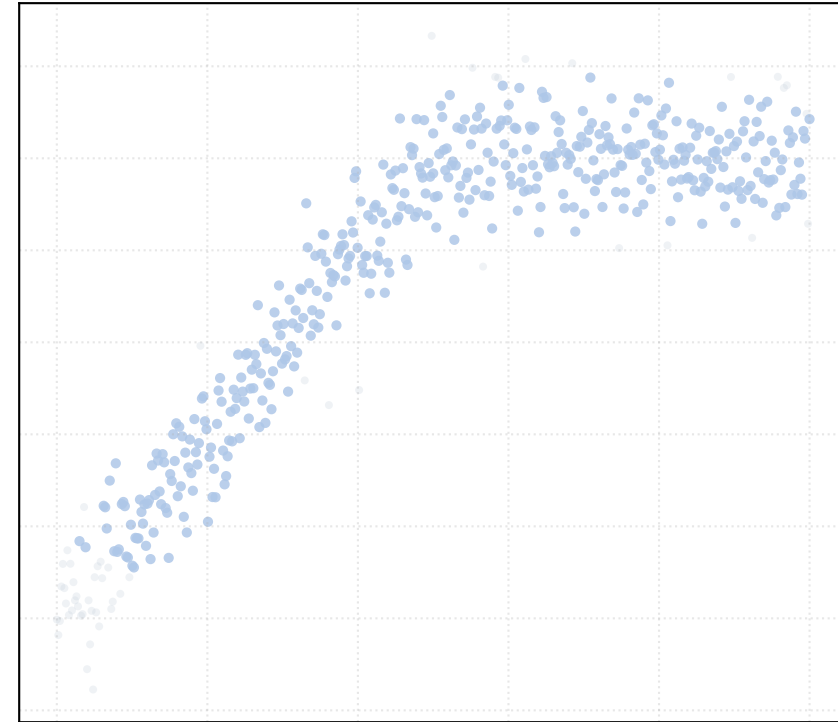
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

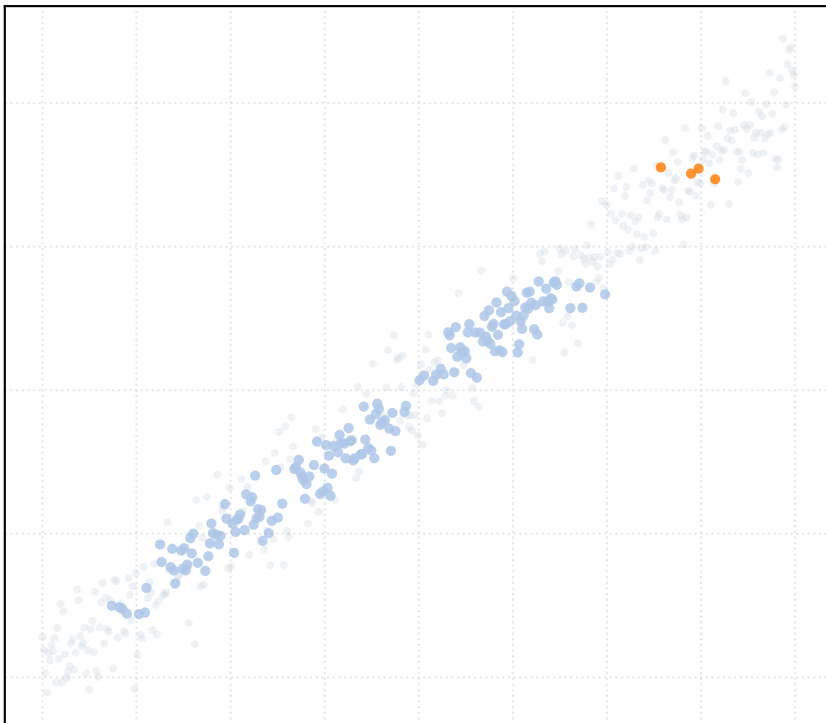


t_percentile = 0.5 [1 Clusters]

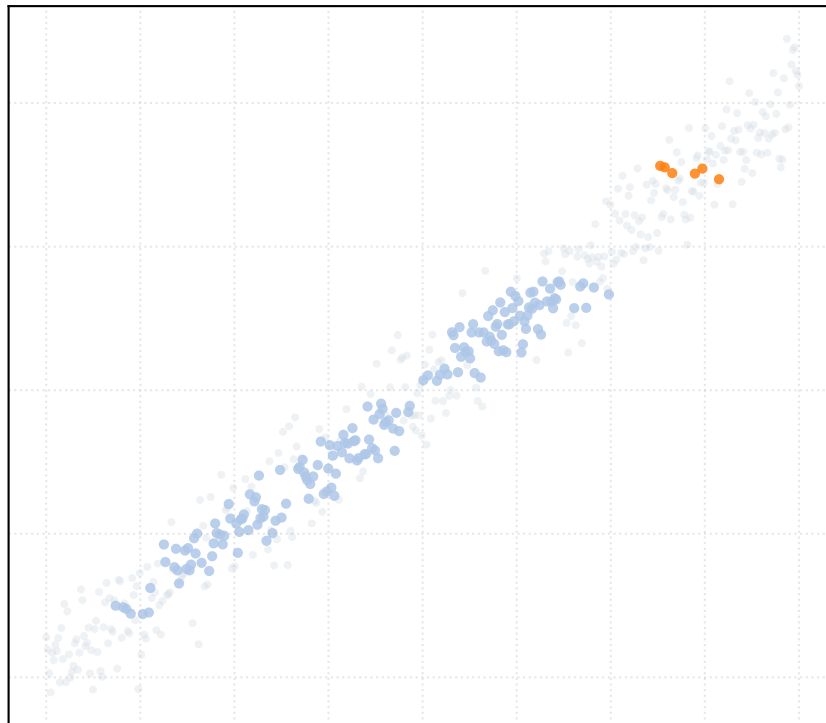


Dataset: linear_noise_high | Seed Percentile Sweep (t_percentile)

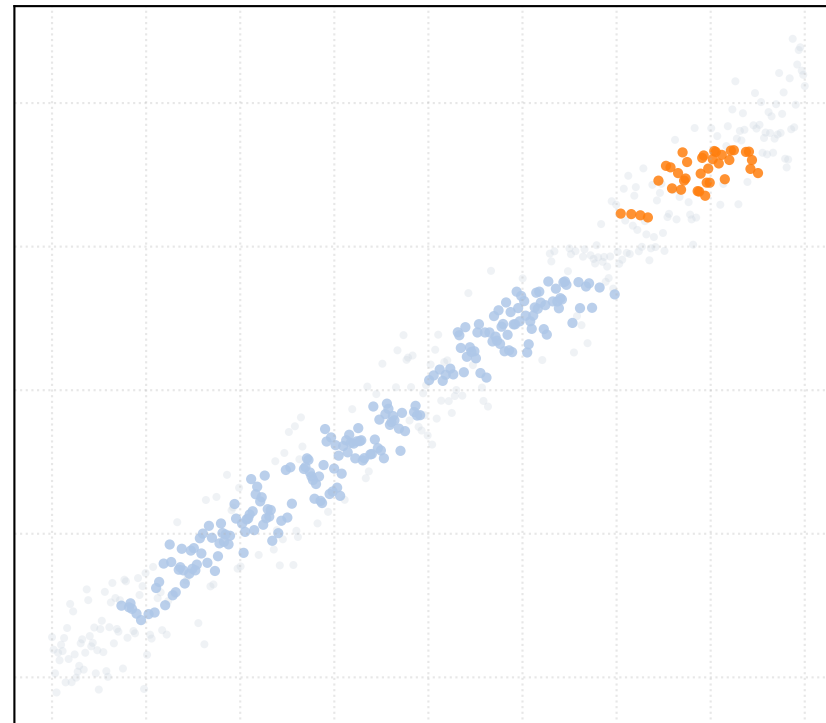
t_percentile = 0.05 [2 Clusters]



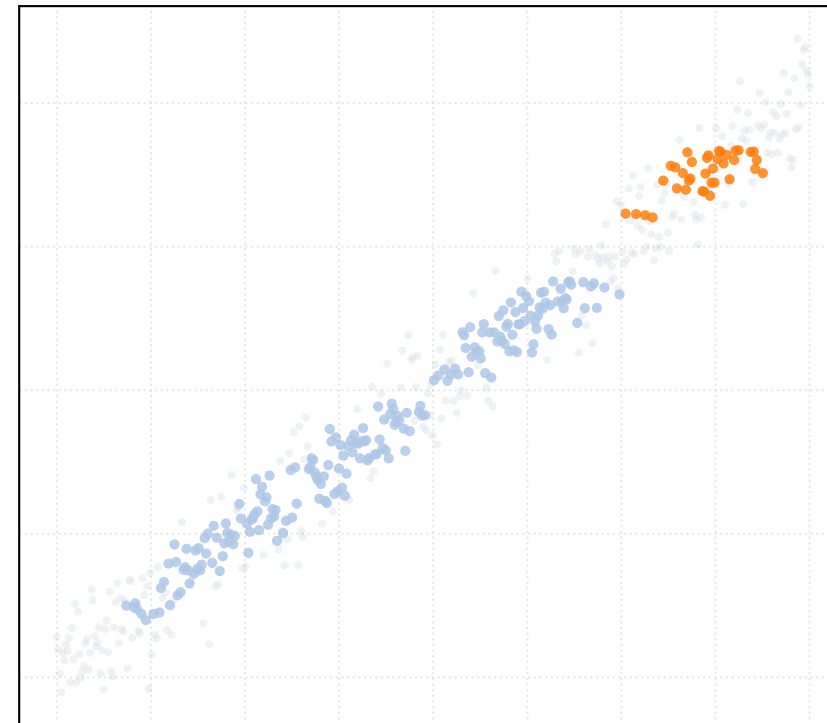
t_percentile = 0.1 [2 Clusters]



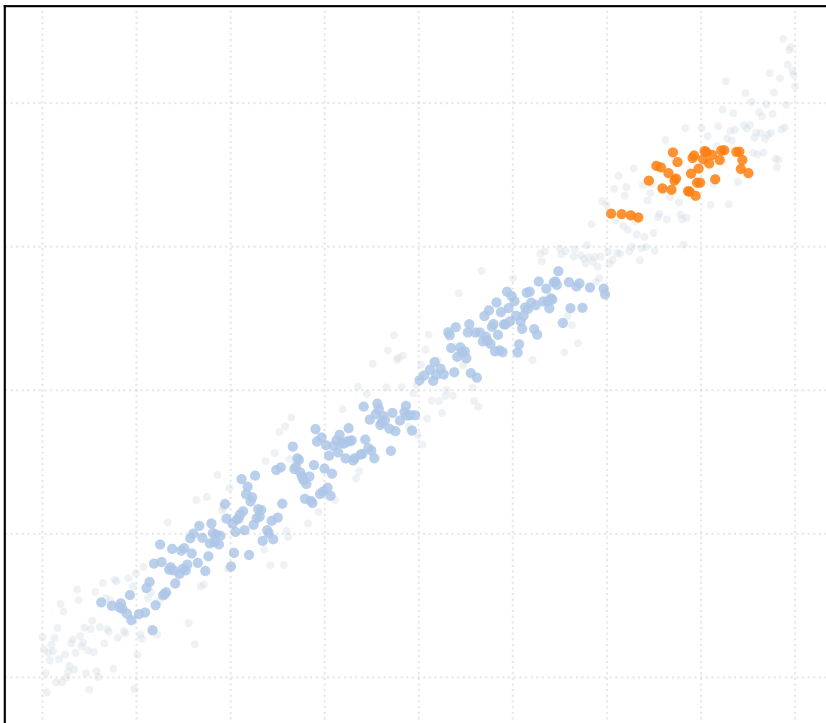
t_percentile = 0.15 (Default) [2 Clusters]



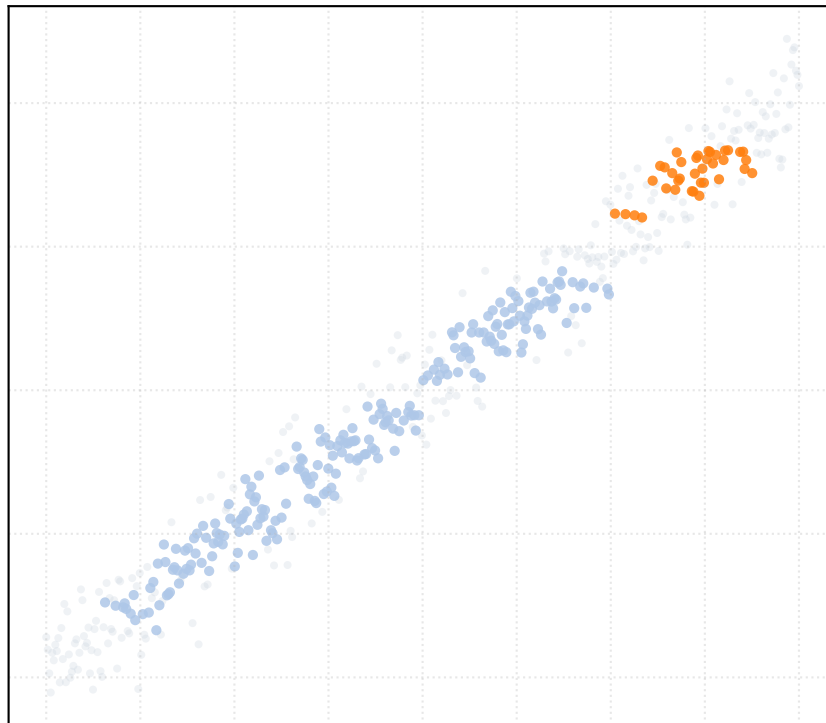
t_percentile = 0.2 [2 Clusters]



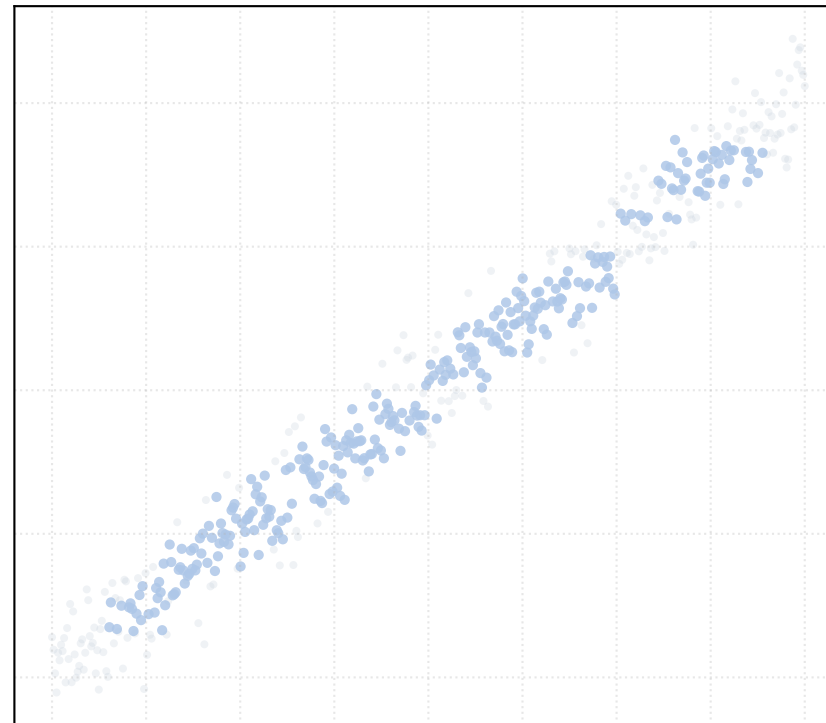
t_percentile = 0.25 [2 Clusters]



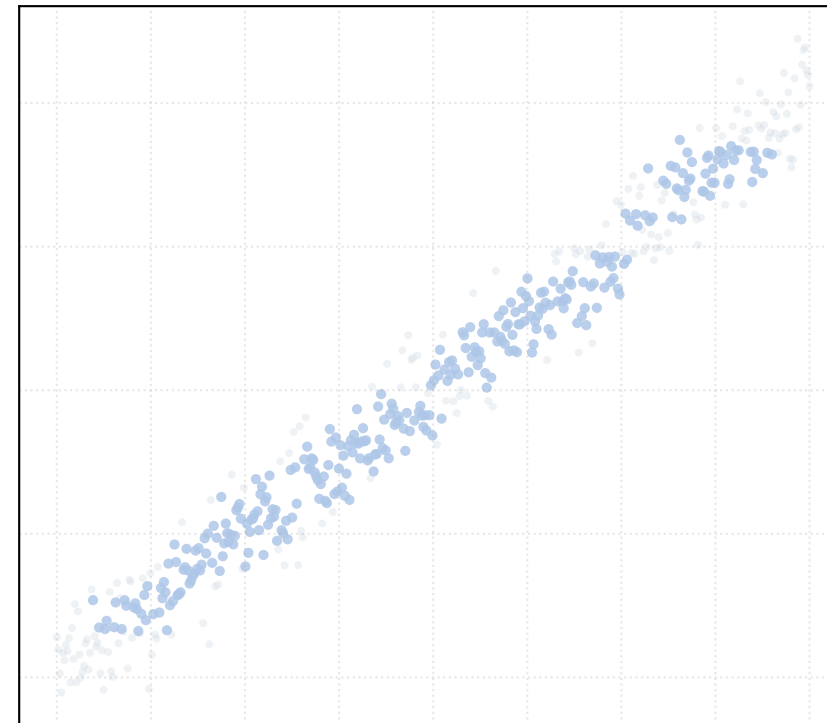
t_percentile = 0.3 [2 Clusters]



t_percentile = 0.4 [1 Clusters]

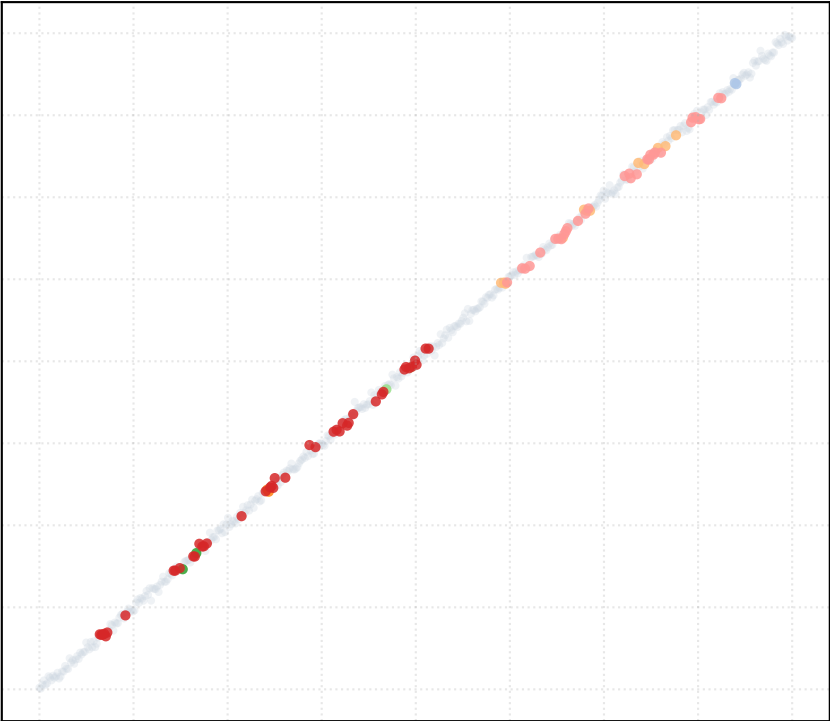


t_percentile = 0.5 [1 Clusters]

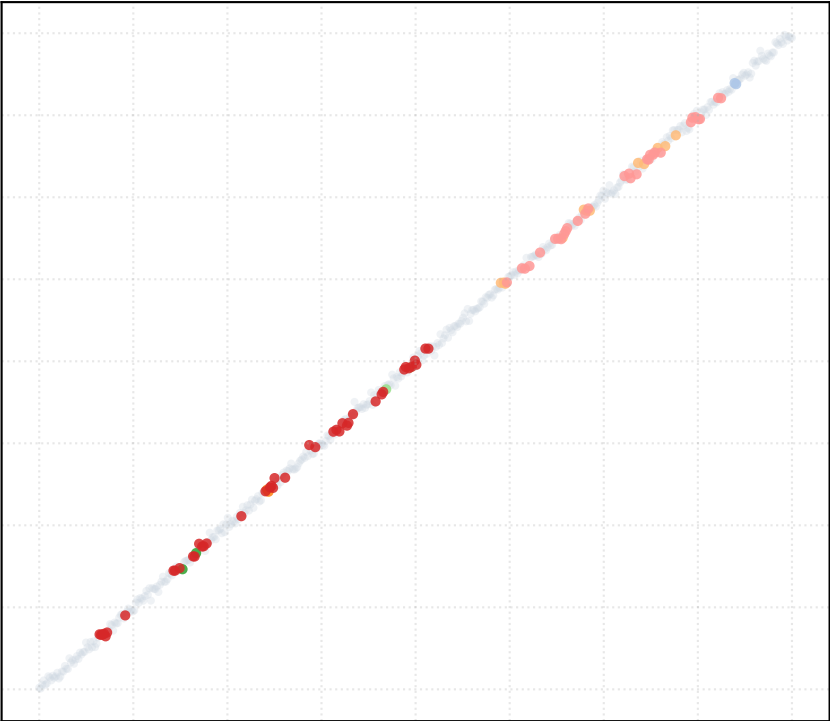


Dataset: linear_noise_low | Seed Percentile Sweep (t_percentile)

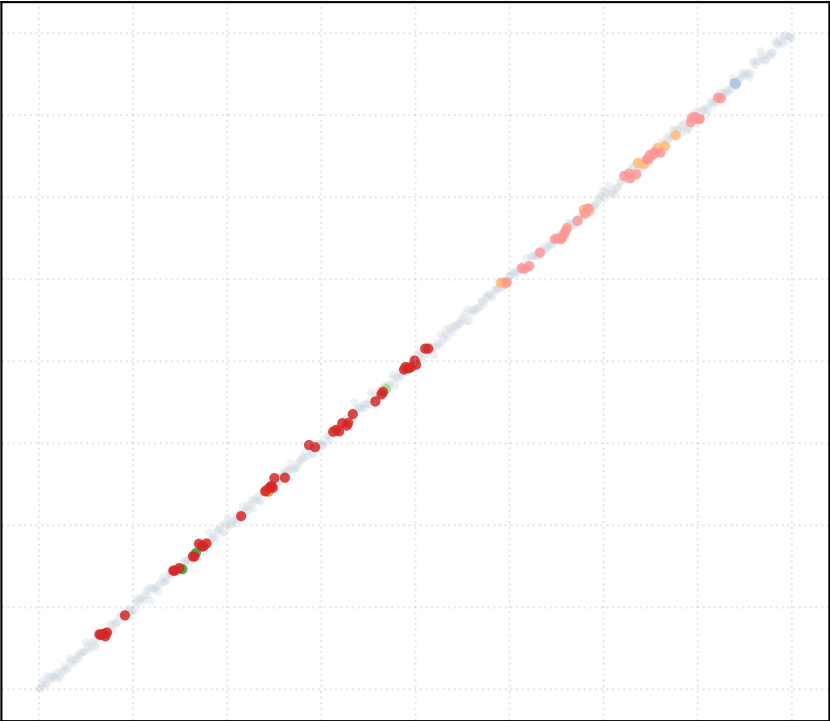
t_percentile = 0.05 [7 Clusters]



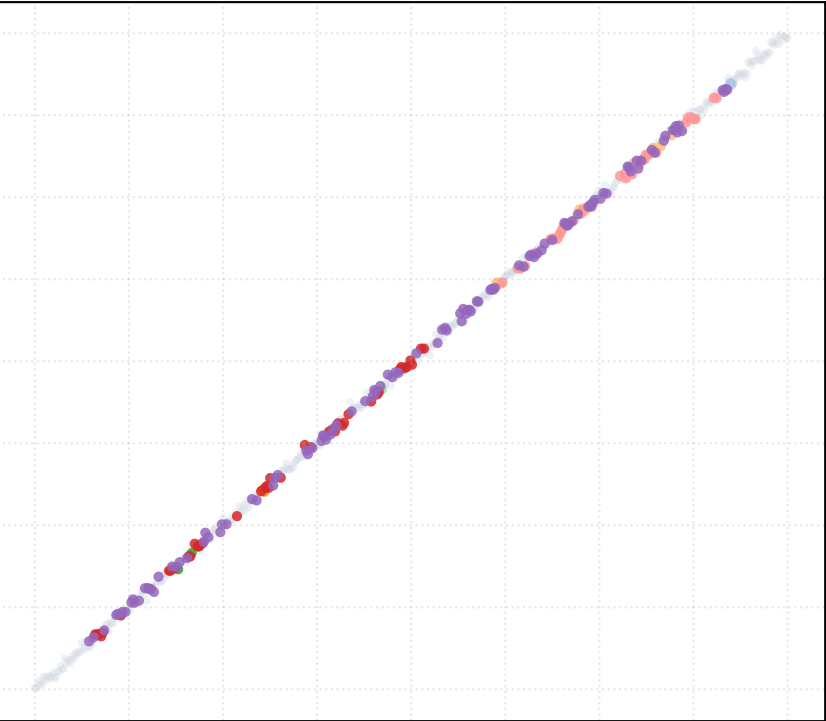
t_percentile = 0.1 [7 Clusters]



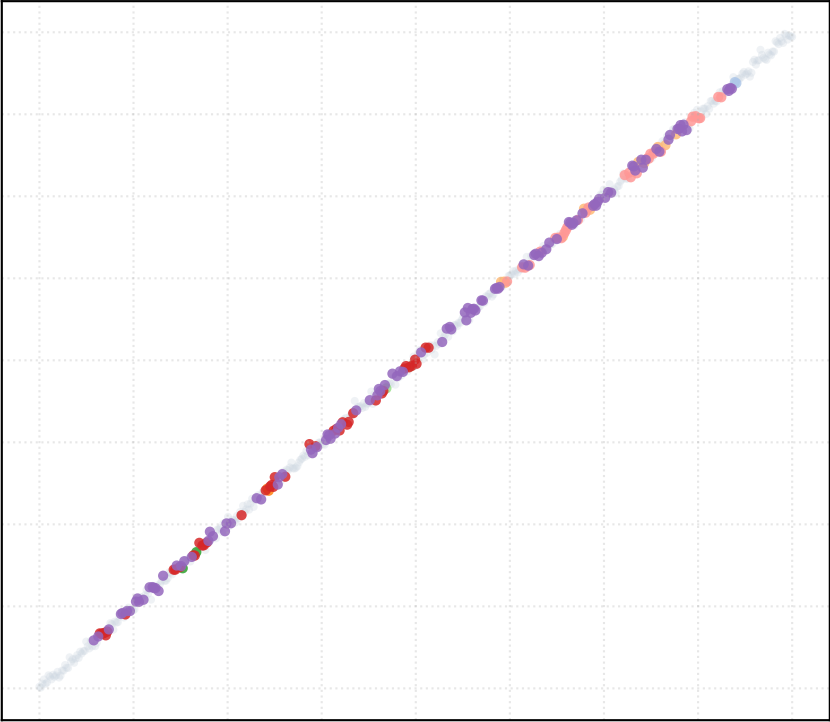
t_percentile = 0.15 (Default) [7 Clusters]



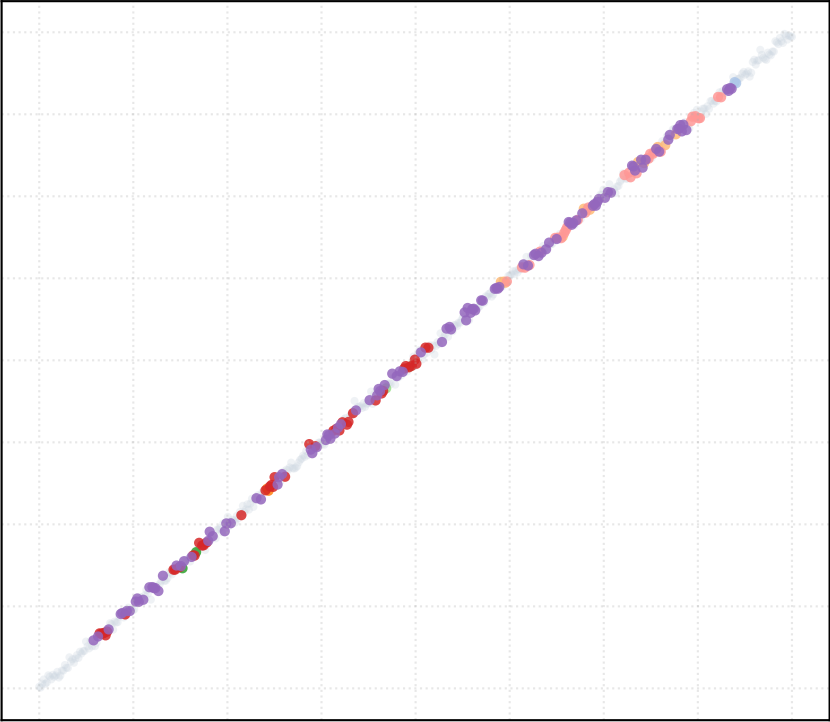
t_percentile = 0.2 [8 Clusters]



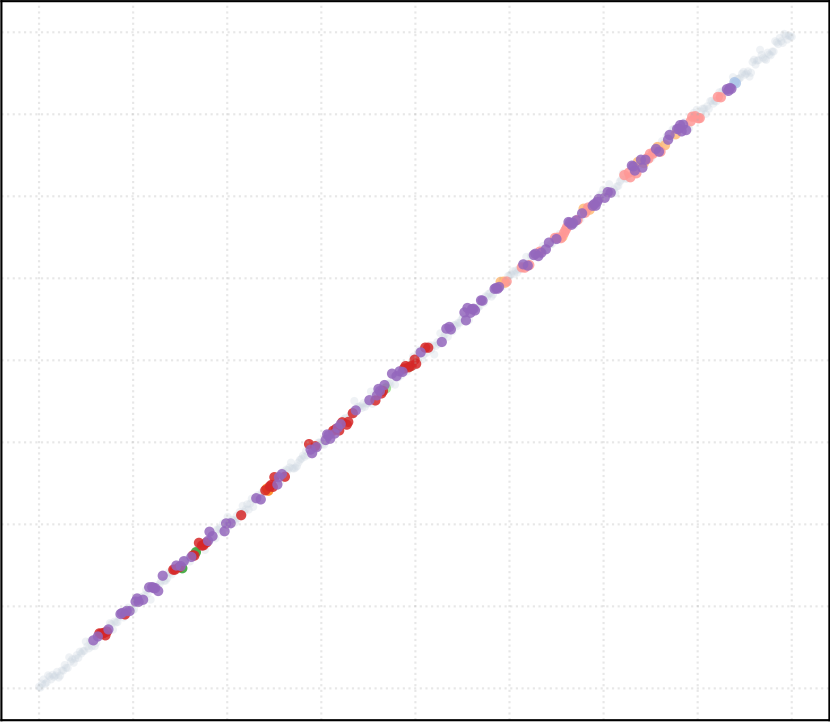
t_percentile = 0.25 [8 Clusters]



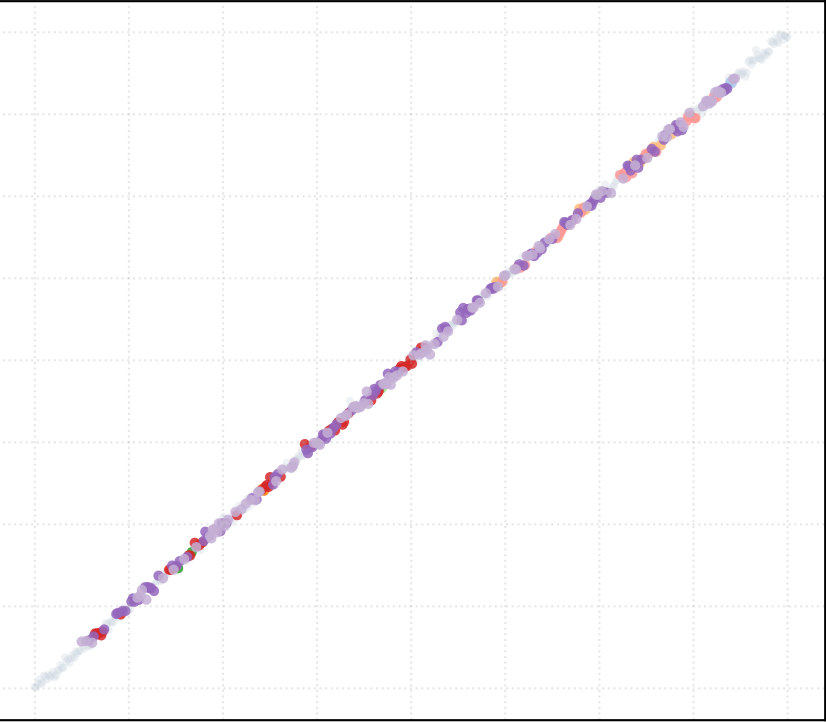
t_percentile = 0.3 [8 Clusters]



t_percentile = 0.4 [8 Clusters]

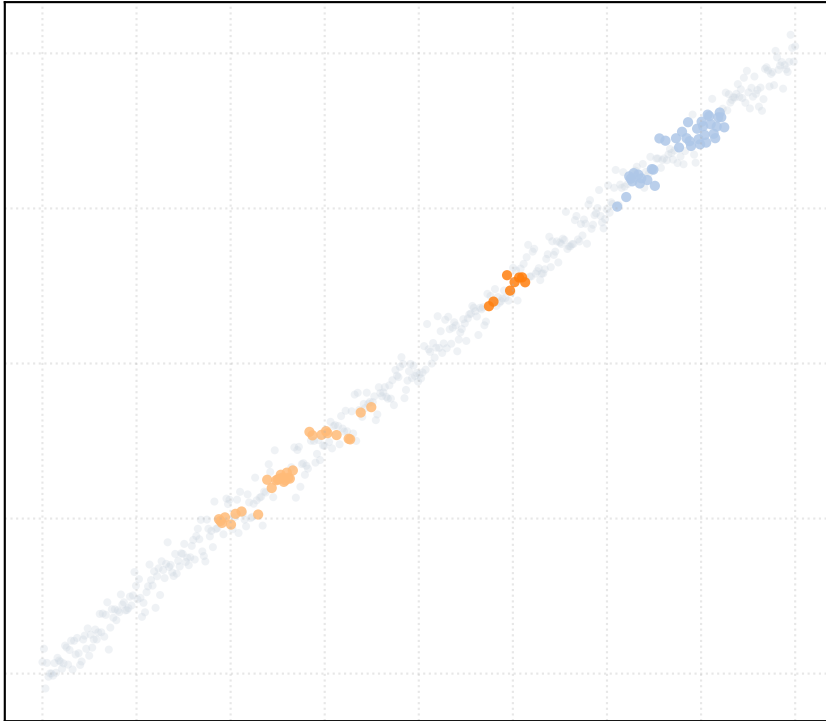


t_percentile = 0.5 [9 Clusters]

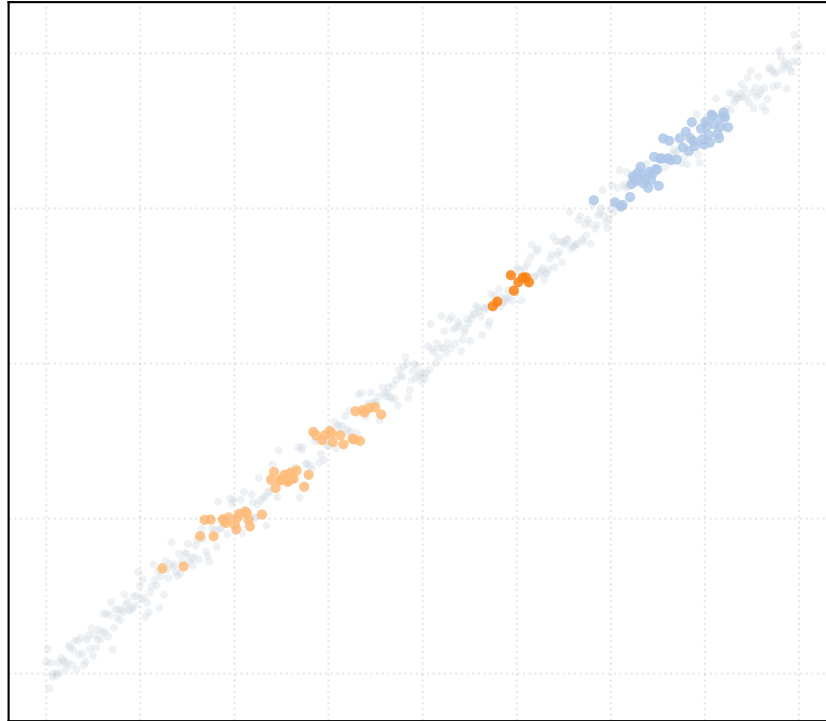


Dataset: linear_noise_medium | Seed Percentile Sweep (t_percentile)

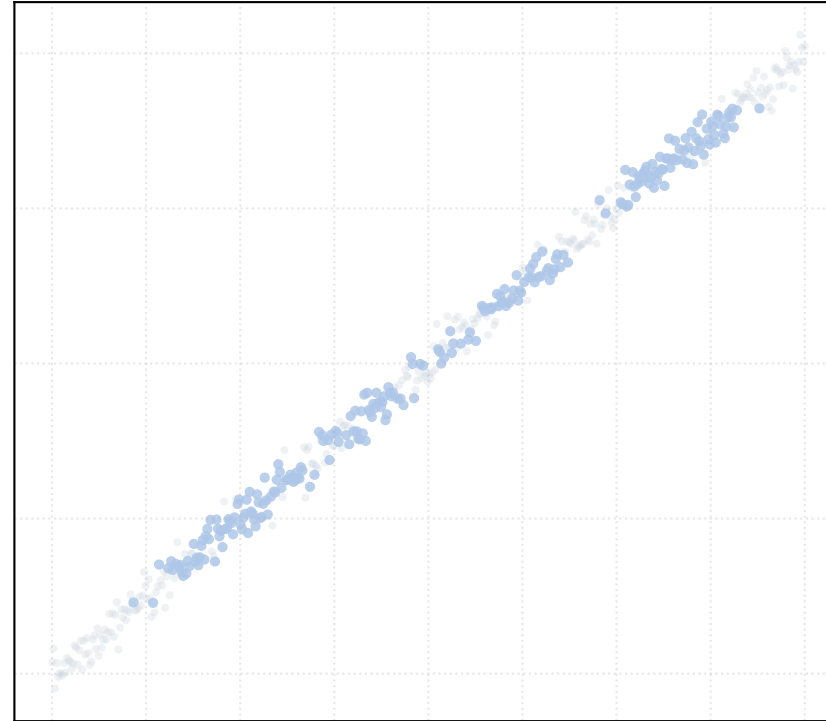
t_percentile = 0.05 [3 Clusters]



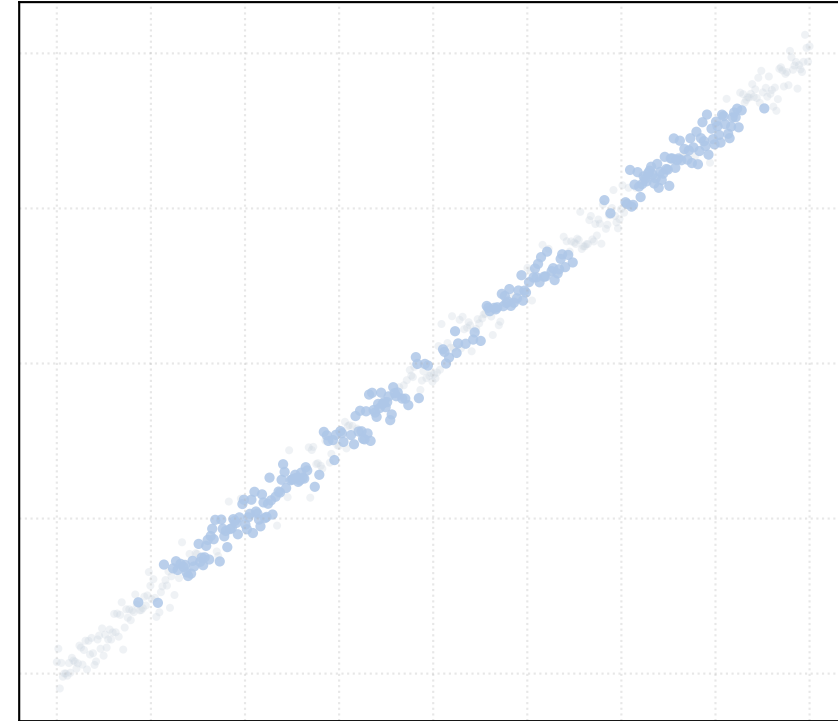
t_percentile = 0.1 [3 Clusters]



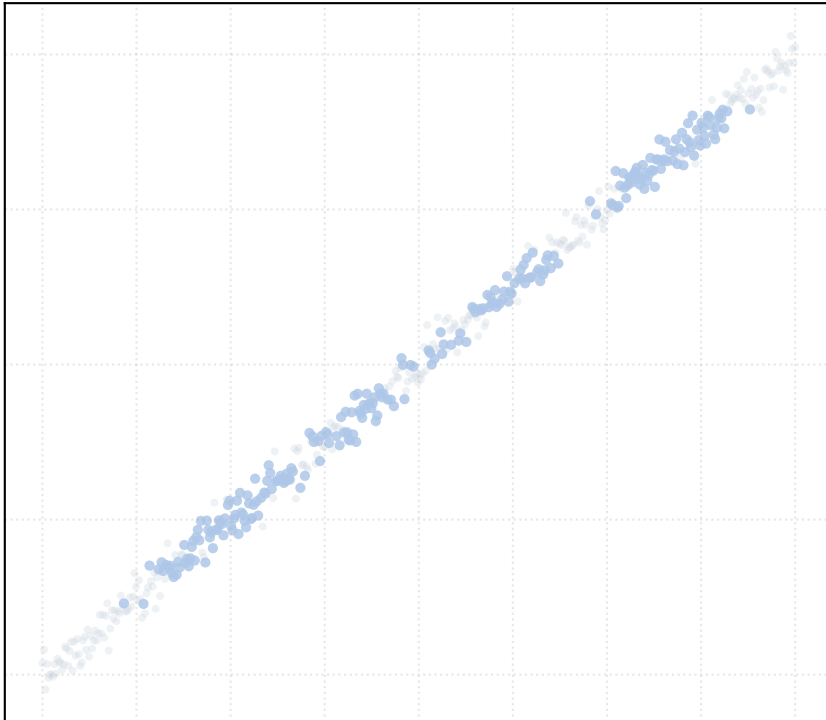
t_percentile = 0.15 (Default) [1 Clusters]



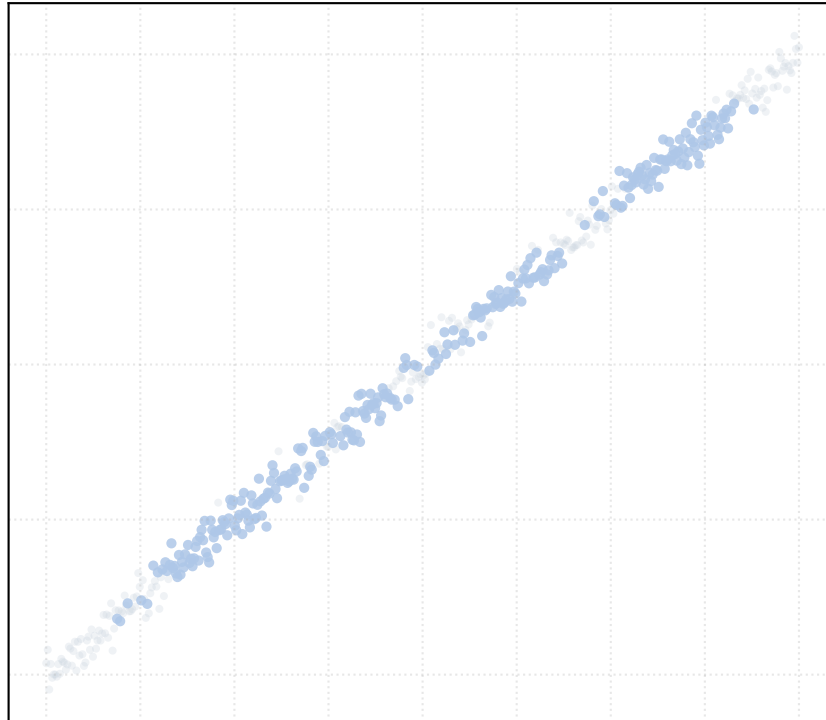
t_percentile = 0.2 [1 Clusters]



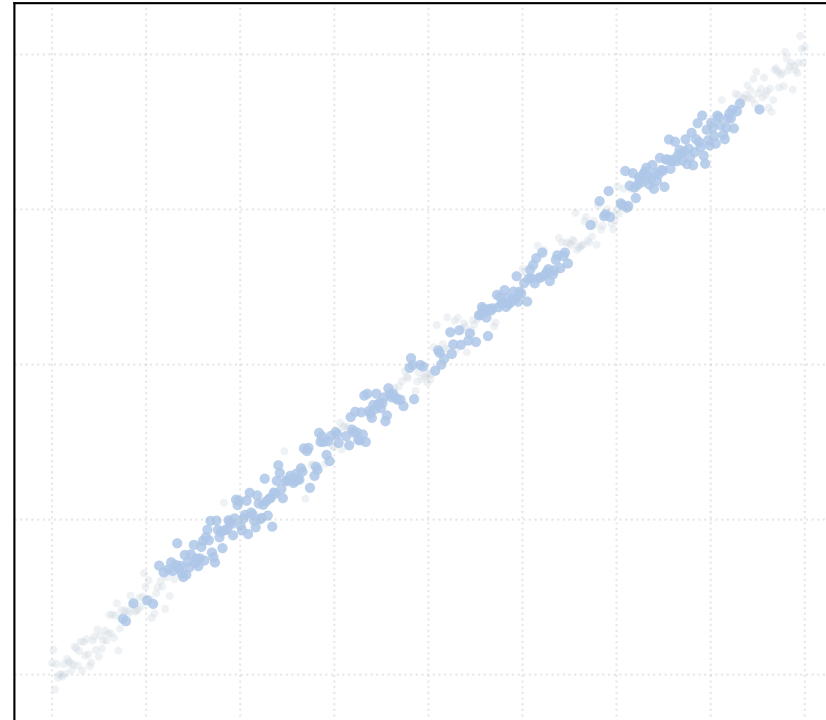
t_percentile = 0.25 [1 Clusters]



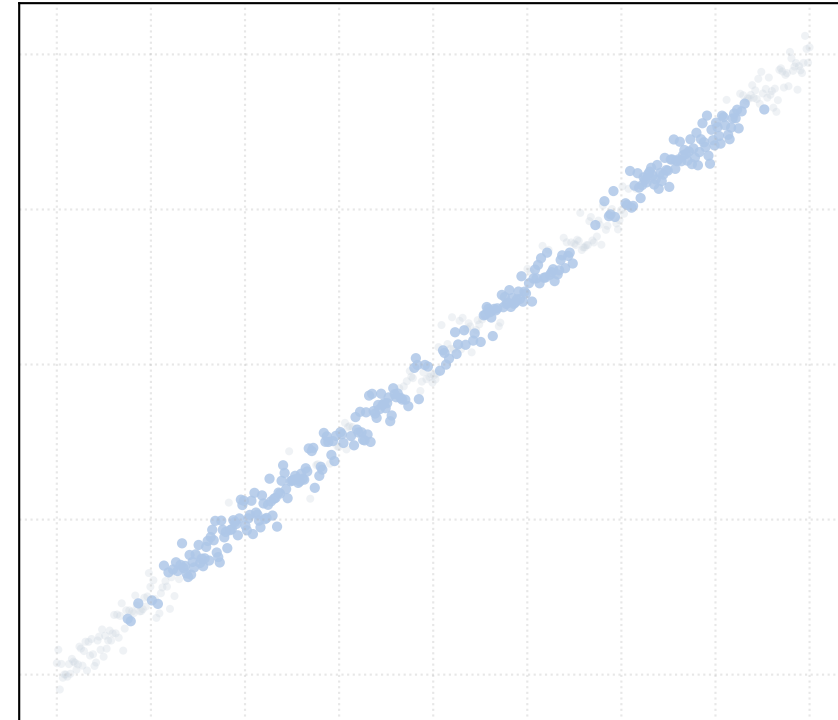
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

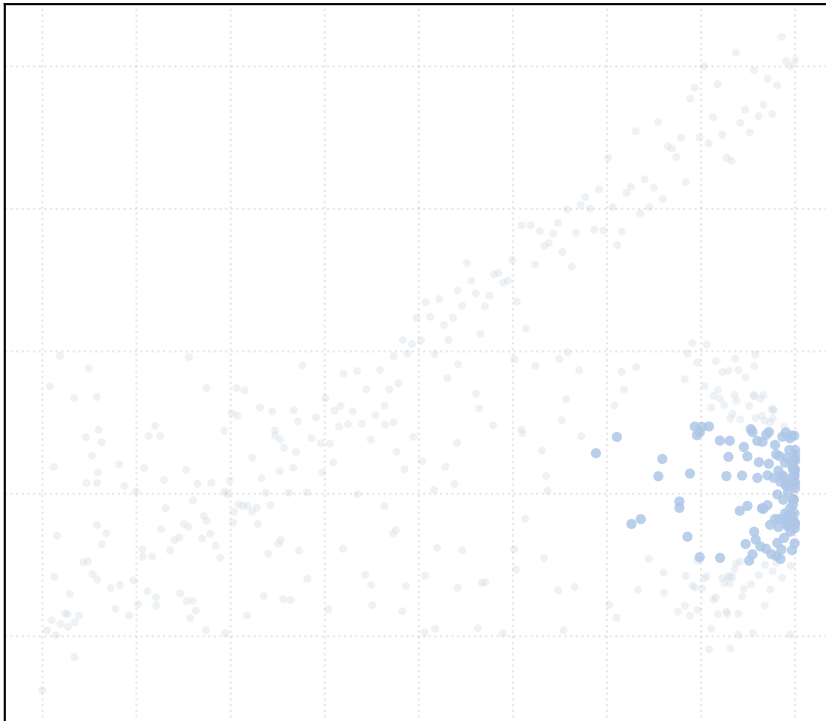


t_percentile = 0.5 [1 Clusters]

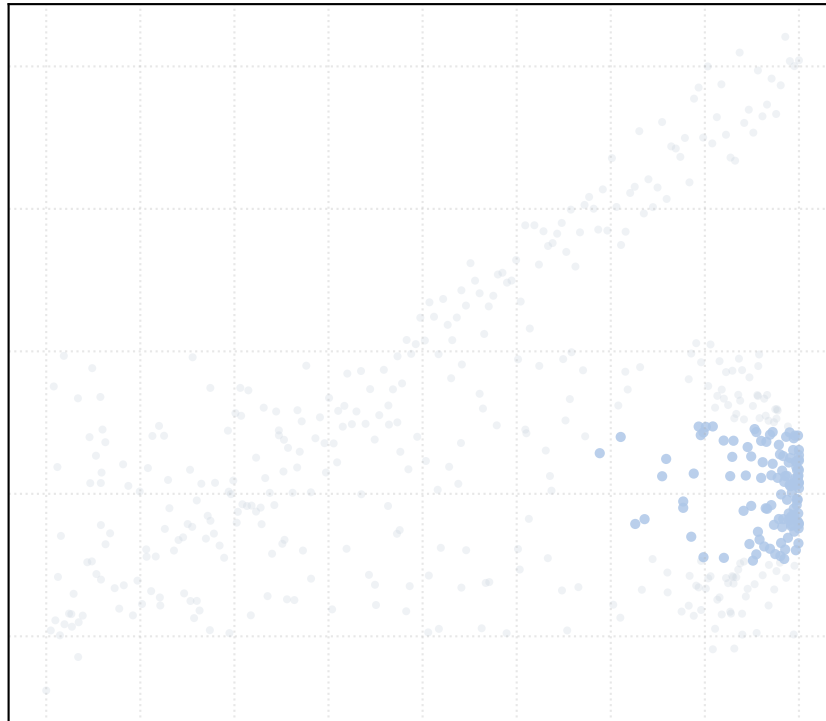


Dataset: mixed_generators_noise_high | Seed Percentile Sweep (t_percentile)

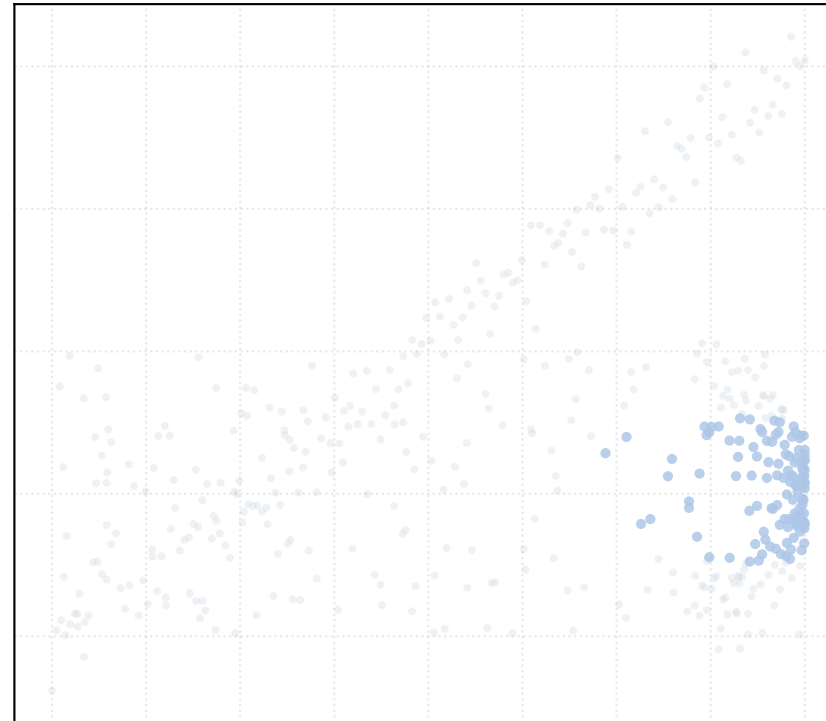
t_percentile = 0.05 [1 Clusters]



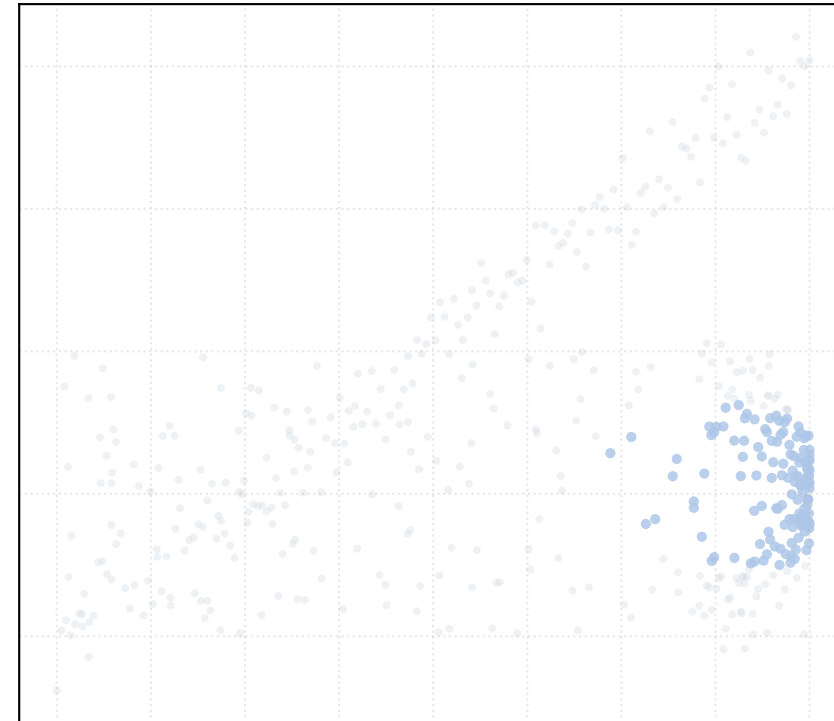
t_percentile = 0.1 [1 Clusters]



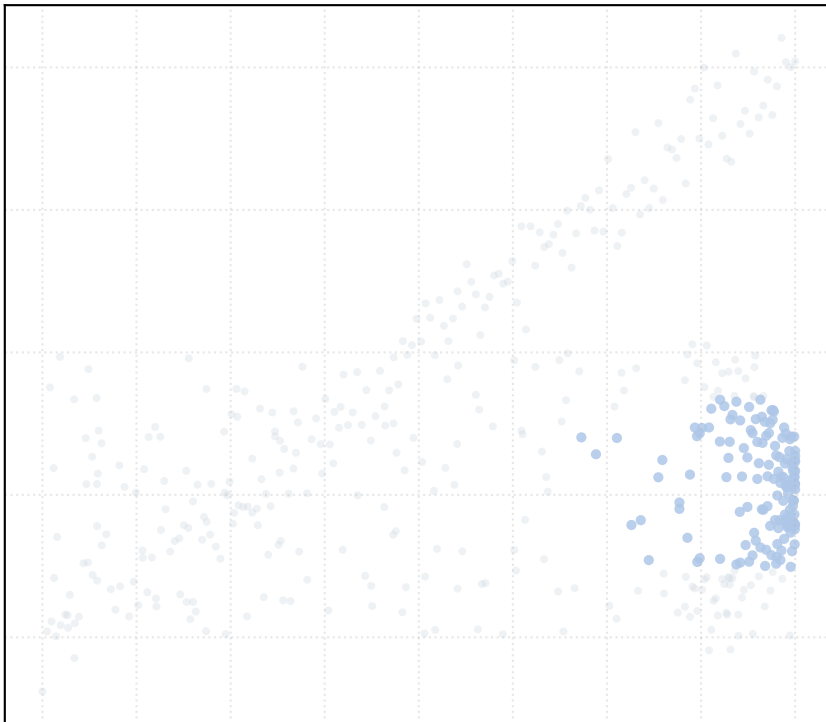
t_percentile = 0.15 (Default) [1 Clusters]



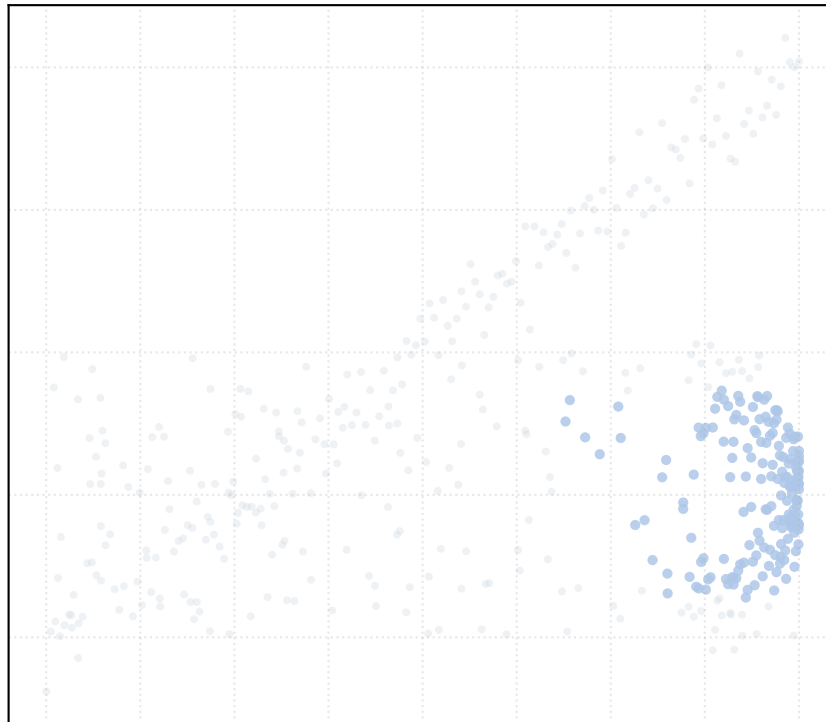
t_percentile = 0.2 [1 Clusters]



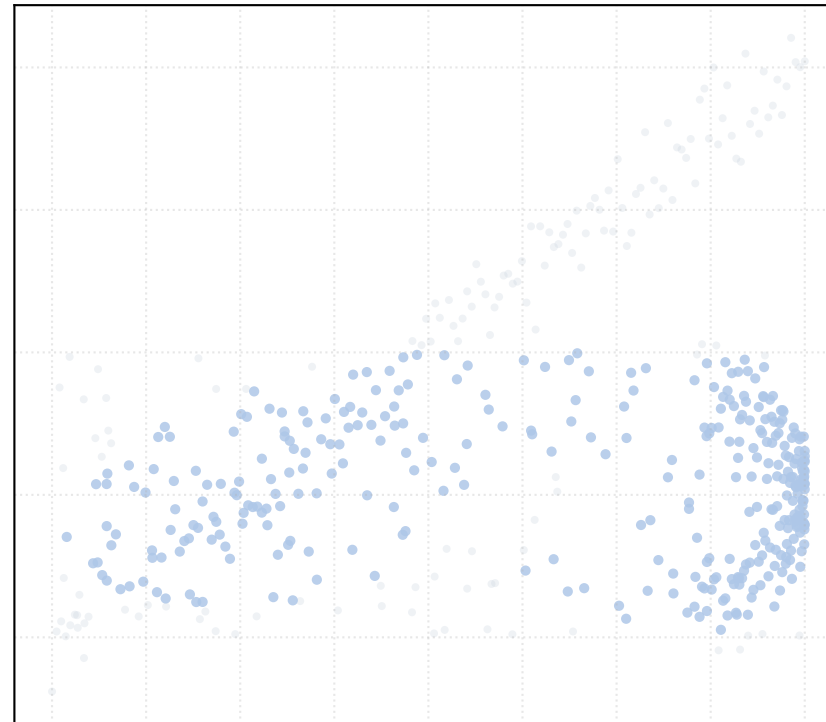
t_percentile = 0.25 [1 Clusters]



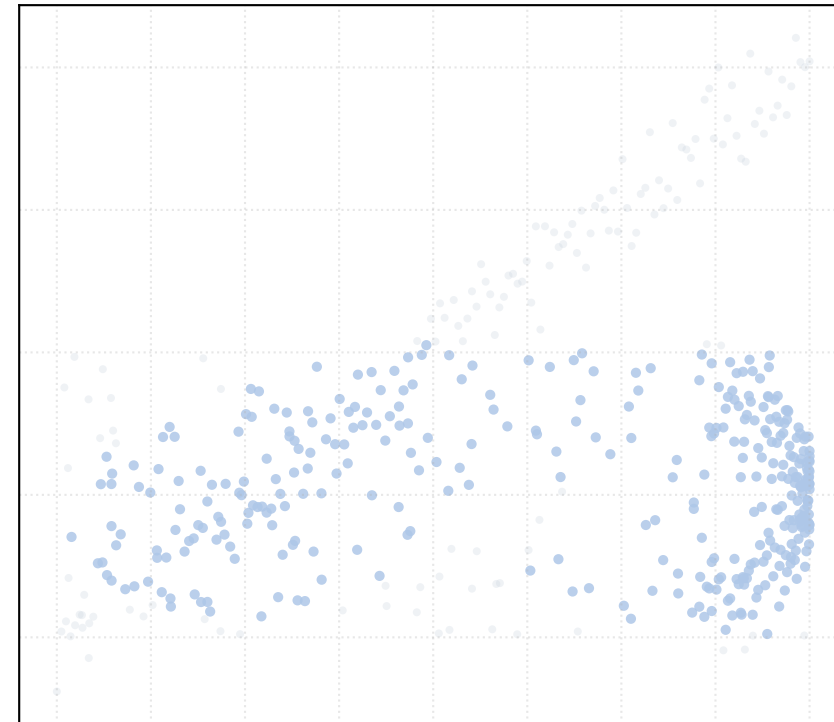
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

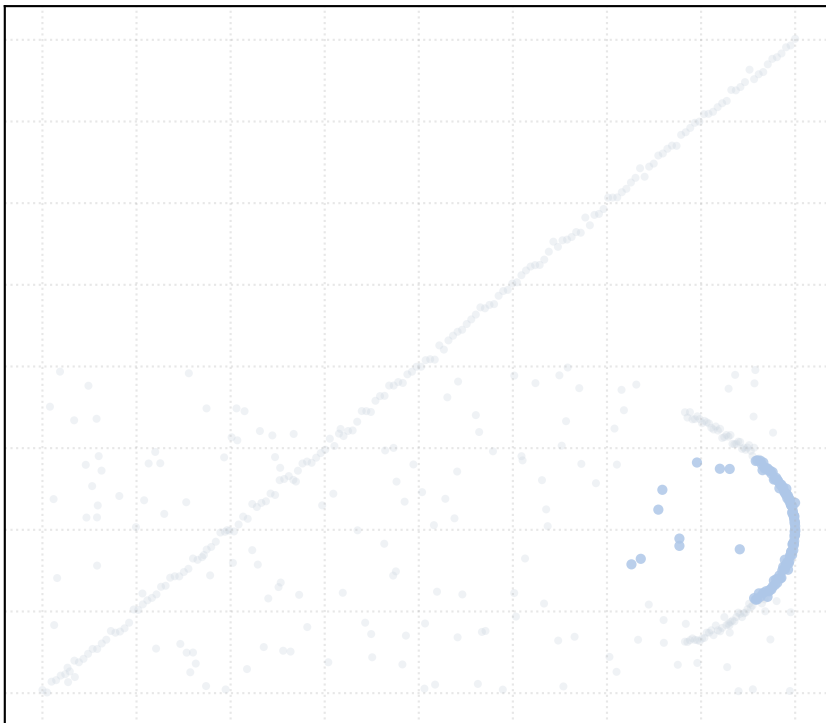


t_percentile = 0.5 [1 Clusters]

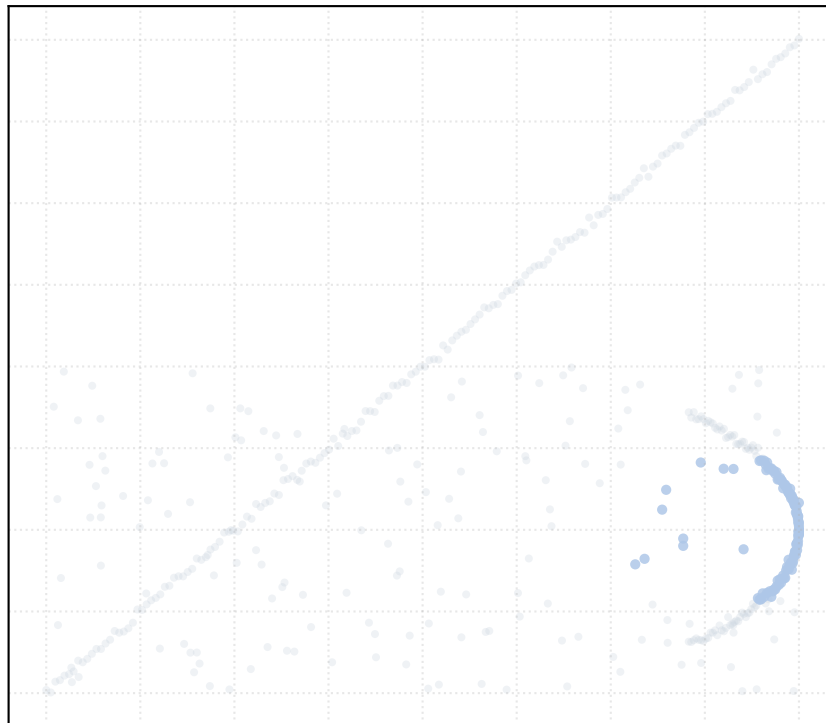


Dataset: mixed_generators_noise_low | Seed Percentile Sweep (t_percentile)

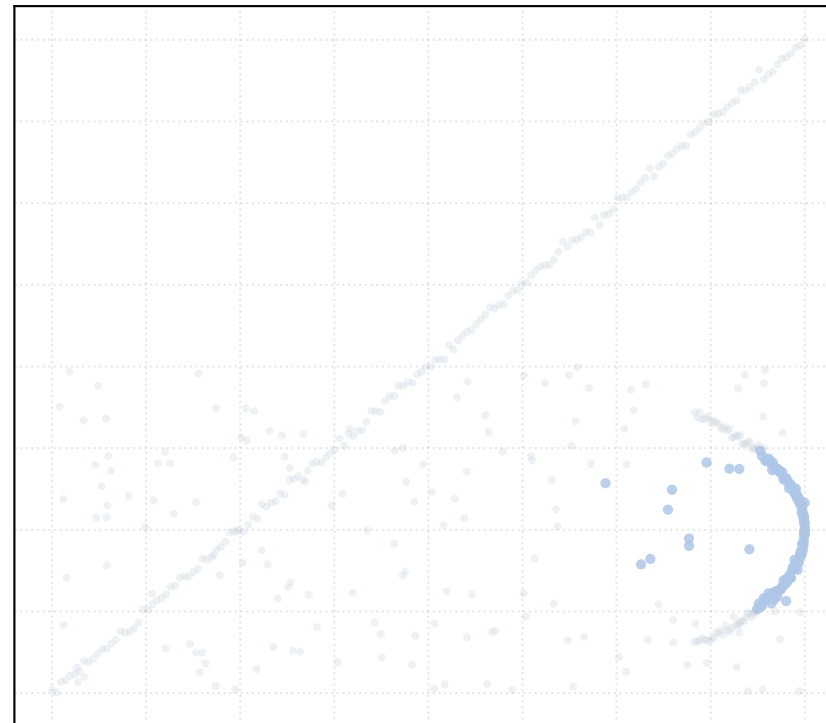
t_percentile = 0.05 [1 Clusters]



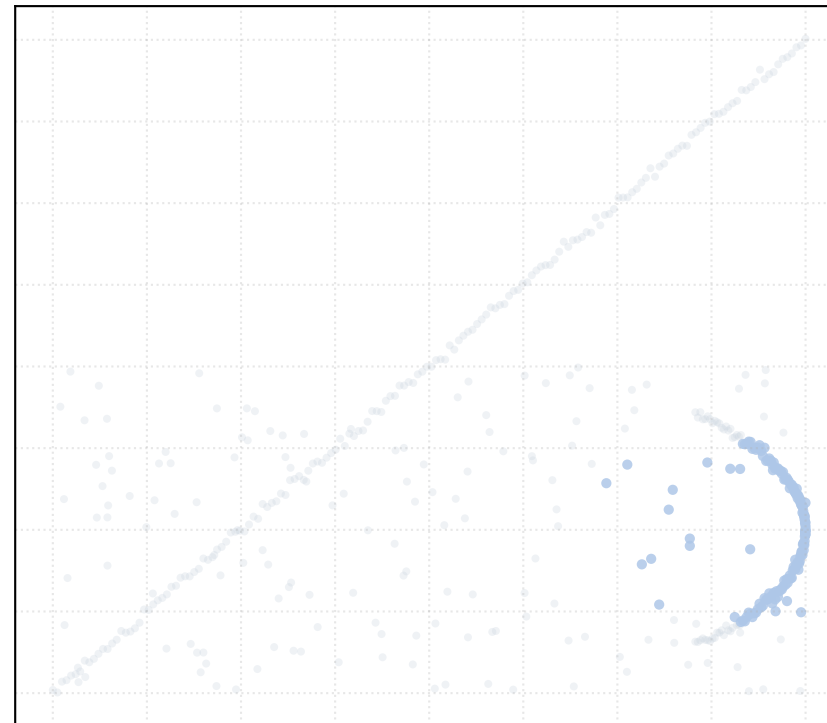
t_percentile = 0.1 [1 Clusters]



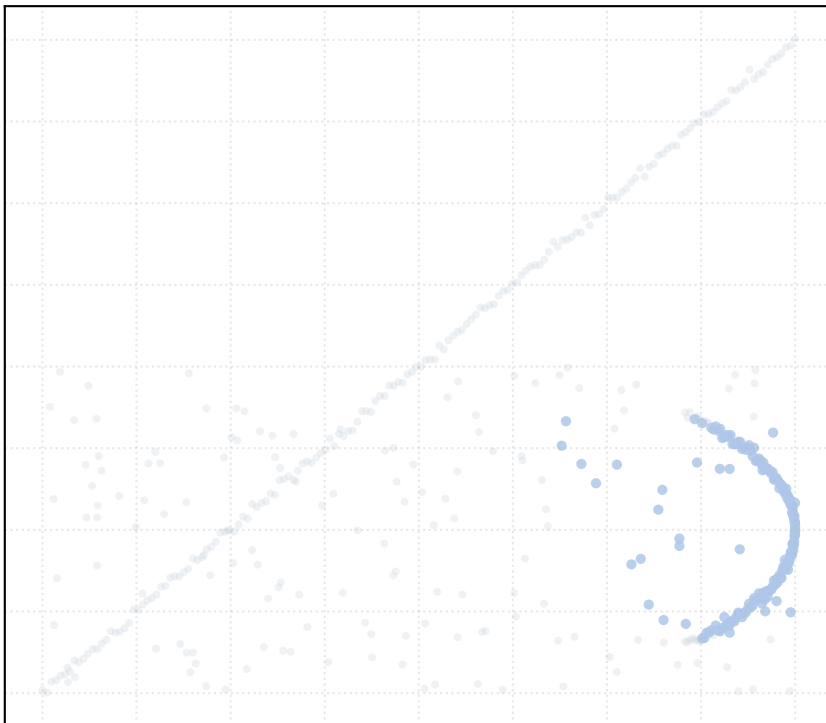
t_percentile = 0.15 (Default) [1 Clusters]



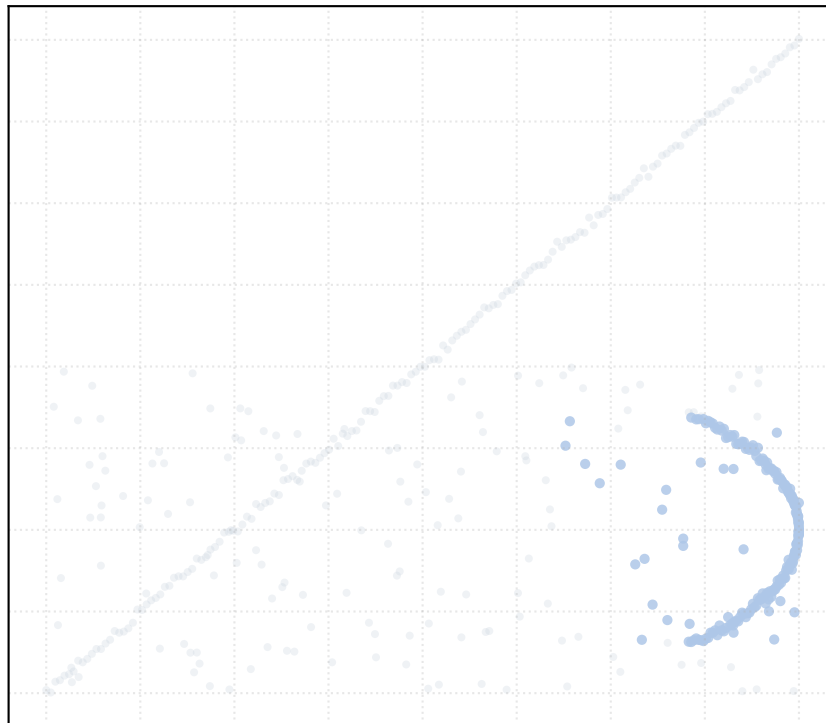
t_percentile = 0.2 [1 Clusters]



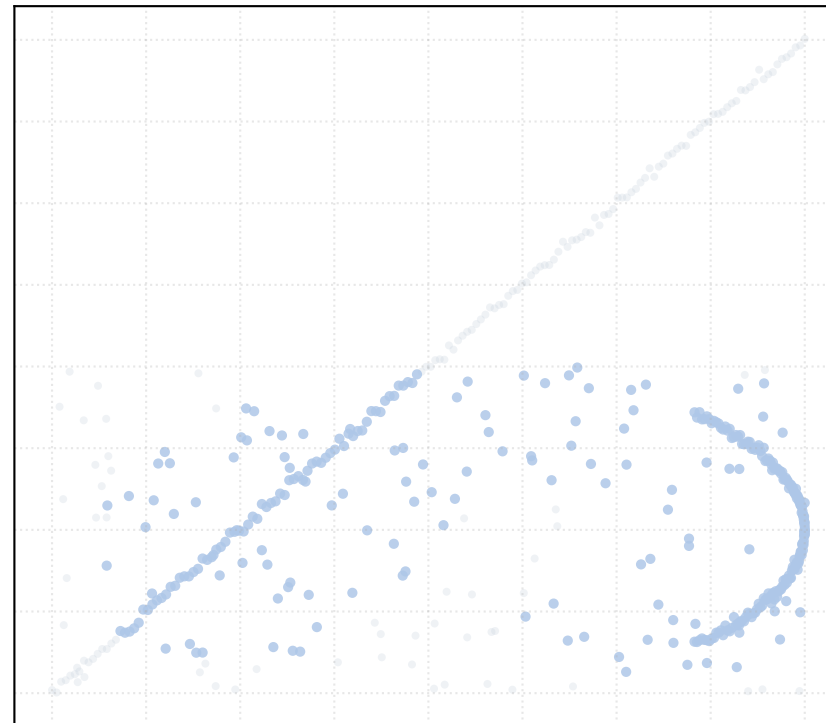
t_percentile = 0.25 [1 Clusters]



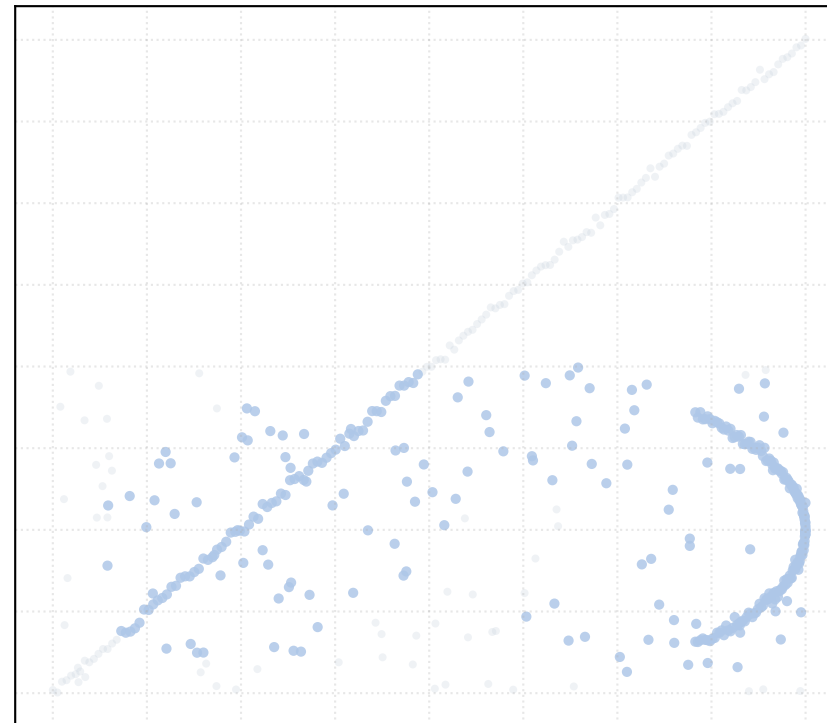
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

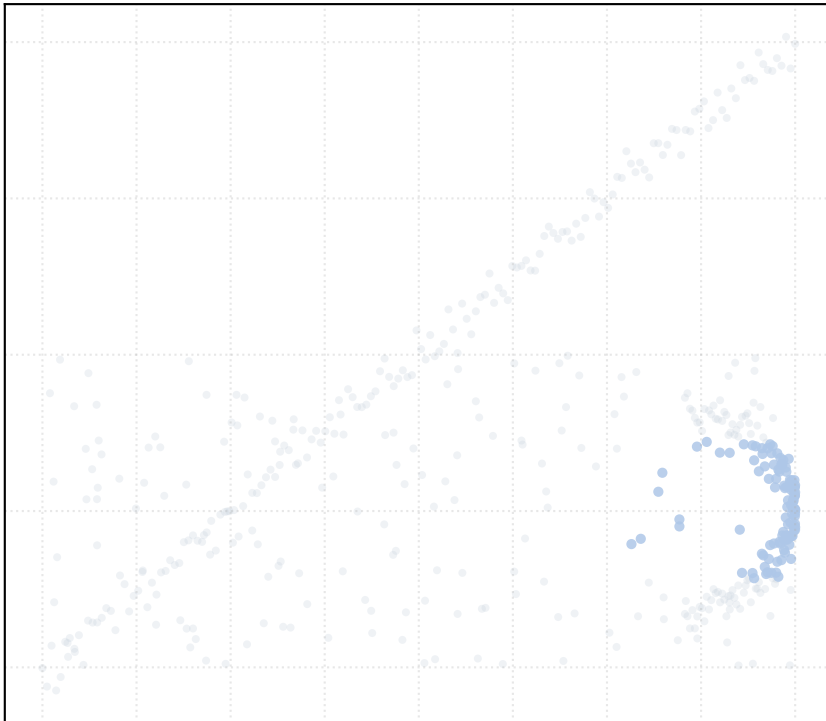


t_percentile = 0.5 [1 Clusters]

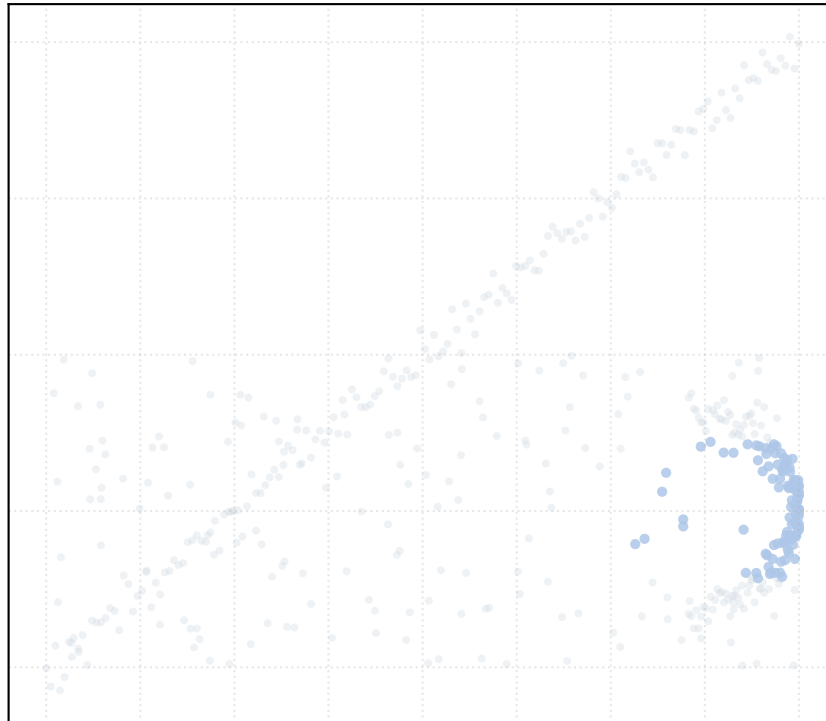


Dataset: mixed_generators_noise_medium | Seed Percentile Sweep (t_percentile)

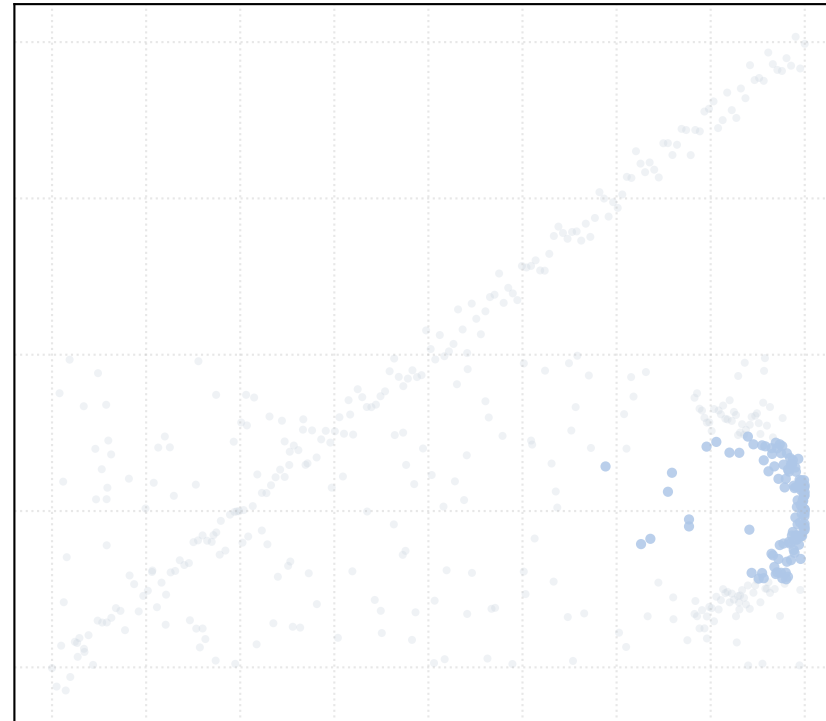
t_percentile = 0.05 [1 Clusters]



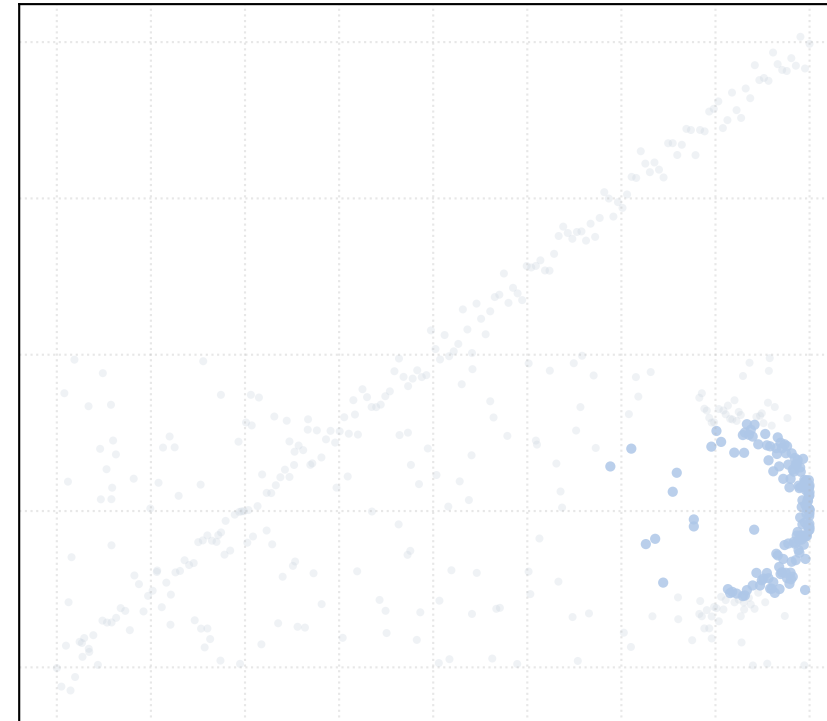
t_percentile = 0.1 [1 Clusters]



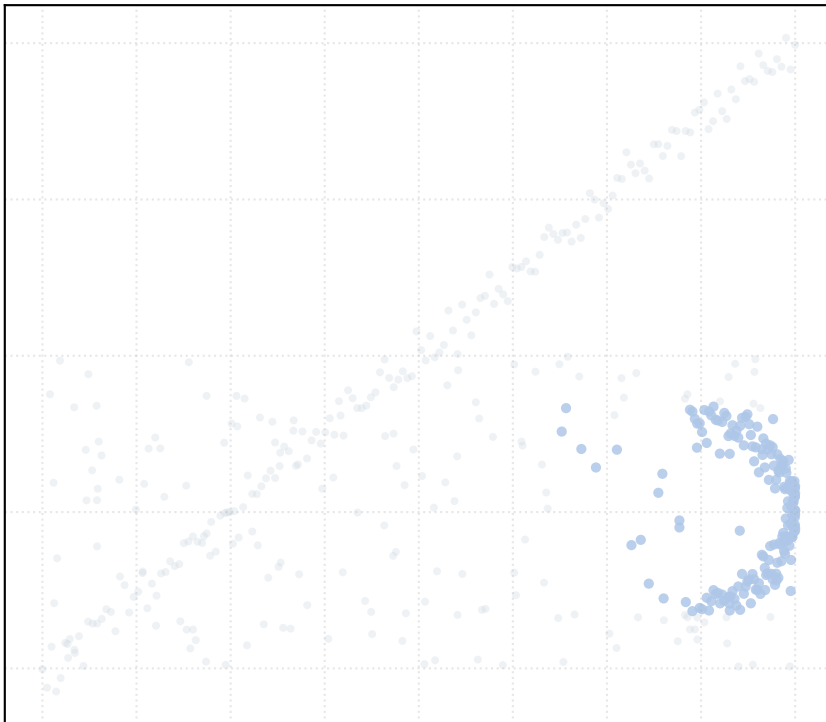
t_percentile = 0.15 (Default) [1 Clusters]



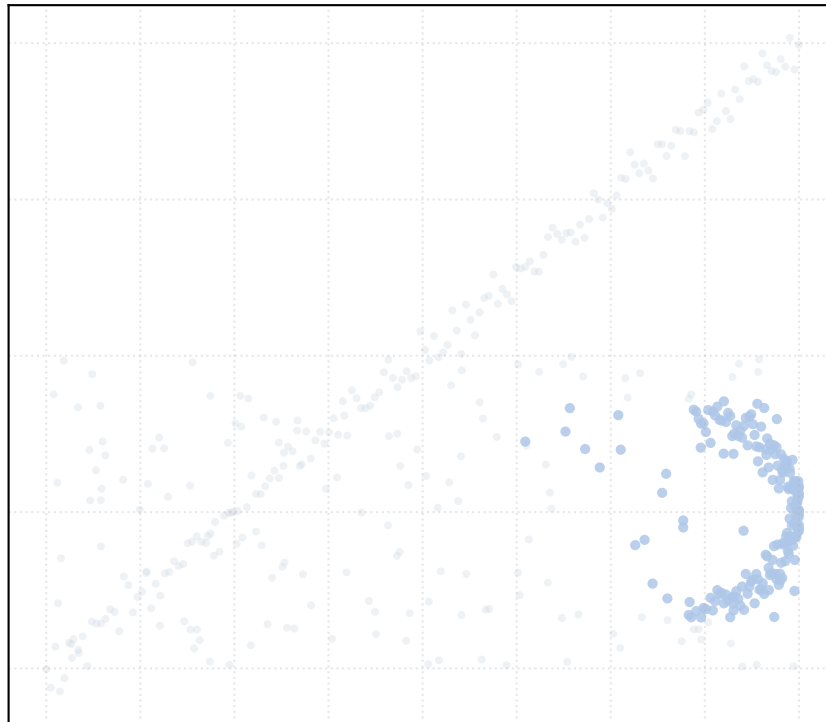
t_percentile = 0.2 [1 Clusters]



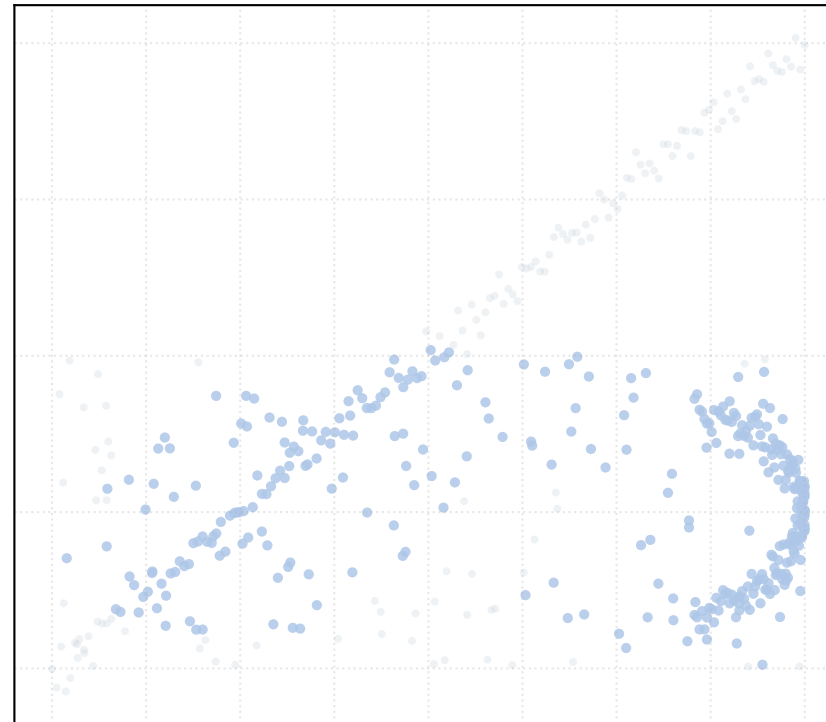
t_percentile = 0.25 [1 Clusters]



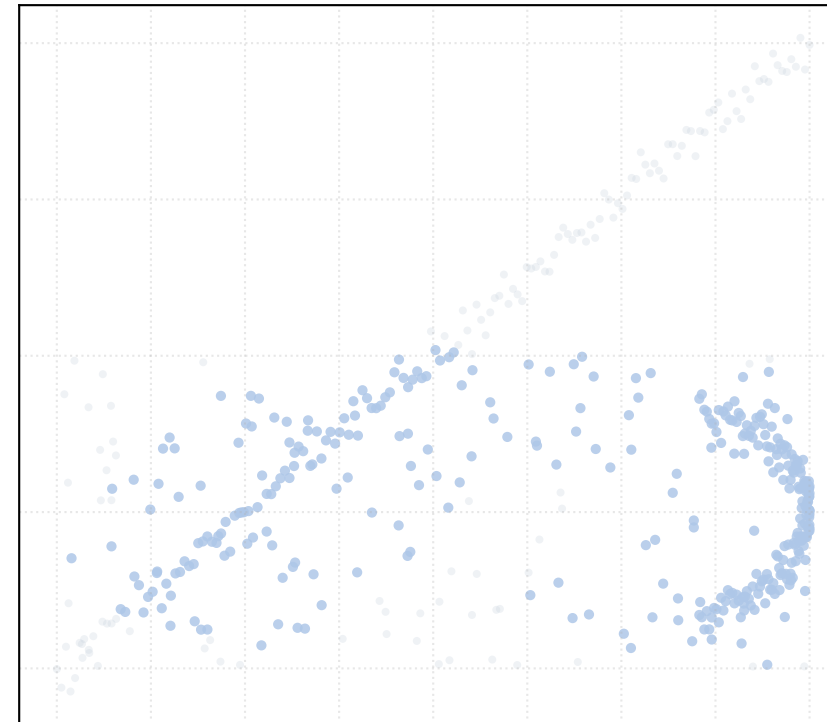
t_percentile = 0.3 [1 Clusters]



t_percentile = 0.4 [1 Clusters]

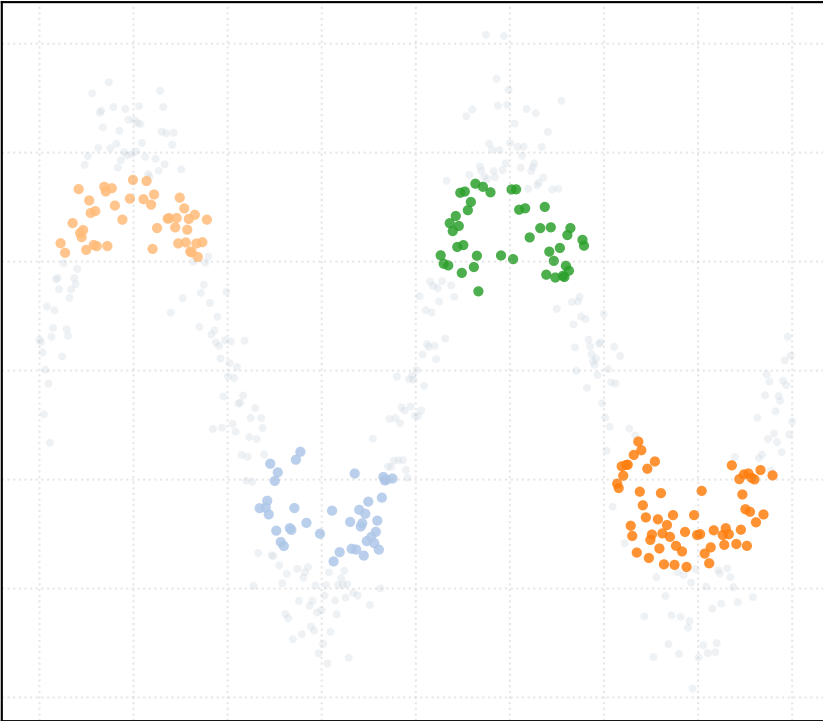


t_percentile = 0.5 [1 Clusters]

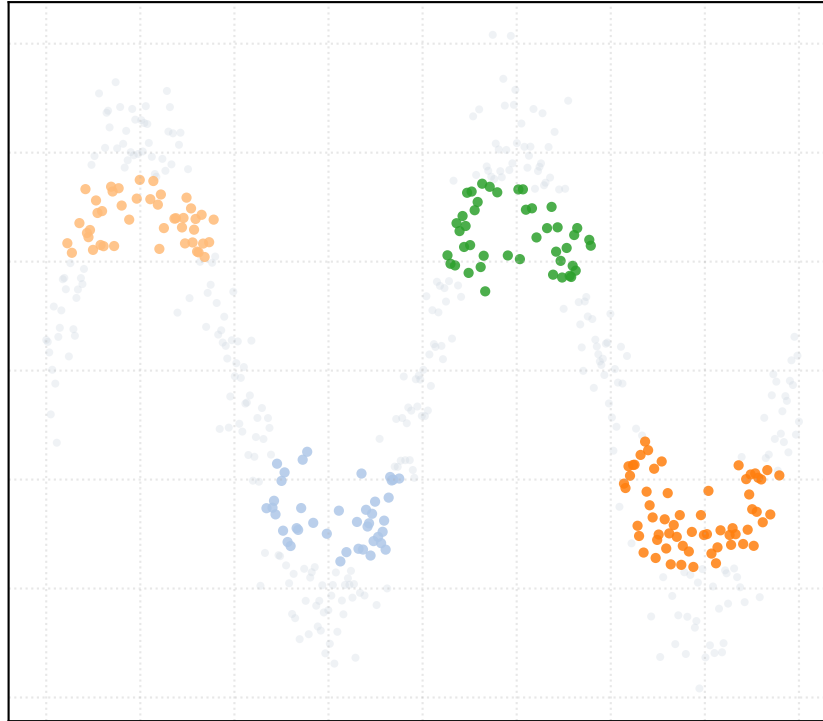


Dataset: s_curve_noise_high | Seed Percentile Sweep (t_percentile)

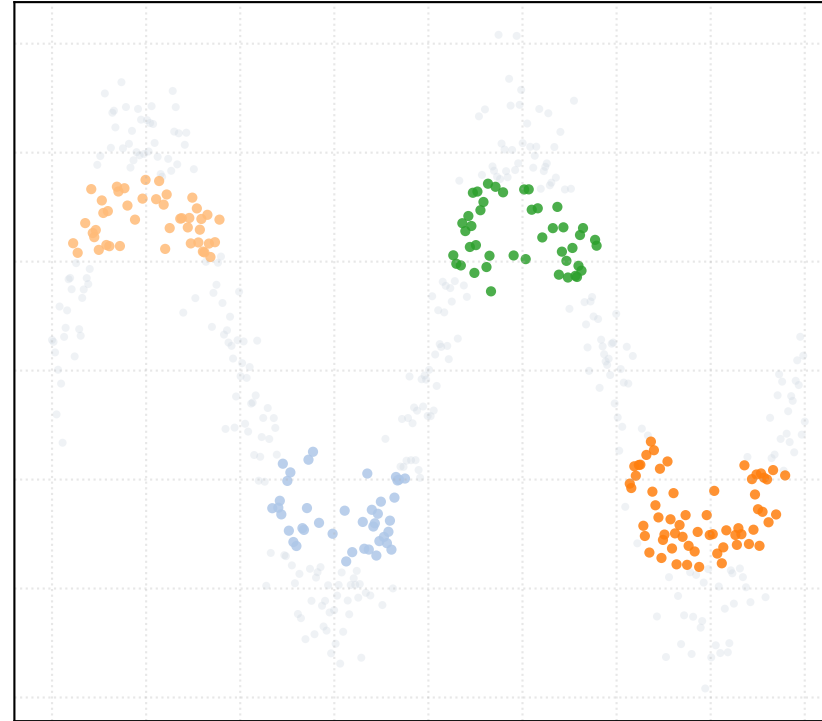
t_percentile = 0.05 [4 Clusters]



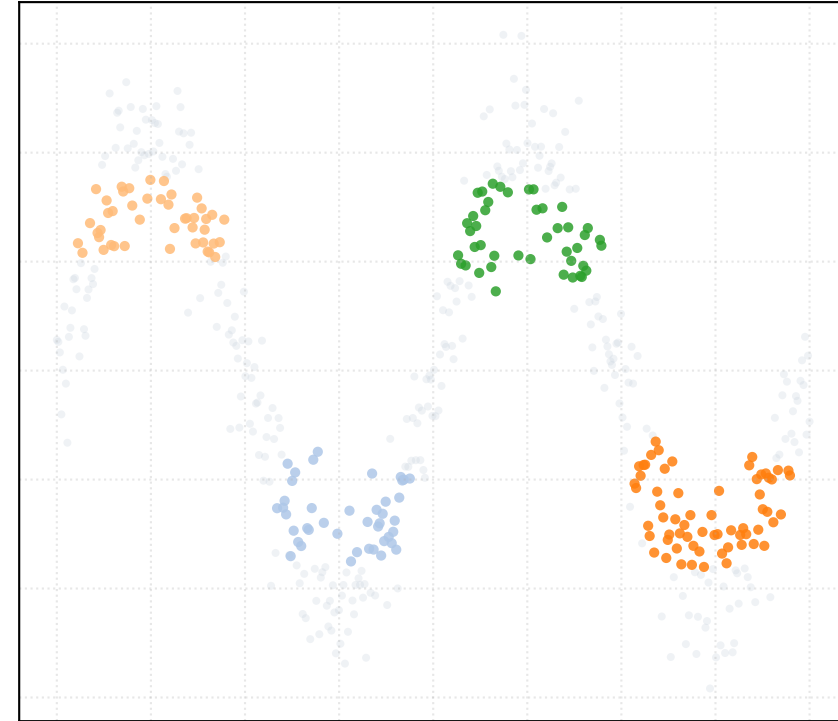
t_percentile = 0.1 [4 Clusters]



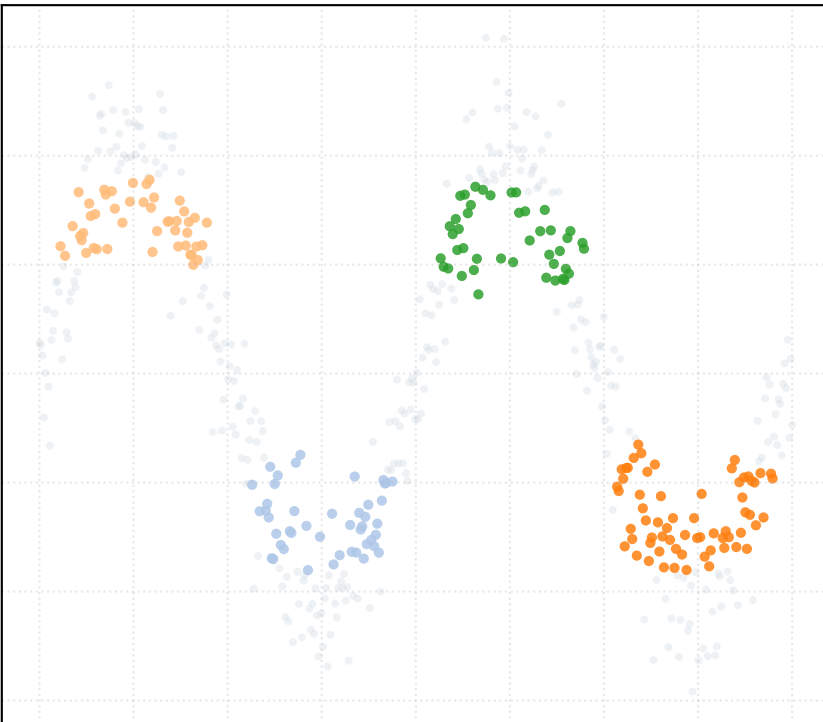
t_percentile = 0.15 (Default) [4 Clusters]



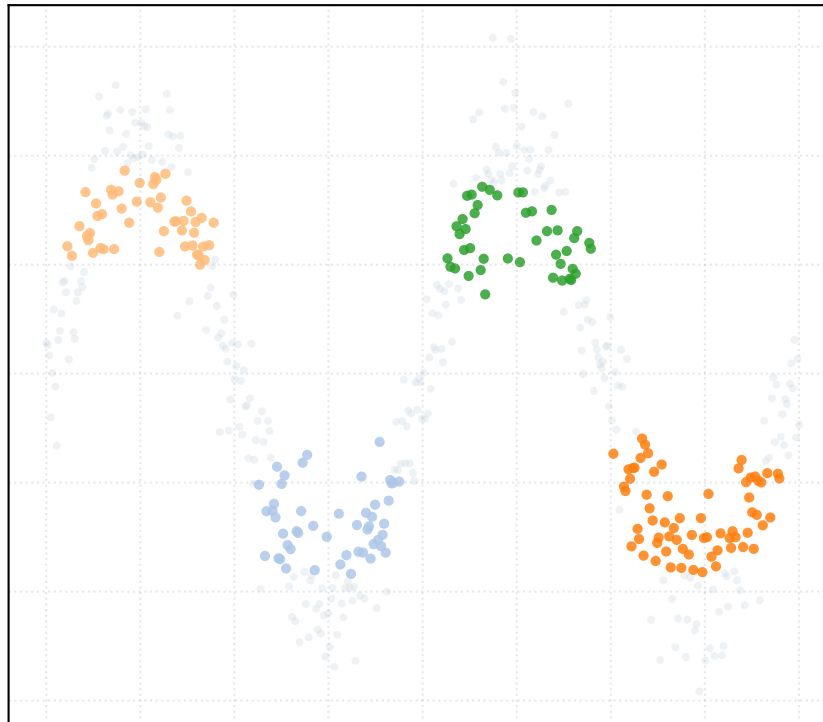
t_percentile = 0.2 [4 Clusters]



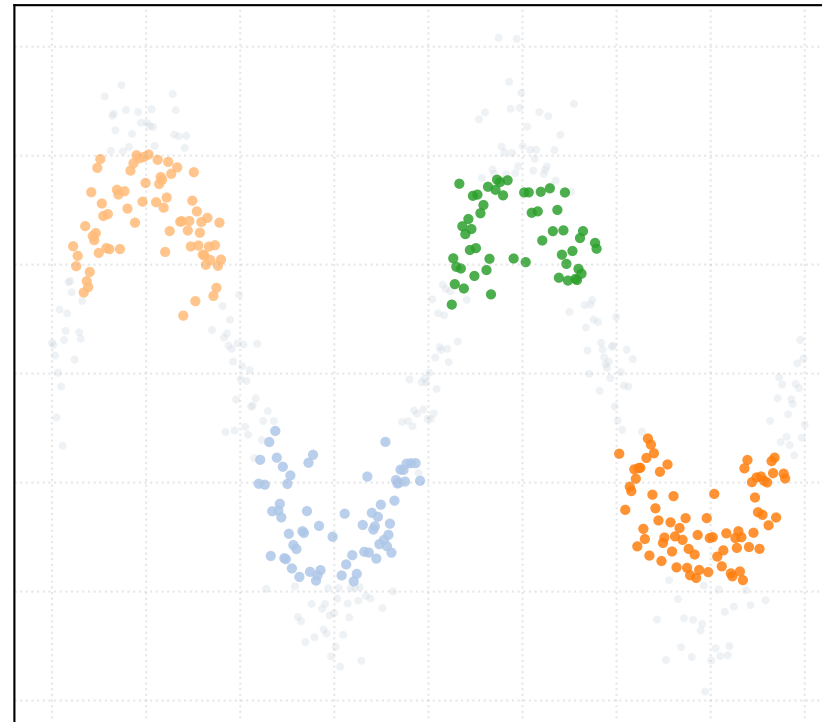
t_percentile = 0.25 [4 Clusters]



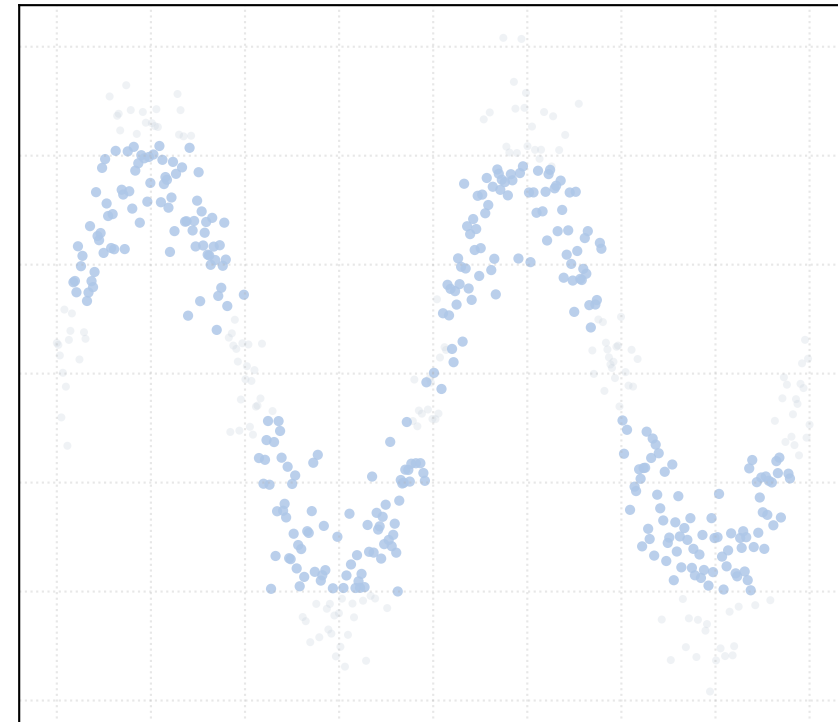
t_percentile = 0.3 [4 Clusters]



t_percentile = 0.4 [4 Clusters]

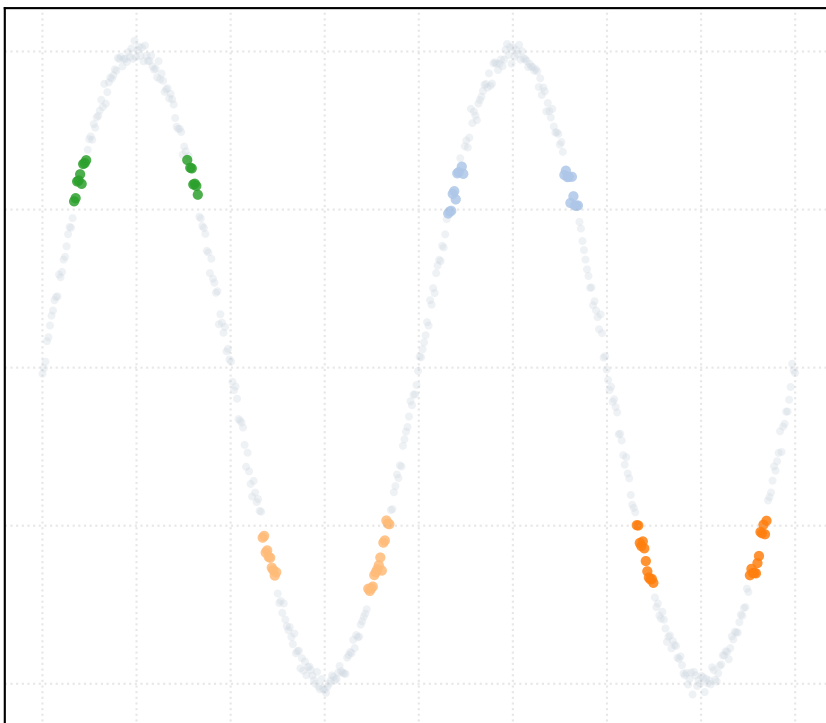


t_percentile = 0.5 [1 Clusters]

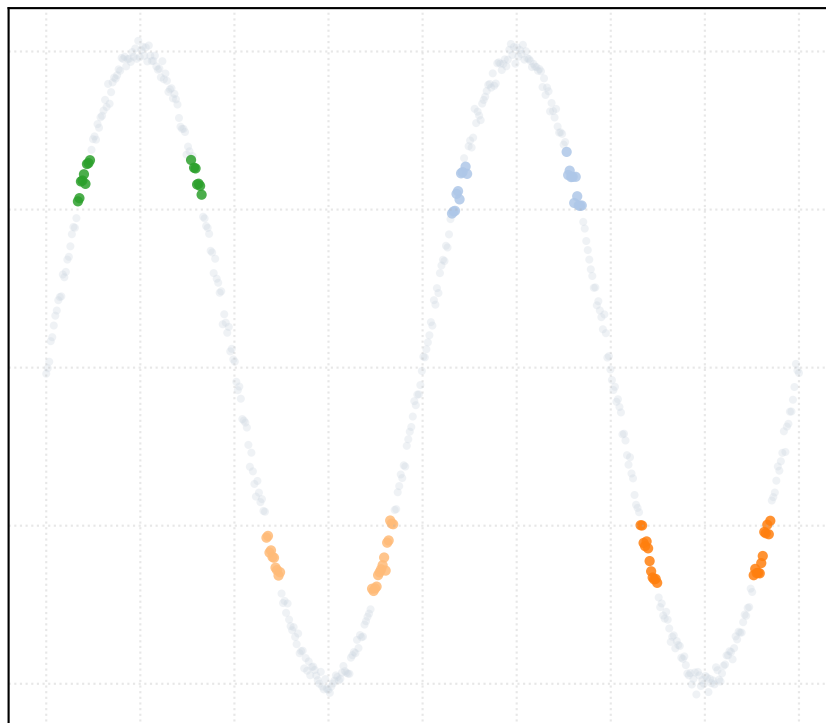


Dataset: s_curve_noise_low | Seed Percentile Sweep (t_percentile)

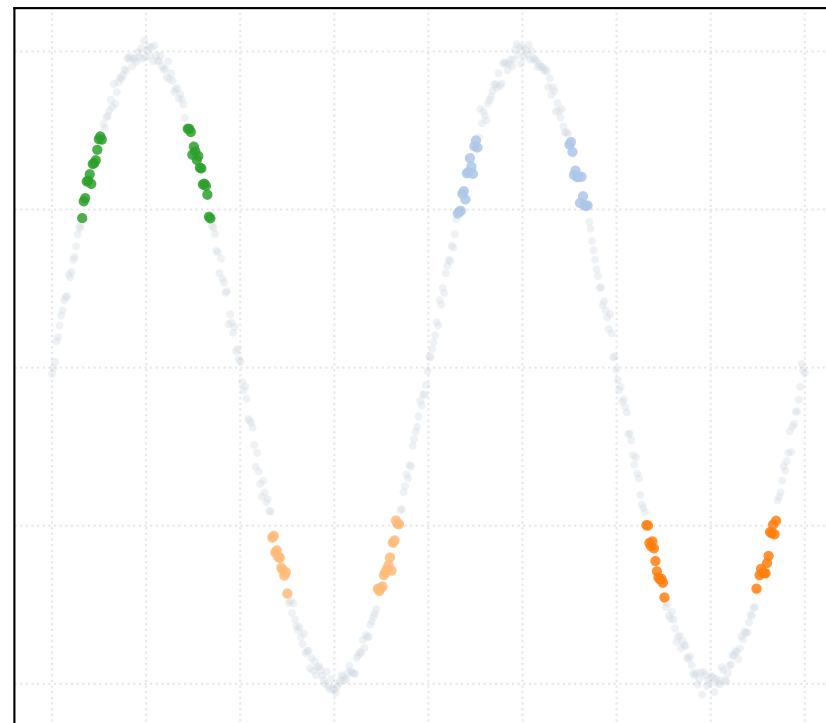
t_percentile = 0.05 [4 Clusters]



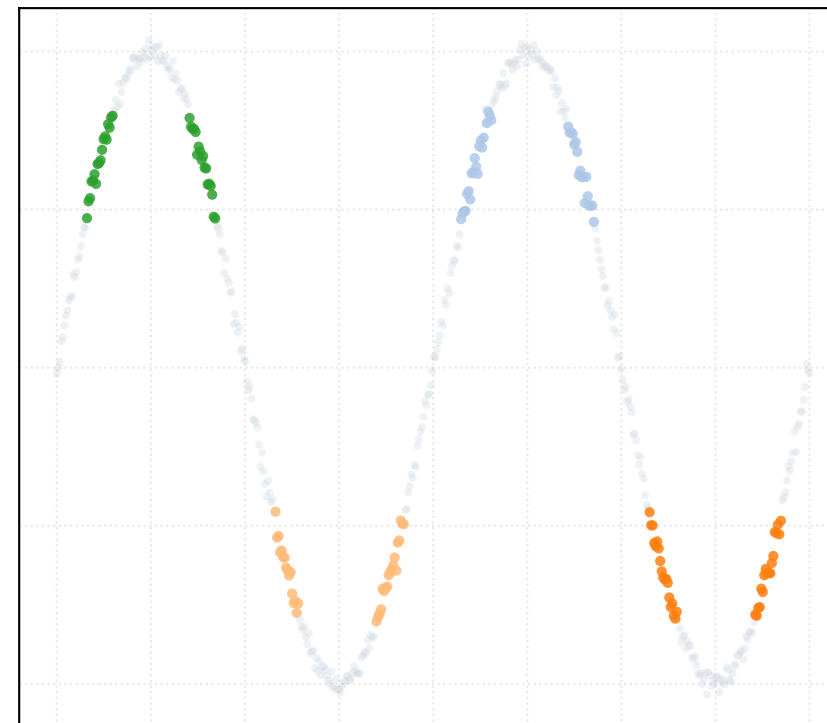
t_percentile = 0.1 [4 Clusters]



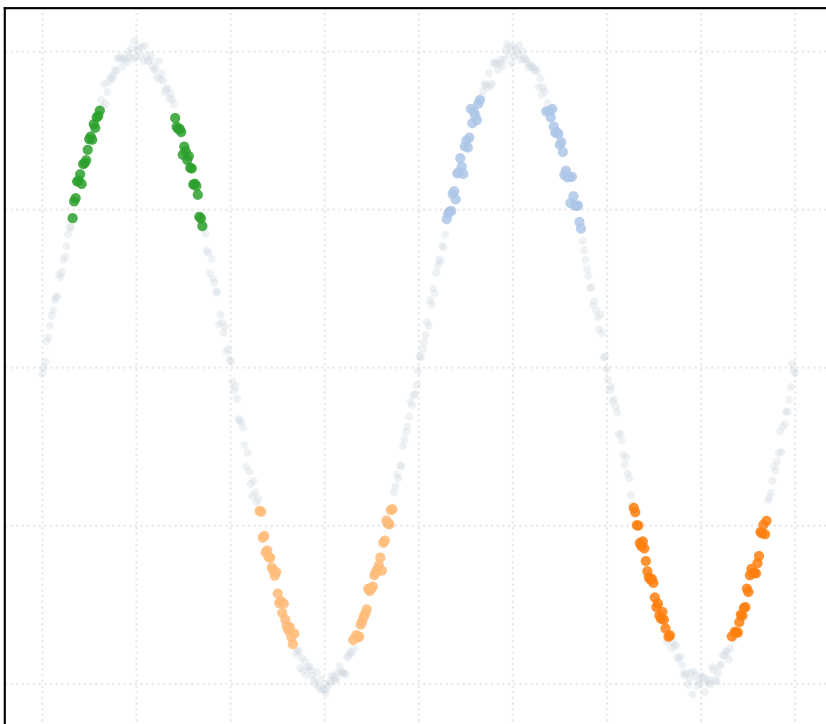
t_percentile = 0.15 (Default) [4 Clusters]



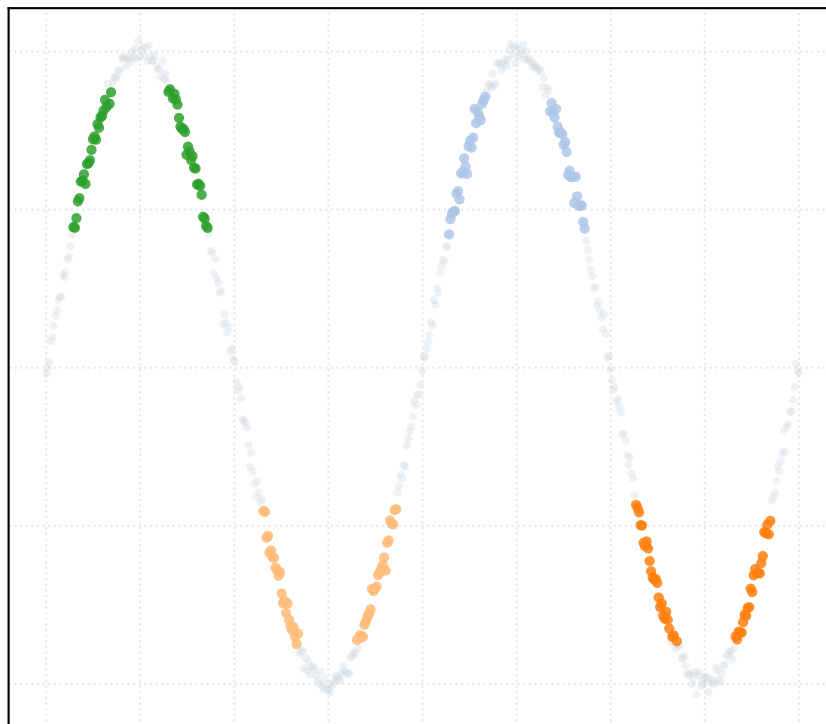
t_percentile = 0.2 [4 Clusters]



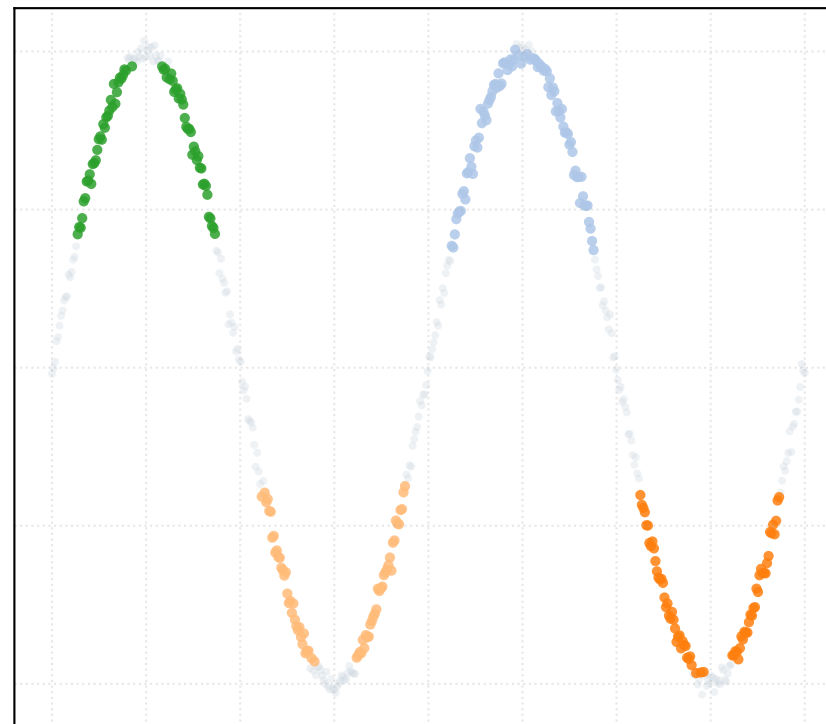
t_percentile = 0.25 [4 Clusters]



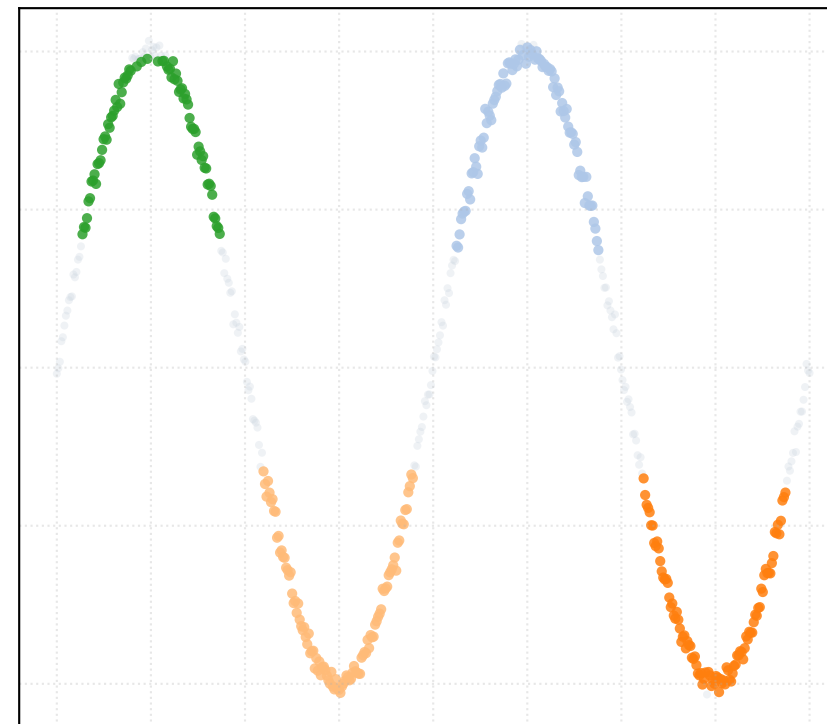
t_percentile = 0.3 [4 Clusters]



t_percentile = 0.4 [4 Clusters]

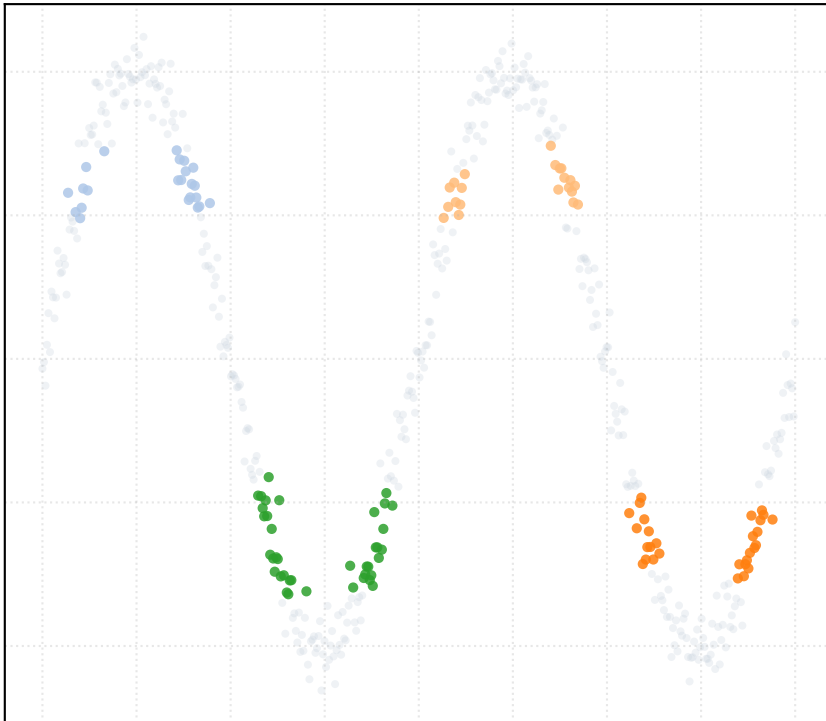


t_percentile = 0.5 [4 Clusters]

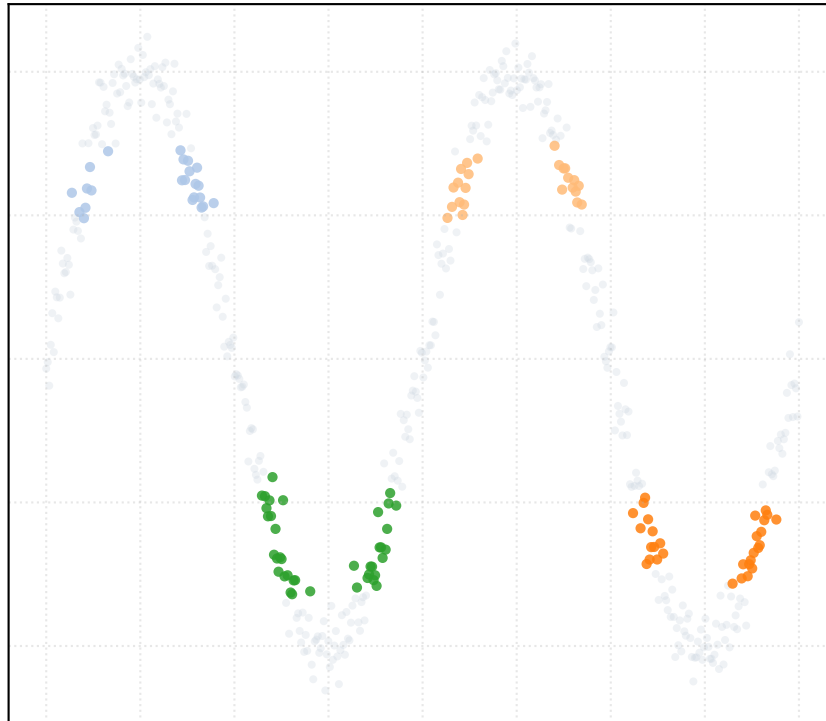


Dataset: s_curve_noise_medium | Seed Percentile Sweep (t_percentile)

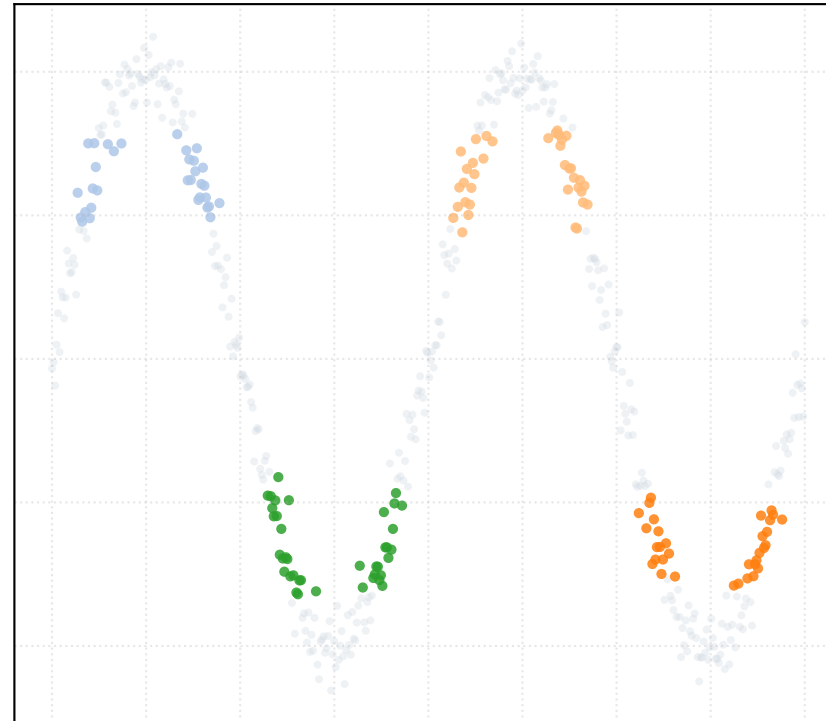
t_percentile = 0.05 [4 Clusters]



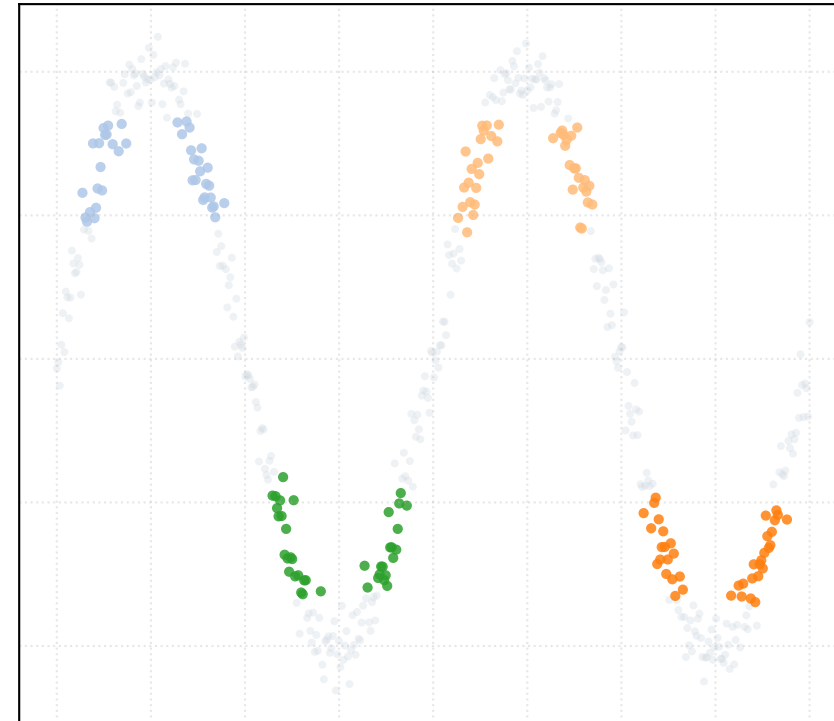
t_percentile = 0.1 [4 Clusters]



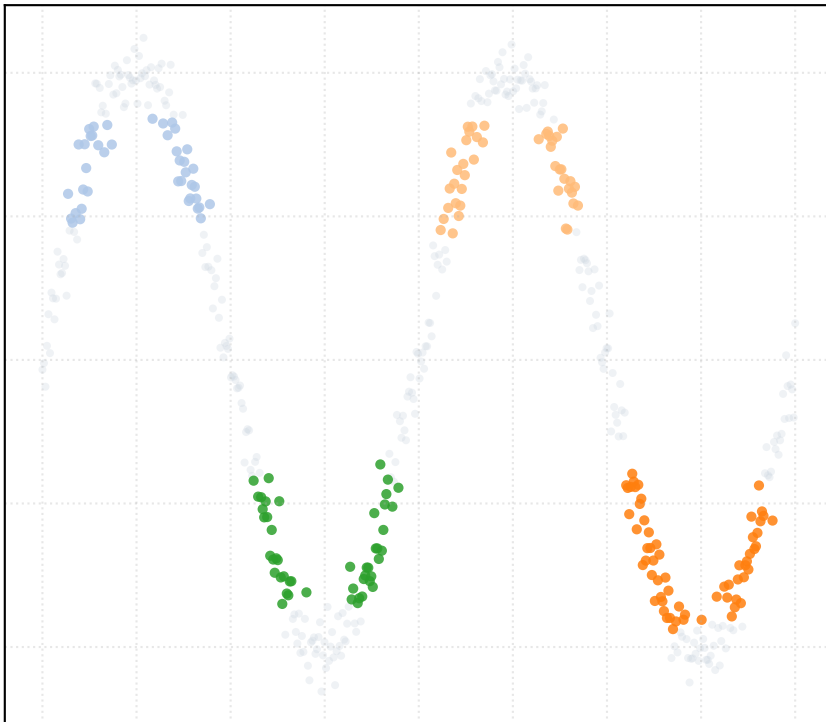
t_percentile = 0.15 (Default) [4 Clusters]



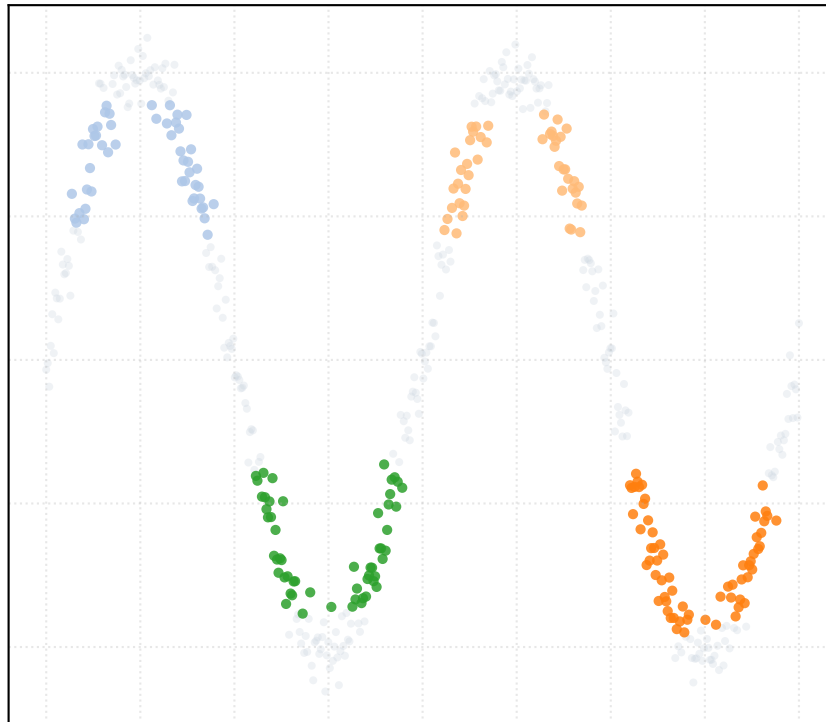
t_percentile = 0.2 [4 Clusters]



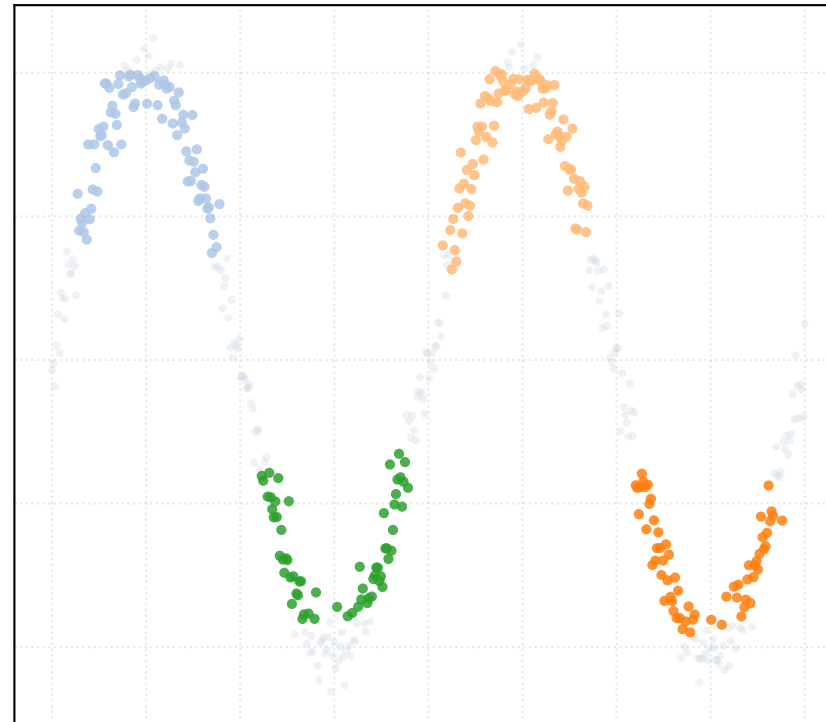
t_percentile = 0.25 [4 Clusters]



t_percentile = 0.3 [4 Clusters]



t_percentile = 0.4 [4 Clusters]



t_percentile = 0.5 [4 Clusters]

